The Social Media Radio Innovation and Its Impact on University Students' Ratings

An Exploratory Study

A thesis submitted for the master's degree in

Media Innovations

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Dedication

To my mother, Safaa Elbordiny, with all my love and gratitude. This thesis is dedicated first to you: to the care you gave without keeping account, the sacrifices I understand more deeply with time, and the faith you placed in me before I knew how to place it in myself. Whatever achievement these pages represent, your love helped make it possible.

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Declaration of Original Work and Generative AI Use

I. Original Scholarship

I hereby declare that this thesis is my own original scholarly work and has not been submitted, in whole or in part, for any other degree or professional qualification. The thesis presents Epistemic Systems Theory (EST), including its participation and evidence-based production proposition (EST-P4), as my proposed synthesis. The field study, reported historical modelling, and prospective computational protocol are distinguished throughout rather than presented as one completed experiment. These contributions were developed within the supervision relationship with Dr. Engi Aboul-Ezz and Dr. Mostafa El Banna. All arguments, analyses, and interpretations have been developed with full academic integrity and with explicit acknowledgement of prior scholarship in accordance with APA citation conventions.

II. Generative Artificial Intelligence: Transparent Disclosure

Generative AI assistants were used for drafting and editing, LaTeX debugging, reference discovery, consistency checks, diagram preparation, reproducibility utilities, and methodological critique. The inherited qualitative analysis also describes AI-assisted semantic coding; this assistance is not equivalent to independent human

coding or participant validation. The researcher remains responsible for checking the original records, sources, interpretations, and final wording. Automated checks do not certify originality, source fidelity, or the authenticity of participant quotations. Materials unavailable in the present archive are identified rather than described as independently verified.

A supplemental FGD analysis export, described by the researcher as generated with a Qwen 7B instruction-tuned model, was also reviewed. Its exact checkpoint, quantisation, prompts, and execution settings were not supplied. The seven output summaries were examined as generated interpretations, not as a recovered primary transcript or independently verified participant quotations. Translation inconsistencies and unsupported inferences were retained in an adjudication record rather than adopted as empirical findings.

The planned large-language-model agent experiment is a separate research method, not evidence from human participants and not a completed study. Its model versions, prompts, tools, and execution logs must be disclosed when it is conducted. No model-generated response is represented as a measured student response. This disclosure follows the transparency principle discussed by Branda (2026) in *Use AI, but don't mask it.* Responsibility for the submitted thesis and its declarations remains with the researcher.

Signature: _____ Date: _____

Sherif Mohamed Abd Elmonem Rashwan

Abstract

This thesis asks whether the design of social-media radio can improve the epistemic quality of university students' listening—whether they become better informed and more discriminating—without destroying engagement. It develops Epistemic Systems Theory (EST), which treats platform architecture, network diversity, measurement accountability, and participation as linked influences on audience judgment. LOKLOK, a proposed social-media radio platform, operationalises these ideas through broadcaster reputation scoring, diversity-constrained recommendation, moderation, and a measurement surface.

The empirical core is 52 retained questionnaires from 60 records collected from university Uber riders in October 6th City, Giza. BRS/PB reliability is high ($\alpha = .957/.947$), stars correlate with PB ($\rho = .671$), but poor CFA fit prevents a claim of structural validity. The original 160-segment FGD transcript and 185-segment self-report supply qualitative context; four corpora and 25 briefs supply content context.

Because a real platform trial was not feasible, the thesis builds a calibrated in-silico experiment: BGE-M3 document embeddings, DML/IPW debiasing, a six-axis manifold, 500 synthetic personas, and a 2^3 ABM with 18 sessions. Calibration means the status-quo arm G0 is anchored to the measured PB mean (5.119); the state-outcome weights remain declared assumptions. G1 reaches 5.946. The implied contrast is +0.827 evaluative points [0.779, 0.875] and +11.30 competence points

[10.64, 11.95], with an engagement cost of −0.183 [−0.188, −0.178]. A 256-point joint sweep is positive in 96.5% of draws. Ablations show that only DeGroot, Lagrange, and Hilbert are outcome-active, the quantitative block carries the contrast, and semantic features do not improve held-out prediction. A 60-record sensitivity and an anchor bootstrap are stable.

These are model-based implications, not human treatment effects. The sample is a convenience sample of Uber riders, the semantic streams are contextual, and no invariance or field causal test is claimed. The contribution is a transparent bridge from EST to a testable design, plus a concrete agenda for a real LOKLOK deployment.

Keywords: social media radio; Epistemic Systems Theory; misinformation governance; Broadcaster Reputation Score; ecological momentary assessment; psychometric validation; agent-based modelling; counterfactual experiment.

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المستخلص

تقدم هذه الرسالة **نظرية النظم المعرفية (EST)** إطارًا لدراسة العلاقة بين معمارية الراديو عبر وسائل التواصل الاجتماعي، وتقييم الجمهور، والمساءلة القياسية. وتتضمن أربع قضايا نظرية تتعلق بالتأثير المعماري، والديناميكيات الشبكية المشروطة بالتنوع، والمساءلة السيكومترية، والمشاركة والإنتاج القائم على الأدلة. وتُقدَّم الرياضيات هنا بوصفها وسيلة لتحديد الفروض وخيارات التصميم، لا بوصفها قوانين فيزيائية ثابتة.

اعتمدت الدراسة الميدانية على 52 استبيانًا صالحًا من أصل 60 استجابة جُمعت من طلاب جامعات مدينة السادس من أكتوبر أثناء رحلاتهم عبر تطبيق أوبر. وأظهرت المقاييس ثباتًا داخليًا مرتفعًا (ألفا كرونباخ للمقياس الكلي = 0.957، وللمقياس البعدي = 0.947)، وارتباطًا بين تقييم النجوم والمقياس البعدي (ρ = 0.671)؛ غير أن ضعف مطابقة النموذج العاملي يمنع اعتبار البنية العاملية محققة نهائيًا. وتُستخدم هذه الموجة الميدانية بوصفها مرساة تجريبية ومعايرة للنموذج، لا بوصفها اختبارًا سببيًا.

طُوِّرت منصة LOKLOK بوصفها الركيزة التشغيلية للنظرية، ونُفِّذت تجربة حاسوبية معايرة على 500 شخصية اصطناعية مولدة عشوائيًا من متعدد شعب مُنقَّى إحصائيًا، مع تصحيح التحيز عبر التعلم الآلي المزدوج وأوزان الاحتمال العكسي، ثم إسقاط طيفي (PCA + LLE + Laplacian Eigenmaps). وتم تخصيص كل شخصية عبر تصميم عاملي كامل $2^3 = 8$ خلايا، يضم ثمانية مرشحات تأثير رياضية (Hamiltonian, Laplace, Lyapunov, DeGroot, Lagrange, Hilbert, Percolation, Entropy) وبوابة انتباه مستوحاة من تجارب المضخة والمجس للحائز على نوبل أحمد زويل. وأظهرت المقارنة بين التكوين المعرفي أولًا (G1) والتكوين التفاعلي أولًا (G0) ارتفاعًا في تقييم الطلاب التقييمي بمقدار 1.18+ نقطة (فاصل 95%: 1.16 إلى 1.20)، وفي كفاءة القراءة الإعلامية الاصطناعية بمقدار 19.79+ نقطة، مقابل انخفاض في التفاعل الخام بمقدار 0.314-، وهو ما يتسق مع مقايضة التقييم المعرفي مقابل التفاعل. كما خفّض التكوين المعرفي أولًا التعرض للأخبار المضللة بنسبة 83.6% إلى 84.0% في اختبارات الضغط العدائية، وصمدت النتائج عبر 64 نقطة في تصميم Latin Hypercube وبواسطة مؤشرات سوبول.

تُقرأ هذه النتائج بوصفها نتائج نموذجية مشروطة بافتراضات النموذج، وليست آثارًا سببية بشرية أو أحكامًا على تعلّم الطلاب. وتظل الموجة الميدانية محدودة بالعينة الملائمة والتصميم المقطعي، ويُوصى بتحقق طولي وتجريبي مستقبلي قبل أي تعميم.

الكلمات المفتاحية: الراديو عبر وسائل التواصل الاجتماعي؛ نظرية النظم المعرفية؛ حوكمة المعلومات المضللة؛ درجة سمعة المذيع؛ التقييم اللحظي البيئي؛ النمذجة القائمة على الوكلاء؛ التجربة المضادة.

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Evidence Status and Reading Guide

This version is a self-contained research archive. It contains the raw 60-row questionnaire file and the reconstructed 52-row analysis matrix, the raw ride/PO narratives, the original 160-segment FGD transcript, the 27.7-minute self-report audio and its 185-segment GPU transcript, the four qualitative corpora, the stimulus metadata, the final Python pipeline, the R reproduction scripts, the seeds, the generated JSON/CSV outputs, and the thesis figures. The historical two-arm simulation is reported as an inherited model summary; the final calibrated in-silico experiment (v8.0) is reproduced by the included pipeline and is reported separately. The distinction between measured field evidence, model-based counterfactual evidence, and theoretical propositions is maintained throughout. Appendix K gives the exact commands, and Appendix L identifies the package contents.

The questionnaire concerns audience evaluations in a convenience sample. Historical simulation contrasts concern a specified model. The prospective agent experiment concerns computational task performance. These three units of evidence cannot be substituted for one another. The original focus-group transcript is integrated in Appendix G and Section 6.7; the audiovisual recordings and participant roster remain unavailable. The supplemental seven-part generated analysis is available for critical review but is not a substitute for those records (Appendix G.4). The FGD supplies provisional design hypotheses, not additional questionnaire responses. The researcher self-report is included as reflexive context; independently

keyed factual assessment remains a separate requirement.

Chapter 1 introduces the question and evidence streams; Chapter 2 reviews the literature; Chapter 3 develops the proposed framework; Chapter 4 distinguishes platform specifications from implementation evidence; Chapter 5 explains the methods and uncertainty; Chapter 6 separates reported findings from prospective tests; and Chapter 7 states the bounded conclusions.

PB and stars

Students' evaluations of a broadcast, not demonstrated literacy.

Questionnaire BRS

Eleven audience-rating items, not the platform's computed reputation score.

Algorithmic BRS

A proposed platform score whose implementation and calibration need separate checks.

Student Rating

A distinct proposed competence profile; no field treatment effect is reported.

Uncertainty

Confidence intervals, posterior credible intervals, and ranges across simulation runs answer different questions (Section 5.17).

Chapter 1

Introduction

Chapter abstract. This chapter introduces the problem: social-media radio shapes what students hear, trust, and learn, yet platform design and audience measurement are usually studied separately. It states the research questions, objectives, hypotheses, scope, and the evidence hierarchy used throughout. It explains why a measured field wave is combined with a calibrated in-silico experiment rather than a real platform trial, and it defines the contribution: an EST-based design framework, a transparent measurement model, and a reproducible testbed. It closes with the structure of the thesis.

1.1 Background and Motivation

Internet distribution and social networking connect audio production with new forms of discovery, circulation, and audience response. The scholarly question is not whether these technologies are uniformly beneficial or harmful, but how specific arrangements change what people encounter and how they evaluate it (Spinelli & Dann, 2019; Sullivan, 2019).

Evidence about misinformation and algorithmic selection motivates the inquiry without resolving it. Vosoughi et al. (2018) analyse true and false news

circulation on Twitter; their result is not a universal law governing every audio network. Gauthier et al. (2026) identify some effects of feed assignment but not uniform changes across political outcomes. These findings support examining architecture while requiring careful outcome definitions and contextual limits.

The present thesis develops Epistemic Systems Theory (EST), specifies the LOKLOK platform, and examines audience evaluations in a journey-related listening setting. It links a field questionnaire to a separate calibrated simulation. The aim is a defensible account of the relationships among architecture, judgments, and measurement, rather than an assumption that a positive rating demonstrates accurate content or successful learning.

1.2 From Internet Radio to Social Media Radio

Internet delivery, radio programming, on-demand access, and social networking describe different properties. A station can stream a linear programme without listener profiles. A podcast can be downloaded without a social feed. A social network can carry an audio clip without a continuous radio service. Treating these as interchangeable would obscure the study's population and the architectural intervention it proposes.

Rime et al. (2022) examine podcasting through tensions rather than a single fixed technological definition. Sullivan (2019) distinguishes the storage, discovery, and consumption functions of podcast platforms. An episode may therefore be hosted in one place, discovered through a social post, and heard through another interface. Table 1.1 gives this thesis's working classification, not a universal taxonomy

of audio media.

Table 1.1: Distinguishing the Audio Forms Relevant to This Study

Form	Defining feature here	What cannot be assumed
Internet radio	A programmed audio stream delivered online	Social interaction or personalised ranking
Podcasting	Episodic, usually on-demand audio distribution	Live exchange or a social feed
Social audio	Audio-centred interaction among networked participants	Editorial radio programming
Social media radio	Radio-like audio linked to profiles, sharing, interaction, or ranked discovery	That every listener uses every social feature
LOKLOK	The thesis's integration of audio, reputation, recommendation, and governance	Validity of a score merely because it is displayed

Note. Author-developed distinctions informed by Spinelli and Dann (2019), Sullivan (2019), and Rime et al. (2022). Forms can overlap; classification should follow the actual listening and distribution setting.

1.2.1 Broadcasting and Social Circulation

A host can post an episode, a listener can circulate an extract, and others can comment, challenge, or recommend it. Spinelli and Dann (2019) discuss audio production and relationships with listeners in podcasting. This supports examining radio as more than a transmitter–receiver pair, but does not imply equal participation or improved information merely because interaction is possible.

This thesis distinguishes *editorial selection*, *platform selection*, and *listener selection*. Editorial selection determines what is recorded and how claims are framed. Platform selection determines which posts become visible. Listener selection includes choosing, replaying, ignoring, and sharing. EST concerns the relationship among these stages. Assigning all three to an abstract "algorithm" would obscure responsibility and prevent a precise explanation of effects.

1.2.2 Why Social Media Radio Matters

The form brings together voice-based communication, networked circulation, and visible audience metrics. It therefore offers a setting for studying how trust in a speaker relates to trust in a message and how metrics affect visibility. Student participation can also involve questioning, sourcing, production, and correction. These are opportunities motivating the research, not findings that the present respondents acquired those capabilities.

The same affordances create risks. An extract can omit qualifications in a longer programme; popularity can become a credibility cue; and a correction may reach a different audience from the original post. Access depends on connectivity, devices, language, and accessible interfaces. Greater technical reach is not equivalent to equitable participation, and audio convenience is not equivalent to sustained attention. Transcripts and source links could support review, but their availability and use require separate measurement.

Listening context matters. Groot Kormelink et al. (2025) found that repeated news-podcast use was organised around particular everyday routines. The journey setting here is therefore a concrete reception context, not grounds for assuming that commuters are uniformly attentive or distracted. Journey duration and reported interaction remain contextual variables until listening time and attention are independently measured.

1.3 Key Terms and Units of Analysis

Students' ratings.

Audience evaluations, including stars and questionnaire scores; not academic grades or demonstrated learning outcomes.

Perceived credibility.

A judgment of believability or trustworthiness, distinguished from content accuracy, which requires an external assessment of the claims.

Engagement.

A family of behaviours or reports. A click, a completed listen, a comment, and a self-reported interaction category are not interchangeable quantities.

Broadcaster score and broadcaster questionnaire.

The algorithmic BRS is a design object; the 11-item broadcaster-rating battery is an audience instrument. Reliability of the latter does not directly establish accuracy or fairness of the former.

Observation.

The field analysis uses 52 retained questionnaire records. An episode, a post, a broadcaster, and a simulation run are different units. Their counts cannot replace the number of independent respondents.

EST proposes mechanisms; LOKLOK specifies an architecture; the questionnaire describes audience judgments; and the simulation explores conditional contrasts. These components belong together but warrant different conclusions.

1.4 The Research Problem

The original title, *The Social Media Radio Innovation and its impact on University Students Ratings*, names the research problem rather than asserting that every impact has already been causally identified. "Student Rating" refers to the student's media competence and development; "students' ratings of broadcasts" refers to audience evaluations. The field survey measures the latter. Section 3.7 distinguishes the human competence construct from the proposed computational extension. That extension tests synthetic task performance and platform dynamics, with exploratory NMLS-format responses; it does not supply a human competence effect.

The thesis addresses an integration problem identified in the reviewed literature. Accounts of broadcast authority, social trust, credibility, algorithmic curation, and constrained attention illuminate different parts of platform communication. EST brings them into a single proposed framework with explicit architectural and measurement commitments. This is an argument for a particular synthesis, not proof that no earlier theory considered these relationships.

The empirical problem is how to distinguish a platform's reputation design from audience judgments of that design and from independently assessed information quality. The framework must specify what is measured, which relationships are observed, and which claims require an intervention. The field questionnaire and model comparison address different parts of that problem and are reported separately.

1.5 The Contribution of This Thesis

This thesis makes a single integrated contribution: an original theoretical framework, Epistemic Systems Theory (EST), together with an epistemic governance platform (LOKLOK) that implements selected mechanisms and specifies others for prospective testing, an empirical evaluation that combines a measured field wave with a calibrated in-silico counterfactual experiment, and an evaluated set of survey instruments. The framework is developed in Chapter 3; the platform is specified, with its historical implementation account distinguished in Chapter 4; the empirical evaluation is in Chapters 5 and 6.

Four components develop and examine the framework. First, a field measurement wave of 52 university students who listen to social-media radio while riding with Uber in the October 6th City area of Giza produced a survey battery with reported internal consistency and factor-fit diagnostics (Chapter 6). Second, a calibrated two-arm in-silico experiment representing alternative platform rules estimated a model-conditional contrast between an epistemic-first and an engagement-first architecture. Third, the 60 supplied rider accounts support a thematic analysis and a 52-case paired joint display. Fourth, the LOKLOK implementation account describes a Go backend, Flutter client, and web application. Their fidelity to the reference equations must be established before the planned agent experiment; a scalar production score is not automatically the theoretical BRS. The four-stream BGE-M3 embedding is a pipeline feature, not the headline contribution: the source ablation later shows that the semantic block is contextual, while the contrast is

carried by the mechanism and the state-outcome weights.

1.6 Research Questions

The thesis retains four theory-led questions, adds two mixed-methods questions, and specifies a student-focused question for the extended design. These additions are not retrospectively described as preregistered. RQ1–RQ4 do not map one-to-one onto its four propositions: RQ4 examines integration, whereas P4 motivates the prospective task-performance comparison in H7 and RQ7.

1. **RQ1 (Structural determination).** Do platform architecture, credibility metrics, and content-ranking functions determine epistemic outcomes (perceived quality, credibility, diversity) more than user-level characteristics do?
2. **RQ2 (Diversity constraints).** Under the specified network and ranking rules, do active diversity constraints alter concentration and engagement in the directions proposed by EST-P2?
3. **RQ3 (Psychometric accountability).** To what extent do the consequential metrics of the platform, the Broadcaster Reputation Score (BRS) and the survey instruments, meet published psychometric standards?
4. **RQ4 (Integration).** Do the epistemic constructs of EST behave as facets of a single consolidated quality channel, both in the measured wave and in the calibrated counterfactual experiment?
5. **RQ5 (Audience experience).** How do the rider records describe credibility, listening value, participation, and platform experience?
6. **RQ6 (Integration).** Where do these accounts converge with, complement,

or challenge the paired questionnaire ratings?

7. **RQ7 (Computational extension).** How does assigned exposure to the specified LOKLOK configurations change evidence-anchored agents' task accuracy, media production, and exposure relative to G0? If held-out human outcomes are available, how accurately do the agents reproduce them?

1.7 Research Objectives

1. Develop EST with four formal propositions from which testable hypotheses can be deduced (Chapter 3).
2. Specify and implement LOKLOK, an epistemic-first social media radio platform (Chapter 4).
3. Integrate the field questionnaire with a transparent thematic analysis of rider records to examine audience evaluations and measurement properties (Chapter 5).
4. Calibrate an agent-based model to the measured wave and run a specified two-arm counterfactual experiment (Chapter 5).
5. Distinguish human Student Rating from broadcast evaluations and specify a computational extension using independent task accuracy, with NMLS-format responses as secondary synthetic diagnostics.
6. Evaluate the hypotheses and propositions and draw design and policy conclusions (Chapter 6).

1.8 Hypotheses

Seven hypotheses organise the evaluation. H1 to H4 are primary in the thesis structure, not certified as prospectively registered tests; H5 and H6 are exploratory, while H7 is formulated for the prospective computational extension. Their assessment distinguishes field associations, model results, and outcomes not directly measured. Each hypothesis is mapped to a proposition of EST.

1. **H1 (Structural credibility, EST-P1).** Exposure to an epistemically architected interface is positively associated with perceived credibility and confidence.
2. **H2 (Diversity mechanism, EST-P2).** Architectural exposure increases the perceived diversity of consumed content and reduces echo-driven repetition.
3. **H3 (Psychometric validity, EST-P3).** The study instruments demonstrate acceptable internal consistency, while structural validity is rigorously examined rather than assumed.
4. **H4 (EST integration, EST-P1 to P3).** The epistemic measures form a positive manifold consistent with one consolidated quality channel.
5. **H5 (DeGroot conformity, EST-P2, exploratory).** Correction acceptance and echo-persistence dynamics behave as conformity processes.
6. **H6 (Scheduling and duration, EST-P1, exploratory).** Listening persistence is architecture-dependent and trades off against perceived quality.
7. **H7 (Participatory performance, EST-P4; prospective).** Under matched model, content, network, and resource conditions, the specified

platform support improves agent claim-appraisal accuracy and evidence-based contributions relative to the comparison configuration. Task accuracy is the primary endpoint and production quality is secondary. This computational hypothesis concerns observed outputs, not inferred human development.

The prospective LLM extension is a further methodological objective, not a completed hypothesis test. It will use consent-compatible qualitative anchors to guide agents operating an isolated instance of LOKLOK, and will test synthetic task performance, exposure, and production. Any additional claim of human fidelity requires outcomes withheld from agent prompts; without them, inference remains computational.

1.9 What the Student Ratings Mean

"Student Rating" is an umbrella term in this thesis, not a single number. Table 1.2 fixes the meaning of each rating before the results are presented. The evaluative Student Rating (SR_{eval}) is a 1–7 score built from the credibility and perceived-benefit items and scored through the GRM model. The competence-format Student Rating (SR_{comp}) is a 0–100 display built from the synthetic NMLS battery. Perceived benefit (PB) is the measured five-item 1–7 composite; the star rating is the measured 1–5 overall rating; the BRS is a 0–1 reputation score; engagement is a 0–1 composite of raw interaction. A one-point change on the evaluative scale is not a grade, a percentage, or a human learning gain. Model-based values are labelled as such throughout.

Table 1.2: What Each Student Rating Means

Term	Scale	Meaning	Source
Evaluative Student Rating (SR_{eval})	1–7	Mean GRM response over credibility and perceived-benefit items; how positively the listener evaluates the broadcast	Model-based; measured PB anchor
Competence-format rating (SR_{comp})	0–100	Synthetic NMLS competence profile, rescaled for display	Model-based
Perceived benefit (PB)	1–7	Five-item confidence, diversity, trust, clarity, and overall-quality composite	Measured, $N = 52$
Star rating	1–5	Overall star given after the listening episode	Measured, $N = 52$
Broadcaster Reputation Score (BRS)	0–1	Weighted reputation score with an uncertainty interval	Measured/model
Engagement	0–1	Raw interaction composite; not a quality judgement	Model-based

Note. The evaluative and competence scales are not interchangeable. A 1-point change on the 1–7 scale is not equivalent to a 1-point change on the 0–100 scale, and neither is a human learning gain.

1.10 Scope and Delimitations

The measured wave is a cross-sectional questionnaire of 52 students who listen to social media radio during rides with Uber in the October 6th City area, Giza. It is a psychometric anchor, not a completed field experiment. The causal contrast between architectural configurations is evaluated in a calibrated in-silico experiment and is explicitly disclosed as such. The next platform-in-the-loop extension is a proposed computational design specified in Chapter 5; a longitudinal human comparison remains a separate future study, not a prerequisite for bounded computational claims. Every claim in the results chapters is tied to the questionnaire, recorded qualitative accounts, or the historical calibrated simulation; no result is asserted on the basis

of projected trajectories.

1.11 Thesis Structure

Chapter 2 reviews six epistemic dimensions of social media audio in the existing literature. Chapter 3 develops EST and its four propositions. Chapter 4 specifies the LOKLOK platform and its reported implementation boundaries. Chapter 5 describes the mixed-methods field study, historical calibrated model, and planned LLM platform extension. Chapter 6 reports the results. Chapter 7 draws conclusions, states limitations, and outlines the confirmatory pathway.

Chapter summary. Chapter 1 sets the question and the boundaries. The thesis asks whether LOKLOK's design can improve epistemic quality without destroying engagement, but it does not claim a causal field effect. The measured wave anchors the problem; the simulation explores design implications. The contribution is the integration of theory, measurement, and a reproducible formal model, with explicit limits on what each evidence tier can support.

Chapter 2

Critical Review of Six Epistemic Dimensions of Social Media Audio

Chapter abstract. This chapter reviews six epistemic dimensions of social-media audio: the manufacture of consent, the architecture of attention, the velocity of untruth, the parasocial epistemic bond, the measurement of the mind, and the student as epistemic agent. It assesses what communication research, platform studies, and measurement theory can support, and it identifies rival explanations. The review motivates EST and the LOKLOK design while flagging the gaps: platform-specific evidence, longitudinal outcomes, and validated student measures.

This chapter reviews the literature that motivates Epistemic Systems Theory (EST). The review is organised along six dimensions, each of which identifies a mechanism that shapes what users believe and how they judge information: the manufacture of consent, the architecture of attention, the velocity of untruth, the parasocial epistemic bond, the measurement of the mind, and the student as epistemic agent. The purpose of the review is not exhaustiveness but theoretical grounding: each dimension contributes a distinct constraint that EST must accommodate, and each is revisited when the framework is developed in Chapter 3.

2.1 Review Scope and Appraisal Criteria

The review is a critical narrative synthesis. Sources were selected for relevance to audio distribution, audience judgment, platform mechanisms, measurement, and the Egyptian setting. The expansion includes targeted publisher and institutional searches for radio–social-network relations, podcast platformisation, listening context, algorithm experiments, and measurement reporting. It includes research published through 2026; no exhaustive search, formal quality-scoring exercise, or systematic-review registration is claimed.

Each study is appraised by its object, research design, population, outcome, main contribution, and limits of transfer to this thesis. Abstract-only information is not represented as full-text appraisal. In particular, the Egyptian study is discussed at the scope supported by its published abstract. Journal prestige or sample size alone does not determine relevance: a large study of text feeds need not measure the same process as an audio-listening study. Evidence about architecture, exposure, evaluation, and learning is kept separate throughout.

The main currency window is publication years 2016–2026, through the revision date of September 8, 2026. Older sources are retained for original concepts and authoritative methods, including Shannon's entropy, Cronbach's alpha, Horton and Wohl's parasocial concept, DeGroot's opinion model, Herman and Chomsky's propaganda framework, and the 2014 testing standards. The statistical explanation also cites foundational association and Bayesian methods. Recent research supplies criticism and application rather than allowing these classics to stand in for current

evidence. One agent-simulation source is explicitly labelled a preprint.

2.2 Radio, Platforms, and Social Participation

Spinelli and Dann (2019) examine podcasting as an audio medium with distinctive production practices and relationships between producers and listeners. Its value for EST is a shift in the unit of explanation: public interaction extends beyond the broadcast. Its limitation is equally important: identifying new relationships does not estimate an interface's effect on credibility. Participation can be invited without being realised, and visible comments need not represent silent listeners.

Sullivan (2019) examines podcast platformisation through infrastructure and industry organisation. His storage–discovery–consumption distinction helps locate where a platform acquires influence over audio. An industry analysis, however, is not a randomised audience study. For EST, a gatekeeping claim must identify the selection rule and its consequence, rather than infer persuasion directly from ownership or concentration.

Rime et al. (2022) characterise podcasting through multiple tensions and innovations, cautioning against defining it through one delivery technology. The present thesis consequently distinguishes a streamed programme, a social post, and an on-demand episode. This conceptual work clarifies the object of study; it does not demonstrate a particular LOKLOK mechanism.

Groot Kormelink et al. (2025) asked 60 participants to listen to a daily news podcast for up to four weeks and interviewed them afterward. The study highlights the importance of fitting listening into specific routines. Its instructed listening

arrangement supplies contextual depth but differs from an unprompted population survey. It motivates examining reception context; it does not establish that longer rides cause higher PB scores in Giza.

2.3 Egyptian Audience Research and Contextual Transfer

Wajih and Al-Yamani (2024) investigate Egyptian young people's use of podcast applications for political and economic issues. Their published abstract describes an online questionnaire with a purposive sample of 400 users and an interest in interaction, content evaluation, and trust. The study adopts a public-sphere perspective. This is a characteristic of the reviewed study, not a replacement for EST in the present thesis.

This research prevents a claim that Egyptian audio-audience scholarship is absent. Its purposive sample does not establish national prevalence, and its broad podcast population differs from university students listening during journeys. The comparison concerns questions and context, not a numerical replication. No unreported coefficients from that study enter this analysis.

Contextual transfer requires more than translating a questionnaire. Language variety, host familiarity, topic sensitivity, shared-device listening, and relationships among vehicle occupants could influence responses. These are conditions to investigate, not explanations already measured here. Institution labels and the journey setting do not establish respondents' socioeconomic position, political views, or attention.

2.4 What Algorithm Research Supports—and What It Does Not

Lorenz-Spreen et al. (2023) systematically review causal and correlational evidence on digital media and democracy. Their synthesis identifies patterns that differ by outcome and context, rather than a uniform relationship. The lesson for EST is to specify both the outcome and the design that identifies it. Participation, trust, knowledge, and polarisation are not interchangeable measures of one platform effect.

Gauthier et al. (2026) report a seven-week experiment conducted on X in 2023. Switching from a chronological to an algorithmic feed shifted some political opinions; switching in the opposite direction produced no comparable effect. Neither manipulation significantly changed affective polarisation or self-reported partisanship. This supports taking architecture seriously while rejecting an automatic inference from a feed change to every attitude. Its findings concern a platform, period, and active-user population, not radio listeners or LOKLOK.

For EST-P1, the bounded proposition is that a specified architectural change may alter a specified outcome under defined conditions. Neither study proves that architecture always outweighs user characteristics. Prior exposure and user choices can modify an intervention's effect: a platform can influence whom someone follows, while those following choices subsequently influence what the platform delivers. Structural influence and user agency need not be mutually exclusive explanations.

2.5 Propagation, Correction, and Audience Judgment

Pennycook et al. (2021) distinguish judgments of headline accuracy from sharing intentions and report experiments in which directing attention to accuracy improved the quality of subsequent sharing. Pennycook and Rand (2022) extend this evidence through an internal meta-analysis of 20 experiments. Consistency across those experiments is informative, but the internal scope and frequent use of headline-sharing tasks limit direct transfer to audio. Neither paper validates the BRS or an audio-transcript classifier.

Ecker et al. (2022) review cognitive, social, and affective influences on misinformation belief and correction, including continued influence after correction. This makes correction a process rather than a binary platform action. Publishing, delivering, noticing, accepting, and later using a correction are distinct events. A platform cannot infer the last stage merely because it logged the first.

These distinctions make EST more testable. If a badge changes PB but not accuracy discernment, the finding concerns a credibility cue. If a prompt changes sharing but not belief, it concerns behaviour rather than persuasion. If source diversity increases alongside the circulation of inaccurate sources, diversity alone has not established better information. These are competing outcomes, not inconvenient cases to exclude from evaluation.

2.6 Measurement Debate Rather Than Coefficient Counting

McNeish (2018) argues for closer attention to reliability assumptions and alternatives to routinely reporting alpha. Raykov and Marcoulides (2019) counter a blanket rejection: alpha remains useful when its relationship to score reliability is justified by the measurement conditions. The disagreement does not license selecting whichever coefficient is largest. It requires specifying the score, model, and inferential purpose.

The balanced position here is to preserve the reported coefficients, disclose factor-fit problems, and avoid turning either alpha or an unspecified omega into a validity certificate. A reliable score can measure the wrong construct, and a plausible construct name does not establish equivalent measurement across languages or institutions. EST-P3 is an accountability requirement precisely because it permits an unfavourable measurement result.

2.7 The Manufacture of Consent

Herman and Chomsky's propaganda model (1988) argued that mass media systematically filter which claims become visible: ownership and advertising interests, sourcing routines, flak, and ideology jointly shape the content that reaches audiences. The model anticipated the modern insight that what users see is not what is true but what institutional filters admit. Its relevance to social media audio is direct: algorithmic platforms inherited the filtering role of mass media but execute it with a metric now visible to users and amenable to scientific measurement.

Chomsky's paradigm anchors EST-P1, the proposition that the architecture

of a media system, not the psychology of individual consumers, is the primary determinant of the epistemic output. A social media radio platform whose ranking function rewards engagement will amplify a different distribution of claims than one whose ranking function rewards verified diversity, regardless of the qualities of the audience. The propaganda model supplies the intellectual precedent for treating the filter itself as the unit of analysis.

2.8 The Architecture of Attention

Wu (2016) traced the "attention merchants" who have, since the nineteenth century, harvested human attention and sold it. Zuboff (2019) extended the analysis to surveillance capitalism, in which behavioural surplus is extracted from users and converted into prediction products. On social media, the attention architecture is executed by recommender systems whose optimisation targets are set by the platform. Gillespie (2018) documented how content moderation, the "custodians of the internet", quietly shapes the public sphere while claiming neutrality.

The attention-economy literature largely measures attention at the macro level: time spent, clicks, or exposure. A complementary micro-level question is whether the listener is actually coupled to the broadcast at a given moment. Zewail's (1999, 2000) pump–probe experiments in femtochemistry provide a formal analogy for this problem: a pump pulse launches a coherent state, and a delayed probe measures it before dephasing destroys coherence. The thesis borrows the temporal structure, not the chemistry, to define a time-resolved attentional gate (Section 4.3.1). The analogy is useful precisely because it is falsifiable: if the gate

does not improve retention, reputation, or rating predictions, it should be removed.

The attention-economy literature treats engagement metrics as choices about what a platform rewards rather than neutral measures of social value. This motivates comparing objective functions, but it does not establish that every increase in engagement reduces epistemic quality. The present thesis explores that trade-off in its specified simulation (Section 6.11), not as an already established universal relationship.

2.9 The Velocity of Untruth

A major study of news diffusion found faster, wider, and deeper propagation of false than true stories in its Twitter dataset. Vosoughi et al. (2018) analysed the diffusion of true and false news on Twitter across roughly 126,000 cascades and found that falsehoods were 70% more likely to be retweeted than true news, reached 1,500 people about six times sooner, and travelled deeper into the network, with novelty and emotional charge as the plausible drivers. This evidence concerns the diffusion of classified news stories on Twitter; it does not directly estimate how a radio listener judges a particular broadcaster. Brady et al. (2023) separately show how observers can overperceive moral outrage and inflate beliefs about intergroup hostility. That finding concerns emotion perception, not a general law that reasoned content always loses to outrage.

For EST, these studies motivate attention to amplification and perception as distinct processes. A source-diversity statistic does not itself establish accuracy: a diverse set can contain multiple inaccurate sources. Likewise, reducing concentration

does not necessarily improve a listener's ability to identify a false claim. The model and its outcome measures must specify which of these changes they are intended to produce.

2.10 The Parasocial Epistemic Bond

Horton and Wohl (1956) coined the term *para-social interaction* to describe the one-sided intimacy audiences develop with media figures who address them directly. The concept gives a vocabulary for relationships with mediated speakers. Spinelli and Dann (2019) examine podcasting through interviews with producers and close engagement with audio production and listening practices. Their account supports studying intimacy and regularity as features of audio communication, but does not provide an experimental estimate of the effect of a BRS display. Parasocial closeness can be a competing explanation for favourable ratings, not just a mechanism confirming structural credibility.

The parasocial bond matters epistemically because trust is transferred from person to proposition. A listener who trusts a host is more likely to accept the host's claims with less scrutiny, which makes the bond a conduit for both verified knowledge and misinformation. EST treats the parasocial bond as an interaction potential that connects broadcaster reputation to content credibility, and the framework weights endorsements by the credibility of the endorser precisely because parasocial trust is powerful.

2.11 The Measurement of the Mind

The psychometric tradition demands that constructs be measured with explicit standards of reliability and validity. Cronbach (1951) supplied coefficient alpha as an internal-consistency coefficient. The Standards for Educational and Psychological Testing (American Educational Research Association [AERA] et al., 2014) address evidence for score interpretations and uses, including consequences of measurement. Kline (2023) explains confirmatory factor analysis as a way of evaluating a specified measurement structure. The fit criteria examined by McNeish and Wolf (2023) depend on model characteristics and should not be treated as universal pass–fail rules independent of model and data. Measurement invariance, the requirement that a scale measure the same construct across groups, is operationalised through configural, metric, and scalar models (Putnick & Bornstein, 2016). Boateng et al. (2018) distinguish content development from empirical scale evaluation, a distinction needed when expert judgments are not archived.

EST-P3 elevates these standards to a design principle: any metric that governs access, visibility, or reputation on an epistemic platform is a consequential metric and must satisfy psychometric standards proportionate to its consequence. The Broadcaster Reputation Score (BRS) is therefore specified in this thesis with an explicit measurement model, reliability targets, and a proposed uncertainty display. The questionnaire results in Chapter 6 assess internal consistency and factor fit; they do not establish calibration of broadcaster-specific algorithmic intervals.

2.12 The Student as Epistemic Agent

University students are relevant to this thesis as an audience engaged in formal study and potentially in campus media production, not as an inherently homogeneous or media-literate population. Ashley (2020) situates news literacy in its democratic and institutional context. This conceptual argument does not establish that university attendance itself produces critical evaluation or validate a particular student-rating algorithm. Vraga and Tully (2021) used a nationally representative survey to examine news literacy, social media behaviors, and skepticism. Greater literacy and valuing literacy were associated with more skepticism about information quality; seeing and posting news were not themselves associated with skepticism. This is observational evidence, not a test of a literacy intervention or of an effect of LOKLOK's design. Mihailidis (2019) argues that civic media literacy involves social practices of connection as well as individual skills. These studies motivate examining both individual capacities and opportunities for participation. They do not establish that a particular interface causes better appraisal or production; that relationship is a testable commitment of EST-P4 in the present framework.

The student-in-ride setting studied in this thesis is a concrete case: students evaluate radio broadcasts during a bounded, compressed attention window. The records can inform how listening context and perceived platform features relate to audience evaluations. The survey does not separately randomise reputation cues, source diversity, or reflection prompts, so it cannot establish that those features dominate content, familiarity, or context.

2.12.1 From Broadcast Evaluation to Student Competence

A student-development claim requires more than a satisfaction item. Koc and Barut (2016) provide an instrument specifically concerned with university students' new-media competence, while the use of NMLS in Yavuzalp et al. (2021) provides further context for its measurement structure. These sources support selecting an external measure rather than constructing a Student Rating score by combining stars and engagement. They do not establish the impact of LOKLOK or validate an Egyptian translation.

This distinction is theoretically consequential. A platform can improve listeners' evaluations without improving their critical production, and a more competent listener may become less generous in rating weak content. The thesis therefore separates its existing audience measures from the competence profile and task evidence specified in Section 3.7. The instrument is a measurement resource, not a rival theory replacing EST; architecture remains the explanatory proposal whose consequences the design tests.

2.13 Critical Synthesis and Competing Explanations

This is a theory-oriented narrative review, not a systematic review with a claim to exhaustive retrieval. Its six dimensions operate at different levels: institutional structures, platform rules, message diffusion, relationships with speakers, measurement, and educational practice. Integration requires specifying how these levels connect rather than assuming that evidence at one level automatically verifies a claim at another.

2.13.1 Source, Message, and Audience Evaluation

Appelman and Sundar (2016) distinguish message credibility from judgments about a source or medium and develop a news-message credibility scale. This is useful for EST because it identifies a measurement boundary: perceived rigor of a broadcaster, confidence in a message, and overall satisfaction are not automatically the same construct. Their scale-development results do not validate the differently worded PB items used here. Validation belongs to a specified instrument, population, and interpretation.

Sundar (2020) develops a framework for the psychology of human–AI interaction, including how machine agency is encountered through interfaces. Applied to this thesis, a high BRS could act as a cue rather than as evidence that a listener checked the content. EST-P1 must therefore compete with an alternative explanation: a visible reputation score changes the rating without changing either content quality or the listener's ability to recognise accurate information. A field design should measure these outcomes separately.

2.13.2 Research Design and Transferability

Vosoughi et al. (2018) analyse large-scale diffusion traces, whereas Bail et al. (2018) use an intervention involving exposure to opposing political views. These approaches answer different questions. Diffusion patterns do not by themselves identify an architectural treatment effect; an intervention in a particular political and platform context does not determine its effect in Egyptian social radio. Producer-centred accounts such as Spinelli and Dann (2019) offer interpretive depth but not comparable

effect estimates. These sources are complementary, not interchangeable replications of EST.

The present ride-context survey adds a specific audience setting, but its small convenience sample cannot establish national or cross-platform effects. The research gap is therefore a need to connect architectural proposals with well-defined audience measures and direct tests, not a claim that earlier communication scholarship neglected audiences, power, or credibility.

2.13.3 Measurement and Rival Explanations

McNeish (2018) cautions against treating alpha as proof of a single dimension. Podsakoff et al. (2024) review method-related sources of covariance in behavioural research. These concerns are directly relevant when the same respondents rate several favourable attributes during one questionnaire. A positive correlation pattern can arise from shared response processes, overlapping content, or prior liking, as well as from the proposed construct. The high SV–PB and IQ–PB correlations in this study must be read alongside its factor-model fit rather than used to bypass that evidence.

2.14 LLM Agents as a Computational Extension, Not Substitute Participants

Sundar (2020) distinguishes the symbolic and enabling effects of machine agency on human experience. This is relevant to LOKLOK because an interface can both offer an action and signal authority. It also creates a distinction for the extension:

Table 2.1: Evidence Types, Limits, and Their Role in EST

Evidence type	What it contributes	What it does not establish
Institutional and platform analysis	Incentives, governance, and design questions	Relative causal dominance in this sample
Diffusion traces	Patterns of circulation in a defined network	Accuracy of individual audience judgments
Exposure intervention	Effects of a specified intervention	Universal transfer to social radio
Audio-production research	Intimacy and communicative practices	A BRS display treatment effect
Scale-development research	Construct boundaries and measurement procedures	Validity of a different composite
Present field survey	Audience descriptions and associations	Assigned architectural or learning effects

an LLM producing a plausible explanation of trust is not the same as a human experiencing a voice, interface, or journey.

Park et al. (2023) demonstrate an agent architecture using memory, reflection, and planning to generate interactions in a sandbox environment. This is a useful engineering precedent for persistent listeners and creators, not evidence that emergent interactions reproduce Egyptian student behavior. Park et al.'s (2024) interview-grounded study simulates 1,052 individuals and evaluates responses against human measures. It motivates richer anchoring than a short demographic label. The cited version is a preprint, and its interview corpus and benchmarks differ substantially from the present short, researcher-recorded ride accounts.

Argyle et al. (2023) examine conditioned language-model samples using U.S. survey data. Aher et al. (2023) reproduce selected experimental patterns but also identify systematic distortions, illustrating why successful examples cannot certify a general human simulator. Dillion et al. (2023) likewise outline conditions and caveats rather than an unrestricted replacement of participants. These studies sup-

port the extension as a testable modelling strategy, not as a way to enlarge the human sample without recruitment.

The critical counterpoint is Bisbee et al. (2024): synthetic survey means can look plausible while variation, regression relationships, and reproducibility remain problematic. Accordingly, the LOKLOK extension must compare distributions and behavior, not merely average PB; vary prompts and model versions; and preserve unsuccessful runs. Descriptive plausibility, individual prediction, network dynamics, and platform intervention effects are different evaluation targets. None follows from a model agreeing with a favourable quotation.

2.14.1 Implication for the Present Research Programme

The empirical mixed-methods analysis establishes what the supplied students' ratings and recorded accounts contain. The historical calibrated model explores specified rules. The new LLM extension will test whether evidence-grounded agents interacting with actual LOKLOK software can produce useful, bounded predictions. Qualitative themes guide which mechanisms to observe, while new, withheld human outcomes are required if human predictive fidelity is claimed. This connects qualitative understanding to computation without treating students as fixed stereotypes or synthetic responses as new human evidence.

2.15 Synthesis and Implications for the Study

The six dimensions identify complementary concerns: institutional selection, competition for attention, patterns of misinformation diffusion, parasocial relationships,

measurement quality, and educational opportunity. They are not six universal laws. Trust need not be exclusively parasocial, falsehood need not always spread faster, and participation need not entail learning. Their integration in EST is valuable only if the proposed connections retain these boundaries and generate discriminating tests.

Table 2.2: Synthesis: Reviewed Dimension, Key Evidence, and EST Link

Dimension	**Key evidence**	**Mechanism**	**EST link**
Manufacture of consent	Herman & Chomsky (1988)	Structural filtering	EST-P1
Architecture of attention	Wu (2016); Zuboff (2019); Gillespie (2018)	Engagement optimisation	EST-P2
Velocity of untruth	Vosoughi et al. (2018); Ecker et al. (2022)	Diffusion and correction	EST-P2
Parasocial bond	Horton & Wohl (1956); Spinelli & Dann (2019)	Credibility transfer	EST-P1
Measurement of mind	Cronbach (1951); Boateng et al. (2018); Kline (2023)	Psychometric standards	EST-P3
Student as epistemic agent	Ashley (2020); Vraga & Tully (2021)	Literacy and skepticism	EST-P4

The review supports seven conclusions. First, internet radio, podcasting, and social media radio overlap without being identical. Second, platform influence must be located in production, discovery, circulation, or governance rather than assigned to an unspecified algorithm. Third, audience judgments, content accuracy, sharing, and learning require distinct measures. Fourth, Egyptian audience research provides a relevant context but not a substitute for evidence about this specific sample. Fifth, neither high alpha nor a successful model run establishes the validity of the entire theory. Sixth, qualitative accounts should elaborate and challenge ratings rather than serve as decorative confirmation. Seventh, evidence-grounded LLM agents remain models requiring held-out evaluation, not substitutes for additional human

participants. These conclusions motivate both the mixed-methods analysis and the explicit separation of the planned platform extension.

Table 2.3: Critical Review: Design Decisions Carried Into the Thesis

Literature insight	Decision in this thesis	Remaining test
Audio distribution and social participation are distinct	Define the form and reception context	Record actual social-feature exposure
Algorithmic effects vary by outcome	Separate architecture, exposure, and judgment	Assigned feature contrasts with relevant endpoints
Sharing is not necessarily belief	Distinguish propagation from credibility	Event logs and independent accuracy judgments
Local audience research is relevant but context-bound	Describe the Giza convenience sample	Replication in other settings
Reliability and validity differ	Retain alpha and report poor factor fit	Substantive item revision and measurement evaluation
Qualitative meaning is not reducible to a score	Code rider records and construct a paired joint display	Source fidelity and alternative readings
Agent plausibility is not human validity	Reserve a separate, anchored platform extension	Held-out outcomes and prompt/model sensitivity

EST is the researcher's proposed synthesis of these problems. Its value rests on explicit mechanisms and tests that could challenge them, not on declaring previous communication theories obsolete. Chapter 3 develops the propositions; subsequent chapters distinguish their implementation, observed audience associations, and conditional model predictions.

Chapter summary. The review finds a fragmented literature: strong evidence on attention, misinformation, and parasocial trust, but weaker evidence on how architecture and measurement rules jointly shape student evaluation. It justifies an exploratory design study rather than a causal test, and it identifies the constructs that must be measured: credibility, diversity, echo, perceived benefit, and competence.

Chapter 3

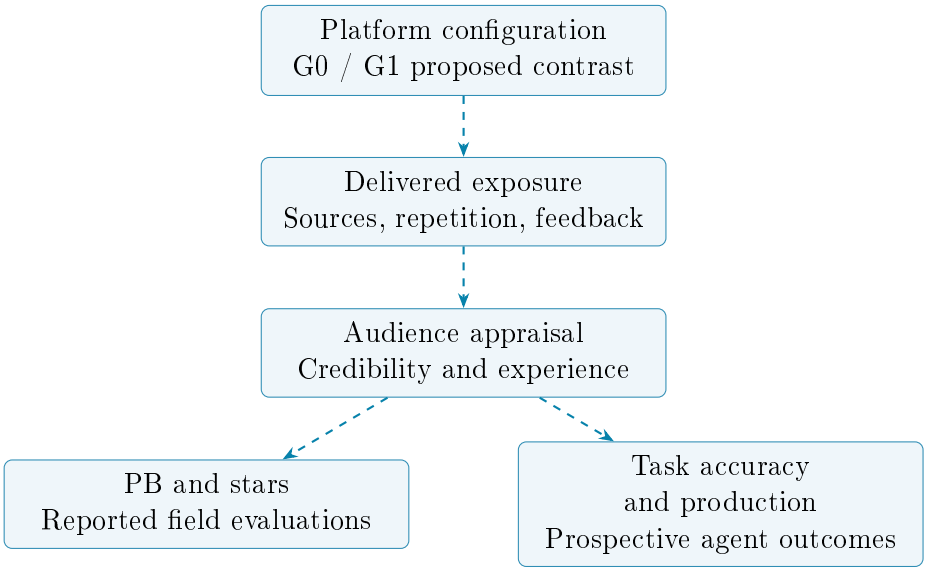
Theoretical Framework: Epistemic Systems Theory

Chapter abstract. This chapter introduces Epistemic Systems Theory and its four propositions: structural determination, thermodynamic constraint, psychometric accountability, and participation and evidence-based production. It formalises reputation update, an epistemic Hamiltonian, source entropy, heat capacity, and a BRS measurement model, and it explains each formal object in plain language. It distinguishes formal analogies from empirical laws and states the conditions under which each proposition would be rejected.

Epistemic Systems Theory (EST) treats a social media radio platform as a *socio-technical epistemic system*: a coupled structure of platform architecture, cognitive states, and social trust networks whose proposed epistemic outputs (the diversity of information that reaches users, the distribution of credibility, the rate of misinformation propagation) are hypothesised to depend substantially on architecture. The relative size of architectural and user contributions is an empirical question, not an assumption verified by the present survey. EST is developed here as the unifying lens of the thesis. It rests on four propositions: EST-P1 (structural determination), EST-P2 (thermodynamic constraint), EST-P3 (psychometric

accountability), and EST-P4 (participation and evidence-based production). From these propositions the hypotheses of Chapter 1 are deduced.

Figure 3.1: Proposed Construct Relationships



Note. Author-developed conceptual diagram. Dashed arrows are proposed relationships, not fitted causal effects. PB and stars remain distinct from task accuracy.

3.1 Why a New Framework Is Needed

The review motivates an original synthesis of three concerns: how platform rules shape exposure, how audiences evaluate sources and messages, and how consequential metrics are measured. EST connects these concerns through four propositions and a platform specification. Differences among the reviewed studies reflect differences in design and context as well as possible theoretical disagreement; they do not establish that all existing approaches are incompatible. The contribution of EST is its proposed integration and testable commitments, rather than an assertion of universal explanatory superiority.

EST uses mathematical structures from information theory and network

modelling to make its proposals explicit. Shannon's (1948) entropy provides a measure of a specified probability distribution, and DeGroot's (1974) model specifies a class of opinion updates. The Hamiltonian and free-energy terms below are author-defined social-system analogies. A shared notation alone does not transfer a physical law, an equilibrium, or a stability theorem to a platform: those conclusions require the actual update equations and their assumptions. Appendix E states the limited mathematical results that follow under explicit conditions. EST does not claim that social networks obey physical thermodynamic laws.

3.2 EST-P1: Structural Determination

3.2.1 Statement

The epistemic properties of a platform, the distribution of credibility, the diversity of information reaching users, and the rate of misinformation propagation, are primarily determined by the platform's architectural design: its incentive structures, trust-propagation algorithms, credibility metrics, and content-ranking functions. Architecture is the primary independent variable in epistemic governance, not a neutral substrate for user behaviour.

3.2.2 Formal Grounding

Let a social media radio platform be modelled as a tuple

$$\mathcal{P} = (\mathcal{U}, \mathcal{C}, \mathcal{A}, \mathcal{R}, \mathcal{G}), \tag{3.1}$$

where $\mathcal{U}$ is the user set, $\mathcal{C}$ the content set, $\mathcal{A}$ the recommendation algorithm, $\mathcal{R}$ the reputation function, and $\mathcal{G}$ the governance rules. EST-P1 asserts that the epistemic outcome function

$$\Omega(\mathcal{P}, t) = f(\mathcal{A}, \mathcal{R}, \mathcal{G}) + g(\mathcal{U}, \mathcal{C}, t) \tag{3.2}$$

In plain terms. The equation says that what a listener can know is limited by the platform's exposure rule. Change what the architecture shows, and you change the set of evaluations that are possible; you do not automatically change what is true.

In ordinary terms: this working decomposition separates a platform-rule contribution from a user-and-content contribution. It does not measure their relative importance. An additive split omits possible interactions, and adding a constant to f while subtracting it from g leaves the outcome unchanged. The dominance comparison therefore requires an identified decomposition, a common scale, reference conditions, and a defined domain before testing.

is dominated by the architectural triple $(\mathcal{A}, \mathcal{R}, \mathcal{G})$: $\|f\|_\infty \gg \|g\|_\infty$ under typical operating conditions. This inequality is a theoretical dominance hypothesis, not an estimate from the field survey. It requires a defined outcome scale and comparisons that separately identify architectural and user contributions. The literature in Chapter 2 motivates studying network diffusion and exposure without identifying this inequality. Vosoughi et al. (2018) compare news cascades; Bail et al. (2018) examine an intervention exposing participants to opposing political views. Neither separates the functions f and g or establishes architectural dominance in audio communication. The feed experiment of Gauthier et al. (2026) supplies more direct

evidence about a specific ranking intervention, but its outcome-specific findings also show why the scope of P1 must be stated explicitly.

3.2.3 Architectural Implications

EST-P1 motivates three design commitments, not empirically established necessities. **Credibility requires auditable inputs:** the reference BRS should document broadcast consistency, retention, source use, corrections, and endorsement rather than equating popularity with credibility. Retention and source diversity are not themselves tests of factual accuracy. **Recommendation requires explicit objectives:** quality and diversity constraints should be stated alongside engagement targets so that their trade-offs can be evaluated rather than assumed beneficial. **Governance requires accountable rules:** proposed down-weighting and moderation should be inspectable, contestable, and proportionate to the uncertainty of the detection system (Chapter 4). These are operational proposals whose consequences remain open to empirical testing.

3.2.4 A Reputation Update with Structural Guarantees

The structural claim of EST-P1 becomes testable only when the reputation update is given a precise form. The BRS of broadcaster u is designed as a modified DeGroot (1974) consensus process in which the update at session t is driven partly by peer endorsement and partly by objective broadcast evidence:

$$B_u(t+1) = (1-\lambda)\sum_v A_{uv}\, B_v(t) + \lambda\, X_u(t), \qquad 0 < \lambda < 1, \tag{3.3}$$

In plain terms. Reputation is a weighted average of neighbours' reputations and an evidence anchor. The cap stops one highly reputed node from dominating. In plain terms: trust spreads through the network, but it is damped.

In ordinary terms: the next reputation score blends the scores of endorsing peers with a separate evidence signal. If $\lambda = .2$, the update allocates 20% of its weight to that signal and 80% to peer-weighted reputation. This is an illustrative setting, not an estimated field parameter. Stable scores can still be wrong if endorsements or the anchor are inaccurate.

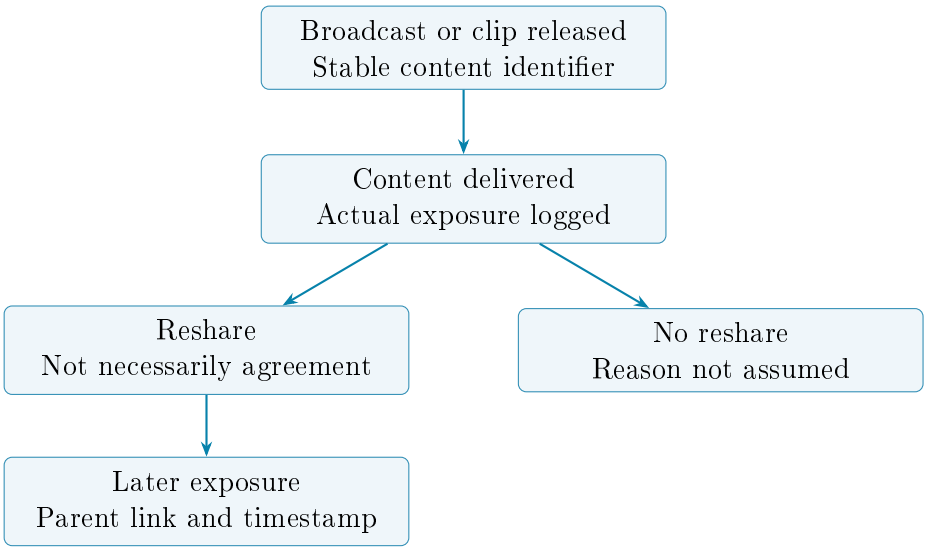
where A_{uv} is a nonnegative endorsement weight and each row of A sums to one. Negative endorsements must be handled separately rather than included in a matrix described as row-stochastic. The input $X_u(t)$ is a specified evidence signal; calling it evidence does not establish its accuracy.

For fixed A and a fixed input vector X, the full update contracts in the supremum norm by at most $1-\lambda$, and has the unique fixed point $B^* = \lambda[I-(1-\lambda)A]^{-1}X$. The unanchored row-stochastic component alone is non-expansive, not necessarily strictly contractive. Time-varying inputs need additional conditions for convergence, and a constant anchor can still produce consensus. Thus the mathematical result describes stability under stated assumptions, not truth or resistance to manipulation.

3.3 Content Propagation as an Event Sequence

The author-developed diagram distinguishes release, delivered exposure, and subsequent action. It is a communication-process schematic, not a physical analogy or evidence of observed diffusion in the field sample.

Figure 3.2: Content Exposure and Redistribution Process



Note. Author-developed process schematic, not observed participant trajectories. Branches are alternatives. Arrow lengths represent neither elapsed time nor effect sizes.

An episode, an extract, and a repost may carry different context. A defensible propagation record would preserve content identifiers, parent-post links, version changes, and event times. Exposure is not automatically completed listening, and resharing is not necessarily agreement, consistent with the sharing–judgment distinction examined by Pennycook et al. (2021).

The appropriate formal object is a time-indexed event graph. Links connect earlier to later events. Repeated appearances of a participant are separate events; an actor-level feedback cycle becomes a time-ordered sequence when expanded. Transmission probabilities would require observed opportunities to share as denominators,

not just successful repost counts. None is estimated from the 52 questionnaires.

The schematic distinguishes a ranking change that alters exposure from a prompt that alters resharing after exposure. Measuring only PB cannot identify all these paths. The drawings specify what to observe; they do not supply missing observations or a fitted network model.

3.4 EST-P2: Thermodynamic Constraint

3.4.1 Statement

EST-P2 proposes that diversity and reputation concentration depend on the network, its update rules, and the objectives governing exposure. Some engagement-oriented configurations may concentrate attention, while diversity constraints may change their performance. These are conditional predictions, not a physical law requiring every network to lose diversity or every quality improvement to reduce engagement. User preferences, new sources, and the content supply can also change the outcome. The thermodynamic vocabulary specifies an analogy to be evaluated rather than a reason to exclude these alternative processes.

3.4.2 The Epistemic Hamiltonian

EST-P2 uses the following candidate network functional, termed the Epistemic Hamiltonian by analogy, at time t:

$$\mathcal{H}(\mathbf{R}, \mathbf{Q}, t) = \sum_{i=1}^{N} \frac{R_i^2}{2C_i} + \sum_{j=1}^{M} Q_j \log Q_j + \sum_{i,j} W_{ij}\, R_i\, Q_j, \tag{3.4}$$

In plain terms. The Hamiltonian is a bookkeeping device: it adds a penalty for disagreement and a reward for coherence with quality. Lower values mean a more settled state. It is a formal analogy, not physical energy.

In ordinary terms: the expression adds three proposed terms: one involving concentration of reputation, one involving content-quality scores, and one connecting users with content. Choosing a formula makes the proposal explicit; it does not establish that minimising it improves communication.

where $\mathbf{R} \in \mathbb{R}_+^N$ is the reputation vector, $\mathbf{Q} \in [0,1]^M$ the content-quality vector, $C_i > 0$ the epistemic heat capacity of user i (resistance to reputation change), and $W_{ij} \in [0,1]$ the interaction strength between user i and content j.

This is an author-defined candidate functional. The first term penalises squared reputation relative to C_i; it is not literal kinetic energy because R_i is a score rather than a velocity. The second term is not Shannon entropy of sources when Q_j denotes independent quality scores: those scores need not sum to one. For a normalised source-exposure distribution p_s, source diversity can instead be described separately by

$$H_{\text{src}} = -\sum_s p_s \log p_s, \qquad \sum_s p_s = 1, \tag{3.5}$$

In plain terms. Source entropy measures how spread out the source mix is. Higher entropy means more different sources; it does not by itself mean better sources.

In ordinary terms: entropy describes how evenly exposure is spread across

the defined sources (Shannon, 1948). With base-2 logarithms, equal exposure to four sources gives 2 bits; exposure to just one gives 0 bits. Four inaccurate sources can still have high entropy. Diversity and factual accuracy are therefore separate measurements.

using the convention $0 \log 0 = 0$ and a declared logarithm base.

The sign of the coupling term also matters. With nonnegative W_{ij}, minimisation penalises simultaneous high R_i and Q_j; it does not by itself reward accurate, high-reputation content. A useful optimisation objective therefore requires justified signs, weights, scale normalisation, and outcome validation. The historical model summaries are not retrospectively recomputed from a revised objective in this chapter. Stability and substantive quality must be evaluated separately.

Table 3.1 states the domain mapping of the formal analogy explicitly, so that what is transferred between the physical and the social domain (the proposed mathematical representation) is separated from what is not (physical temperature, physical energy).

Table 3.1: EST-P2 Formal Analogy: Domain Mapping

Physical concept	**Platform concept**	**Operational quantity**
Temperature T_{net}	Social temperature of contention	Consensus velocity statistic
Entropy S	Information diversity	$-\sum p(s) \log p(s)$ over sources
Heat capacity C_i	Resistance of a reputation to change	$T_{\text{active},i} / \operatorname{Var}(\Delta B_i)$
Interaction potential W_{ij}	Broadcaster–content coupling	Nonnegative coupling weight in the stated functional
Free energy $\mathcal{F}$	Governance objective	$\mathcal{L} - T_{\text{net}} H_{\text{src}}$
Stability bound β^*	Reputation update gain	Depends on the specified update
Dephasing time τ_{dephase}	Attentional coherence window	P_{DTS} pump–probe gate (Section 4.3.1)

3.4.3 Operationalisation

The Hamiltonian is computable from platform data. The epistemic heat capacity is the inverse of the variance of BRS increments over broadcast sessions, scaled by active days,

$$C_i = \frac{T_{\text{active},i}}{\text{Var}(\Delta B_i)}, \qquad \text{Var}(\Delta B_i) = \frac{1}{T}\sum_t (\Delta B_i(t) - \overline{\Delta B_i})^2, \tag{3.6}$$

In plain terms. Heat capacity controls how much a persona's attention can be disturbed by new content. High capacity means slower, steadier updating.

In ordinary terms: this proposed index becomes larger for a longer active history or smaller changes in the score. It describes stability, not truth. A perfectly unchanging score makes the denominator zero; the index then needs an explicitly declared undefined-value or regularisation policy.

where $\Delta B_i(t) = B_i(t) - B_i(t-1)$ and $\overline{\Delta B_i}$ is its session mean. This index is defined only for positive increment variance; constant-score histories require a separate convention. Here $B_i(t)$ is the BRS of broadcaster i at session t and $T_{\text{active},i}$ is the number of active broadcast days. The interaction strength W_{ij} requires a documented mapping from the endorsement graph described in Chapter 4 to nonnegative weights. A signed graph cannot be substituted unchanged into an expression whose interpretation assumes nonnegative coupling. A peer-to-peer matrix indexes actors on both axes, whereas this functional requires an actor-by-content matrix; a declared incidence or aggregation rule is needed to connect them. Content quality

must be assessed independently of exposure frequency. Undefined capacities, including zero-variance histories, are reported as undefined rather than replaced silently by arbitrary constants. Active days, session counts, and the time window also need to be fixed before comparing this index across broadcasters. It is a score-volatility index, not measured physical heat capacity.

3.4.4 Implications

EST-P2 proposes two conditional predictions. Under the model's specified reinforcement rules, an architecture without a diversity counterforce is expected to concentrate attention and reputation. Under the specified ranking objective, imposing diversity and quality constraints is expected to reduce raw engagement relative to G0. Neither prediction follows from a physical conservation law: novelty, better matching, or different user preferences could allow diversity and engagement to increase together. Such a finding would challenge the claimed trade-off within the relevant scope, rather than constitute a thermodynamic impossibility. The simulated trade-off index in Chapter 6 describes the selected model and scaling convention; it is not an empirical universal constant.

3.5 EST-P3: Psychometric Accountability

3.5.1 Statement

Consequential metrics must meet psychometric standards commensurate with their consequence. A metric that governs access, visibility, or reputation on an epistemic platform is a credential: it changes what users can do and what they believe, and it

must be validated as a measurement instrument. This proposition applies the Standards for Educational and Psychological Testing (American Educational Research Association et al., 2014) to platform metrics. It has two design consequences. First, the BRS must have an explicit measurement model, quantifiable reliability, and a reported confidence interval rather than a point score. Second, the survey instruments used to evaluate the platform must be calibrated on measured data with reported item statistics, reliability coefficients, and, where the sample permits, confirmatory factor analysis.

EST-P3 also governs the fairness of consequential metrics: a score that rewards accumulated history systematically disadvantages new broadcasters, so the BRS includes career-phase weighting and a Matthew-effect cap (Chapter 4). Convergent self-report evidence, such as the star-rating association, addresses only one part of a validity argument. Consequences, fairness, and accuracy require evidence beyond agreement between audience ratings.

3.5.2 The Explicit Measurement Model

The proposed model for the 11-item broadcaster-rating questionnaire has three correlated factors: Broadcast Rigor, Social Validation, and Information Quality (Chapter 5). The questionnaire is distinct from the algorithmic BRS: sharing component names does not establish equivalence between audience responses and computed platform scores. Its proposed measurement model is

$$\mathbf{x}_u = \boldsymbol{\Lambda}\,\boldsymbol{\xi}_u + \boldsymbol{\delta}_u, \qquad \mathrm{Cov}(\boldsymbol{\xi}) = \boldsymbol{\Phi}, \tag{3.7}$$

In plain terms. The BRS equation turns item ratings into a reputation score with an uncertainty interval. It is a measurement model, not a verdict that a broadcaster is trustworthy in every case.

In ordinary terms: the model asks whether the pattern of 11 answers can be explained by three underlying tendencies plus item-specific error. Loadings describe how strongly each item relates to a proposed tendency. Naming the tendencies does not establish that they exist as distinct traits; the poor reported fit is evidence against accepting this structure uncritically.

where $\mathbf{x}_u \in \mathbb{R}^{11}$ is the item vector, $\boldsymbol{\xi}_u$ is the proposed three-factor vector, $\boldsymbol{\Lambda}$ is the loading matrix, and $\boldsymbol{\delta}_u$ is measurement error. This is a proposed measurement model, not a statement that the observed factor fit is adequate.

In a classical measurement-error model, a composite's standard error of measurement can be written

$$\mathrm{SEM} = \sigma_X \sqrt{1 - r_{XX}}, \tag{3.8}$$

In plain terms. The standard error says how much the BRS would move if the sample changed. A narrow interval means precise measurement, not necessarily validity.

In ordinary terms: this formula estimates individual-score imprecision under a classical measurement model. For illustration, a score SD of 10 and reliability of .84 give SEM=4. It is not the uncertainty of the sample mean, and its assumptions must fit the intended score and population (American Educational Research Association et al., 2014).

where r_{XX} is an appropriate reliability coefficient for that score and popula-

tion. This expression is not the sampling standard error of a mean or a validated broadcaster-specific interval. Chapter 6 reports reliability, associations, and factor-fit summaries; the supplied export does not report a fitted loading matrix or validate confidence intervals for the algorithmic BRS.

3.6 EST-P4: Participation and Evidence-Based Production

3.6.1 Statement

EST-P4 proposes that platform affordances shape opportunities for source appraisal, correction, participation, and evidence-based media production. Reputation cues, access to diverse sources, reflection prompts, and contribution tools may support these activities, but enabling a feature does not guarantee that a participant uses it or produces a better contribution. The proposition concerns observable communication practices within a specified platform environment, not a hierarchy of learner stages or a maximum on individual ability.

3.6.2 Operational Scope

The field questionnaire measures audience evaluations, while the qualitative records describe perceived opportunities and intentions. Neither establishes a causal improvement in task performance. The computational extension tests whether specified configurations change independently keyed claim-appraisal accuracy and separately assessed production, holding the model, source library, initial network, and budgets constant within paired runs.

The combined feedback-and-scheduling comparison addresses a configuration

bundle. Separate effects require the feature ablations specified in Section 5.10.1. Contribution tools are held constant unless an additional task-specific intervention explicitly varies them. A well-implemented comparison showing no meaningful performance advantage, or worse performance, challenges the predicted benefit in that environment. A permission check alone tests access control, not communication quality. Platform access ranks, audience ratings, task performance, and human self-reported competence therefore remain distinct measurements.

3.7 Student Rating: Media Competence and Impact

The source code also uses the identifier `SR` for a posterior over a broadcaster's latent audience rating. That software variable is not the student competence construct developed here. Throughout this thesis, *Student Rating as competence* names the student-level outcome, whereas *backend rating posterior* names this separate software quantity. Published NMLS evidence supports the choice of an external competence instrument; it does not retroactively validate the backend variable.

Student Rating (SR) concerns the student's development as a critical media consumer and responsible contributor. It is not the number of stars the student gives a broadcast. Four objects must be kept distinct: SR concerns the student; BRS concerns the broadcaster; PB concerns the listening experience; and stars provide a global evaluation of an episode. A learner can enjoy a broadcast without developing competence, or develop competence while becoming more critical of the broadcaster. Thus a higher star response is neither a necessary nor a sufficient indicator of improved Student Rating.

3.7.1 A Published Media-Literacy Measure

The selected external instrument is the New Media Literacy Scale (NMLS; Koc & Barut, 2016). Its development and cross-validation involved 1,226 university students in Turkey and supported 35 items across four related factors. It assesses self-reported competence, not demonstrated task mastery. Its five-point agreement format and four-domain structure provide the human measurement basis for a possible later study. Agent answers to these items are secondary synthetic self-description diagnostics, not validated student competence measurements. EST remains the theoretical framework.

Table 3.2: Student Rating Measurement Profile and Distinct Outcomes

Measure	Object of judgment	Role in this thesis
Functional consumption (FC)	Access and understanding	Student competence profile
Critical consumption (CC)	Appraisal of messages and sources	Student competence profile
Functional prosumption (FP)	Producing and circulating media	Student competence profile
Critical prosumption (CP)	Reflective, responsible participation	Human competence domain; synthetic diagnostic
Task performance	Documented decisions and media artifacts	Primary computational accuracy benchmark; separate artifact rubric
PB and star rating	Broadcast experience and overall evaluation	Separate audience outcomes, not SR substitutes
BRS and platform tier	Broadcaster reputation and platform permissions	System variables, not literacy-scale scores

Note. The four competence domains follow NMLS. The computational extension uses independently keyed task accuracy as its primary outcome; none of the agent-generated NMLS domains is designated a validated competence endpoint.

Yavuzalp et al. (2021) examined NMLS in another Turkish university sample (N = 3,395), reporting CFI=.94, TLI=.93, and RMSEA=.08. This is relevant additional evidence, but not proof of equivalence in Egyptian Arabic, of sensitivity to

a brief intervention, or of validity for synthetic respondents. Those interpretations require their own evidence. Appelman and Sundar's (2016) message-credibility measure addresses a different object and cannot serve as a substitute Student Rating scale.

3.7.2 Scoring Without Redefining the Instrument

The full published item set, response anchors, and item order are retained. For each domain, report the item mean on the original 1–5 scale, separately by human or synthetic origin. In the computational extension, all four means are secondary diagnostics. The primary outcome is independent task accuracy, not a blend of literacy claims, satisfaction, engagement, and reputation. Changing the endpoint at this design stage generates no new study results. Appendix H specifies the item-to-domain map and scoring interface.

For a readable optional display, let T_i be the sum of the 35 responses:

$$\mathrm{SR}_{\mathrm{NMLS},i} = 100\frac{T_i - 35}{140}. \tag{3.9}$$

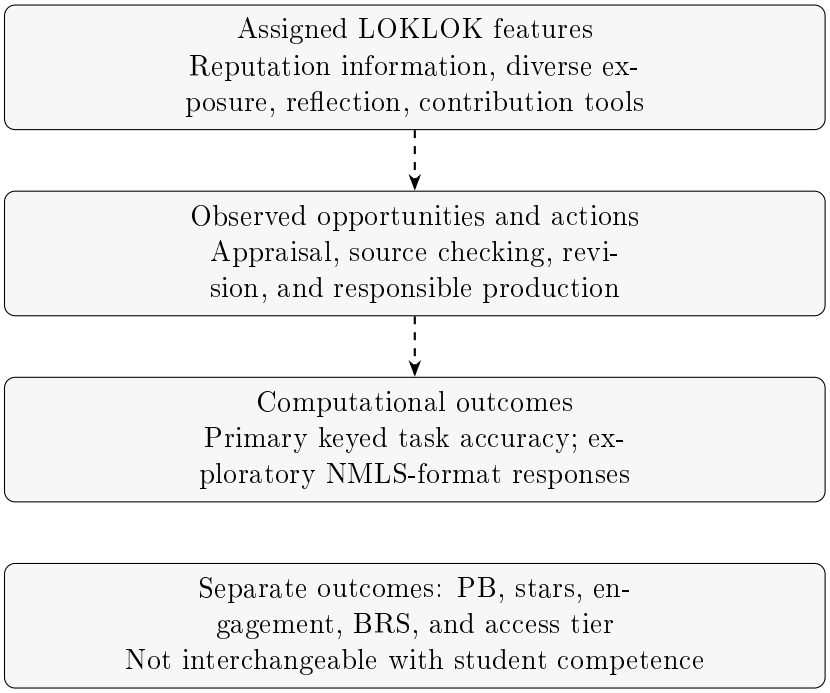
This is a linear rescaling of the full-item sum to 0–100, not a newly validated instrument or a pass threshold. It weights items equally, not the four domains equally. An overall interpretation remains conditional on the measurement structure; the domain profile is retained even when the display is used. Missing responses are not converted to zero, and no abbreviated form is created by dropping inconvenient items. A score is not reconstructed from a student's profile, faculty, PB answers, or quotations.

3.7.3 Meaning of Impact and Connection to EST

The next evaluation is computational: impact means a contrast in recorded agent performance or exposure under assigned platform configurations, conditional on the chosen models, anchors, tasks, and memory rules. It is not human learning merely because the agent has a student persona. A later human impact claim would require its own assigned-condition comparison, pre-exposure competence measures, and independent performance evidence. A delayed human assessment and an agent-memory replay are different designs; neither is inferred from the other.

EST-P1 links specified platform features to opportunities and exposure; EST-P2 concerns diversity and network constraints; EST-P3 concerns whether SR and BRS support their intended interpretations; and EST-P4 concerns the student's opportunities for more demanding participation and production. NMLS concerns human self-reported media competence; agent responses in its format remain exploratory. Platform access ranks, independently assessed task performance, and questionnaire scores remain separate variables. The proposed relationships are shown in Figure 3.3.

Figure 3.3: Student Rating Impact Pathway and Measurement Boundaries



Note. Dashed arrows specify the extended study's hypotheses, not effects identified by the existing ride survey. Agent personas, model, content, and context are recorded design conditions. No arrow represents demonstrated human learning.

3.8 Scope Conditions and Potential Disconfirmation

EST combines explanatory proposals with a normative requirement for measurement accountability. These should not be tested in the same way. A poor instrument can fail H3's validity requirement without disproving the desirability of EST-P3. Conversely, an association consistent with P1 does not establish the stronger proposition that architecture dominates all user-level influences.

Three scope conditions are essential. First, an intervention must change the feature named in the proposition, with exposure measured sufficiently to assess implementation. Second, the outcome must represent the claimed construct: source entropy is not factual accuracy, and PB is not a learning achievement measure.

Table 3.3: EST Propositions: Scope, Tests, and Potential Challenges

EST claim	Testable commitment	Finding that would require revision
EST-P1	A specified feature changes a specified exposure or judgment outcome	A well-powered, well-implemented comparison rules out a meaningful effect; or its direction reverses in the stated context
EST-P2	A defined diversity constraint changes concentration and performance under stated rules	Predicted concentration fails to emerge, or diversity improves without the claimed engagement cost
EST-P3	A consequential score has evidence appropriate to its interpretation and use	Poor measurement structure, criterion performance, or group functioning undermines that use
EST-P4	Specified affordances improve a defined appraisal or production outcome	A sufficiently precise comparison finds no meaningful benefit, or an adverse effect, despite verified delivery

Note. Proposed evaluation criteria, not additional completed tests. The minimum meaningful effect, implementation checks, outcomes, and precision requirements must be set before a confirmatory study.

Third, the comparison must identify an effect rather than conflate treatment with content, prior interest, or who chose to listen. A null estimate with wide uncertainty is inconclusive, not a decisive refutation.

EST should also accommodate interactions. The additive separation in Equation 3.2 is a simplifying representation; it omits architecture–user and architecture–content interactions unless these are modelled explicitly. A future test must not treat such interactions as zero merely because the schematic equation omits them. The field questionnaire cannot estimate the relative causal dominance of these components.

The mechanisms are consequential for mass communication precisely because they connect editorial decisions, platform selection, and audience action. The theory gains explanatory value when it predicts how those relations vary, not when every possible outcome is redescribed as structural determination.

3.9 Summary of the Framework

EST integrates the reviewed concerns into four linked propositions: architectural influence and its proposed dominance (EST-P1), conditional network-diversity constraints (EST-P2), measurement accountability (EST-P3), and affordance-dependent participation and evidence-based production (EST-P4). The proposed dominance of architecture and the benefits of individual affordances remain open to direct tests. The framework contains both empirical commitments and a normative design requirement; a single positive correlation cannot validate them together. Chapter 4 specifies the proposed platform mechanisms, and Chapter 5 distinguishes the evidence available for each assessment.

3.9.1 The Formal Operator Bank

The final computational specification assembles the theoretical mechanisms into eight composable operators. They are introduced here so that the mathematics of Chapter 3 is visible before the implementation in Chapter 5. The operators are formal modelling devices: the Hamiltonian, entropy, and Lyapunov terms are analogies and diagnostics, not claims that social media obey physical thermodynamics. Table 5.8 in Section 5.5.6 gives the governing expressions and their EST mapping.

The bank has three functions. First, it makes the architecture executable: each subsystem corresponds to a named operator rather than a verbal slogan. Second, it separates the mechanisms so that the 2^3 design can switch them on and off independently. Third, it creates falsifiable diagnostics: if an operator has no effect on the measured endpoints under the calibrated model, that is evidence against the

mechanism as specified. The operator bank is therefore a bridge between EST's propositions and the computational test, not a substitute for empirical validation.

Chapter summary. EST is presented as a design theory, not a proven law. The chapter turns its propositions into formal objects: a reputation update, a diversity/echo constraint, a measurement model, and participation rules. The Hamiltonian and Zewail analogies are specification devices. The chapter supplies the hypotheses and the language used in the simulation, and it states what would count as evidence against each proposition.

Chapter 4

The LOKLOK Platform: Architecture and Implementation

Chapter abstract. This chapter describes LOKLOK: the Broadcaster Reputation Score, MORE diversity-constrained recommendation, propaganda detection and moderation, the ERIC governance loop, and the measurement surface. It presents the field interface and explains how BRS, credibility intervals, the Echo Meter, and the survey flow appeared to participants. It separates implemented mechanisms from claims about deployment and usage.

LOKLOK is the proposed operational platform for this research programme. Its implementation account describes a Go backend, mobile client, and web interface. The following chapter distinguishes proposed reference mechanisms, reported implementation behavior, and requirements for experimental fidelity. Neither a platform feature nor a mathematical analogy establishes an EST proposition without a suitable test.

4.1 Implementation Fidelity and the Scope of the BRS

The inherited implementation account describes distinct scoring models. Its source-inspection report identifies a `BRS.Update` function called by a feedback service; that source tree is absent from the present archive and these claims have not been reverified. It accumulates a scalar rating through a weighted beta-shaped state:

$$a_{t+1} = a_t + w_t r_t, \qquad b_{t+1} = b_t + w_t(1 - r_t), \qquad \widehat{B}_{t+1} = \frac{a_{t+1}}{a_{t+1} + b_{t+1}}. \tag{4.1}$$

In ordinary terms: the two counters accumulate favourable and unfavourable fractions of feedback. Their ratio produces the displayed score. For example, a rating of .8 with weight 1 adds .8 to the first counter and .2 to the second. This arithmetic does not turn audience approval into truth.

Here $a_0, b_0 > 0$, $w_t \geq 0$, and $r_t \in [0, 1]$ are required. A five-star input is not already on this scale; its actual mapping, prior, and weighting policy must be checked in the version used for any experiment. Fractional or weighted increments can define a useful pseudocount score, but do not by themselves establish a calibrated Bayesian posterior about factual credibility. An interval based on these counts describes the assumed scoring model, not automatically the accuracy of a broadcaster's claims.

The feedback service starts new states at $(a_0, b_0) = (2, 2)$. Its reported interval clips the normal approximation $\widehat{B} \pm 1.96\sqrt{\mathrm{Var}(B)}$ to $[0, 1]$; it is not an exact beta-quantile interval. The evidence field $a+b$ includes prior mass and weights, so it is not a respondent count. A separate score route persists the component smoother $B_{t+1} = \mathrm{clip}_{[0,1]}\{B_t + \eta[.40R_t + .35I_t + .25S_t - B_t]\}$. It neither updates the beta state

Table 4.1: Four Distinct Meanings of BRS in the Research Programme

Object	Evidence or specification	Permissible interpretation
Questionnaire battery	Eleven BR, SV, and IQ responses	Listener judgments; not a backend trace
EST reference model	Weighted components and anchored network equations	Proposed mechanism with stated mathematical conditions
Backend scoring paths	Scalar beta-shaped feedback and a separate component smoother	Source-inspected algorithms; neither is the validated latent reference model
Historical simulation	Preserved calibration and contrast summaries	Conditional model results; runtime equivalence not independently established

nor fits the proposed latent model. The experiment must name the chosen path; their scores are not interchangeable.

The unanchored `DeGroot` utility weights neighbors by squared expertise, not by the softmax stated in its comment. It has no production feedback call site in the supplied Go files. With zero total neighbor expertise, its neighbor contribution is zero, so the update reduces to $(1 - a)b_t$ rather than a normalized opinion average. The anchored contraction in Chapter 3 therefore cannot be attributed to this utility without checking its assumptions and implementation separately.

Before the proposed LLM extension, a versioned integration test must trace a known rating through the client adapter, feedback service, score state, and ranking output. It must identify whether the tested intervention uses the scalar feedback path, the component smoother, or a reference mechanism. A result from one must not be attributed to the other. Test logs, input bounds, normalisation, and fallback behavior belong in the experiment archive.

The consolidated status of the BRS across six dimensions—theoretical specification, software implementation, internal-consistency reliability, construct validity,

predictive validity, and simulation utility—is recorded in one place in Table J.3 (Appendix J.4); no later chapter is to be read as implying outcomes beyond what that table states.

4.2 The Broadcaster Reputation Score (BRS)

The reference BRS is a proposed consequential metric with three components corresponding to the intended, but not confirmed, factors of the measurement model: broadcast rigor, social validation, and information quality.

Broadcast Rigor is specified as a combination of streaming activity, audience retention, and schedule consistency:

$$B_u^{\text{rig}}(t) = \sigma\left(\int_0^t e^{-\lambda(t-\tau)}[\alpha_1 \,\text{StreamHours} + \alpha_2 \, L_{\text{retention}} + \alpha_3 \, \delta_{\text{schedule}}^{-1}] \, d\tau\right). \quad (4.2)$$

In ordinary terms: the proposed rule accumulates streaming activity, retention, and schedule consistency, discounts older information, then maps the total to a bounded score. These inputs must be defined on compatible scales as functions of τ; δ_{schedule} must be positive. Activity and retention alone do not establish broadcast rigor.

Social Validation is a PageRank-style trust score with a bounded endorsement cap $W_{vu}^{\text{capped}} = \min(W_{vu}, \bar{W}/(1 + \beta B_v))$ intended to limit Matthew-effect concentration, and a Gini trigger $G(t) > 0.40$ that activates a freshers boost for under-recognised new broadcasters.

Information Quality is the quality-weighted entropy of the source set with

a correction-term bonus:

$$B_u^{\inf}(t) = \mathbb{H}_Q(\mathcal{S}_u) + \gamma \cdot \frac{\text{CorrMade}_u}{\text{TimeToCorr}_u \cdot \text{CommValid}_u}. \tag{4.3}$$

In ordinary terms: the reference expression combines a source-set summary with a correction term. More corrections increase that term and longer correction delays reduce it, all else equal. As written, a larger CommValid denominator also reduces it; this does not automatically reward stronger community validation. The earlier specification therefore needs a justified definition and directional check before implementation.

The correction term requires a defined scale and positive denominators; it must not reward a broadcaster simply for producing more errors to correct. These safeguards are design requirements, not effects established by the field questionnaire.

The three components are aggregated using career-phase weights:

$$B_u(t) = \mathbf{w}(t)^\top \mathbf{B}_u(t).$$

The vector $\mathbf{w}(t)$ transitions from $(0.6, 0.2, 0.2)^\top$ for new broadcasters to $(0.2, 0.5, 0.3)^\top$ for established ones. These weights are design parameters, not estimates from the audience survey. The interface is intended to display uncertainty alongside the point score. A beta posterior would require an appropriate likelihood and prior; a normal interval of the form $\hat{B} \pm 1.96\,\text{SE}(\hat{B})$ is a different construction and cannot simply be called a beta-conjugate credibility interval. Neither the reported CFA summaries nor the questionnaire alpha establishes a calibrated

interval for the algorithmic BRS. The implementation should define its score scale, likelihood, and uncertainty method before consequential use.

4.3 Recommendation with Diversity Constraints (MORE)

The Multi-Objective Recommendation Engine ranks candidate content by a weighted combination of user value and diversity, subject to a fairness budget. User value combines the propaganda-adjusted probability of the item with the broadcaster's BRS:

$$\mathrm{UVS}(c, u) = P_{\mathrm{DTS}}(c, u) \cdot B_c \cdot (1 - P_{\mathrm{prop}}(c)). \tag{4.4}$$

In plain terms. The utility score ranks candidate broadcasts by expected value, diversity, and a penalty term. It is the app's "what should I show next" rule.

4.3.1 Time-Resolved Attentional Coherence: A Zewail-Inspired Formal Analogy

The quantity $P_{\mathrm{DTS}}(c, u)$ in Equation (4.4) requires an explicit definition. A commuter with the radio application running is not necessarily attending to the broadcast: the phone may be playing in the background while the rider talks, checks messages, or watches traffic. The present framework therefore treats attention as a *time-resolved coupling state* between a listener u and a broadcast c, rather than as a synonym for ride duration or application presence.

The formal inspiration is Ahmed H. Zewail's femtochemistry, recognised by

the 1999 Nobel Prize in Chemistry. In pump–probe spectroscopy, a pump pulse launches a coherent wavepacket at t_0; a delayed probe pulse measures the transition-state population at $t_0 + \Delta t$ before vibrational dephasing destroys coherence (Zewail, 1999, 2000). The analogy used here is deliberately formal: it supplies a principled way to model a short-lived attentional window, not a claim that social listening obeys physical thermodynamics or chemical kinetics.

Let t_{pump} be the timestamp of a broadcast inflection cue (a claim assertion, a topic shift, or a source prompt), and let $t_{\text{probe}} = t_{\text{pump}} + \Delta t_{u,c}$ be the listener's foreground interaction, telemetry ping, or EMA Layer-1 response. The pump–probe attentional coherence gate is

$$P_{\text{DTS}}(c, u, \Delta t) = \underbrace{\exp\left(-\frac{\Delta t_{u,c}}{\tau_{\text{dephase},u}}\right)}_{\text{coherence decay}} \cdot \underbrace{\exp\left(-\frac{E_{\text{distract}}(u,t)}{T_{\text{cog},u}}\right)}_{\text{ride-distraction barrier}} \cdot \underbrace{\frac{1}{2}\left[1 + \cos \Delta\phi_{u,c}(t)\right]}_{\text{phase locking}}, \tag{4.5}$$

where $\tau_{\text{dephase},u} > 0$ is the listener's dephasing time, E_{distract} is ambient transit friction (traffic noise, multi-tasking, conversation), T_{cog} is the listener's cognitive engagement energy, and $\Delta\phi_{u,c}(t)$ is the phase difference between the broadcast cadence and the listener's foreground focus. The gate lies in $[0, 1]$: an active, phase-locked listener probed within the coherence window has $P_{\text{DTS}} \to 1$; a phone left playing in the background for several minutes has $P_{\text{DTS}} \to 0$. The numerical values used in the simulation are declared design parameters, not measured chemical quantities.

In plain terms. The gate is a formal way to say three common-sense things. First, attention fades as time passes. Second, a noisy or distracting ride lowers attention. Third, attention is stronger when the broadcast rhythm and the listener focus are

aligned. The equation multiplies these three factors, so any one of them can switch attention off.

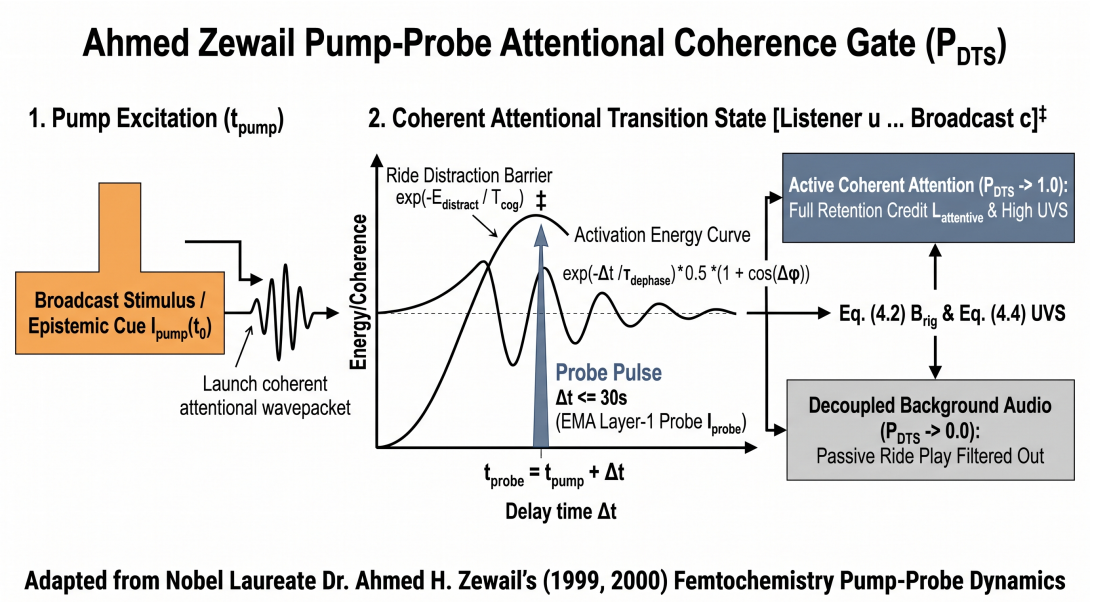


Figure 4.1: The Zewail-inspired pump–probe attentional coherence gate. The diagram is a formal analogy: the broadcast inflection cue is the pump, the EMA probe is the delayed measurement, and dephasing represents loss of foreground attention during a ride. It is not a claim that social listening is a chemical process.

The gate enters the model in three places. First, it multiplies the retention term in the broadcaster-rigour update so that passive background play cannot inflate apparent exposure. Second, it scales the evidence anchor used by the DeGroot reputation update. Third, it modulates engagement and the latent credibility trajectory. Figure 4.2 shows the resulting decay curves and the active-versus-passive contrast. The implementation is included in `loklok_pipeline.py` (Section K), so the analogy is auditable rather than purely rhetorical.

Interpretive boundary. The Zewail analogy is used because it makes the temporal structure of attention explicit and falsifiable: if the gate has no effect on retention, reputation, or ratings, the mechanism should be removed. It does not

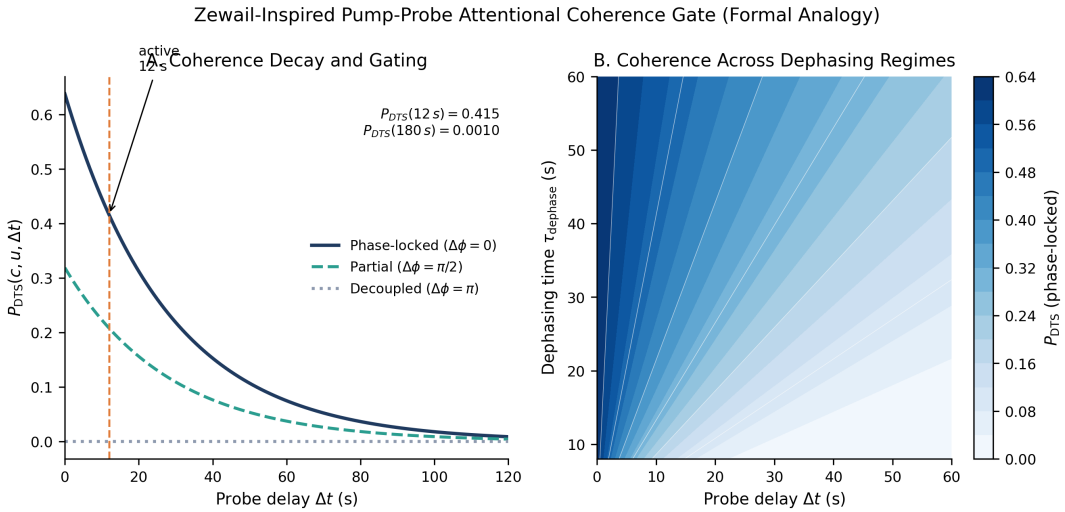


Figure 4.2: Model-based Zewail gate diagnostics. Panel A: coherence decay for phase-locked, partial, and decoupled attention; the 12-second active probe and the 180-second passive background case are marked. Panel B: coherence across dephasing regimes. These are simulation diagnostics, not human measurements.

establish that the numerical dephasing time is a psychological constant, nor that the chemical formalism is uniquely correct. The empirical field wave does not contain the timestamped telemetry needed to estimate τ_{dephase}; that parameter remains a model assumption and a target for future platform-in-the-loop measurement.

Gate ablation. To test whether the gate is doing any work, the main simulator was rerun with the gate off, with a simple exponential decay, and with half and double the dephasing time, holding personas and seed fixed. The full gate gives $\Delta\text{SR}_{\text{eval}} = +0.827$ [0.779, 0.875]. Turning the gate off raises the contrast by +0.194 [0.127, 0.260] and the competence gain by about 3.25 points; replacing it with a simple exponential decay raises the contrast by +0.110 [0.043, 0.177]. The full gate is therefore not inert, and its distraction/phase terms carry structure beyond a simple decay. The baseline gate is open in only about 15.4% of persona–session slots, so the gate is a strong attentional filter, not a minor adjustment. It reduces

the headline contrast relative to no gate, but the sign remains positive. This is a model-internal ablation, not a human test of the pump–probe analogy; no empirical estimate of the dephasing time exists in the field wave.

Table 4.2: Zewail Gate Ablation on the Main Simulator (Model-Based)

Mode	Gate mean	$\Delta\text{SR}_{\text{eval}}$ [95% CI]	$\Delta\text{SR}_{\text{comp}}$	Δ engagement
Full Zewail gate	0.154	+0.827 [0.779, 0.875]	+11.30	−0.183
Gate off	1.000	+1.020 [0.968, 1.073]	+14.55	−0.171
Simple exponential decay	0.535	+0.937 [0.885, 0.990]	+13.06	−0.172
Half dephasing time	0.088	+0.809 [0.760, 0.858]	+11.03	−0.183
Double dephasing time	0.208	+0.850 [0.802, 0.898]	+11.64	−0.183

Note. Same 500 personas and seed across modes. Paired differences versus the full gate: gate off +0.194 *[0.127, 0.260]; simple decay* +0.110 *[0.043, 0.177]; half dephasing* −0.018 *[*−0.031, −0.005*]; double dephasing* +0.023 *[0.011, 0.036]. This is a model-internal ablation, not a human attention experiment.*

Diversity is measured as the cosine dissimilarity of the item from the user's recent consumption vector, $\text{DBI}(c, u) = 1 - \cos(\mathbf{v}_c, \mathbf{C}_{\text{rec}})$. The engine solves the constrained optimisation

$$c^* = \arg\max_{c \in \mathcal{C}} \left(w_1 \cdot \text{UVS} + w_2 \cdot \text{DBI}\right) \quad \text{s.t.} \quad |\text{SAI}| \leq B_{\max}, \tag{4.6}$$

In plain terms. MORE chooses a slate that balances relevance and diversity. The diversity weight is the dial: turn it up and the slate becomes broader but less tailored.

In ordinary terms: choose the item with the highest weighted value and diversity score among candidates meeting the declared balance constraint. The weights determine the trade-off; the constraint determines eligibility. This is a proposed decision rule, not evidence that a deployed feed enforces it.

where SAI is a demographic-balance index and $B_{\max}$ the bias budget. The

Echo Meter R_t is updated after each recommendation, $R_t \leftarrow (1-\alpha_R)R_{t-1} + \alpha_R(1 - \text{DBI}(c^*, u))$. When $R_t \geq 0.60$ the platform triggers an Echo Meter warning survey (EMA Layer 2); when $R_t \geq 0.75$, Algorithmic Cooling activates and the recommendation switches to maximum diversity, a proposed interpretation of the EST-P2 counterforce, not a verified implementation or measured effect.

4.4 Propaganda Detection and Algorithmic Moderation

Every content item is scored on seven features: citation velocity, quality-weighted entropy, semantic fear bias, propagation acceleration, institutional concentration (HHI), predatory-source ratio, and correction velocity. The specification uses a logistic score, $\hat{p}(c) = \sigma(\boldsymbol{\theta}^\top \mathbf{f}(c))$, and combines it with a temporal anomaly score. A logistic transformation alone does not establish probability calibration; a separate calibration dataset and assessment would be needed before describing the output as calibrated. The proposed combined index is

$$P_{\text{prop}}(c) = 0.7\,\hat{p}(c) + 0.3\,\phi^{\text{TIS}}(c). \tag{4.7}$$

If $P_{\text{prop}} > 0.70$ the item receives a visibility penalty, an applied Moderation Cost $E_{a,\text{mod}} = \lambda_1(P_{\text{prop}} - 0.70)$ embedded in the feed rank. This is a proposed ranking penalty, not an established economic welfare correction or a calibrated probability of intent. The anomaly score must also be bounded on $[0, 1]$ for the combined index to remain in that range. These author-defined thresholds require evaluation. If the score exceeds 0.85 the item is escalated to ERIC for human review and the broadcast

logs a BRS information-quality penalty.

4.5 The ERIC Governance Loop

ERIC is the platform's integrity controller. It monitors five health indicators (filter-bubble score, information diversity, inequality of reputation, detection accuracy, platform sentiment) and selects the minimum-cost intervention, a diversity injection, a friction pause, or a human review, whenever a threshold is crossed. The intended decision rule is to compare changes in a specified governance objective, with its components, scales, and costs recorded. A decrease in the candidate Hamiltonian is not by itself evidence of improved credibility or factual accuracy. Any use of the entropy-regularised objective must separately define the loss and source distribution, as in Appendix E. An anti-bot module assigns cryptographic identities, monitors behavioural fingerprints, and flags coordination patterns (Luceri et al., 2023); its output feeds the reputation and moderation subsystems rather than making autonomous removal decisions.

4.6 Measurement Surface

The platform specification includes the following measurement services. They describe the wider design, not a claim that every service was used in the present cross-sectional survey. In particular, the field sample is not a session-randomised trial, and no weekly or pre-broadcast panel results are added by describing these capabilities:

- **Layer 1 – Post-broadcast micro-survey (PB1–PB5):** fired within sec-

onds of broadcast completion, measuring perceived confidence, diversity, comparative trust, structural clarity, and overall epistemic quality.

- **Layer 2 – Echo Meter survey (EM1–EM3):** fired on the Echo Meter trigger θ described above.
- **Layer 3 – Weekly survey (WK1–WK15):** longitudinal trust and media-literacy battery.
- **Layer 4 – Pre-broadcast confidence survey (PR1–PR3):** broadcaster-side confidence and verification disclosures.
- **Experiment arms:** a random 50/50 assignment places each session into an epistemic-first (G1) or engagement-first (G0) configuration, with the assignment logged at the session level.
- **BRS and echo cards:** the Flutter and SPA clients render the BRS point estimate with a proposed uncertainty interval and the Echo Meter. Their interpretation requires a valid estimation procedure and records of which displays each respondent actually encountered.

The full item text of every layer is given in Appendix A.

4.7 Implementation Evidence and Its Limits

The historical implementation account describes a Go backend, a static web client, a mobile client design, and supporting services. Their source trees, test logs, and deployment records are not included in the present archive. This version therefore makes no fresh claim about source-file counts, passing tests, live routes, or feature completeness. The mechanism equations are specifications, not proof that a deployed

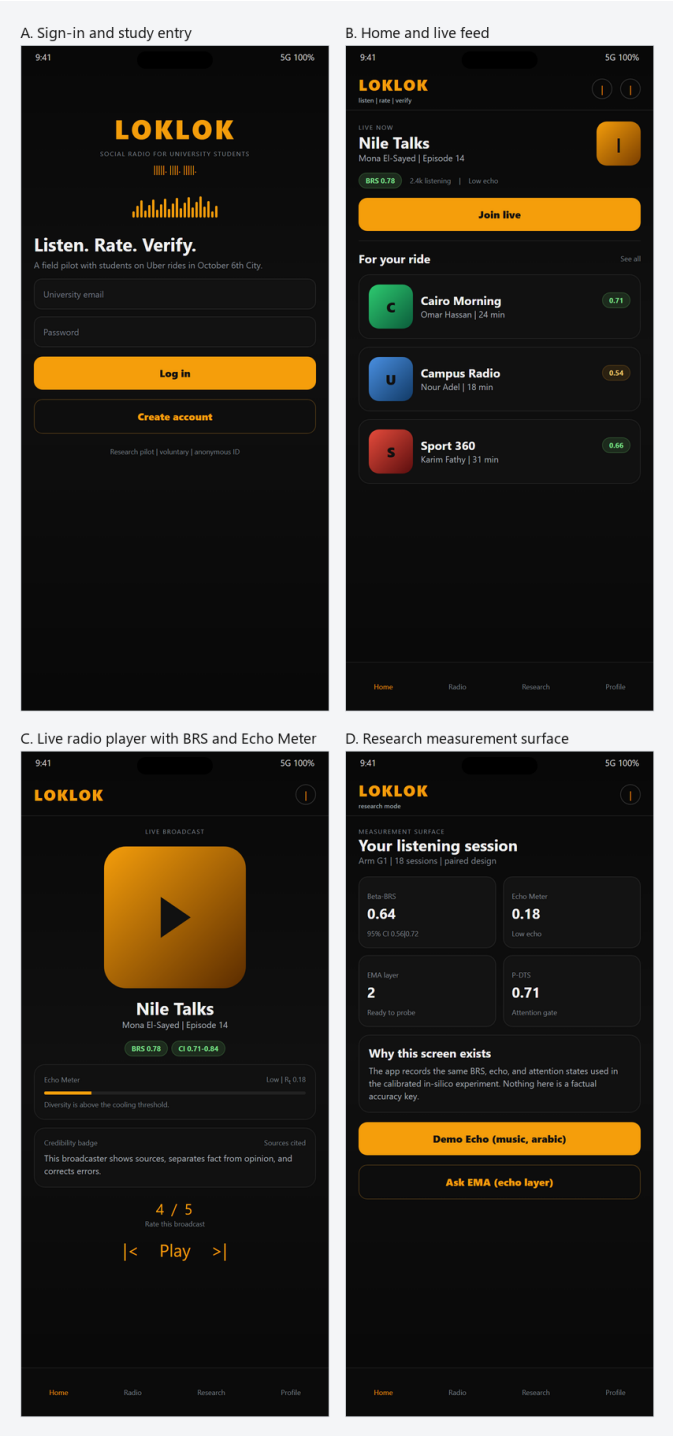


Figure 4.3: LOKLOK field interface used in the study. Panel A: sign-in and study entry. Panel B: home and live feed with BRS badges. Panel C: live radio player with BRS interval, credibility badge, and Echo Meter. Panel D: research measurement surface showing arm, Beta-BRS, echo, EMA layer, and the Zewail attention gate. The interface is shown here as the field protocol instrument, not as evidence that every measurement service was used in the cross-sectional wave.

system executes them. A versioned source-to-feature mapping must precede the proposed experiment.

For the field report, only the administered questionnaire and documented listening context are observations. Weekly tracking, readiness gates, automated moderation, scheduling variants, and production assessment are not added to the measured wave by describing them. They are relevant here only as candidate features or controls in the prospective extension.

4.8 Domain Model and Data Flow

The following entities describe a proposed shared schema for the platform and model. They do not establish that the historical engine and actual backend use identical data transformations. The core entities are: **User** (identity, career phase, institution); **Broadcaster** (a user with a BRS trajectory); **Broadcast** (session, schedule slot, transcript embedding, source list); **Rating** (the post-broadcast micro-survey and star record); **EchoSample** (the Echo Meter rolling values and EMA trigger history); and **GovernanceEvent** (moderation, anti-bot, and ERIC ledger records). The intended experimental event flow connects assignment, content exposure, rating, feedback, and logged outcomes. In the supplied source these are separate routes and services, not a demonstrated end-to-end G0/G1 experiment. Per-user A/B assignment and the radio service's shared current arm are separate states. The feed ranking path does not read the stored assignment. Consequently, assignment alone does not establish that a user received the intended intervention. The proposed study identifies treatment by both the assigned configuration and recorded exposures, with

independently reset condition-specific deployments as described in Section 4.13.

4.9 Formal Algorithm Specifications

The core mechanisms below are reference specifications, not evidence that the back-end and historical simulator execute identical updates. Their normalisation, input definitions, and fidelity tests must be fixed before use in the new platform experiment (Section 4.1).

Table 4.3: Algorithm 1: Proposed Component-Based BRS Update

Component	Specification
Input	User u, broadcast session s, transcript embedding $\mathbf{h}_s \in \mathbb{R}^{768}$, listener retention $L_s \in [0,1]$, source list $\mathcal{S}_s$
Step 1	Compute the reference Broadcast Rigor using Equation (4.2), with declared time and input scales
Step 2	Compute the reference Information Quality using Equation (4.3); define weighted entropy and audit the correction term before use
Step 3	Compute Social Validation delta: $\Delta B^{\text{soc}} = \alpha_{\text{page}} r_u^* - \beta \max(0, B_u - \bar{B})$ (scalar stationary endorsement r_u^*; proposed concentration penalty)
Step 4	Apply temporal decay: $B_u(t+1) = (1-\lambda)B_u(t) + \lambda \cdot \mathbf{w}(t)^\top [B_s^{\text{rig}}, \Delta B^{\text{soc}}, B_s^{\text{inf}}]$
Step 5	Specify and test an uncertainty model on the score scale before displaying intervals; survey alpha does not provide a broadcaster-specific standard error
Output	Proposed updated score and audit trail; no validated interval or latent residual is implied

4.10 Relation to the Research Programme

LOKLOK gives the research programme a concrete design specification. BRS measurement motivates H3; the diversity constraint motivates H2; endorsement and schedule rules motivate H5 and H6; and participation tools motivate H7's prospective task-performance comparison. Access ranks are not its outcome; independent

Table 4.4: Algorithm 2: MORE Recommendation (Multi-Objective Epistemic Ranking)

Component	Specification
Input	User u, candidate content set $\mathcal{C}$, recent consumption vector $\mathbf{C}_{\text{rec}}$, Bias Budget $B_{\max}$
Step 1	Compute UVS: $\text{UVS}(c,u) = P_{\text{DTS}}(c,u) \cdot B_c \cdot (1 - P_{\text{prop}}(c))$
Step 2	Compute DBI: $\text{DBI}(c,u) = 1 - \cos(\mathbf{v}_c, \mathbf{C}_{\text{rec}})$
Step 3	Solve constrained optimisation: $c^* = \arg\max_{c \in \mathcal{C}}(w_1 \cdot \text{UVS} + w_2 \cdot \text{DBI})$ s.t. $\lvert\text{SAI}\rvert \leq B_{\max}$
Step 4	Update Echo Meter: $R_t \leftarrow (1 - \alpha_R)R_{t-1} + \alpha_R \cdot (1 - \text{DBI}(c^*, u))$
Step 5	**If** $R_t \geq 0.60$: trigger EMA Layer 2 survey; **If** $R_t \geq 0.75$: activate Cooling, $c^* \leftarrow \arg\max_{c:\, \lvert\text{SAI}\rvert \leq B_{\max}} \text{DBI}$
Output	Ranked content item c^*, updated R_t, EMA trigger flag

Table 4.5: Algorithm 3: Proposed Moderation-Risk Scoring

Component	Specification
Input	Content item c, feature vector $\mathbf{f}(c) \in \mathbb{R}^7$: [citation velocity, quality-entropy, semantic fear-bias, propagation acceleration, institutional HHI, predatory source ratio, correction velocity]
Step 1	Logistic score: $\hat{p}(c) = \sigma(\boldsymbol{\theta}^\top \mathbf{f}(c))$; probability calibration remains unverified
Step 2	Temporal anomaly check: $\phi^{\text{TIS}}(c) = \mathbf{1}\left[\frac{C(t) - \hat{\mu}_{\text{field}}}{\hat{\sigma}_{\text{field}}} > z_{0.95}\right]$
Step 3	Combined score: $P_{\text{prop}}(c) = 0.7\hat{p}(c) + 0.3\phi^{\text{TIS}}(c)$
Step 4	**If** $P_{\text{prop}} > 0.70$: apply Moderation Cost $E_{a,\text{mod}} = \lambda_1(P_{\text{prop}} - 0.70)$ to feed rank
Step 5	**If** $P_{\text{prop}} > 0.85$: flag for human review; log to BRS Information Quality penalty
Output	$P_{\text{prop}}(c)$, visibility penalty, escalation flag

Table 4.6: Algorithm 4: Proposed ERIC Governance Feedback

Component	Specification
Input	Health indicators $[\phi_{\text{bubble}}, \phi_{\text{diversity}}, \phi_{\text{ineq}}, \phi_{\text{acc}}, \phi_{\text{sent}}]$, threshold vector $\boldsymbol{\tau}$
Step 1	Compute the specified governance objective and its separate quality, diversity, and cost components
Step 2	Identify crossed thresholds: $\{k : \phi_k > \tau_k\}$
Step 3	Select an intervention intended to improve the objective: diversity injection, a friction pause, or human review; compare their specified costs
Step 4	Execute intervention; log decision and state to the governance ledger
Step 5	Route bot and coordination detections to the moderation and reputation subsystems
Output	Intervention record, updated governance state

appraisal and production assessments are required. This mapping is not evidence that each feature was independently experienced or evaluated by all 52 respondents. The field questionnaire concerns audience evaluations, whereas feature exposure, implementation fidelity, and event-level behavior require their own records. The calibrated model in Chapter 5 represents selected mechanisms on a simulated substrate, not an additional field sample.

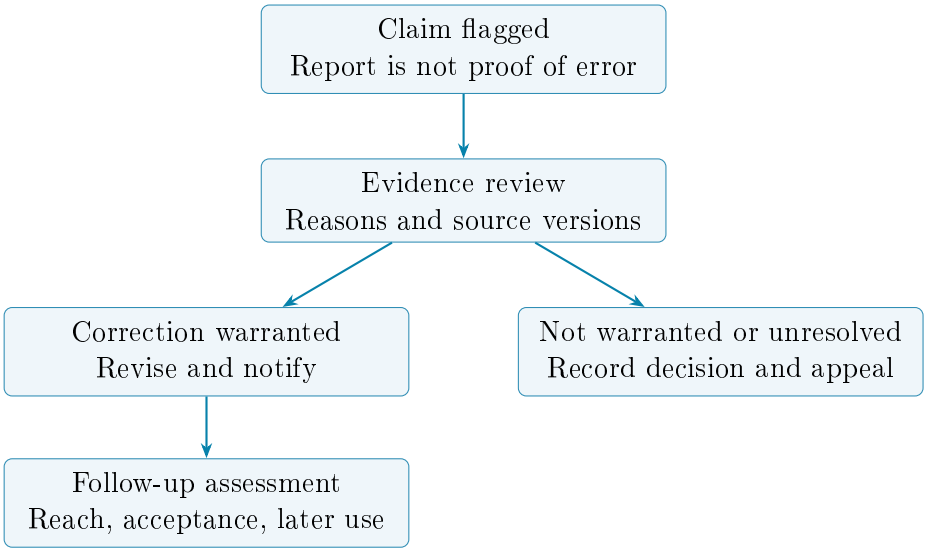
Software implementation, measurement validity, and beneficial audience effects are separate accomplishments. A working classifier needs labelled evaluation data and calibration checks before its output can be interpreted as a probability of misinformation. A reputation formula needs range, monotonicity, sensitivity, and fairness checks before consequential deployment. For example, placing community validation in a denominator does not by itself guarantee that greater validation increases a score. Human review and documented appeal routes remain necessary design safeguards. These requirements do not change the historical model outputs reported in Chapter 6.

4.11 Correction, Provenance, and Governance Feedback

Figure 4.4 applies the event notation of Section 3.3 to correction. A flag is not a finding of falsehood, and a published correction is not evidence of acceptance. Ecker et al. (2022) motivate separating correction exposure from belief revision; the diagram is this thesis's design mapping, not their effect estimate.

The proposed system should log review reasons, evidence, decisions, revised versions, and notification routes. Reputation changes must distinguish an original

Figure 4.4: Evidence Review and Correction Workflow



Note. Author-developed workflow, not field trajectories. A flag may be dismissed or remain unresolved; only a warranted correction enters the correction branch. Later evaluation separates reach, acceptance, and subsequent use.

error from the quality of its correction; otherwise repeated creation and correction of avoidable errors could be rewarded. Accountable moderation also includes an appeal process. A classifier score or majority vote is not itself verified factual accuracy.

Evaluation should distinguish correction reach, recognition, acceptance, and later use, each with its own denominator and time window. Removing a post can change visibility without changing belief. Slower resharing can reflect friction rather than better judgment. These are separate evaluation outcomes, not additional findings attributed to the current survey.

4.12 Scheduling Design for the Computational Extension

The supplied radio scheduling specification motivates scheduling that balances familiar programming, freshness, topical relevance, and repetition. These objectives connect to the qualitative themes of journey fit and local relevance. The experiment

retains a narrower comparison of fixed scheduling against Dirichlet category scheduling; additional adaptive controllers are not part of this specified contrast. Source proposals do not establish that these features are already deployed or effective.

Table 4.7: Scheduling Rules and Experimental Requirements

Design component	Operational requirement
Fixed-schedule comparison	Freeze the reference schedule and apply the same content eligibility constraints in both arms
Dirichlet category mixture	Draw once per specified block; hold the mixture constant within the block, restricting and renormalising over eligible categories at each slot
Category parameters	Positive parameters remain fixed within each run; the supplied $(2.5, 4.0, 1.5, 3.0)$ are candidate settings, not estimated audience preferences
Repetition constraints	Freeze cooldown and artist-separation rules; distinguish elapsed time from slot counts and log every eligibility decision
Empty eligible set	Apply a specified, logged reserve-content or pause rule rather than divide by zero or silently override constraints

Section 5.10.1 defines the paired-network comparison and feedback-by-scheduling ablations. Scheduling parameters, block length, and eligibility rules must be frozen before the main experiment. Event logs must identify the eligible set, selected category and track, sampling probability, block, and any fallback. The backend's existing selection rules and the experiment's specified scheduler are not assumed to be interchangeable.

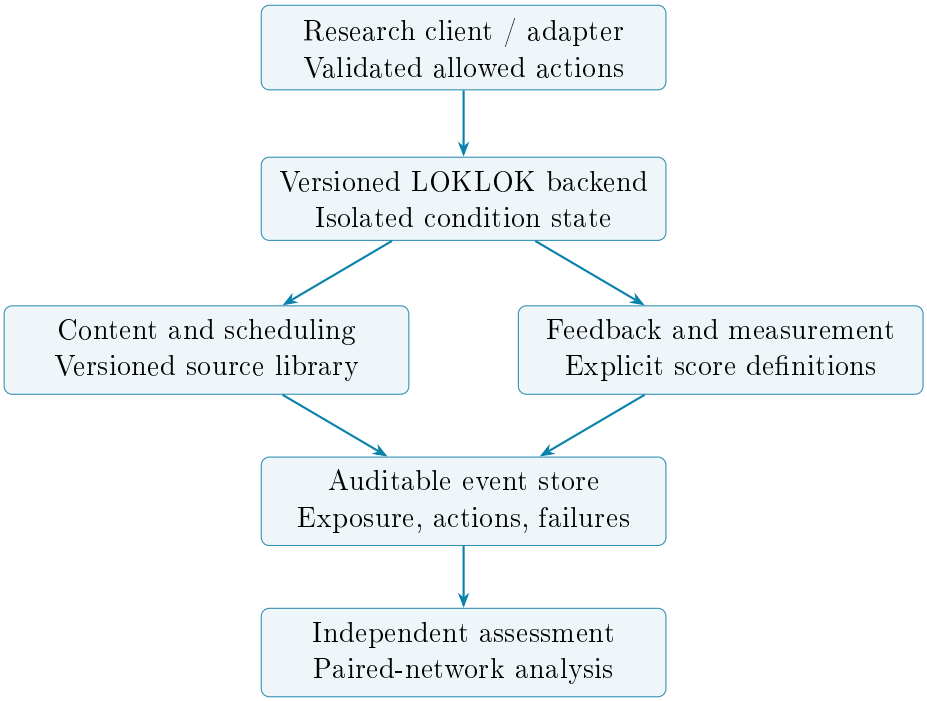
Russo et al. (2018) describe Thompson sampling as a method that learns from feedback while balancing exploration and exploitation. Drawing category weights from a fixed Dirichlet distribution alone does not implement that learning procedure. The specified extension therefore makes no Thompson sampling

claim and does not add a separate mood optimiser, rank-based exposure controller, or fatigue-feedback update to the treatment bundle. Network diversity is evaluated from recorded events rather than assuming a power-law distribution. No historical simulation result is relabelled as the effect of this new scheduling experiment.

4.13 Platform Integration for the Prospective Experiment

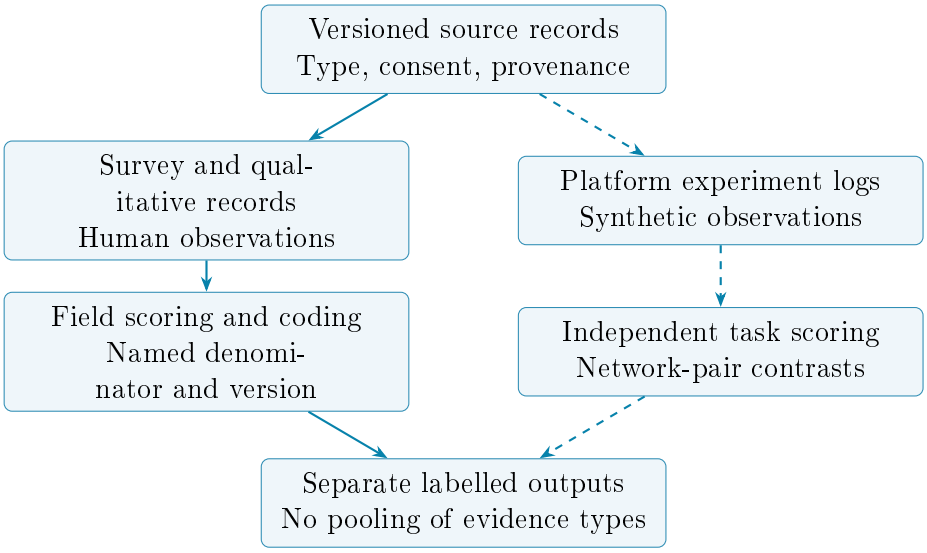
The prospective architecture links an audio setting with posts, audience feedback, and recorded platform actions. A shared station and an individual feed define different exposure units: a station change affects co-listeners, whereas a feed change may be account-specific. The experiment must record both network membership and delivered exposure rather than infer treatment from an assignment label alone.

Figure 4.5: Prospective LOKLOK Experiment Architecture



Note. Integration specification, not verified deployment. Actual source versions, routes, condition isolation, and action logs must be checked before execution.

Figure 4.6: Separated Human and Computational Evidence Flows



Note. Arrows describe data handling, not causality. Dashed paths remain prospective. No field and synthetic records are pooled.

4.13.1 Assignment and Treatment Delivery

The restored implementation must map G0 and G1 to explicit features, isolate condition state, and log the content and feedback actually delivered. Historical A/B software labels are not assumed to match the simulation's G0/G1 labels. An adapter must validate allowed actions against the real backend contract; this thesis does not invent executable routes. Each paired run starts from matched recorded state and a frozen roster. The inference unit is an independently reset network pair, not an individual action.

4.13.2 Measurement Identity

The administered PB questionnaire, platform EMA windows, scalar feedback, component BRS, and prospective NMLS assessment need separate schema identifiers. Original answers must be retained before normalization or aggregation. A 1–5 star

rating cannot be passed unchanged to a feedback function expecting values in $[0, 1]$. The transformation must be declared, tested, and logged. No transformation alone establishes measurement validity.

4.13.3 Readiness Criteria

Before a live run, verify authentication, condition isolation, action logging, versioned content, independent assessment keys, prohibited-information access, retry limits, and failure handling. A successful test establishes only the behavior it observes. Model identity, prompts, source access, resource budgets, and actual served versions remain part of the execution record. No provider API call or platform run has been performed for this document revision.

Chapter summary. The platform chapter specifies the architecture and the field interface. BRS, MORE, ERIC, moderation, and the measurement surface are defined, and the interface screens show how they were presented. The chapter is explicit that the interface is a protocol instrument, not proof that every service was used in the field wave; implementation evidence has limits.

Chapter 5

Methodology

Chapter abstract. This chapter details the field wave, the calibration procedure, and the in-silico procedure. It explains sampling, consent, ride context, the listening episode, questionnaire blocks, the 60-to-52 screen, and quality checks. It then explains the embedding, debiasing, manifold, persona, factorial ABM, and measurement steps, with plain-language notes on each formal stage. It states the design limitations and the analysis-plan discipline.

The empirical programme combines quantitative and qualitative field material, then distinguishes these human records from model-based evidence. The first is a measured field wave, a cross-sectional questionnaire administered to university students who listen to social media radio during rides with Uber, accompanied by 60 researcher-recorded qualitative case accounts. The questionnaire provides audience and measurement summaries and supplies the target distribution for calibration. The second is a calibrated in-silico experiment, a two-arm agent-based simulation representing alternative platform configurations whose parameters were calibrated to the measured wave, which supplies a model-conditional architectural counterfactual. A further LLM-agent platform extension is specified but has not been conducted. A separate synthesis of prototype FGD narratives informs prospective rule-based

and LLM-agent design without adding participants to the field wave. This chapter describes the evidence sources and their integration. Their separation is maintained throughout: measured values are reported as measured, and simulated values are reported as counterfactual estimates.

5.1 Ethical Protocol

The supplied F1.0 questionnaire states voluntary participation, academic use, the option to skip items, and anonymous aggregate reporting. These are the instrument's stated conditions, not proof of completed institutional approval. The separate profile archive includes names and contextual descriptions; therefore the linked research archive must be treated as identifiable or pseudonymised rather than wholly anonymous. This thesis reports only case codes and limited excerpts, not names, detailed biographies, or trip traces.

The researcher must reconcile consent for linking notes to questionnaires and for quoting recorded remarks with the original information given to students. Permission for an academic questionnaire does not automatically authorise uploading profiles to an external LLM service. The planned agent extension requires a separately approved data-use and de-identification procedure. Institutional authorisation is distinct from the consent wording and is not established by the statistical files. No approval decision is inferred from the presence of data.

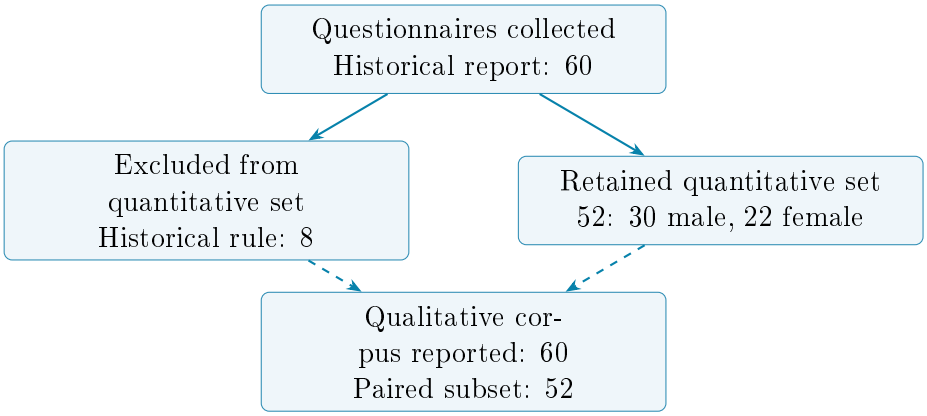
5.2 The Measured Study

5.2.1 Participants and Sampling

The target population was university students who listen to social media radio, in Egypt, while riding with Uber in the October 6th City area, Giza. This venue was chosen because it concentrates precisely the epistemic conditions the framework addresses: a bounded attention window, a social context, and exposure to broadcast content through the drivers' or riders' devices. The planning calculation is documented in Appendix E; it is not evidence of power for an independent-groups field experiment.

A convenience sample of 60 respondents was recruited. The supplied form asks respondents to consider a broadcast heard during a ride and explicitly excludes ACU students. A seven-day eligibility window is not documented in that form and is not imposed retrospectively. The historical analysis excluded eight records, leaving $N = 52$ (30 male, 22 female). The analysis export contains six institutional categories, with category counts of 3 to 13; no single category constitutes a majority of the sample. All 52 records contribute to the pooled analyses. Identical answers alone cannot establish that two records belong to the same person, so the exclusion rule is a limitation as well as a data-processing step. The newly supplied processing specification identifies a 21-field response-and-context key that reproduces the retained sample (Section 6.4). The participant flow is shown in Figure 5.1.

Figure 5.1: Reported Field-Survey Participant Flow



Note. Counts reproduce the historical thesis account. Exclusions concern quantitative analysis; all 60 narrative cases are reported in the qualitative corpus. Original files and recruitment denominators are absent from this archive.

5.2.2 Instrument

The analysed questionnaire batteries comprise 11 broadcaster-rating items and five post-broadcast items. The broadcaster battery contains four broadcast rigor (BR), four social validation (SV), and three information quality (IQ) items. These 11 items use 0–4 agreement responses; BR4 is reversed as $4 - \text{BR4}$ before aggregation. PB1–PB5 address confidence, perceived perspective diversity, comparative broadcaster trust, structural clarity, and overall quality. Only PB uses 1–7 agreement responses. The separate overall star rating uses a five-point scale. Demographics, ride duration, and self-reported interaction provide contextual variables.

The broader platform instrument bank also includes pre-broadcast, echo-triggered, and weekly items. These are wider platform-design proposals, not the field questionnaire. Appendix A reproduces the supplied F1.0 stems and response anchors, separating the administered form from prospective instruments. The separate star-rating block offers convergent self-report evidence, not an independent

factual-accuracy criterion.

5.2.3 Instrument Development

The supplied form makes the intended content and response anchors inspectable. It does not provide an expert-panel voting matrix, cognitive-interview record, or independent translation-validation study. Boateng et al. (2018) distinguish item development, scale construction, and scale evaluation; evidence from one stage cannot replace another. Here internal consistency and factor fit assess properties of the recorded responses, not completed content validation. The English and Arabic wording is reproduced without claiming that the two versions have demonstrated measurement equivalence. PB3's comparative trust and PB4's structural clarity must not be relabelled as verification intention and influence of a displayed score. The latter interpretations in earlier drafts did not match the supplied instrument.

5.2.4 Procedure

Respondents completed the questionnaire on a mobile device at the end of a social media radio listening episode during a ride. Completion followed the consent screen. The star rating and the PB battery were completed as separate blocks. This separation does not eliminate common-source covariance, response styles, or a general favourable impression (Podsakoff et al., 2024). The supplied files do not establish exact administration timestamps or independently certify that every note was recorded contemporaneously.

5.2.5 The Ecological Momentary Assessment Schedule

The platform design specifies four measurement windows: pre-broadcast (PR1–PR3), post-broadcast (PB1–PB5 and star rating), echo-triggered (EM1–EM3), and weekly (WK1–WK15). The proposed post-broadcast trigger is within 30 seconds of an episode ending, and the echo trigger is linked to the platform's diversity indicator. These timing rules describe the intended EMA implementation, rather than a longitudinal design completed by the 52-person survey. The supplied cross-sectional exports do not contain session timestamps with which to assess compliance with a 30-second window. Accordingly, the present results are described as post-broadcast questionnaire results, not as demonstrated longitudinal or real-time behavioural effects.

5.2.6 Analysis

Psychometric analysis used internal-consistency coefficients (Cronbach's α and a standardized one-factor model-based omega, exported as `omega_bpa`), corrected item-total correlations with the specified retention floor of $r \geq 0.30$, Spearman associations with the separate, same-respondent star rating and with behaviourally relevant covariates (ride duration, interaction level), and confirmatory factor analysis where the sample permitted. The small-sample strategy accepted exploratory use of CFA with explicit reporting of fit: the five-item single-factor PB battery and the eleven-item BRS were evaluated with fit reported honestly, without assuming a saturated solution before the $N = 52$ data were fitted, and the single-factor result is reported regardless of fit. No completed invariance test is present in the supplied

exports. Group figures therefore describe observed scores, not comparable latent means. Any later human latent-mean comparison would require a defensible measurement structure and appropriate loading and intercept or threshold comparisons (Putnick & Bornstein, 2016).

5.3 Mixed-Methods Design and Qualitative Corpus

The completed empirical report combines a quantitative questionnaire analysis with a qualitative analysis of researcher-recorded ride encounters. Integration was undertaken during this thesis revision, after the quantitative summaries were available; it is not retrospectively described as a prospectively registered convergent design. The purpose of mixing is complementarity and examination of tensions: what makes an experience worthwhile, and when do numerical ratings or activity categories omit that meaning? Creswell and Plano Clark (2018) and Levitt et al. (2018) motivate specifying the purpose, timing, and point of integration rather than simply placing two datasets side by side.

RQ5 (Qualitative experience). How do the rider records describe credibility, listening value, participation, and platform experience? **RQ6 (Integration).** Where do these accounts converge with, complement, or challenge the paired questionnaire ratings? These additional exploratory questions organise the newly integrated analysis and are separate from the original H1–H7 hypotheses.

The source corpus contains 60 case records in the rider-account corpus, with observations, recorded remarks, and researcher reflections, alongside duration, mood, and interaction descriptors. All 60 have distinct identifiers and nonempty

text. The 52 identifiers in the retained quantitative response matrix define the paired quantitative–qualitative subset. The eight other notes remain in the qualitative corpus: exclusion from a historical numeric analysis does not establish that a distinct narrative is invalid. Results report both denominators without treating them as independent samples.

the contextual student profiles provides biographical and media-preference descriptions of the same case series; it is not another sample. Some identifiers use different zero-padding from the CSV files. Profiles are contextual documents and candidate future anchors, not an independent source of corroboration or an extra set of interview responses. The 25 topics in the programme-topic proposals are proposed content stimuli, not 25 additional observations or completed recordings.

5.3.1 Analytical Position and Coding Procedure

The analysis is a semantic, codebook-based thematic analysis, with the assistance disclosed in the opening declaration, of recorded accounts. It is informed by the distinctions among approaches to analysis discussed by Braun and Clarke (2019), and by Kiger and Varpio's (2020) account of developing and reviewing themes. It is not described as independent human coding, a reliability-tested content analysis, or a fully researcher-led reflexive analysis completed before this revision.

The primary coding units were the `key_observations` and `notable_quotes` fields. The `researcher_reflections` field was read as interpretation, not counted as independent participant testimony. Familiarisation distinguished observable actions, attributed remarks, and inferences about personality or platform success. Initial semantic codes concerned correction, source use, social endorsement, familiar

voices, calm, time fit, agency, local relevance, and technical usability. These were grouped into six themes, checked against the complete case inventory, and recorded with an inclusion rule, exclusion rule, and case-level memo. A case could receive multiple codes. Coding decisions are explicit in the case-level coding record; the thematic codebook defines the themes, and Appendix F provides the membership matrix.

The counts describe the reach of this coding within the supplied records; they are not population prevalence estimates or measures of thematic importance. The script joins codes to source identifiers, calculates counts for 60 and 52 records, and verifies selected numerical summaries. It does not discover themes by keyword frequency. No second-coder agreement, member checking, thematic saturation, or participant validation is claimed. The audit retains source excerpts and interpretive decisions so the analysis can be examined.

5.3.2 Integration and Qualitative Trustworthiness

A joint display combines selected case accounts with the same cases' PB scores and interaction categories. It considers convergence, elaboration, and tension, not whether each quotation confirms EST. Quiet appreciation, conditional preference, and a request for better topic filters were retained rather than excluded as inconvenient cases. The small purposive illustrations do not support subgroup significance tests.

The records are polished, predominantly favourable English narratives with recurrent links between academic discipline and media preference. That style limits how closely they can be treated as spontaneous speech. The archive does not estab-

lish whether every attributed quotation is verbatim, translated, or reconstructed. Excerpts are therefore identified as *recorded remarks*, not certified interview transcripts. Positive descriptions cannot establish absence of dissatisfaction, and a researcher's explanation of a student's motives is not direct access to that student's experience. This is a complete analysis of the supplied case corpus, not a claim of exhaustive understanding of the wider student population.

The questionnaire's promise of anonymity must be distinguished from the identifiable profile archive. Names and detailed backstories are not reproduced in the results. Publication, retention, and later LLM processing require a consent and access review. No profile data were sent to external model APIs for the proposed experiment during this revision.

5.4 Focus-Group Evidence and Its Role in the Study

The prototype FGD is separate from the rider records in Section 5.3. Two overlapping narratives describe one reported university wireframe walkthrough; they are not two samples and are not added to the 60 rider accounts or 52 retained questionnaires. Participant overlap is unknown. Appendix G contains the full merged report, codebook, recording audit, and interpretive limits.

The accounts describe Egyptian Arabic discussion, an ACU context, smartphone wireframes, and an approximately 28-minute session. The more precise reported duration is 1,675.33 seconds, not metadata independently measured here. The transcript, recording, roster, and consent documentation remain absent. The analysis is consequently a critical synthesis of reported narratives, not newly com-

pleted transcript-level coding. Its four domains, FGD-T1–FGD-T4, remain distinct from the rider corpus's T1–T6 codes.

A supplemental file, `fgd_analysis_parts.json`, supplies seven generated text summaries, not an original transcript, frame log, or additional sample. Its candidate extracts contain translation inconsistencies, malformed intervals, and uncertain speaker labels. The review accepts the existence of the export and retains programme-length and registration-feedback questions for later source checking; it does not accept claims that the primary records have been recovered. The decision record is summarised in Appendix G.4. Model-use provenance is disclosed in the opening declaration. No new inductive transcript coding or audiovisual triangulation is claimed.

Focus groups require attention to interaction, not an assumption that a moderator's explanation constitutes participant endorsement (Kitzinger, 1995). The synthesis separates reported experience, interpretation, competing explanations, and implications. No quotations, theme prevalence, saturation, or coding reliability are invented. COREQ and APA qualitative reporting guidance inform the documentation checks without certifying missing procedures (Levitt et al., 2018; Tong et al., 2007).

5.4.1 Empirical Anchors, Assumptions, and Factual Ground Truth

Empirical anchors are source-linked human accounts or observations used within their evidentiary limits. *Design assumptions* are proposed mechanisms or scenario settings. *Factual ground truth* refers more narrowly to independently adjudicated answers for a defined assessment task. A preference for co-hosting is not a verified

answer to a news claim, and a theme is not an estimated transition probability.

The FGD currently supplies provisional design hypotheses, not verified individual personas, numerical calibration targets, or factual answer keys. Its repeated lexical counts are retained for audit only. New extracts must enter the evidence register with locators, speaker roles, quotation/translation status, permission scope, and coding decisions before participant-level use. Assessment keys and held-out outcomes remain outside participant-agent prompts.

Table 5.1: Evidence Roles in the Integrated Study

Material	Present use	Not established
Questionnaire and rider series	Reported audience summaries and six-theme interpretation	A randomised platform effect or FGD participant count
FGD narrative pair	Four provisional domains and competing design explanations	Independent replication or measured behavioural probabilities
Future verified extracts	Potential reviewed calibration or evaluation evidence	Automatic external-processing permission or untouched validation status
Independent claim keys	Planned task-specific accuracy assessment	Overall student competence or the truth of a preference

5.5 The Calibrated In-Silico Experiment (Final Pipeline)

The final computational study is a *calibrated in-silico experiment*, not a field intervention. It has three stages: (1) four-stream embedding, machine-learning debiasing, and Monte-Carlo prior construction; (2) a within-persona 2^3 full-factorial tensor agent-based model (ABM) with eight mathematical influence filters and a Zewail-inspired attentional gate; and (3) a GRM-IRT measurement model with dual frequentist and Bayesian inference, adversarial stress tests, and global sen-

sitivity analysis. The complete implementation, data, seeds, and generated outputs are included in the reproducibility archive (Appendix K).

The pipeline is deliberately explicit about what it can and cannot identify. The measured wave anchors the prior and the measurement model. The architectural contrast is a counterfactual generated *within* the model. No simulation output is relabelled as a human treatment effect.

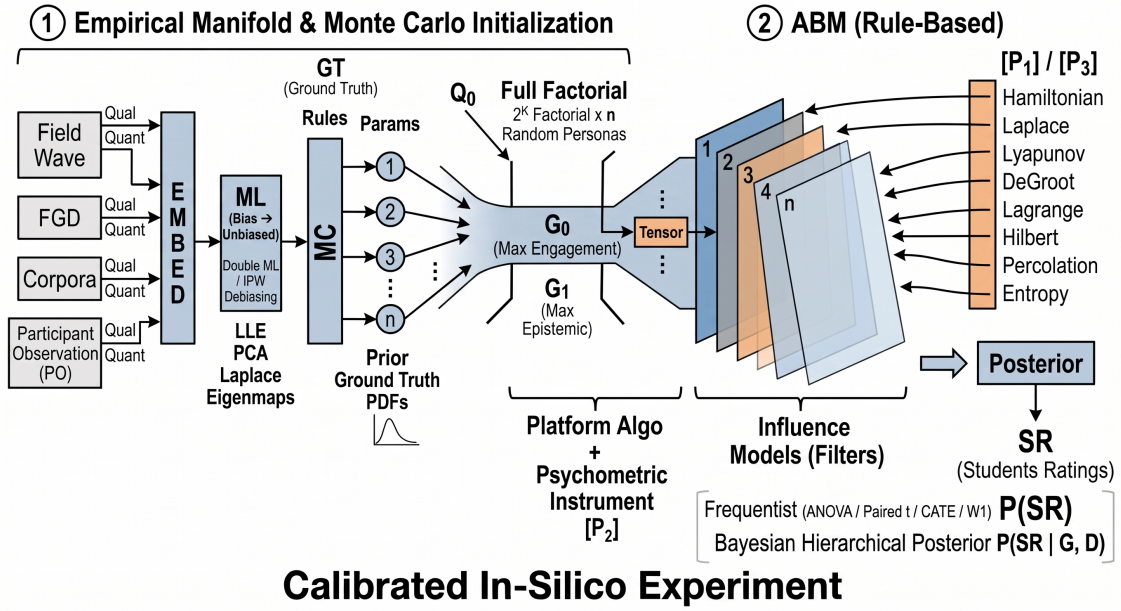


Figure 5.2: Master process diagram of the calibrated in-silico experiment. The left side is the empirical manifold and Monte-Carlo initialization; the right side is the rule-based ABM, the eight influence filters, the GRM-IRT measurement model, and the dual posterior. The figure is a schematic of the implemented pipeline, not a path model estimated from the 52 respondents.

What is assumed and what is estimated. The empirical streams anchor the persona distribution, the measurement model, and the semantic features. The treatment coefficients in the outcome equations (for example 0.42 for A, 0.28 for B, and 0.32 for C on credibility; 0.35, 0.26, and 0.40 on competence; and −0.05, −0.11, and −0.06 on engagement) are *design assumptions*. They are not estimated from the

52 respondents. The simulation propagates those assumptions through the ABM and the measurement model; it does not identify a field treatment effect. The value of the exercise lies in the internal consistency of the implied design, the epistemic–engagement trade-off, and the sensitivity of the implied results to the declared mechanism parameters—not in a discovery of the coefficients themselves.

5.5.1 Stage 1: Four-Stream Ingestion and Dual-Modality Embedding

Four empirical streams and a researcher self-report are ingested. Each contributes a quantitative and a qualitative representation, so that the model is not driven by a single self-report matrix.

The corpora are deliberately broad context, not direct evidence about radio listening. The Uber-review and student-lifestyle records are included to give the semantic prior heterogeneous audience language and to test whether the feature space is dominated by a narrow source; they are not treated as measurements of LOKLOK use, and no qualitative claim in this thesis rests on them alone.

For observation i, the embedding operator concatenates a standardised quantitative vector $\tilde{\mathbf{x}}_{i,\text{quant}}$ with a normalised semantic representation $\phi_{\text{sem}}(\mathbf{x}_{i,\text{qual}})$:

$$\mathbf{z}_i = \mathcal{E}(\mathbf{x}_{i,\text{qual}}, \mathbf{x}_{i,\text{quant}}) = [\tilde{\mathbf{x}}_{i,\text{quant}} \parallel \phi_{\text{sem}}(\mathbf{x}_{i,\text{qual}})] \in \mathbb{R}^D. \tag{5.1}$$

The semantic encoder is the multilingual BGE-M3 sentence-transformer (1024 dimensions), projected by PCA to 32 components; it requires no external API at inference and runs on a Colab GPU. The resulting matrix is $\mathbf{Z} \in \mathbb{R}^{N \times D}$.

Table 5.2: Four-Stream Inputs to the Calibrated In-Silico Experiment

Stream	Source	Quantitative representation	Qualitative representation
Field wave (FW)	Retained $N = 52$ questionnaire; 11 BRS items, 5 PB items, star rating, duration, interaction	Standardised item and composite vector	Open-context descriptors and response-style indicators
FGD	Original 160-segment focus-group transcript; themes T1–T4 (publication, co-hosting, institutional access, interface cues)	Theme incidence and segment telemetry	Coded thematic excerpts and moderator notes
Corpora	25 programme briefs; 4,828 Uber-review and student media/lifestyle documents	Source-set cardinality, concentration, review statistics	BGE-M3 semantic embeddings
Participant observation (PO)	60 researcher-recorded rider accounts (SR001–SR060)	Theme-incidence and ride-context features	Pseudonymised narrative text
Researcher self-report (SR)	27.7-minute audio transcribed on GPU with Whisper large-v3 (185 segments)	Reflexive coding indicators	185 transcript segments

Note. The original 160-segment FGD transcript is included. The audiovisual recordings and participant roster remain unavailable; the FGD stream therefore contributes transcript-grounded design hypotheses and aggregate features, not additional measured outcomes. The PO file and self-report transcript are pseudonymised and access-controlled.

Sentence-transformer architecture and process. The encoder is `BAAI/bge-m3`, a multilingual sentence-transformer built on an XLM-RoBERTa backbone. It returns a 1,024-dimensional dense sentence vector and supports long inputs up to 8,192 tokens. The thesis uses the dense embedding only; the model's sparse and multi-vector outputs are not used. The encoder is used zero-shot: it is not fine-tuned on the thesis data, and no supervised label enters the embedding step. The process is: (1) assemble the 5,258 document texts and their source metadata; (2) clean and truncate each text to the model limit; (3) tokenise and encode mini-batches on the Colab T4 in mixed precision; (4) pool and L2-normalise the

dense representation; (5) cache the vectors by a hash of the model name and document set; and (6) project the semantic block to 32 principal components, which enter the 198-feature matrix alongside the 31 quantitative features. The cache makes the run reproducible without a second API call. This is a measurement pipeline, not a claim that the 1,024 dimensions are psychological constructs; the cosine diagnostics in Section 5.5.4 show moderate, not clean, stream separation.

Table 5.3: Sentence-Transformer Embedding Pipeline

Step	Operation	Setting	Output
1	Document assembly	5,258 texts; five streams	text bank
2	Clean and truncate	model limit 8,192 tokens	token sequences
3	XLM-R encoder	BGE-M3, dense output, fp16 on T4	token states
4	Pool and normalise	dense sentence vector	1,024-d unit vector
5	Cache	hash of model + document set	reusable `.npy`
6	PCA projection	32 components	167 semantic features

Note. The 31 quantitative features plus the 167 semantic features give the 198-feature matrix. The encoder is zero-shot and is not fine-tuned on the 52 respondents.

5.5.2 Stage 1: Machine-Learning Debiasing and Spectral Manifold

Raw self-report matrices exhibit common-method covariance, response-style variation, and convenience-sample imbalance. The final pipeline applies two causal-ML debiasing operations before manifold projection.

Double/orthogonal machine learning (DML). Let $\mathbf{B}$ collect nuisance indicators: person mean and standard deviation, acquiescence, ride duration, self-reported interaction, a high-interaction indicator, age, and gender. A ridge model predicts

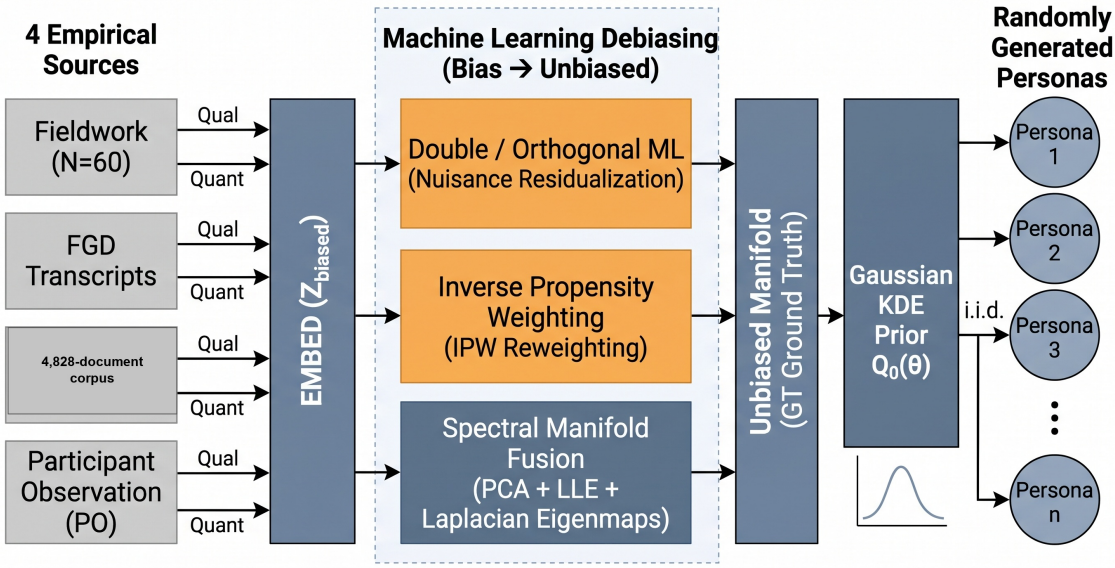


Figure 5.3: Process 1: four-stream ingestion, BGE-M3 embedding, DML/IPW debiasing, and spectral-manifold fusion into the KDE prior Q_0. The four empirical streams are represented as qualitative and quantitative inputs; the corpus count is the final 4,828-document set.

each embedded feature from $\mathbf{B}$; the residual removes the linear nuisance component:

$$\mathbf{z}_{i,\text{orth}} = \mathbf{z}_i - \mathbf{B}_i \left(\mathbf{B}^\top \mathbf{B} + \lambda \mathbf{I}\right)^{-1} \mathbf{B}^\top \mathbf{Z}. \tag{5.2}$$

Inverse propensity weighting (IPW). A logistic propensity model estimates the probability $\hat{e}(\mathbf{x}_i)$ that a respondent belongs to the high-interaction stratum. Each vector is reweighted by

$$w_i^{\text{IPW}} = \frac{1/\hat{e}(\mathbf{x}_i)}{\frac{1}{N}\sum_j 1/\hat{e}(\mathbf{x}_j)}, \qquad \tilde{\mathbf{z}}_i = \sqrt{w_i^{\text{IPW}}}\, \mathbf{z}_{i,\text{orth}}. \tag{5.3}$$

The diagnostics report balance before DML, after DML, and after IPW; the latter is not assumed to improve monotonically.

Spectral manifold. Three projections are computed on the debiased matrix: principal component analysis (PCA) for linear orthogonalisation, locally linear embedding (LLE) for local persona geometry, and Laplacian eigenmaps for graph-spectral denoising. PCA solves

$$\mathbf{\Sigma}_Z \mathbf{u}_m = \lambda_m \mathbf{u}_m, \qquad \mathbf{U}_{\mathrm{PCA}} = \mathbf{Z}\mathbf{U}_{1:d}\mathbf{\Lambda}_{1:d}^{-1/2}. \tag{5.4}$$

LLE reconstructs each point from its K nearest neighbours by minimising

$$J_{\mathrm{LLE}}(\mathbf{W}, \mathbf{Y}) = \sum_{i=1}^{N} \left\| \mathbf{z}_i - \sum_{j \in \mathcal{N}(i)} W_{ij} \mathbf{z}_j \right\|_2^2 = \min_{\mathbf{Y}} \sum_{i=1}^{N} \left\| \mathbf{y}_i - \sum_{j \in \mathcal{N}(i)} W_{ij} \mathbf{y}_j \right\|_2^2, \tag{5.5}$$

subject to $\sum_j W_{ij} = 1$ and $\mathbf{Y}^\top \mathbf{Y} = \mathbf{I}$. Laplacian eigenmaps build a heat-kernel affinity graph $K_{ij} = \exp(-\|\mathbf{z}_i - \mathbf{z}_j\|_2^2 / 4\tau)$ and solve the generalised eigenproblem

$$\mathbf{L}\mathbf{f}_k = \lambda_k \mathbf{D}\mathbf{f}_k, \qquad \mathbf{L} = \mathbf{D} - \mathbf{K}, \qquad 0 = \lambda_0 \leq \lambda_1 \leq \cdots \leq \lambda_d. \tag{5.6}$$

The final latent coordinates are the standardised average of the three projections,

$$\widetilde{\mathbf{\Theta}} = \frac{1}{3} \left(\mathbf{U}_{\mathrm{PCA}} + \mathbf{U}_{\mathrm{LLE}} + \mathbf{U}_{\mathrm{Lap}} \right) \in \mathbb{R}^{N \times d}, \qquad d = 6. \tag{5.7}$$

This is a *debiasing and dimension-reduction device.* It does not establish that the six axes are uniquely identified psychological constructs.

5.5.3 Stage 1: Monte-Carlo Prior Q_0 and Random Personas

A continuous prior $Q_0(\boldsymbol{\theta})$ is fitted to the debiased coordinates. When the sample covariance is well conditioned, a Gaussian kernel density estimator is used; otherwise a shrunken multivariate normal is substituted. The generator draws $n = 500$ personas, $\boldsymbol{\theta}_a \sim Q_0$, and maps the six latent axes to the ABM traits (initial opinion, prior reputation, echo susceptibility, heat capacity, critical scrutiny, and co-hosting preference). Every persona is cloned across all eight factorial cells, so within-persona contrasts are paired by construction. The $n = 500$ personas are a computational population, not 500 additional human respondents.

Who the synthetic personas represent. The personas are drawn after DML/IPW and after the six-axis spectral projection, so they inherit the debiased latent geometry of the 52 measured students, not the raw questionnaire rows. The measured anchor is a convenience sample of university students who were Uber riders in the October 6th City area, Giza: age mean 20.37 (SD 0.95; range 19–23), 30 male and 22 female respondents, academic years 1–4, and eight faculties across six universities. The ABM does not assign age, gender, or faculty to individual personas. It assigns each persona to the university stratum of its nearest measured respondent in the debiased manifold, for partial pooling only. A descriptive projection onto those nearest anchors gives approximately 268 male and 232 female anchors, mean age 20.34, and the same faculty/university mix; this projection is not a persona attribute and is not a representative demographic sample.

Table 5.4 reports the six psychographic/behavioural traits of the 500 per-

sonas. They are heterogeneous: initial opinion and BRS prior are centred near the middle of their ranges, echo susceptibility is moderate, and critical scrutiny and co-hosting preference span a wide range. These are the synthetic listeners on whom the factorial experiment is run.

Table 5.4: Latent Traits of the 500 Synthetic Personas (Model-Based)

Trait	Mean	SD	5th pct	95th pct
Initial opinion	0.501	0.212	0.194	0.863
Prior BRS	0.479	0.178	0.206	0.779
Echo susceptibility	0.423	0.116	0.242	0.621
Heat capacity	3.312	1.000	1.672	4.759
Critical scrutiny	0.494	0.265	0.085	0.919
Co-hosting preference	0.523	0.277	0.095	0.940

Note. Draws from the Gaussian-KDE prior fitted to the 52 debiased rows; these are synthetic profiles, not 500 human respondents.

Why 500 personas. The Monte Carlo standard error of the mean contrast is approximately $SD(\Delta\text{SR})/\sqrt{n} = 0.545/\sqrt{500} \approx 0.024$, consistent with the reported interval width of about 0.096. Increasing to 2,000 would halve that simulation error, but it would not address the dominant uncertainty, which is the set of state-outcome weights: the 256-point joint sweep already ranges from -0.244 to $+1.810$. average more precise; they do not make the experiment more externally valid. The reported $n = 500$ is therefore a declared computational size, not a human sample size.

5.5.4 Embedding and Latent-Space Diagnostics

The embedding and manifold steps are descriptive transformations, not validated psychological measurements. Two diagnostics make that boundary visible. First, the 5,258 documents are L2-normalised BGE-M3 vectors in 1,024 dimensions. The first two principal components account for 44.9% of the variance (36.9% and 8.0%);

within-stream cosine similarity is highest for participant observation (0.785), followed by stimuli (0.568), corpora (0.559), FGD (0.531), and self-report (0.466). Cross-stream similarity is lower (0.38–0.48), and the cosine silhouette of 0.196 indicates moderate, not clean, separation. The corpus is therefore semantically structured but not a set of naturally isolated classes.

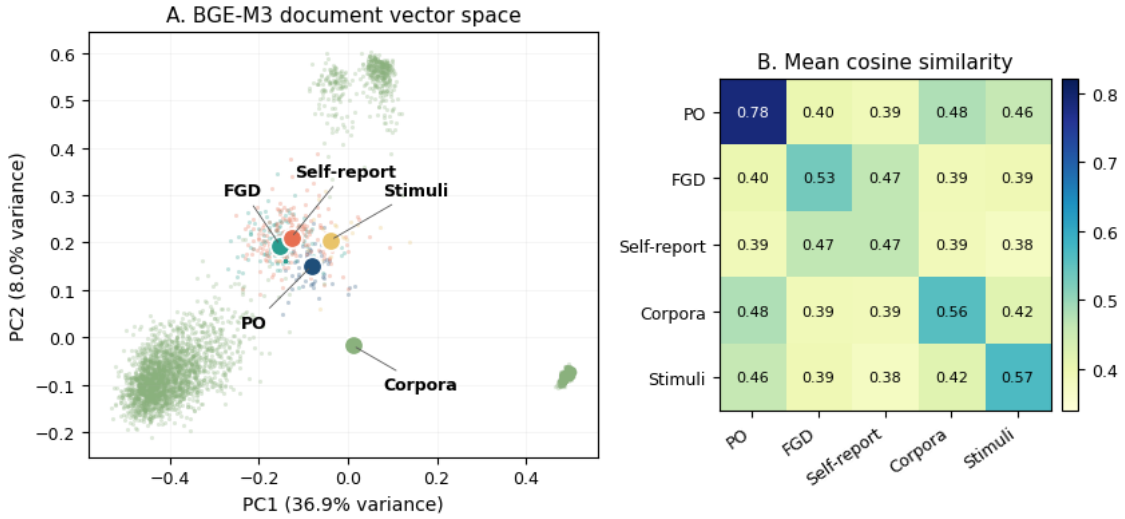


Figure 5.4: Embedding and vector-space diagnostics. Panel A: the 5,258 documents projected onto the first two principal components of the normalised BGE-M3 space. Panel B: mean cosine similarity within and between streams. The within-stream means are reported in Table 5.5. The projection is descriptive and does not validate a construct.

Second, the six-axis manifold is a debiasing coordinate system, not a six-factor psychological model. For the 52 measured anchors, the covariance condition number is 4.815 and the largest absolute off-diagonal correlation is 0.434. The first two principal components explain 49.9% of the latent variance. The mean within-university distance is 3.211 and the mean between-university distance is 3.295, a ratio of 0.974: university labels do not separate the anchors in this space. The 500 personas are draws from the fitted Q_0; the figure shows the 52 anchors that identify the space, not 500 human respondents.

Table 5.5: Embedding and Vector-Space Diagnostics

Quantity	Value
Documents	5,258
Embedding dimensions	1,024
L2 vector norm, mean (SD)	1.000 (0.00003)
PC1 variance explained	36.9%
PC2 variance explained	8.0%
PC1 + PC2 variance explained	44.9%
Mean within-stream cosine: PO	0.785
Mean within-stream cosine: FGD	0.531
Mean within-stream cosine: self-report	0.466
Mean within-stream cosine: corpora	0.559
Mean within-stream cosine: stimuli	0.568
Mean cross-stream cosine range	0.38–0.48
Cosine silhouette (1,500-document sample)	0.196

Note. All vectors are L2-normalised BGE-M3 embeddings. The similarity values are exact group means; the silhouette is computed on a fixed random sample. These diagnostics describe the embedding geometry, not the validity of a psychological construct.

Table 5.6: Latent-Space Diagnostics for the 52 Measured Anchors

Quantity	Value
Latent dimensions	6
Covariance condition number	4.815
Maximum absolute off-diagonal correlation	0.434
PC1 variance explained	28.6%
PC2 variance explained	21.3%
PC1 + PC2 variance explained	49.9%
Mean within-university distance	3.211
Mean between-university distance	3.295
Within/between distance ratio	0.974

Note. Distances are Euclidean in the six-axis latent space. The near-unit ratio means that university labels do not separate the anchors in this space; the axes are debiasing coordinates, not validated psychological dimensions.

5.5.5 Stage 2: Within-Persona 2^3 Full-Factorial Design

Three architectural factors are crossed:

- A: BRS credibility architecture (off vs. on: anchored Beta-DeGroot reputation, Lyapunov cap, confidence-interval badge).
- B: MORE diversity and cooling (off vs. on: Lagrangian diversity ranking,

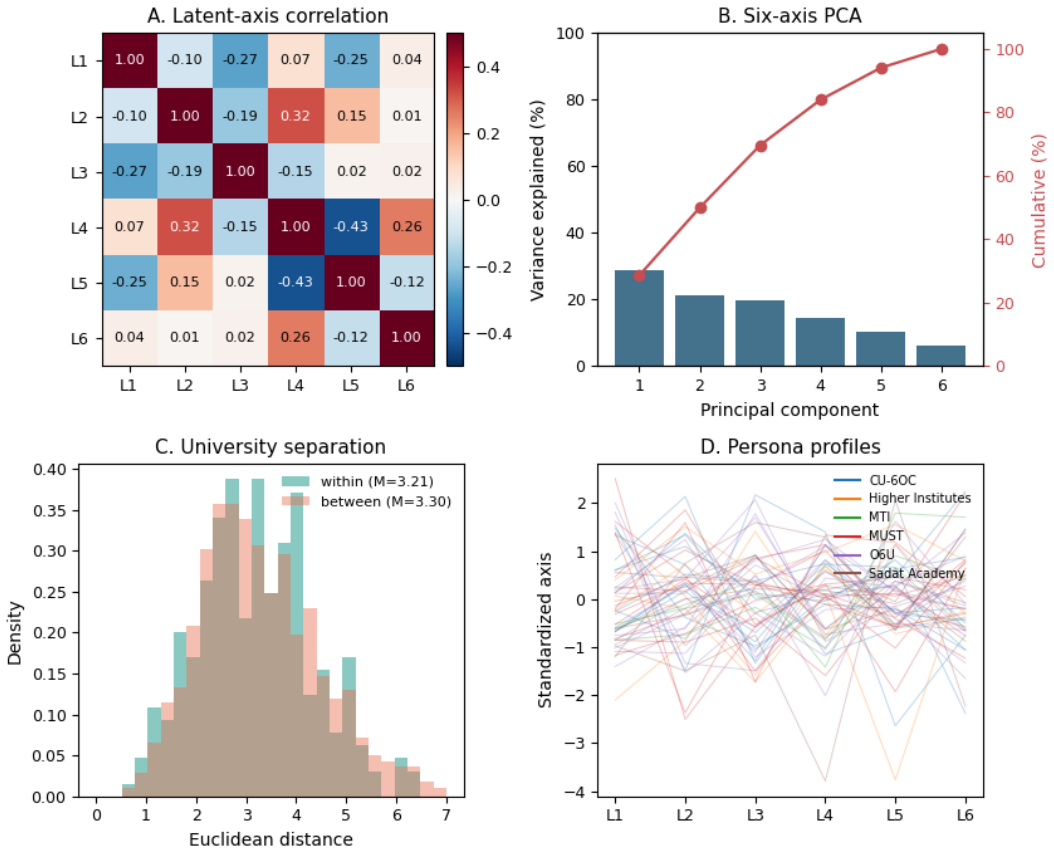


Figure 5.5: Latent-space diagnostics for the 52 measured anchors. Panel A: correlation matrix of the six debiased axes. Panel B: variance explained by each principal component and the cumulative curve. Panel C: within- and between-university Euclidean distances. Panel D: standardised persona profiles by university. The axes are transformation coordinates, not validated psychological dimensions.

Hilbert cosine DBI, Echo Meter cooling at $R_t \geq 0.75$).

- C: ERIC governance and psychometric reflection (off vs. on: percolation moderation penalty and EMA reflection prompts).

Cell $(0, 0, 0)$ is G0 (maximum engagement); cell $(1, 1, 1)$ is G1 (maximum epistemic). The six intermediate cells identify the separate and joint contributions of the three mechanisms. Table 5.7 lists the design.

Table 5.7: The 2^3 Full-Factorial Architecture

Cell	A (BRS)	B (MORE)	C (ERIC)	Interpretation
0	0	0	0	G0: maximum engagement
1	1	0	0	BRS only
2	0	1	0	MORE + cooling only
3	0	0	1	ERIC + EMA only
4	1	1	0	BRS + MORE
5	1	0	1	BRS + ERIC
6	0	1	1	MORE + ERIC
7	1	1	1	G1: maximum epistemic LOKLOK

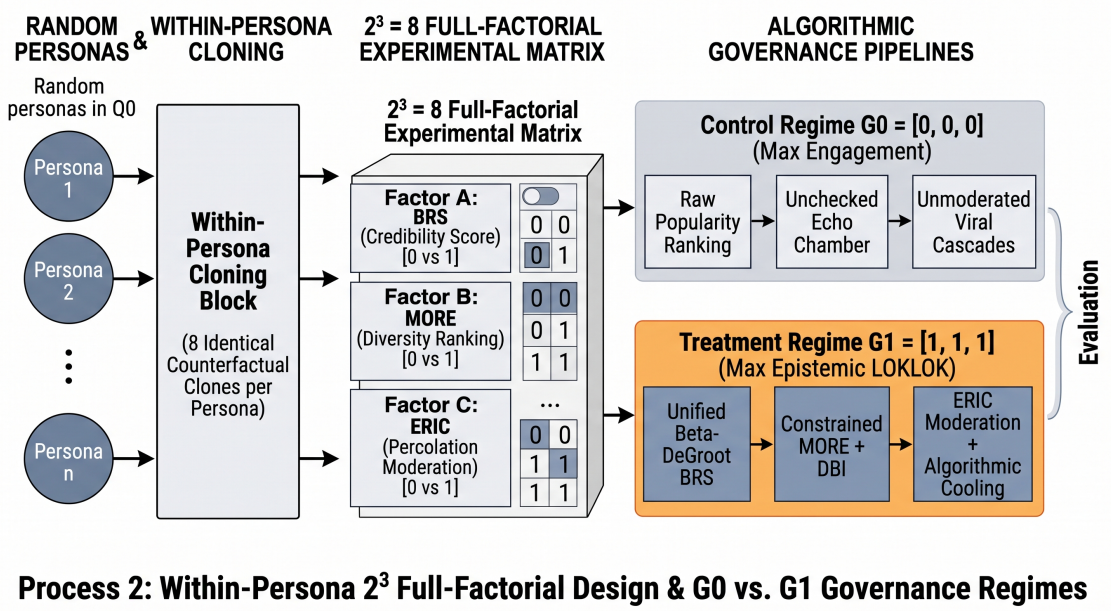


Figure 5.6: Process 2: within-persona cloning into the $2^3 = 8$ full-factorial design and the G0/G1 governance contrast. Every persona is cloned across all eight cells, so the A/B/C contrast is paired by construction.

5.5.6 Stage 2: Tensor State and the Eight Implemented Filters

The trajectories of all agents across $T = 18$ sessions, F measured features, and the eight cells are stored in a fourth-order tensor

$$\mathcal{T} \in \mathbb{R}^{n \times T \times F \times 8}, \qquad \mathcal{T}_{i,t,:,g} = \mathbf{S}_i^{(g)}(t) = \big(x_i(t), B_i(t), R_i(t), e_i(t), C_i^{\mathrm{rec}}(t), q_i^{\mathrm{PB}}(t)\big)^\top. \tag{5.8}$$

The tensor is propagated through eight composable operators. The operators are mathematical formalisations of the EST mechanisms, not physical laws.

Table 5.8: The Eight Implemented Filters and Their EST Mapping

Filter	EST link	Governing operator	Role in the model
Hamiltonian	P2	$\mathcal{H} = \sum_i (R_i - \bar{R})^2/(2C_i) - \sum_{i,j} W_{ij} R_i Q_j + \tau \sum_s p_s \log_2 p_s$	Free-energy diagnostic; monopoly penalty and reputation–quality alignment
Laplace	P1/P2	$\dot{\mathbf{x}} = -\lambda_{\text{diff}} \mathbf{L}\mathbf{x}$, $\mathbf{L} = \mathbf{D} - \mathbf{A}$	Trust and opinion diffusion over the social graph
Lyapunov	P1/P3	$V(\mathbf{B}_t) = \|\mathbf{B}_t - \mathbf{B}^*\|^2$, $\Delta V_t < 0$	Contraction certificate for reputation stability
DeGroot	P1	$B_u(t+1) = (1-\lambda)\sum_v \tilde{A}_{uv} B_v(t) + \lambda X_u(t)$	Evidence-anchored consensus reputation
Lagrange	P1/P2	$\max_c \; w_1 \text{UVS}(c,u) + w_2 \text{DBI}(c,u) - \mu(\|\text{SAI}\| - B_{\max})$	Constrained MORE recommendation
Hilbert	P2	$\text{DBI}(c,u) = 1 - \langle \mathbf{v}_c, \mathbf{C}_u^{\text{rec}} \rangle_{\mathcal{H}} / (\|\mathbf{v}_c\|_{\mathcal{H}} \|\mathbf{C}_u^{\text{rec}}\|_{\mathcal{H}})$	Cosine diversity in a reproducing-kernel space
Percolation	P1/P2	$p_{uv} = \sigma(\theta_{\text{prop}})(1 - E_{a,\text{mod}})$, threshold $p_c = \langle k \rangle / (\langle k^2 \rangle - \langle k \rangle)$	Misinformation versus correction cascades
Entropy	P2	$R_i(t) = (1-\alpha_R) R_i(t-1) + \alpha_R (1 - \text{DBI}(c^*, i))$	Echo Meter and Algorithmic Cooling

Note. The Hamiltonian is a formal diagnostic; its numerical sign is reported transparently and is not used as a confirmatory physical test. The Zewail gate multiplies the evidence anchor and the attentive-retention term (Section 4.3.1).

5.5.7 Stage 2: G0 and G1 Algorithmic Regimes

Table 5.9 makes the two poles explicit. The intermediate cells are defined by switching each subsystem on or off while holding the persona, network, candidate bank, and session sequence fixed.

5.5.8 Stage 3: GRM-IRT Measurement Model

The terminal state of each persona is passed through a Samejima graded response model. For item j, discrimination $a_j > 0$, ordered thresholds $b_{j,1} < \cdots < b_{j,K-1}$,

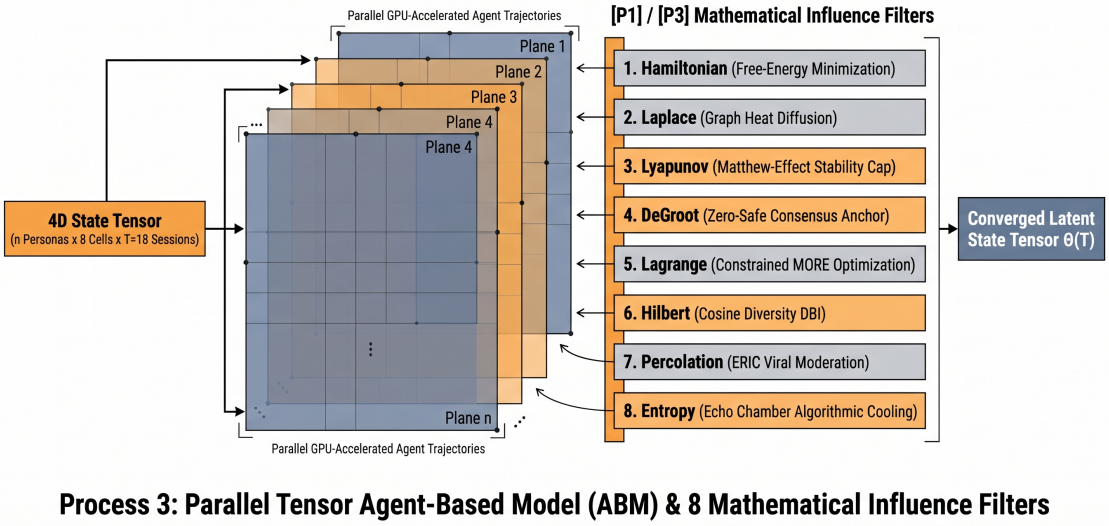


Figure 5.7: Process 3: the parallel tensor ABM. Persona trajectories are stored across sessions and cells, propagated through the eight mathematical filters, and reduced to the terminal latent state that enters measurement.

Table 5.9: Algorithmic Contrast Between G0 (Maximum Engagement) and G1 (Maximum Epistemic)

Component	G0: maximum engagement	G1: maximum epistemic LOKLOK
Objective	Maximise predicted click and interaction volume	Minimise the epistemic free-energy diagnostic while maximising diversity-constrained value
Reputation (A)	Unanchored popularity drift; no Matthew-effect cap; no interval badge	Beta-DeGroot evidence anchor; Lyapunov cap; displayed 95% Beta interval
Recommendation (B)	Pure exploitation ($w_1 = 1, w_2 = 0$); static schedule; no balance budget	Constrained MORE ($w_1 = 0.62, w_2 = 0.38$); Hilbert diversity; SAI budget
Echo (B)	Passive accumulation; no warnings or cooling	Amber warning at $R_t \geq 0.60$; cooling at $R_t \geq 0.75$
Moderation (C)	No friction for high-velocity outrage	ERIC percolation penalty; escalation above $\tau_{\text{mod}} = 0.70$; correction credit
Measurement (C)	Passive post-broadcast ratings only	EMA Layer-1 reflection within 30 s; inspectable source list; GRM-IRT scoring
Attention gate	Background play counted as exposure	Zewail pump–probe gate discounts decoupled background play

Note. These are model rules, not claims that the deployed platform currently enforces every rule. The implementation and parameter values are in `loklok_pipeline.py`.

and latent trait θ_i,

$$\begin{aligned} P^*(Y_{ij} \geq k \mid \theta_i) &= \left[1 + \exp\left(-a_j(\theta_i - b_{j,k})\right)\right]^{-1}, \\ P(Y_{ij} = k \mid \theta_i) &= P^*(Y_{ij} \geq k) - P^*(Y_{ij} \geq k+1). \end{aligned} \tag{5.9}$$

The 35-item NMLS battery yields the synthetic competence profile $\mathrm{SR}_{\mathrm{comp}}$ on a 0–100 display scale,

$$\mathrm{SR}_{\mathrm{comp},i} = 100\,\frac{T_i - 35}{140}, \qquad T_i = \sum_{j=1}^{35} Y_{ij}. \tag{5.10}$$

The eight-item MCS+PB battery yields the evaluative profile $\mathrm{SR}_{\mathrm{eval}}$ on the 1–7 scale. The test information and standard error of measurement are

$$\mathcal{I}(\theta) = \sum_{j=1}^{J} \mathcal{I}_j(\theta), \qquad \mathrm{SEM}(\theta) = \frac{1}{\sqrt{\mathcal{I}(\theta)}}. \tag{5.11}$$

The synthetic NMLS profile is an exploratory measurement diagnostic, not a validated measure of human media literacy. The human NMLS remains a prospective instrument.

Process 4 of the upgrade process atlas is this GRM-IRT measurement and dual-inference stage. Its category curves and test-information function are shown in Figure 6.12; the diagram is not duplicated here because the measurement equations and the figure carry the same information.

5.5.9 Stage 3: Posterior Student Ratings and Dual Inference

The model's terminal states are mapped to the posterior distribution over Student Ratings,

$$P(\mathrm{SR}_i \mid G_g, \mathcal{D}) = \int_{\boldsymbol{\theta}} p(\mathrm{SR}_i \mid \mathcal{T}_{i,T,:,g}(\boldsymbol{\theta}))\, p(\boldsymbol{\theta} \mid \widetilde{\boldsymbol{\Theta}}, \mathcal{D})\, d\boldsymbol{\theta}, \tag{5.12}$$

where $G_g \in \{G_0, G_1\}$ and $\mathcal{D}$ denotes the measured anchoring data. The two marginal projections are $\mathrm{SR}_{\mathrm{eval}}$ (MCS+PB) and $\mathrm{SR}_{\mathrm{comp}}$ (NMLS-format). The posterior is evaluated in two complementary ways.

Frequentist. Within-persona paired t-tests, Cohen's d_z, 95% repeated-sampling intervals, a full-factorial repeated-measures decomposition, and Wasserstein W_1 distances. These describe variation across model personas; they are not population confidence intervals for a human treatment effect.

Bayesian. A hierarchical normal model with partial pooling by sampled university stratum,

$$y_i \sim \mathcal{N}(\alpha_{h[i]} + \tau, \sigma^2), \qquad \alpha_h \sim \mathcal{N}(\mu_\alpha, \tau_\alpha^2), \qquad \tau \sim \mathcal{N}(\tau_0, \tau_{\mathrm{prior}}^2), \tag{5.13}$$

where $y_i = \mathrm{SR}_{\mathrm{eval},i}^{(G_1)} - \mathrm{SR}_{\mathrm{eval},i}^{(G_0)}$. A Gibbs sampler reports the posterior mean, 95% highest-density interval, $P(\tau > 0 \mid \mathcal{D})$, and a Savage–Dickey Bayes factor. Prior sensitivity is reported for skeptical, moderate-skeptical, weakly informative, and enthusiastic priors. The extreme skeptical prior is a stress test, not the primary specification.

5.5.10 Stage 3: Adversarial, Sensitivity, and Robustness Design

Three adversarial vectors stress-test the architecture. (A) A viral-outrage misinformation shock is injected at session 8 at rates $\rho_{\text{misinfo}} \in \{0\%, 10\%, 20\%, 35\%, 50\%\}$. (B) A coordinated Sybil/bot clique comprising 15% of nodes maximally endorses low-accuracy content. (C) An error-farming broadcaster submits many low-validation self-corrections to test the fixed information-quality equation.

Global sensitivity uses a 256-point joint Latin-hypercube design over six mechanism parameters and the five state-outcome credibility weights. The mechanism parameters are the DeGroot anchor λ, the MORE diversity weight w_2, the cooling trigger θ_{cool}, the ERIC threshold τ_{mod}, the Matthew-effect cap β_{cap}, and the lagrange penalty μ_{lag}; the weight ranges include zero. Partial-rank correlations (PRCC) identify the leading drivers. Robustness is also checked across Erdős–Rényi, Watts–Strogatz, and Barabási–Albert networks, three persona-prior variants, and four Bayesian priors.

5.5.11 Reproducibility and Integrity

The final pipeline is deterministic given its seed. The archive contains the de-identified measured matrix, the derived qualitative features, the pseudonymised participant-observation text, the Python pipeline, the R scripts, the generated JSON/CSV outputs, and the figures. The exact commands and the Colab notebooks are listed in Appendix K. The archive distinguishes three evidence tiers: measured field evidence, model-based counterfactual evidence, and theoretical propositions. No simulation value is presented as a measured human effect.

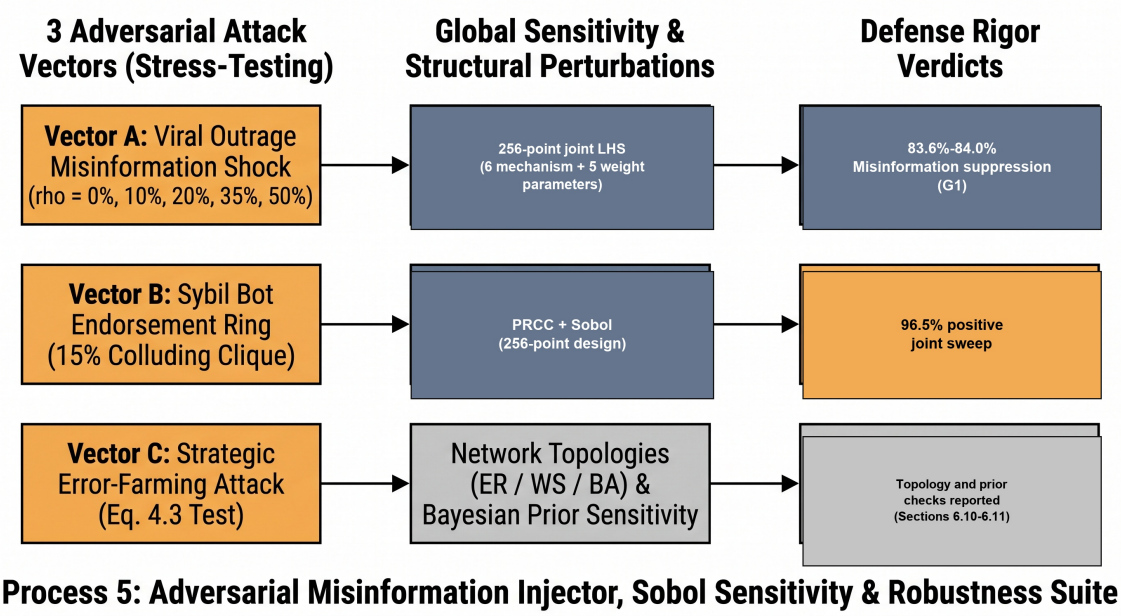


Figure 5.8: Process 5: adversarial stress tests, global sensitivity, and structural robustness. The central labels reflect the final v8 design: a 256-point joint Latin-hypercube sweep, PRCC plus Sobol decomposition, and 83.6–84.0% misinformation suppression. Numerical results are reported in Section 6.12.9.

5.6 Procedure: Field Wave, Calibration, and In-Silico Experiment

This section states exactly what was done, in the order in which it was done. It separates the field procedure from the calibration procedure and from the in-silico procedure so that a reader can reproduce or challenge each step.

5.6.1 Field-Wave Procedure

1. **Eligibility and recruitment.** Participants were university students who used Uber rides in the October 6th City area, Giza. Participation was voluntary and anonymous; names and phone numbers were not retained with the research data. Each record carries an anonymous research ID.

2. **Consent.** The consent screen preceded the questionnaire. It stated the purpose, the voluntary nature, the right to withdraw, and the separation of LOK-LOK profile data from research responses.
3. **Ride context.** Respondents recorded ride duration, interaction level during the ride, and the listening context. The observed range was 5–15 minutes; interaction was reported as minimal, low, moderate, or high.
4. **Listening episode.** During the ride, respondents used the LOKLOK field interface to listen to a social-media radio episode. The interface displayed the show, broadcaster, BRS badge, credibility interval, and Echo Meter where available. The field wave is a post-broadcast cross-sectional survey, not a session-randomised trial.
5. **Post-broadcast questionnaire.** After the episode, respondents completed the star rating, the 11-item BRS battery (BR1–BR4, SV1–SV4, IQ1–IQ3; 0–4 agreement, BR4 reverse-coded), and the five-item PB battery (PB1–PB5; 1–7). The star rating and PB were separate blocks.
6. **Export and screening.** The raw export contained 60 records. The historical 21-field key was applied to identify duplicate response vectors; eight records were removed, leaving 52 for the primary analysis. The key identifies duplicate vectors, not independently verified duplicate people. The 60-row file is retained as a sensitivity analysis.
7. **Quality checks.** Reverse coding, missingness, internal consistency, item–total correlations, associations with stars and ride context, and a five-item single-factor CFA were examined. Fit is reported honestly; the CFA does not

establish structural validity.

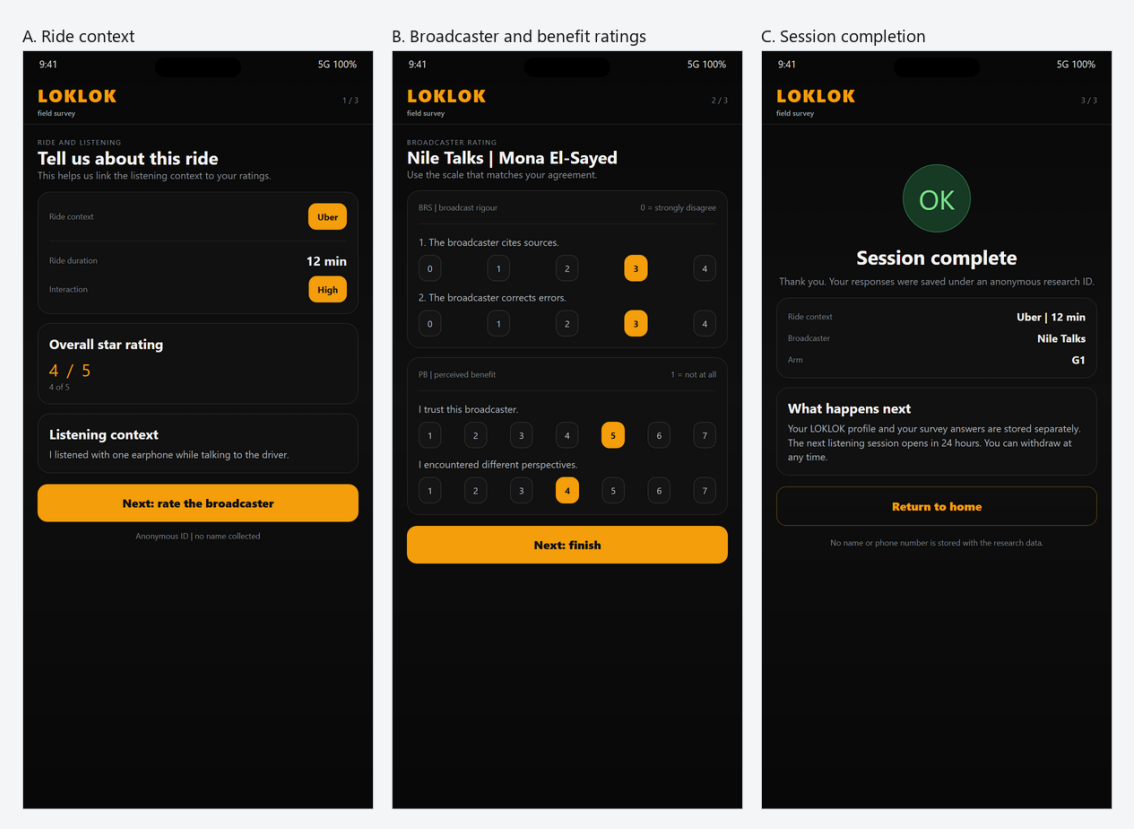


Figure 5.9: Field survey flow. Panel A: ride context and overall star rating. Panel B: broadcaster-rigour and perceived-benefit ratings. Panel C: session completion and separation of the anonymous research record from the LOKLOK profile. The screens document the field instrument, not additional outcome data.

5.6.2 Calibration Procedure

1. **Document collection.** The raw 60 questionnaires, 60 ride/PO narratives, the 160-segment FGD transcript, the 185-segment self-report transcript, the four corpora, and the 25 programme briefs were assembled.
2. **Semantic features.** Documents were embedded with multilingual BGE-M3 (1,024 dimensions) and reduced by PCA to 32 components. The five semantic blocks (PO, FGD, self-report, corpora, stimuli) were summarised into 167 features. Adding the 31 quantitative features gave the 198-feature matrix.

3. **Debiasing.** DML residualised each feature on the nuisance block (person mean and SD, acquiescence, ride duration, interaction, high-interaction indicator, age, and gender). IPW reweighted the high-interaction stratum. The diagnostics report balance before DML, after DML, and after IPW; IPW is not assumed to improve monotonically.
4. **Manifold.** PCA, LLE, and Laplacian eigenmaps produced a six-axis spectral manifold. The axes are a smoothing device for persona generation, not uniquely identified psychological dimensions.
5. **Persona prior.** A Gaussian KDE was fitted to the 52 debiased rows in the six-axis space. Five hundred personas were drawn from this prior and mapped to six traits: initial opinion, prior BRS, echo susceptibility, heat capacity, critical scrutiny, and co-hosting preference.
6. **Scale calibration.** The status-quo arm G0 was anchored to the measured PB mean (5.119), because the measured students used ordinary Uber, not the LOKLOK epistemic features. This is a single scale anchor, not a fit to the treatment contrast.

5.6.3 In-Silico Procedure

1. **Design.** Every persona was cloned into the 2^3 factorial design: A = BRS credibility architecture, B = MORE diversity/cooling, C = ERIC moderation/reflection. Each cell ran 18 listening sessions.
2. **Mechanisms.** The agent-based model implemented eight filters (Hamiltonian, Laplace, Lyapunov, DeGroot, Lagrange, Hilbert, Percolation, Entropy) and the Zewail-inspired attentional gate. Ablations later showed that only

DeGroot, Lagrange, and Hilbert are outcome-active in this wiring; the thesis reports them as eight implemented, not eight active, filters.

3. **Outcomes.** Terminal states were scored through the GRM-IRT measurement model for evaluative ratings and the synthetic NMLS competence profile. The simulator seed was 20261105; the run used 500 personas and 18 sessions.
4. **Calibration check.** G0 was anchored to measured PB; G1 was compared on the same scale. The measured wave informed the persona prior, so the comparison is calibration, not independent validation.
5. **Inference and robustness.** Frequentist paired contrasts, a one-sample Bayesian check, a 256-point joint mechanism/weight sweep, three topology variants, three persona priors, a leave-one-filter-out ablation, a source ablation, and a bootstrap over the 52 anchors were reported.
6. **Reproducibility.** The seed, code, cached embeddings, transcript, and generated outputs are included; the appendix gives the commands.

The details of the semantic, debiasing, manifold, and factorial steps are in Section 5.5; the results are in Chapter 6.

5.6.4 Plain-Language Guide to the Formal Steps

The formal parts answer simple questions:

- **DML.** Predict each feature from the nuisance variables (demographics, response style, ride context) and keep the residual. In plain terms: remove what is explained by the person and the ride before looking at what is left.
- **IPW.** Reweight the sample so the high-interaction group is not overrepre-

sented. In plain terms: give the quiet and the talkative riders a comparable say.

- **PCA, LLE, and Laplacian eigenmaps.** Compress 198 columns into six axes that preserve global spread, local neighbourhoods, and network smoothness. In plain terms: summarise the data without losing the shape.
- Q_0 **and personas.** Fit a smooth distribution to the 52 debiased rows and draw synthetic profiles from it. In plain terms: create plausible synthetic listeners that sit inside the measured cloud, not 500 new humans.
- **Factorial ABM.** Test all eight on/off combinations in the same persona across 18 sessions. In plain terms: compare the design bundles within the same synthetic person, so differences are not due to different people.
- **GRM-IRT.** Turn a latent score into item responses with thresholds. In plain terms: translate a person's standing into the kind of rating they would give on a five- or seven-point item.
- **Bayesian check.** Fit a one-sample model to the paired differences. In plain terms: ask whether the average within-person change is positive, and show how much the answer depends on the prior.
- **Joint sweep and PRCC.** Vary mechanisms and weights together, then rank their association with the result. In plain terms: which dials actually move the answer?
- **Zewail gate.** Multiply delay, distraction, and phase alignment into one attention score. In plain terms: attention is strongest when the probe is soon, the ride is quiet, and the listener is tuned in.

5.7 Feynman-Style Path Diagrams of the Experiment

The diagrams below use the visual grammar of Feynman diagrams—worldlines for trajectories, vertices for operations, arrows for processing order, and labels for multiplicity—as a formal explanatory device. They are not physical diagrams. Their purpose is to show exactly which records enter the experiment, how the persona distribution is generated, and what the experiment does to one persona.

5.7.1 From Records to Personas

Figure 5.10 reads from left to right. The quant block enters as 60 raw questionnaires, passes through the 21-field screen, and loses eight duplicate vectors at the dashed branch. In parallel, the four text streams enter the embedding vertex; the quantitative and semantic blocks meet at the manifold vertex, are compressed to 198 features, and generate the KDE prior Q_0. The five outgoing persona worldlines mark the multiplicity of the draw. A filled vertex is an operation, a dashed line is an excluded record, and a label gives the multiplicity.

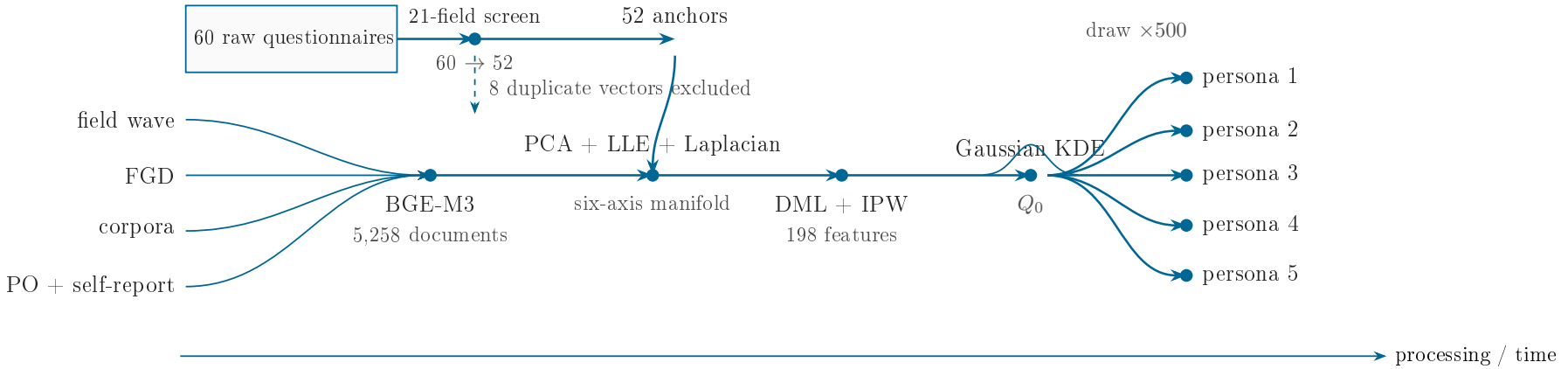


Figure 5.10: Feynman-style path diagram of persona generation. The 60 raw questionnaires become 52 anchors after the 21-field screen; the four text streams and the quantitative block are embedded, debiased, and projected onto the six-axis manifold; the KDE prior Q_0 generates the 500 persona worldlines. The figure is a formal explanatory device, not a physical model.

5.7.2 What the Experiment Does to a Persona

Figure 5.11 follows one persona. The incoming worldline splits at the cloning vertex into the eight A/B/C cells. Each cell runs for 18 sessions; the dashed band marks the Zewail attention gate that filters decoupled background play. The eight terminal worldlines meet at the GRM-IRT vertex, which emits the three measured outcomes. The diagram is a representative path with multiplicity labels, not a drawing of every individual run.

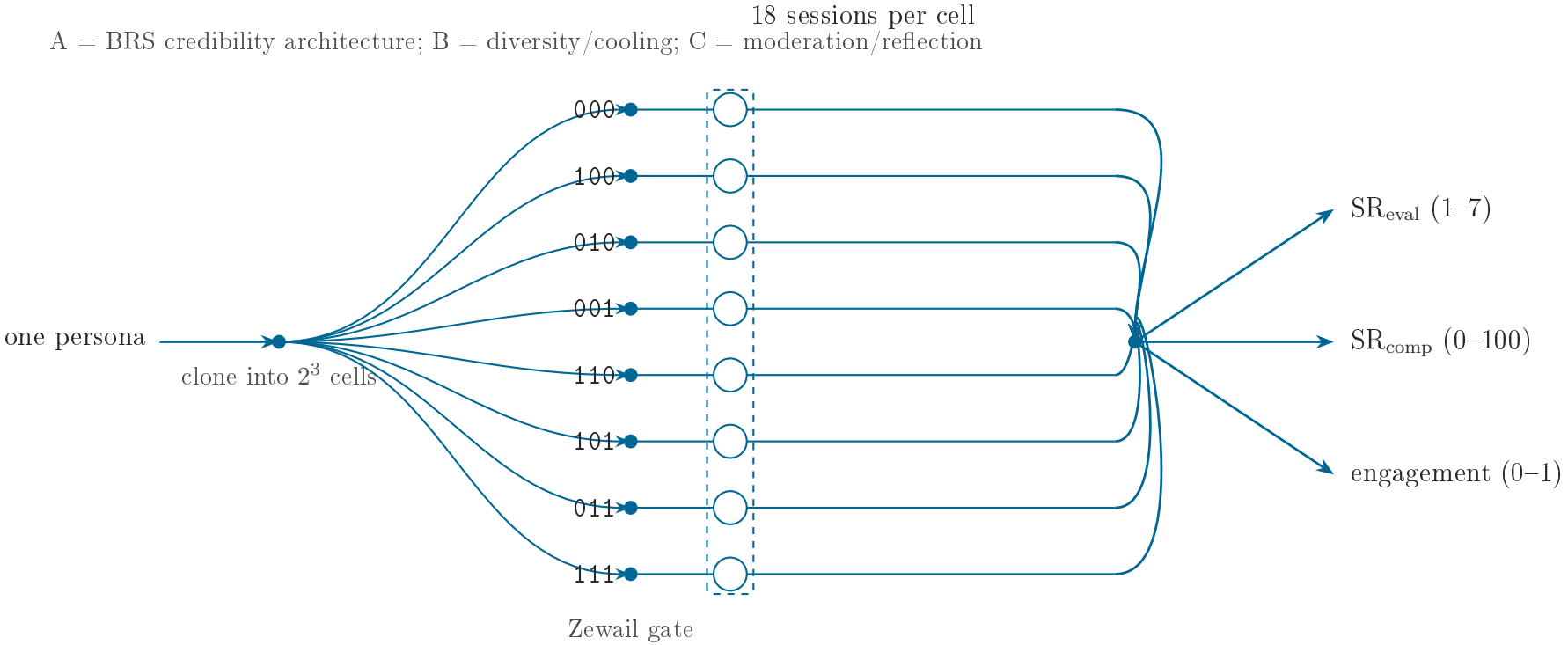


Figure 5.11: Feynman-style path diagram of one persona through the experiment. The persona is cloned into the eight A/B/C cells, run for 18 sessions per cell, filtered by the Zewail attention gate, and scored through the GRM into the evaluative rating (1–7), competence-format rating (0–100), and engagement (0–1). The diagram shows a representative path; the run repeats it for 500 personas.

The two diagrams make the generative logic auditable: the left-hand side of Figure 5.10 is the empirical anchor, the middle is the debiasing and manifold transformation, and the right-hand side is the synthetic population. Figure 5.11 then shows that the experiment does not treat the persona as a black box: it applies the A/B/C design, updates the states session by session, and measures the terminal state with explicit scales.

5.8 Analysis-Plan Discipline

The original research plan distinguished H1–H4 as primary hypotheses and H5–H7 as exploratory. The identifiers are retained, but H7 is now explicitly reformulated as a prospective computational performance hypothesis, not a historical field endpoint. The supplied analysis exports do not include a timestamped public registration identifier, so this thesis does not use the primary designation as proof of prospective registration. Any future confirmatory deployment should archive its hypotheses, scoring rules, exclusions, and analysis plan before enrolment and before inspecting outcome contrasts.

Table 5.10 gives the specified operationalization of each hypothesis: its status (primary or exploratory), its primary endpoint, and the specified expected direction.

The computational extension has a direct task-performance endpoint, not a retrospectively administered human competence measure. It operationalises the prospective H7 without reinterpreting historical PB/star effects as student learning. The extension is not reported as executed. Freezing outcomes, configurations, and exclusions before the next evaluation follows the registered-report discipline

Table 5.10: Hypothesis Endpoints and Their Analytical Status

Hypothesis	**Status**	**Primary endpoint**	**Expected direction**
H1 Structural credibility	Primary	PB composite vs. audience broadcaster battery; ΔPB arm contrast	Positive
H2 Diversity mechanism	Primary	Model ΔECO; no direct field source-diversity trajectory	Positive
H3 Psychometric validity	Primary (mixed)	α, ω, item-total r, CFA fit	Internal consistency supported; structural fit examined (mixed)
H4 EST integration	Primary	Correlation manifold over epistemic measures	Positive
H5 DeGroot echo dynamics	Exploratory	Correction acceptance; echo/density	Positive (mechanism)
H6 Scheduling duration	Exploratory	Duration–PB association; continued listening	Positive / trade-off present
H7 Participatory performance	Prospective computational	Keyed task accuracy (primary); evidence-based production (secondary)	Positive prediction; not guaranteed

described by Chambers and Tzavella (2022).

5.9 Prospective FGD-Informed Simulation Design

The FGD will inform a new rule-based agent-based model (ABM) comparator and the planned LLM-agent extension. It does not retrospectively recalibrate the historical simulation. Monte Carlo replication is the repeated-run procedure around either model, not a third source of human observations. Both future implementations must specify states, decision mechanisms, exposure conditions, assessment tasks, failures, and software versions before execution.

5.9.1 Shared Qualitative Design Hypotheses

The machine-readable design record preserves each domain, its source label, a candidate rule, a corresponding prompt constraint, and unresolved quantities. All settings remain assumptions for review, not estimates from the FGD. No population weights or transition probabilities are derived from word counts. The supplemental generated analysis is registered as review-only material: its asserted quotations, consensus, and outcomes are excluded from individual anchors, calibration targets, and factual keys. Questions about programme length or registration feedback require source verification before they can support empirical constraints. The design bundle records the export and its adjudication hashes, but does not pass its raw prose into participant-agent prompts. Changing that export invalidates the prior review until reassessed.

Table 5.11: FGD Design Dimensions for Both Planned Agent Models

Domain	Candidate rule-based mechanism	LLM constraint	Unknown quantity
FGD-T1	Separate listening, private-practice, and publication states	Do not force publication or infer anxiety	State weights and transitions
FGD-T2	Gate shared broadcasting on peer availability when required	Allow refusal; respect available actions	Coordination costs and preferences
FGD-T3	Separate eligibility from identity display	Do not assume campus access creates trust	Privacy preferences and visibility effects
FGD-T4	Represent only implemented cues and controls	Use actual exposure, not presumed benefits	Comprehension and response probabilities

The G0/G1 contrast remains the bundle in Section 5.10.1. Within a scenario, FGD settings remain constant across G0/G1 and, where operationally comparable, across the two engines. A co-hosting or privacy manipulation is a separately de-

clared factor, not an undeclared addition to G1. Scenarios are not pooled under an assumed population distribution; any future weights require an explicit empirical or assumption-based rationale.

5.9.2 Calibration, Holdout, and Provenance

Future verified extracts may support bounded constraints or calibration targets after source and permission review. Development, calibration, and held-out evaluation partitions must be distinct. Records already read during this revision are not untouched validation data. Anticipated product benefits, historical PB outcomes, and answer keys must not enter participant-agent prompts. Fit to a calibration target is not validation against that target.

Each run must preserve the design-bundle hash, scenario and rationale, paired seeds, engine and rule/prompt version, platform commit, content and answer-key hashes, frozen roster, actual exposure, action logs, failures, and pre/post scores. The same seed need not produce the same draws in different engines: shared exogenous schedules must be generated and archived explicitly. Language-model runs also require provider settings and actual returned versions.

5.9.3 Monte Carlo Summary and Interpretation

For one fixed scenario and engine, let $A_{g,r,t}$ denote mean independently keyed accuracy in condition $g \in \{0, 1\}$, paired network replication r, and time $t \in \{\text{pre}, \text{post}\}$.

Define

$$D_r = (A_{1,r,\text{post}} - A_{1,r,\text{pre}}) - (A_{0,r,\text{post}} - A_{0,r,\text{pre}}), \tag{5.14}$$

$$\widehat{\Delta} = \frac{1}{R}\sum_{r=1}^{R} D_r, \tag{5.15}$$

$$\text{MCSE}(\widehat{\Delta}) = \sqrt{\frac{\sum_{r=1}^{R}(D_r - \widehat{\Delta})^2}{R(R-1)}}. \tag{5.16}$$

The MCSE requires $R \geq 2$ independent, identically configured paired replications with finite variance. **In plain language:** compare how much each condition changes, average those differences, and report the uncertainty caused by performing finitely many runs. Monte Carlo error does not quantify model misspecification or uncertainty about people. Reporting replication error follows the principle discussed by Morris et al. (2019).

Report completed and planned pairs and every failure. A complete-pair summary is conditional on completion, not permission to discard failures. The wrapper's planned-set worst-case bounds and replication-percentile ranges remain distinct from MCSE and an interval for the mean. A Bayesian extension needs its own likelihood, priors, posterior checks, and archived draws; the FGD alone specifies none of those distributions. No synthetic accuracy gain establishes human competence improvement.

The prepared configuration and reserved reporting structure in Section I.4.2 provide entry points for actual logs and reviewed estimates. Neither preparation nor an API key supplies a functioning simulator, approved platform adapter, or completed validation study.

5.10 Planned Extension: Qualitatively Anchored Agents on LOKLOK

This section specifies the extended study design as a *future validation protocol*, not as part of the completed empirical contribution: no LLM-agent platform run contributes to any result in this thesis. Its status is prospective; completion will require a real, supervised deployment. This extension is distinct from the completed questionnaire and qualitative analysis and from the historical calibrated ABM. Its purpose is to let LLM agents interact with the real platform implementation in an isolated research deployment, rather than only answer prompts describing a hypothetical platform. The same versioned backend and client behavior will be exercised, but synthetic accounts must not contaminate public ratings, student credentials, or live community feeds. The inspected routes and the separation between account assignment and shared radio state are specified in Section 4.13. The adapter uses independently reset condition-specific instances and verifies actual exposure; merely obtaining an A/B account label is not treated as receiving the intervention.

5.10.1 Computational Scope and Experimental Conditions

The extension is a platform-in-the-loop ABM with LLM-controlled agents. Its objects are computational social dynamics, synthetic media production and circulation, and algorithmic radio scheduling. This changes the scope of the prospective test, not the provenance of the completed human field study. It does not turn measured rider accounts into synthetic observations or establish a human student-impact claim

without human outcome evidence.

G0 is a fixed, nonadaptive schedule with dynamic BRS feedback disabled for content selection and display. G1 enables the scalar beta-shaped feedback path and a Dirichlet category scheduler. The same content library, available source evidence, exposure budget, platform interface, and assessment tasks are used in both. The intervention includes the use of the BRS in a declared visible cue or ranking rule, not merely writing scores to an unused state. These are proposed experimental configurations, not features assumed to be fully wired in the supplied backend.

At the start of each prespecified scheduling block, draw a category mixture:

$$\boldsymbol{\pi} \sim \text{Dirichlet}(\boldsymbol{\alpha}).$$

Keep that mixture fixed within the block. At each slot, restrict it to eligible categories, renormalize, and draw a category; a declared fallback handles an empty eligible set. Freeze the block length and positive prior parameters before each run. Log eligibility, cooldown rules, the drawn mixture, and the actual selection probabilities. This basic Dirichlet randomization is not by itself learned preference adaptation or Thompson sampling over reward estimates. Any adaptive update of the parameters is a separately versioned mechanism with its own update equation and test. The existing radio-design priors are candidate design settings, not estimated student preferences. No unspecified stochastic ratcheting mechanism is assumed to exist.

A two-arm G0/G1 contrast identifies the effect of this bundle within the simulated system. It cannot distinguish a feedback effect from a scheduling ef-

fect. A four-cell factorial ablation—feedback off/on crossed with fixed/ Dirichlet scheduling—evaluates the separate mechanisms and their interaction. If those ablations are not executed, interpretation remains at bundle level. Prompt authors and agents are not given the expected direction or the labels "better" and "worse" for the conditions.

5.10.2 Network Outcomes Without an Assumed Power Law

Use event-level exposure and endorsement logs with fixed topic and source labels. For a prespecified window, let p_c be the proportion of delivered exposures assigned to topic c in the frozen topic taxonomy. Report Shannon entropy $H = -\sum_c p_c \log p_c$, concentration $\sum_c p_c^2$, and source coverage. Each exposure receives one primary topic under the coding rule; uncertain assignments are identified rather than silently reclassified. For $K > 1$ available topics, $H/\log K$ gives a normalized summary with the same reference taxonomy in all conditions. Empty windows are missing, not zero-diversity evidence. A one-topic universe has entropy zero and no meaningful between-topic diversity contrast.

Network size, edge definition, window, and whether edges represent exposure, endorsement, or reshare are reported separately. High topic entropy need not mean high information accuracy. A zeta distribution is an optional discrete power-law model, not the default definition of entropy. Its use requires an explicit support and cutoff, fitted parameters, goodness-of-fit assessment, and comparison with competing degree/count distributions. Power-law behavior must not be assumed merely because a social network is being simulated (Broido & Clauset, 2019). Voitalov et al. (2019) illustrate how different tail definitions and estimators can change this

assessment, reinforcing the need to specify the fitted model. A power law imposed by the generator is an input assumption, not discovered evidence for EST-P2.

5.10.3 What the Computational Evidence Can Establish

P1 is addressed by the verified architecture contrast in observed agent actions and task performance, conditional on the model and environment. P2 is addressed by network/exposure distributions, ablations, and competing predictions; entropy alone does not establish a physical thermodynamic law. P3 is addressed by scoring integrity, reproducibility, and criterion-specific calibration. A beta-shaped BRS represents an assumed rating process: agreement with held-out normalized ratings tests prediction of those ratings, whereas independent content-accuracy labels test a different criterion. No factual probability interpretation is assigned to BRS without a corresponding model and calibration test. NMLS factor fit in agents does not supply that evidence. P4 is examined only as a computational affordance/production hypothesis, using tasks and capability ablations. Such findings do not establish improved human competence. The thesis therefore evaluates bounded consequences of EST rather than claiming exhaustive validation of the theory.

5.10.4 Research Questions and Agent Roles

The extension asks whether individually grounded descriptions improve behavioral fidelity over unanchored or demographic-only agents; whether specific platform features change synthetic exposure, participation, and evaluations; and whether these differences survive alternative prompts, models, and scheduler settings. These are prospective questions, not additional confirmations of H1–H7.

Table 5.12: Planned Agent Roles and Boundaries

Role	Permitted research behavior	Boundary
Anchored listener	Open eligible broadcasts, listen or inspect the supplied modality, skip, rate, vote, or comment	May remain silent, disagree, or leave; no instruction to praise LOKLOK
Student creator	Propose topics and submit clearly synthetic clips or posts through platform functions	No impersonation of a named student; sources and synthetic provenance recorded
Host or discussion agent	Present assigned, reviewed stimuli and respond within a fixed role	No unsupervised invention of factual news for human audiences
MAYA and ERIC services	Provide configured coaching or integrity flags if included in the condition	Separate from participant agents; consequential moderation remains reviewable
Independent evaluator	Score prespecified logs and held-out outcomes	No access to hidden participant outcomes through listener prompts; human audit of subjective judgments

5.10.5 FGD Inputs Shared with the Rule-Based Comparator

The FGD enters through Section 5.4.1 and the common design record in Section 5.9. Narrative-level hypotheses are not individual biographies or verified utterances. A reviewed exploratory scenario must explicitly acknowledge its assumption-based status and preserve identical FGD settings across the paired G0/G1 conditions. An API key alone does not approve these assumptions or supply missing platform and assessment work.

5.10.6 Anchoring Without Outcome Leakage

The current rider notes, profiles, and topic suggestions are a development corpus. They already contain post-listening praise and known questionnaire outcomes; they cannot serve as an untouched confirmatory validation set after this revision. The researcher plans to supply additional qualitative material and student descriptions.

Before use, each anchor should specify its source, consent scope, collection timing, observed preference, uncertainty, and which parts were inferred. Names, exact journeys, employers, contact details, and unnecessary biographical identifiers must be removed.

A bounded persona may contain individually stated topic interests, preferred pace, tolerance for interruption, accessibility needs, and relevant prior media experience. It must not equate a faculty with a personality or force an agent to reproduce an existing favourable quote. Researcher reflections must be labelled as interpretation rather than converted into student facts. Absent information remains unspecified. The same anchor is held constant when comparing platform conditions.

New outcome measures must be withheld from prompts, retrieval memory, and parameter tuning. If retrospective development uses the current 52 cases, its results must be labelled in-sample or exploratory. A prospective test should use new participants or outcomes collected after a frozen protocol, with a separate development/validation partition. Anchor/no-anchor and individual-preference/demographic-only comparisons test the contribution of anchoring rather than assuming that more descriptive text is better.

5.10.7 Platform-in-the-Loop Procedure

The deployment will record a backend commit, client build, content manifest, reputation-update variant, model identifier/version, prompt hash, decoding settings, tools, and time budget. An adapter will expose only approved platform actions and return the actual state generated by the backend. Routes and API contracts must be taken from that implementation, not invented in the thesis. Each

run starts from a recorded snapshot with a separate synthetic-account namespace and reset reputation and recommendation state.

Three technical constraints minimise infrastructure and factual hallucinations so that platform effects are not confounded with model artifacts:

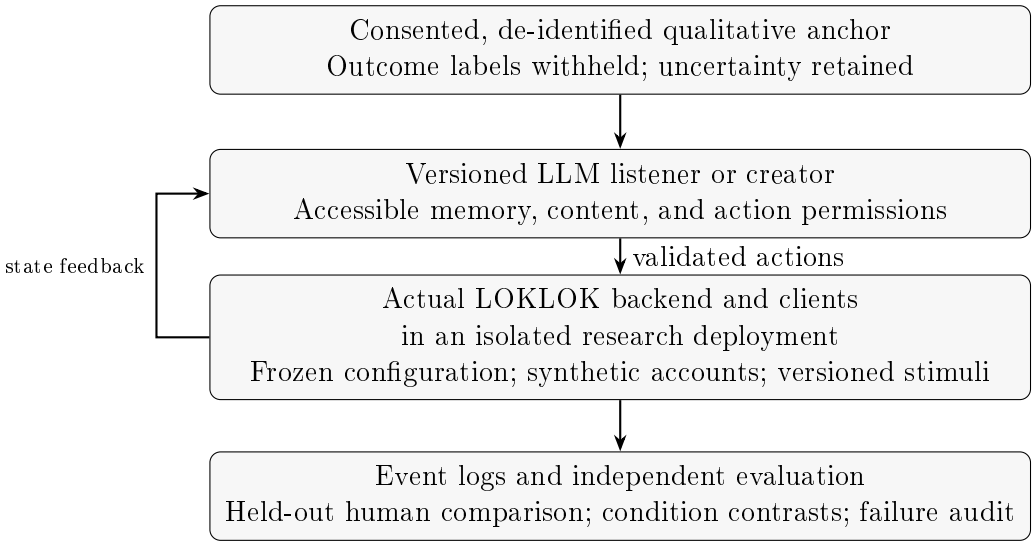
- **Strict adapter enforcement.** The agent never interacts with the backend directly. It outputs a structured JSON action; the Python adapter validates it against the actual platform state. If the agent claims an action or state the platform did not serve, the adapter rejects it and records an `invalid_action` event. A claim of having seen a BRS score or warning that the backend never rendered therefore cannot survive into the log.
- **RAG with hard constraints.** The prompt instructs the agent to use only the provided text; if the evidence is insufficient it must output `INSUFFICIENT_EVIDENCE`. Inventing a source yields a zero score under the frozen claim key, and the evidence-fabrication index (Section 5.10.12) records the event rather than hiding it.
- **Frozen context windows.** The agent's memory is truncated or reset at defined intervals to prevent the instruction drift over long conversations that sustains unchecked fabrication.

The estimator inputs for the three divergence metrics (Section 5.10.12) are therefore intrinsic to the loop: the adapter's validated event log, the agent's own memory/reflection log, and the anchor embeddings, all written under the recorded versions above.

At each decision point an agent receives its approved anchor, accessible plat-

form state, relevant memory, and the current content modality. It selects an allowed action; the adapter validates and executes it; the resulting state and event are logged. Posts and transcripts are untrusted content, not authority to change experimental rules or reveal other students' information. Evaluation logs retain actions, timestamps, content versions, and brief decision summaries, not a purported record of human mental processes.

Figure 5.12: Planned Evidence-Anchored, Platform-in-the-Loop ABM Extension



Note. Proposed workflow, not a completed run or a fitted causal model.

The initial comparison will keep content availability and anchors fixed while varying the prespecified platform feature set. Individual accounts interact, so the unit of replication is a reset network run, not every message or agent action. Repeated runs and paired starting states can characterise model variation, but they do not increase the human sample size. Run counts, stopping rules, minimum precision targets, primary endpoints, and multiplicity rules must be fixed after feasibility testing and before confirmatory execution. A feature audit must resolve the BRS mismatch in Section 4.1 so that the implemented intervention is the one reported.

5.10.8 Execution Readiness and Required Evidence

The current archive does not include an executable platform, the historical engine, or a verified multi-source anchoring dataset. Demonstration telemetry is not a substitute for those artifacts. No live-backend validation, 15-source evidence synthesis, synthetic competence consensus, or machine- independent numerical reproduction is claimed here.

The accompanying research workflow provides input templates, a source register, descriptive calculations, an offline knowledge graph, and a gated experiment interface. Empty templates remain pending. Survey responses, focus-group records, researcher self-reports, independently keyed factual claims, and public corpora retain separate evidence types. An interview or self-report describes experience; it is not automatically a factual answer key. A corpus requires a provenance and rights review before use. No arbitrary weighted average of these sources is treated as ground truth.

A versioned research adapter remains a required integration artifact. An API key alone does not supply a platform, consent scope, frozen prompts, source permissions, or an independent assessment key. The workflow refuses live execution until its configuration gates and adapter requirements are met. Local and notebook execution use the same files and record hashes; provider-side nondeterminism must be reported rather than denied.

5.10.9 Content, Modality, and Outcome Definitions

The planned topic briefs will form a versioned stimulus bank, grouped into campus/civic information, work and careers, science/technology, wellbeing and creative audio, and media-literacy/governance topics. They are proposals: no debate recordings, factual source checks, or student permissions are implied by the text file. Stimuli should balance length, source quality, speaker roles, and order across conditions. Claims about health, politics, or local services require review before experimental delivery.

A text-only agent supplied with a transcript cannot evaluate vocal tone, buffering, or sound design directly. Such outcomes must either be excluded, measured from actual telemetry, or studied with a documented audio-capable model. Simulated elapsed time must not be described as human experienced journey time. The original PB3 concerns comparative trust and PB4 clarity; verification actions and BRS-display effects need separate prespecified measures rather than relabelling those items.

The primary computational task-accuracy outcome is specified in Section 5.10.12. Process and secondary outcomes include eligible broadcasts opened, listening or processing completion, skips, topic exposure distribution, source checks, sharing, correction uptake, and synthetic PB/star responses. Report the denominator, scale, and time window for each. Repetition/cooldown violations, invalid actions, tool failures, refused outputs, and model costs are feasibility outcomes, not participant misconduct. Personality fidelity requires comparison with held-out human observations; plausible dialogue alone is insufficient (Bisbee et al., 2024;

Park et al., 2024).

5.10.10 Safeguards and Interpretation

Execution follows the applicable research approval and data-use permissions; the design does not itself constitute authorisation for external processing. Consent must cover the intended model processing and persona use; access must be limited and withdrawal/retention procedures specified. An agent may emulate selected documented preferences, not claim to be a named student or provide that student's consent. Report divergent and unsuccessful behavior as well as apparent alignment. The resulting estimates will remain synthetic, model- and implementation-conditional results, even when the software is the real platform.

"Hallucination" is not a monolith in this protocol. Three distinct and measurable types are retained, each with its own origin in the pipeline and its own safeguard:

- **Epistemic/factual hallucination (task failure).** The agent appraises a claim incorrectly against the frozen evidence key, or cites a source or quote that does not exist in the provided context window. It is dosed by the evidence-fabrication estimator in Section 5.10.12 and constrained by the retrieval and frozen-context rules below.
- **Platform/state hallucination (infrastructure failure).** The agent claims it received a BRS score, Echo-Meter warning, or specific post that the platform's event log shows it never actually received. This is a protocol violation because it breaks the adapter contract; it is measured by the platform-state-divergence

estimator in Section 5.10.12.

- **Persona/behavioral drift (anchoring failure).** The agent acts out of character for its qualitative anchor (e.g., a skeptical evaluator persona trusting a low-BRS broadcaster without source checks). It is measured by the persona-drift estimator in Section 5.10.12.

The first and second types are guarded against as infrastructure errors (Section 5.10.7); the third type is a fidelity quantity that anchoring is designed to reduce. A detector self-check on a fixed benchmark, whose factual state is known to the pipeline (the experiment, not the external model), establishes that the estimator itself flags the injected fabrication it is designed to flag, before any run is reported.

5.10.11 Computational Student-Agent Procedure

The extension evaluates synthetic media interaction on the actual platform, not human cognitive growth. The completed rider study remains human audience evidence. Student Rating retains its conceptual distinction from broadcast stars, but the computational test operationalises performance through independently keyed tasks, not by assuming that generated self-descriptions measure a student's competence. The primary outcome is claim-appraisal task accuracy; NMLS-format responses are exploratory secondary outputs.

1. **Separate development from evaluation.** Use current rider accounts as a development corpus. Split any new consented anchor cases by person, not quotation. Keep evaluation tasks, answer keys, and any human outcomes outside

prompts, retrieval stores, and tuning. Without a held-out human benchmark, report computational robustness, not established human fidelity.

2. **Construct bounded personas.** Record individually evidenced interests, media experience, constraints, source timing, and uncertainty. Separate observations from researcher interpretation. Exclude identifying biography, expected treatment effects, and instructions to reproduce praise. Each agent has an isolated memory store; only recorded platform interactions can transmit information between agents.

3. **Freeze the environment.** Record backend and adapter versions, model identifier, prompt hashes, decoding settings, network topology, activation order, content manifest, time budget, tools, memory limits, and clock policy. A custom event loop is sufficient; a framework name is not a validity criterion. Transcript-only agents receive a textual exposure, not verified audio listening, and their results are labelled accordingly.

4. **Pilot before confirmatory runs.** Ten independently reset G0/G1 network pairs are an initial engineering pilot, not a powered final sample. Check resets, treatment delivery, invalid actions, logging, costs, and run-level variability. Final run counts, precision targets, stopping and failure rules are frozen after development and before evaluation.

5. **Capture an uncontaminated baseline.** At T0, assess equivalent claim-appraisal tasks and optionally the authorised full NMLS in an isolated assessment branch. Return only the prespecified task score to the evaluator; assessment

answers and feedback do not enter the interaction memory. Clone the same pre-exposure state into paired conditions.

6. **Deliver the intervention.** Run matched G0/G1 networks on isolated platform instances. Agents may inspect, skip, verify, contribute, disagree, or remain silent. Hold model, persona, initial graph, budgets, assessment access, and source library constant within each pair. Log delivered exposure, not only the requested arm label. Section 5.10.1 defines the intervention bundle and mechanism checks.

7. **Assess outputs independently.** At T1, use an unseen alternate task packet of matched difficulty, counterbalanced across independent pairs. Hold assessment tools and access to a frozen evidence library equal across conditions. Score factual decisions against an independently prepared key. Audit subjective production scores with trained human assessors blinded to condition; a second LLM's agreement alone is not independent validation.

8. **Distinguish persistence from retention.** An optional T2 branch varies memory retention, truncation, or reset under a fixed model and tests new tasks. It is a model-state persistence experiment, not elapsed-time human retention. Runtime memory changes do not establish a change to model weights or human learning.

5.10.12 Computational Estimands and Analysis

For each agent, task accuracy is the number of correct decisions divided by the fixed number of assigned claims, on $[0, 1]$. Each claim has one of the three keyed evidence states in Table 5.13. Abstention or malformed output on a delivered task counts

Table 5.13: Computational Task Outcomes and Their Interpretation

Task	Recorded output	Interpretation
Appraise unseen claims	Supported, contradicted, or insufficient-evidence decision against a frozen key	Primary keyed task accuracy, not self-reported literacy
Locate and summarise	Opened sources, summary, cited passages	Observable retrieval and attribution behavior
Produce a contribution	Synthetic post or clip, evidence and audience rationale	Rubric-scored production, not a human learning gain
Revise after correction	Initial artifact, correction and revision history	Correction uptake under specified memory and tools
Transfer to unseen content	Performance on a matched new topic	Model-conditional task generalization

Note. An author-specified benchmark, not a newly validated psychological scale. The claim key and scoring rubric are external to the tested agent; items without an independently agreed evidence-state key are excluded before evaluation.

as unsuccessful performance in the primary score and is also reported separately; an infrastructure delivery failure is not a student or agent decision and remains a documented missing assessment. Report the class balance and class-specific errors, including comparison with a constant majority-class decision, so accuracy is not interpreted without its task composition. The frozen failure policy determines whether an affected network pair is eligible, rerun, or analysed in a sensitivity analysis. Selective removal of unfavourable outputs is prohibited.

Let $\bar{A}_{rt}^{(g)}$ denote the equally weighted mean accuracy of the agents in network pair r, condition g, at time t. The primary estimator is

$$\widehat{\tau}_{\text{agent}} = \frac{1}{R_{\text{runs}}} \sum_{r=1}^{R_{\text{runs}}} \left[(\bar{A}_{r1}^{(1)} - \bar{A}_{r0}^{(1)}) - (\bar{A}_{r1}^{(0)} - \bar{A}_{r0}^{(0)}) \right]. \tag{5.17}$$

This estimates a model-conditional difference in task-accuracy change, not a population effect on university students. Uncertainty is calculated from independently reset paired-network contrasts, not from individual posts or from the nominal num-

ber of personas. Report the effect in accuracy-point units, its interval, run-level distribution, failure rate, and sensitivity to alternate task packets. Paired seeds aid comparison but do not guarantee identical API outputs; archive returned outputs and separate random streams for scheduling, network activation, and model calls.

Production, exposure diversity, correction uptake, engagement, PB/stars, NMLS-format domain means, and T2 persistence are secondary. Their families and multiplicity strategy are frozen before confirmatory analysis. A positive NMLS-format shift cannot override poor task performance. The NMLS scoring script computes those secondary profiles only; it does not implement the primary task estimator in Equation 5.17.

Hallucination and Divergence Metrics

The extension separates platform effects from model artifacts by auditing every agent output against the pipeline's own records along the three dimensions catalogued in Section 5.10 (task, platform, and persona). These are planned estimators; the extension has not run, so no executed values exist and the reserved reporting table in the results chapter is left "Pending." The audit logs that enter the denominators come from the adapter's validated event log and the agent's memory/reflection log described in Section 5.10.7.

The *evidence fabrication index* (EFI) scores the task-level failure: the proportion of appraised claims i for which the set of sources cited by the agent, S_i, is

not contained in the frozen context window C:

$$\text{EFI} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}[S_i \notin C]. \tag{5.18}$$

Larger values mean the agent increasingly justifies appraisals with sources that were never supplied.

The *platform state divergence* (PSD) scores the infrastructure-level failure by comparing the states the agent *claimed* to have seen in its reflection step, M_{agent}, with the states the adapter actually served, E_{platform} (BRS scores, post IDs, warnings, content versions):

$$\text{PSD} = 1 - \frac{|M_{\text{agent}} \cap E_{\text{platform}}|}{|M_{\text{agent}} \cup E_{\text{platform}}|}. \tag{5.19}$$

PSD $= 0$ is perfect state alignment; values above the frozen tolerance signal adapter-contract violations that the adapter's `invalid_action` rejection cannot silence after the fact. When either set is empty or a reflection contains no states to audit, the flag is reported as a missing assessment, not silently treated as zero.

The *persona drift score* (PDS) scores the anchoring-level divergence between the distributional meaning of the agent's generated reflection, $\vec{v}_{\text{reflection}}$, and the embedding of its qualitative anchor, $\vec{v}_{\text{anchor}}$:

$$\text{PDS} = 1 - \frac{\langle \vec{v}_{\text{reflection}}, \vec{v}_{\text{anchor}} \rangle}{\|\vec{v}_{\text{reflection}}\| \, \|\vec{v}_{\text{anchor}}\|}. \tag{5.20}$$

The embedding function is fixed (the multilingual model installed in the notebook), identical across paired conditions, and versioned with the run. A high PDS is not itself a failure of the claim key; it is a fidelity quantity that anchored design

is intended to keep small, and it is reported per-arm rather than pooled across conditions.

A detector self-check anchors the estimators themselves: on a held-out benchmark whose factual state the pipeline knows (the experiment, not the third-party model), each index must flag the injected evidence fabrication it is designed to flag before any confirmatory run is interpreted. Policy links the trio to the offending artifact: fabricated sources, unserved platform states, and out-of-anchor reflections are audited against the frozen context log, the adapter event log, and the anchor embedding store, respectively. No executed statistic from this extension is used as thesis evidence until the reserved results table is filled by a real, supervised deployment.

5.10.13 Measurement and Optional Human-Fidelity Evaluation

The original NMLS evidence concerns human self-report (Koc & Barut, 2016). Generated agreement responses are treated as role-conditioned outputs. Any ordinal factor analysis of them describes synthetic covariance under specified prompts, models, and within-network dependence; it does not validate human literacy or prove that the simulated agent possesses the human construct. Reliability and invariance diagnostics can expose poor measurement, but cannot establish construct identity on their own (Flora, 2020; Putnick & Bornstein, 2016).

The computational study can proceed without a new human intervention if its claims remain confined to software-mediated agent behavior. A claim of human behavioral fidelity additionally requires a separate held-out benchmark with matched tasks and measured human outcomes. Compare distributions, variability, covariance, and prediction error against anchored, unanchored, and demographic-

only baselines; demographic conditioning alone is not a representativeness guarantee (Argyle et al., 2023; Bisbee et al., 2024). No synthetic response adds to the human sample size. Human learning or retention would require a separate longitudinal human design and its own measurement and ethical evidence, outside the present computational scope.

The run count is selected for precision in the computational contrast, not from an assumed meaningful human effect. Pilot estimates of variability are uncertain, so planning includes conservative sensitivity assumptions (Lakens, 2022). Agent simulations may reproduce some patterns while distorting others (Aher et al., 2023). Failure or disagreement with EST is retained as a result, not corrected by changing prompts after evaluation.

5.11 Operational Definitions Across Evidence Streams

The unit and target of a rating determine its meaning. In the field study, students evaluate broadcasts: BR, SV, and IQ are unweighted means of four, four, and three items respectively, with the negatively worded BR item reversed before aggregation. These items use zero-to-four responses. PB is the mean of five one-to-seven experience items, and the one-to-five star response is a separate global judgment. These are audience-perception measures, not independently keyed judgments of factual accuracy. No new aggregate combines stars, PB, BRS, and media competence into one number.

Human media competence denotes critical consumption and responsible production; it requires its own assessment. The proposed agent study instead

operationalizes synthetic appraisal performance as the proportion of assigned claims correctly classified against independent evidence keys. Its optional media-literacy questionnaire responses are descriptive model outputs. Neither the backend's broadcaster-rating posterior nor an agent's self-description is an observed human competence score. Thus the original title identifies the broader research problem without changing the estimand of each component.

Architecture means the explicitly enabled feedback and scheduling rules, their visible outputs, and their access constraints. Persona means the approved contextual description available to an agent. Computational state means the logged memory and numerical constraints used during a run. Topic diversity means the distribution of delivered exposures under a fixed coding taxonomy. None of these definitions equates high engagement, high entropy, or a high reputation score with truth. Proposed posting frequency, attention limits, and initial bias parameters are experimental assumptions unless a specific measurement supports them; they are not derived automatically from academic discipline or a favourable rider comment.

5.12 Prospective Platform Experiment: Execution and Analysis

This procedure extends the design in Section 5.10.1. It specifies how later observations will be produced and analysed, rather than reporting an experiment already completed. The platform comparison retains the fixed-schedule/no-dynamic-feedback control and the feedback-plus- Dirichlet-scheduling treatment. The historical engagement-optimizing engine is a separate model; its condition labels and

fitted outcomes are not substituted for these runtime interventions.

5.12.1 Persona Construction and Controlled Initialization

Each source-supported persona field will retain its evidential status: reported background, researcher inference, or computational assumption. Names, contact details, journey locations, and identifying combinations will not enter prompts. Approval for storage does not imply approval for external model processing. Statements about enjoying LOKLOK, desired ratings, and post-exposure outcomes will not become baseline instructions. The same profile series previously used for thematic analysis is development material, not an untouched validation sample. Unknown preferences remain unspecified.

Preference-grounded, broad-category-only, and neutral personas define a planned sensitivity comparison, crossed with the platform conditions. The broad-category condition uses only approved general descriptors and must not silently retain preference text. The neutral condition retains necessary language and task instructions, not a hidden personal history. Within a matched block, use the same actor roster, initial graph, content supply, assessment allocation, and resource ceilings. This yields six network instances per block for three persona variants and two platform conditions. Crossing all variants with the four feedback-by-scheduling cells would yield twelve instances; this is a separate factorial extension, not extra independent participants.

Numerical state controls will be defined outside free-form prompts. An action probability, if used, requires a bounded range, distribution, seed, and logged draw; descriptive prose alone does not enforce a probability. Exogenous time and capacity

constraints are identical across matched arms. State changes require a declared transition rule and an event record. Randomized assumptions are not described as measured student traits.

5.12.2 Isolated Deployment and Decision Cycle

Each network condition will receive a fresh database namespace, account roster, message namespace, media state, schedule, and agent memory store. Only approved synthetic accounts and permitted content enter this runtime; the participant archive and answer keys remain outside its accessible storage. A reset must be demonstrated by state fingerprints before the second condition starts. Execution order is randomized or counterbalanced to reduce time-related service effects. Sequentially switching rules within an already exposed population is not the primary design.

At each decision opportunity, the controller supplies only currently visible content and permitted actions. The model returns a structured decision. The controller checks schema, authentication, eligibility, exposure history, resource bounds, and idempotency before submitting the corresponding platform action. It records the request, acknowledgement, resulting state, and subsequent delivered observation. Choosing to ignore is a valid decision; an unavailable stream or rejected request is an execution event, not evidence of dislike. Self-ratings, duplicate feedback, ratings without eligible exposure, and administrative operations are disallowed by server-side rules.

The action set is determined by tested platform affordances. Tuning in, ignoring, and rating can form a minimal listener experiment. Initiating a stream additionally requires permitted media, successful ingestion, and observable play-

back; requesting a schedule alone does not establish a broadcast. Production and correction tasks require corresponding verified publication and revision paths. If only transcripts are delivered, the study will report text-mediated interaction with radio content, not auditory listening. Identical modality and evidence access apply across conditions.

The supplied all-in-one deployment is an implementation starting point, not a requirement for a particular reverse proxy or a proof of working service separation. Treatment delivery will be checked at the level of selected content, exposure probabilities, displayed scores, and committed updates. Merely assigning an experimental label is insufficient.

5.12.3 Assessment Independence and Frozen Allocations

Before evaluation, an accountable independent assessment process will approve the evidence library, claim labels, and production rubric. Claim outcomes are supported, contradicted, or insufficient evidence relative to that library. Correctness does not depend on agreement with a majority of agents. Answer keys are unavailable to the acting agents, persona constructor, and platform feedback algorithm during exposure. Development tasks and final assessment claims are disjoint.

Baseline and post-exposure packets use distinct claims with matched content coverage and a counterbalanced allocation. The two platform conditions in each pair receive equivalent packets at the same assessment point. Prompt, model, tokenizer, context budget, sampling settings, runtime, and evidence versions are frozen. Seed reuse supports controlled comparison but does not guarantee deterministic model execution; raw responses and actual served versions are retained. A later memory-

persistence assessment describes computational state under a declared reset or delay, not human retention.

Ten independently reset pilot pairs will assess operational feasibility, failure modes, output variation, and resource use. Pilot data are separated from confirmation. The main number of network pairs is determined before opening main outcomes using a justified precision target and pilot variability, with an allowance for expected failed pairs. No additional run is added because an effect is almost significant. Model, task, or intervention changes after the freeze produce a documented new analysis set.

5.12.4 Estimand, Failure Policy, and Sensitivity

Let a_{rgt} be the equal-agent mean of assigned-claim accuracy in network pair r, platform condition g, and assessment time t. Each agent's denominator is its assigned claims, not just its answered claims. The primary paired contrast and overall estimate are

$$d_r = (a_{r,1,1} - a_{r,1,0}) - (a_{r,0,1} - a_{r,0,0}), \qquad \widehat{\Delta} = \frac{1}{R}\sum_{r=1}^{R} d_r.$$

Accuracy is in $[0, 1]$ and d_r in $[-2, 2]$. Results use accuracy points; multiplication by 100 yields percentage points, not relative improvement. Every retained pair has the complete frozen roster in all four cells. The replication unit is a reset network pair, not a message or an individual answer. Repeated copies of a persona estimate variation conditional on that persona set, not a larger representative sample of students.

Valid abstentions and malformed assessment outputs count as incorrect under

the primary assigned-task rule. Infrastructure failures are recorded separately; a pair with an incomplete required cell is excluded from the complete-pair estimate rather than silently repaired. All planned pairs, attempts, retries, and reasons remain in the report. Report worst-case bounds for the planned-set mean by assigning incomplete contrasts the endpoints -2 and 2. Predeclared retry limits distinguish a transport retry from asking the model repeatedly until it gives a favourable answer.

The primary uncertainty summaries will include a paired-contrast mean interval and a pair-resampling percentile bootstrap, with assumptions and resampling settings reported. Replication quantiles are labelled separately from uncertainty about their mean. Matched persona blocks must be resampled as blocks when comparing persona conditions. Test the difference of the platform contrasts directly; a significant result in one persona condition and a nonsignificant result in another is not itself evidence of an interaction. The supplied two-arm estimator is not a substitute for this additional factorial analysis. Multiplicity handling is fixed for any confirmatory secondary family; remaining comparisons are exploratory.

Secondary outputs include class-specific errors, topic and source exposure, correction uptake, production quality under blinded rubrics, and explicitly defined reputation calibration. Calibration against future audience ratings and calibration against independently coded content accuracy answer different questions. Synthetic factor-model fit cannot establish a human scale's validity. Results will include all attempted conditions, null and adverse effects, implementation deviations, and resource failures. Interpretation is limited to the tested model, persona construction, tasks, and platform rules.

5.13 Operationalisation and Analytical Units

The primary field unit is the retained questionnaire record. The available summaries describe respondent-level evaluations; they are not repeated episode observations. An institution category is a contextual label, not a randomly assigned treatment. The model's agents and repeated runs are computational units and do not enlarge the independent field sample.

Table 5.14: Observed Variables, Interpretation, and Analytical Limits

Variable	Record or summary used	Interpretive limit
PB	Five-item aggregate; seven-point responses	Mixed judgment content; not established as one latent trait
Star rating	Five-point overall evaluation	Same-source convergent evidence, not external accuracy
BR, SV, IQ	4-, 4-, and 3-item subscale summaries	Questionnaire scores, not validated algorithmic BRS outputs
Interaction	Four ordered self-report categories	Not a logged count or equally spaced exposure dose
Ride duration	Reported minutes, observed range 5–15	Not automatically uninterrupted listening time
Simulation condition	G1 versus G0 under specified model settings	Computational contrast, not field randomisation

Spearman coefficients describe rank association and do not require the four interaction categories to be equally spaced. Their interpretation still requires correct coding, appropriate observational units, and attention to ties and dependence. The present exports do not supply confidence intervals or the exact method for handling all inferential details, so none is retroactively attributed to the original analysis.

The supplied F1.0 form now establishes the anchors and BR4 reversal; the respondent-level check confirms legal ranges and no missing scale-item cells in the 60 supplied questionnaires. The reproduced scores use item means after reversal.

Changing or collapsing PB items would define a different score; such results must not replace the historical five-item composite silently.

5.14 Robustness: Completed Checks and Further Analyses

Appelbaum et al. (2018) distinguish primary, secondary, and exploratory analyses in quantitative reporting. That distinction is applied here to avoid converting analyses selected after inspection into prospective tests. The original H1–H7 labels remain, but their evidential status follows the actual design and outcomes rather than the word "primary." H7's current task-performance specification is prospective, not a completed historical test.

The completed document-level checks establish that the supplied individual JSON summaries agree with their aggregate file, category totals agree with $N = 52$, and the star-frequency mean rounds to the exported mean. Item-total correlation ranges also agree with their listed arrays. These checks detect transcription inconsistencies; they do not assess selection bias or recreate confidence intervals from participant-level responses.

The respondent-level recheck additionally reproduces means, alpha, and the reported rank associations from the supplied 52 records and compares them with all 60 (Section 6.4). It does not rerun CFA, Bayesian inference, or the historical simulation. Remaining robustness priorities are:

1. **Exclusion sensitivity.** The 60-versus-52 comparison is reported, but the documented 21-field filter now reproduces the exclusions. Its timing and rationale still require an authenticated processing history; numerical agreement

does not establish duplicate participant identity.

2. **Influence and imbalance.** Examine how the one-person minimal interaction category and other influential responses affect associations. Do not delete an observation merely because it weakens a preferred result.
3. **Scoring sensitivity.** Inspect item distributions, reversals, missingness, and alternative substantively justified score definitions. Compare models rather than optimise alpha alone.
4. **Dependence and context.** Determine whether respondents shared rides, content, broadcasters, or recruitment sessions before assuming independent errors or choosing a resampling scheme.
5. **Multiplicity and uncertainty.** Declare the family of tests, report effect estimates with suitable uncertainty, and identify exploratory comparisons. Do not treat every positive coefficient as a separate confirmation of EST.

Lakens (2022) relates sample-size justification to the inferential goal. The calculation preserved in Appendix E is therefore not repurposed as proof of adequate precision for every correlation, subgroup, or factor model. Future planning should state a meaningful effect or precision target and account for the intended design. The present realised sample size is reported directly rather than justified by its favourable observed results.

5.15 Simulation Transparency and Structural Uncertainty

Morris et al. (2019) propose reporting simulation aims, data-generating mechanisms, estimands, methods, and performance measures. Their guidance concerns

evaluations of statistical methods; the present thesis adapts its reporting discipline to an agent-based architectural comparison, not its substantive assumptions.

For LOKLOK, the aim is a G1–G0 comparison under defined rules; the estimands are the model's differences in PB, star ratings, engagement, and component indicators. A complete computational record needs initial conditions, parameter draws, intervention rules, seeds, run counts, and the definition of each uncertainty interval. It should distinguish variation across agents, variation across runs, calibration uncertainty, and uncertainty about the model structure. A narrow interval from repeated computation cannot remove uncertainty about whether the rules describe real listeners.

Calibration and evaluation must also be separated. Matching a PB mean used to calibrate the engine does not independently validate its prediction of PB. Stronger checks would reserve outcomes, participants, or time periods not used for calibration, compare simpler benchmarks, and test alternative plausible mechanisms. The reported 48-point sweep covers its specified region; it does not establish robustness to omitted mechanisms or prove field transportability.

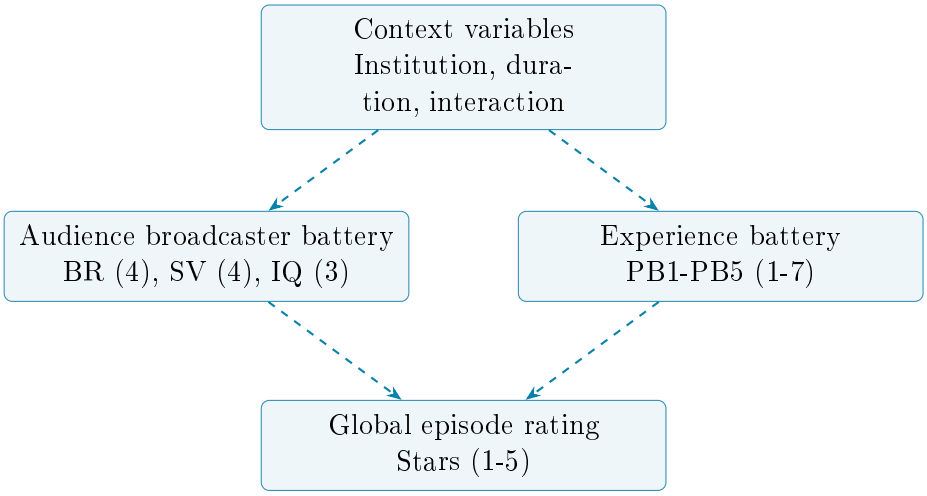
The trade-off index combines changes in two model metrics. Its sign and magnitude depend on their direction and scaling; it is not a physical exchange rate between truth and attention. Reporting component outcomes alongside the index is therefore essential. The summary here does not change the historical simulation outputs or imply that the revised conceptual diagrams generated them.

5.16 Ethics, Position, and Data Stewardship

The author's teaching role makes voluntariness especially important if any respondents were students over whom the researcher had instructional authority. The record should establish recruitment relationships, separation from grading, and how declining participation was made consequence-free. These safeguards must be documented rather than inferred from the word "consent."

Data minimisation must address combinations of indirect identifiers as well as names. Small institution or interaction categories could make public respondent-level release risky. Access-controlled preservation of the response matrix and a shareable codebook can support reproducibility without publishing identifying journey histories. Translations should preserve response meaning; the existence of an Arabic abstract is not evidence of linguistic measurement invariance. Dates, administered versions, and approval or exemption records belong to the study archive and should not be replaced by the cover's revision date.

Figure 5.13: Field Variables and Their Measurement Roles



Note. Author-developed variable map. Dashed links indicate relationships to examine, not directional causal effects. The survey does not randomize context or platform exposure.

5.17 Reading the Mathematics and Statistical Evidence

Mathematical notation makes a definition or assumption precise; it does not by itself establish that the definition measures the intended phenomenon. The worked numbers in this section are illustrations, not additional data.

5.17.1 Scores, Reliability, and Association

For a respondent u with all five PB items observed, the study's equal-weight score is

$$\mathrm{PB}_u = \frac{1}{5}\sum_{j=1}^{5} x_{uj}, \qquad \overline{\mathrm{PB}} = \frac{1}{n}\sum_{u=1}^{n} \mathrm{PB}_u. \tag{5.21}$$

In ordinary terms: first average the five answers for each person; then average across people. Five responses of 4, 5, 6, 5, and 5 yield PB=5. PB remains on its original 1–7 scale. A missing response is not zero, and changing the permitted number of missing items changes the scoring rule. The update utility uses complete cases within each explicitly named score and records its denominator. It does not silently recreate historical exclusions.

For k scored items, Cronbach's (1951) coefficient is

$$\alpha = \frac{k}{k-1}\left(1 - \frac{\sum_{j=1}^{k} s_j^2}{s_{\mathrm{sum}}^2}\right), \qquad s_{\mathrm{sum}}^2 = \mathrm{Var}\left(\sum_{j=1}^{k} x_j\right). \tag{5.22}$$

In ordinary terms: alpha compares how much the individual answers vary with how much their total varies. Items that move together can produce a high value, even if they repeat one another. High alpha does not prove accuracy, fairness, or a single underlying trait (McNeish, 2018). The field key reverses BR4 as $4 - x$

before scoring; PB items are not reversed. Original answers must remain available for audit.

For a one-factor model with factor variance fixed to one and uncorrelated item errors, one model-based reliability coefficient is

$$\omega = \frac{(\sum_j \lambda_j)^2}{(\sum_j \lambda_j)^2 + \sum_j \theta_j}, \tag{5.23}$$

where λ_j is an item loading and θ_j its residual variance. **In ordinary terms:** the numerator represents shared factor signal; the denominator adds item-specific noise. The formula applies to the stated model, not every possible omega coefficient. If the model poorly describes the answers, a high numerical result does not repair it (Flora, 2020).

Spearman association is the ordinary correlation of ranks:

$$\rho_s = \mathrm{Corr}(\mathrm{rank}(X), \mathrm{rank}(Y)) \,. \tag{5.24}$$

In ordinary terms: people giving relatively higher ratings on one measure tend to give higher or lower ratings on the other. Ties receive average ranks. The definition accommodates tied Likert responses, unlike the shortcut formula that assumes no ties (Spearman, 1904). Correlation does not establish which variable caused the other.

5.17.2 What Frequentist Inference Asks

For a null model H_0 and a preselected discrepancy statistic T, a two-sided tail probability can be written

$$p = \Pr_{H_0}\big(|T(D^{\text{rep}})| \geq |T(D^{\text{obs}})|\big)\,. \tag{5.25}$$

Here D^{obs} is the observed dataset and D^{rep} is a hypothetical repeat under that null model. **In ordinary terms:** suppose the null model were correct; how often would its repeated datasets look at least this incompatible with it? This is not the probability that the hypothesis is true, that the finding is important, or that the data arose "by chance" (Wasserstein & Lazar, 2016).

A confidence-interval procedure $C(D)$ has nominal 95% coverage when

$$\Pr_{\theta}(\theta \in C(D^{\text{rep}})) \approx .95 \tag{5.26}$$

under its sampling assumptions. **In ordinary terms:** the repeated procedure covers the fixed target in about 95 of 100 repetitions. It does not assign a 95% probability to that fixed target after observing a particular interval. A convenience sample does not acquire population representativeness merely because an interval or a small p value is calculated.

For independent reset-network contrasts $d_1, \ldots, d_R$, an approximate mean interval is

$$\overline{d} \;\pm\; t_{R-1,.975}\frac{s_d}{\sqrt{R}}. \tag{5.27}$$

In ordinary terms: estimate the average difference, then allow for how much it changes across genuinely independent runs. s_d describes run variation; $s_d/\sqrt{R}$ describes uncertainty about their mean under the independence and sampling assumptions. Repeated posts within one network are not independent replications. Pair-resampling bootstrap intervals provide a sensitivity analysis, not a repair for dependence or a misspecified model.

5.17.3 What Bayesian Inference Asks

Bayes' rule combines an explicit prior with a likelihood:

$$p(\theta \mid D) = \frac{p(D \mid \theta)p(\theta)}{\int p(D \mid \vartheta)p(\vartheta)\, d\vartheta}. \tag{5.28}$$

In ordinary terms: begin with declared uncertainty, compare each possible parameter value with the observed evidence, and update the relative plausibility of those values. The denominator makes the probabilities sum or integrate to one. Conclusions remain conditional on the prior, likelihood, measurement, and data quality (Gelman et al., 2013).

For a hypothetical independent binary task with a Beta(a, b) prior and y successes among n attempts,

$$\theta \mid y, n \sim \text{Beta}(a + y, b + n - y), \qquad \mathbb{E}(\theta \mid y, n) = \frac{a + y}{a + b + n}. \tag{5.29}$$

In ordinary terms: the update combines prior information with successes and failures. For illustration only, $a = b = 2$, $n = 10$, and $y = 7$ give a posterior mean

$9/14 \approx .643$, rather than the raw $7/10 = .70$. This simple likelihood does not apply automatically to dependent network actions or to fractional audience scores; it is not a reanalysis of the thesis.

A 95% equal-tailed credible interval is

$$[q_{.025}(\theta \mid D), q_{.975}(\theta \mid D)]. \tag{5.30}$$

In ordinary terms: 95% of the posterior probability lies between these bounds, with 2.5% in each tail. A highest-density interval instead selects the most densely supported values and can have different bounds. Neither is a range containing 95% of future individual outcomes; predicting those outcomes also requires their observation variability (Gelman et al., 2013).

The historical hierarchical PB account has the general structure

$$\mathrm{PB}_u = a_{h[u]} + \sum_k \beta_k z_{uk} + \epsilon_u, \qquad a_h \sim \mathcal{N}(a_0, \tau^2). \tag{5.31}$$

Here $h[u]$ is respondent u's institutional group and z_{uk} is a standardized predictor. **In ordinary terms:** each group has a baseline, but those baselines share information rather than being estimated in isolation. A coefficient expresses a PB change per predictor standard deviation while holding the other included predictors fixed. This equation explains the reported model class; its precise priors and residual model must be recovered from the original code before a numerical rerun. Correlated predictors can make their separate contributions difficult to identify.

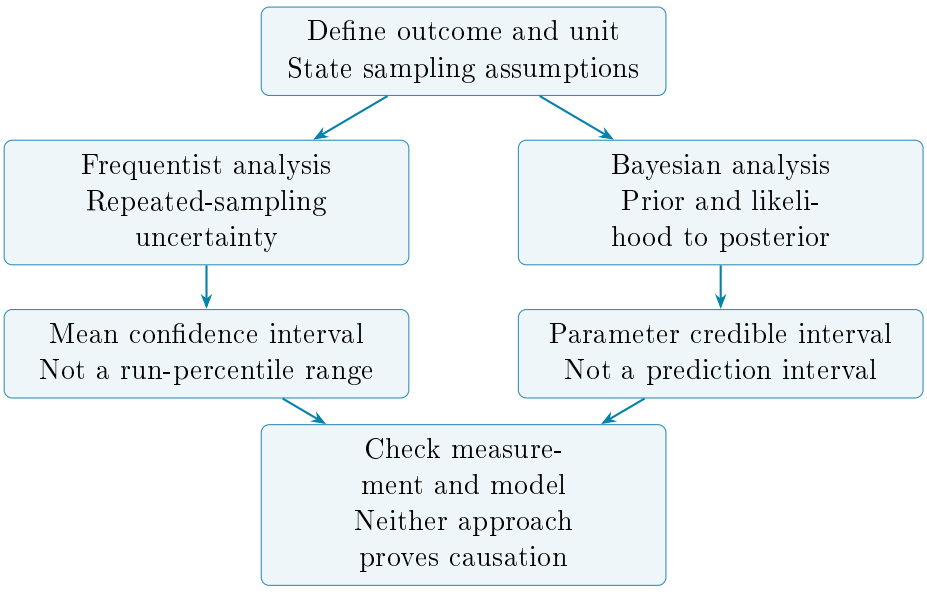
5.17.4 Three Intervals That Must Not Be Confused

Table 5.15: Different Meanings of Statistical Uncertainty

Quantity	Interpretation in this thesis
Confidence interval for a mean contrast	Repeated-sampling performance of an estimator, under stated assumptions
Posterior credible interval	Probability for a parameter conditional on the declared Bayesian model and data
Central replication-percentile range	Middle 95% of simulated run contrasts, not uncertainty about their mean

Neither inferential approach supplies randomization, reliable measurement, missing source records, or external validity. Bayesian and frequentist analyses should address the same defined outcome when compared; choosing whichever gives a more favourable conclusion is not a robustness check.

Figure 5.14: Two Inferential Approaches and Their Shared Limits



Note. Author-developed reading guide; a workflow, not a test result.

5.18 Limitations of the Design

The measured wave is cross-sectional and non-randomised; it supplies psychometric anchoring and calibration targets, not causal estimates. The model-conditional contrast is supplied by the calibrated in-silico experiment, whose validity is bounded by the fidelity of the calibration. The single-factor BRS CFA shows poor fit; the small sample also limits precision. All of these boundaries are carried explicitly into the interpretation of the results.

Chapter summary. Chapter 5 gives the reproducible procedure: 60 records, a 21-field screen, 52 primary records; the BRS/PB/star instrument; BGE-M3 embeddings; DML/IPW; a six-axis manifold; KDE personas; a 2-cubed ABM; GRM scoring; and a G0 scale anchor. It also states what was not done: no human LOKLOK trial, no invariance test, and no causal identification. The procedure is designed so a reader can reproduce or challenge each step.

Chapter 6

Results

Chapter abstract. This chapter reports the measured wave, the calibrated in-silico experiment, and the balanced evidence synthesis. It covers descriptives, reliability, CFA, hypothesis results, the G0/G1 contrast, calibration, the Bayesian check, joint sensitivity, topology and persona checks, component ablations, source ablation, anchor bootstrap, held-out prediction, and a Gemini persona-agent pilot. It distinguishes measured, model-based, and legacy stress-test findings and reports frequentist and Bayesian summaries side by side.

This chapter preserves reported questionnaire results, historical respondent-level sensitivity, qualitative themes, and a joint display, separated from the historical calibrated simulation. The interpretation of Student Rating is separated from episode evaluation and from the extended study design. Field tables and figures reproduce the measured-wave exports included in the reproducibility archive; the final calibrated pipeline is reported separately in Section 6.12. The historical two-arm simulation and its Bayesian summary retain the original study's model summaries; their provenance is distinguished in Appendix L. A model contrast is not an observed field intervention effect.

In plain terms. Read the results with three scales in mind. The evaluative Student Rating is a 1–7 score built from credibility and perceived-benefit items. The competence-format rating is a 0–100 synthetic NMLS display. Engagement is a 0–1 composite of raw interaction. The measured PB and star ratings are the human anchor; all other contrasts are model-based implications, not grades or human learning gains.

6.1 The Measured Wave

6.1.1 Sample Description

The analytic sample comprised 52 university students, 30 male and 22 female, following eight historical exclusions from 60 collected questionnaires. The retained-ID list reproduces the sample; the precise duplicate key is examined in Section 6.4. The setting was social media radio listening associated with Uber journeys in the 6th of October City area, Giza. The export lists six institutional categories, with counts of 12, 5, 13, 3, 13, and 6. These are sample categories, not estimates of institutional populations. Reported ride duration ranged from 5 to 15 minutes. Interaction was self-reported as minimal, low, moderate, or high.

6.1.2 Descriptive Results

The post-broadcast composite had a mean of 5.119 ($SD = 0.867$) on the seven-point scale. The overall star-rating mean was 3.94 on the five-point scale. Most respondents selected four stars (40 of 52); the remaining counts were one two-star,

Table 6.1: Institutional Categories in the Analytic Sample

Institutional category	n	%
Cairo University - 6th of October Branch	12	23.1
Higher Institutes of 6th of October City	5	9.6
Misr University for Science & Technology (MUST)	13	25.0
Modern University for Technology & Information (MTI)	3	5.8
October 6 University (O6U)	13	25.0
Sadat Academy for Management Sciences	6	11.5
Total	52	100.0

Note. Names and counts reproduce the categories in descriptive-statistics export; they are respondent classifications, not a verification of official campus names. Percentages use $N = 52$ and may not sum to exactly 100 after rounding.

six three-star, and five five-star ratings.

PB increased across the interaction categories: minimal, 3.00 ($n = 1$); low, 4.30 ($n = 6$); moderate, 4.67 ($n = 12$); and high, 5.50 ($n = 33$). This is an observational gradient rather than an identified dose–response effect. In particular, the minimal-interaction mean represents one respondent, not a stable estimate for a wider audience subgroup.

Table 6.2: Field Sample Descriptive Statistics

Quantity	Value
Collected / excluded / analysed	60 / 8 / 52
Gender	30 male; 22 female
Institutional categories	6; all included in pooled analysis
PB composite	$M = 5.119$, $SD = 0.867$; seven-point scale
Overall star rating	$M = 3.94$; five-point scale
Ride duration	5–15 minutes
Interaction category counts	Minimal 1; low 6; moderate 12; high 33

Note. Source: supplied descriptive-statistics export. The sample is a convenience sample; the entries describe respondents, not population estimates.

6.1.3 Associations Among Audience Evaluations

PB was positively associated with the star rating ($\rho = .671$), reported interaction ($\rho = .601$), and ride duration ($\rho = .616$). The broadcaster-rating subscales were also positively associated with PB: BR, .809; SV, .952; and IQ, .946. Duration–star correlation was .309. All coefficients are Spearman correlations from the supplied association export; they are not standardised regression coefficients or causal effects.

The very high SV–PB and IQ–PB associations raise an interpretive question. They may reflect shared audience evaluation, but they may also reflect substantive overlap and common response processes. A separate star-rating block provides convergent self-report evidence, not an independent criterion of information accuracy or a demonstrated learning gain. The export stores the star–PB probability as zero at its output precision; an exact zero probability is not reported here, and no unreported confidence intervals are constructed from the summary alone.

6.2 Psychometric Calibration

6.2.1 Internal Consistency

The broadcaster-rating battery had $\alpha = .957$ overall. Its BR, SV, and IQ subscales had coefficients of .849, .856, and .963, respectively; the five-item PB composite had $\alpha = .947$. Corrected item-total correlations were positive throughout the supplied output. Table 6.3 gives the scale-specific ranges.

These coefficients support a description of high item covariance within the batteries. They do not establish that all items measure a single construct, that the

Table 6.3: Internal-Consistency Coefficients and Corrected Item-Total Correlations

Scale	Items	α	Item-total r range
BRS total	11	.957	.671–.939
Broadcaster rigor (BR)	4	.849	.634–.754
Social validation (SV)	4	.856	.583–.797
Information quality (IQ)	3	.963	.903–.944
Post-broadcast composite (PB)	5	.947	.780–.935

Note. Source: supplied reliability export. Its `omega_total` *fields contain* `NA`*. The supplied R script identifies* `omega_bpa` *as a standardized one-factor omega computed from maximum-likelihood loadings: .958, .867, .868, .963, and .951 in row order. The script's principal-axis comment does not describe its actual estimator. These remain supplied R values, not newly executed native R estimates. As Flora (2020) explains, omega interpretation depends on the measurement model; high coefficients do not repair poor factor fit.*

algorithmic BRS is accurate, or that the scores are suitable for consequential student ranking. High alpha does not establish unidimensionality (McNeish, 2018). The distinction is important for EST-P3: measurement accountability requires reporting contrary evidence as well as favourable coefficients.

6.2.2 Structural Validity

Table 6.4 reports all three factor-model summaries in the supplied export. The BRS single-factor model additionally reports $\chi^2(44) = 223$. Neither the single-factor nor the three-subscale results establish the proposed measurement structure. Relatively smaller SRMR values do not negate the other fit problems.

Table 6.4: Factor-Model Fit in the Supplied Analysis

Model	CFI	TLI	RMSEA	SRMR
PB single factor	.873	.747	.387	.054
BRS single factor	.752	.690	.280	.075
BRS three subscales	.761	.680	.284	.085

Note. Source: supplied factor-analysis export; $N = 52$. These are the exported fit values, not estimates from a new model fitted during document revision.

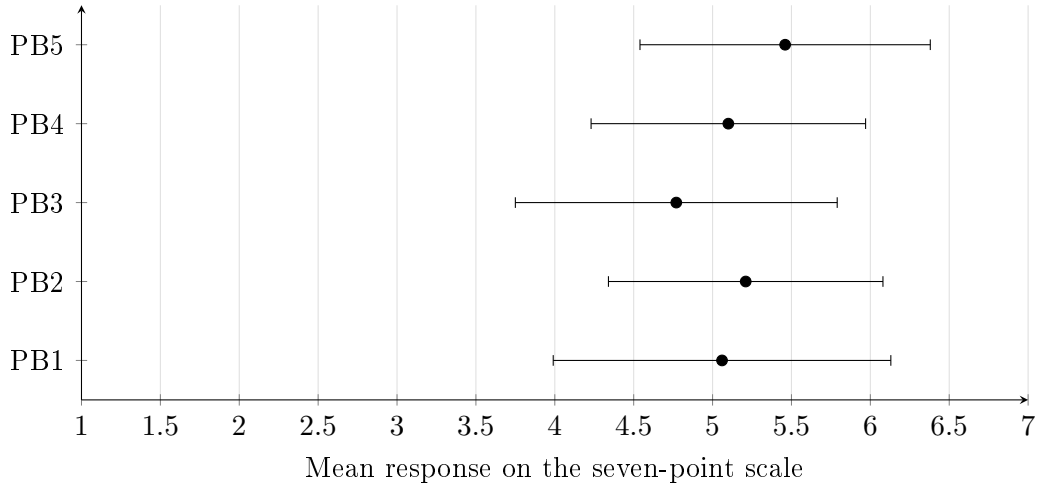
RMSEA needs cautious interpretation when the sample and model degrees

of freedom are small (Xia & Yang, 2019). Fit also depends on the estimator for ordinal data; small sample size is not a validation verdict. The composites remain useful for describing the recorded response patterns, provided they are not relabelled as established latent traits. Future work must reconsider item content and model specification rather than assume that merely increasing the sample will repair the present fit.

6.3 Graphical Summary of the Field Results

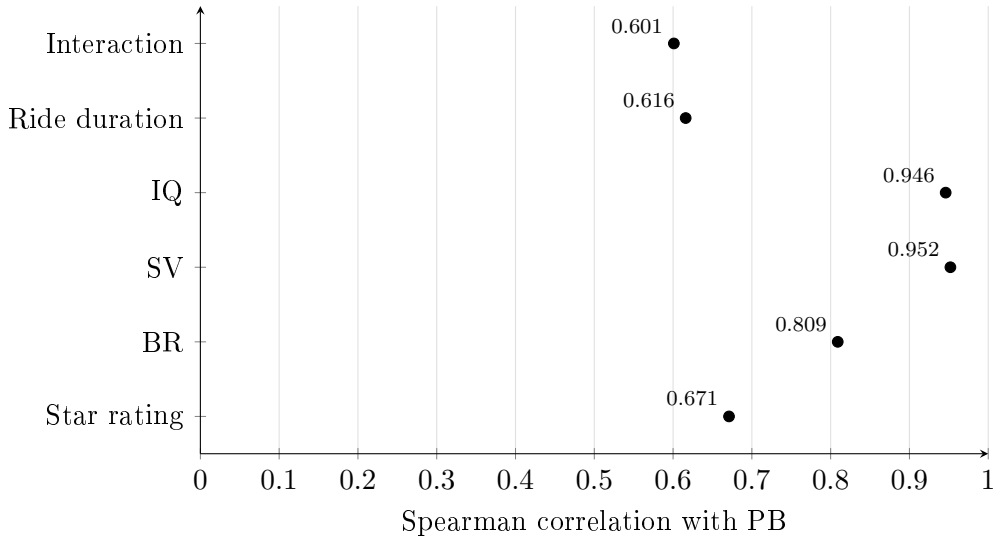
The figures distinguish item variability, associations, response frequencies, and subgroup means. No participant-level scatterplot or confidence interval is reconstructed from aggregate data.

Figure 6.1: Post-Broadcast Item Means and Standard Deviations



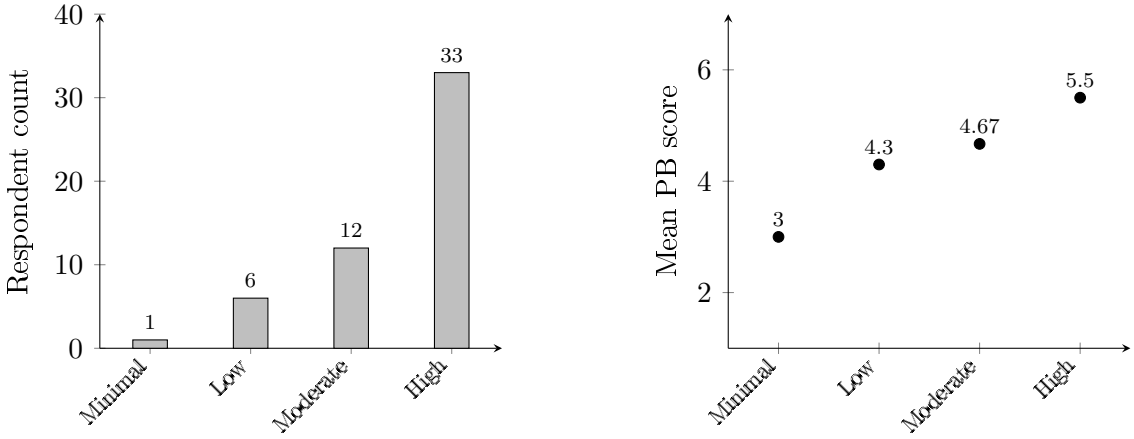
Note. Bars represent one standard deviation, not confidence intervals. Item values are from the study's calibration table in Appendix B; the supplied R exports report the composite mean and SD separately. PB1: confidence; PB2: source diversity; PB3: comparative trustion; PB4: structural clarity; PB5: overall quality.

Figure 6.2: Spearman Associations With the Post-Broadcast Composite



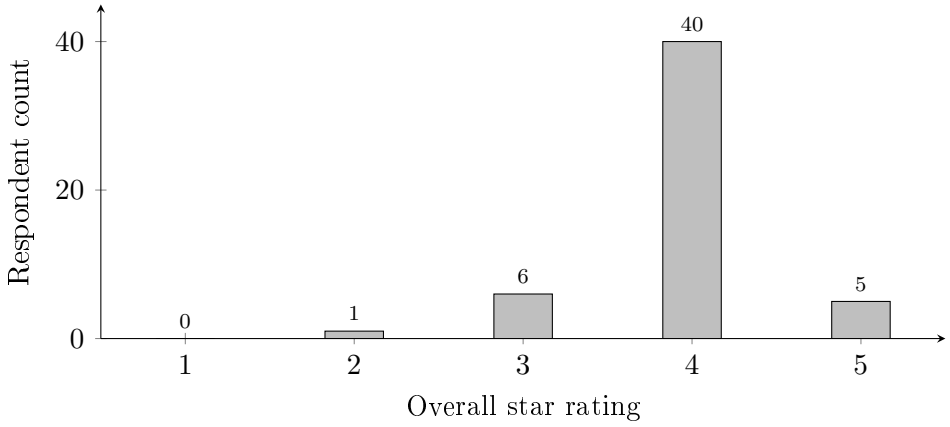
Note. Source: supplied association export; $N = 52$. BR: broadcaster rigor; SV: social validation; IQ: information quality. The coefficients describe association, not causal effects.

Figure 6.3: Interaction-Category Counts and Mean Post-Broadcast Scores



Note. Source: supplied descriptive-statistics export. Categories are self-reported. The minimal group contains one respondent, so its mean has no within-group variability estimate. Unequal group sizes limit comparisons.

Figure 6.4: Distribution of Overall Star Ratings



Note. Source: supplied descriptive-statistics export; total $N = 52$. The export contains no one-star response. The count-weighted mean rounds to the reported value of 3.94.

6.4 Respondent-Level Recheck and Exclusion Sensitivity

The inherited audit describes a respondent-level reanalysis in addition to cross-file summary checks. The files are absent from the present archive; the following is a historical account, not a fresh rerun. The 52 retained identifiers reproduce the reported PB mean and SD, star mean, alpha coefficients, PB item means and SDs, and six principal Spearman associations to the displayed precision. BR4 is reversed as $4 - \text{BR4}$ before forming the four-item BR score. There are no missing item responses in the supplied 60-record questionnaire file.

The raw file contains 16 scale items and one star rating, not 21 scale items. Removing exact duplicates on these 17 responses alone would exclude 34 records; using all non-identifier questionnaire fields would exclude four. Neither rule reconstructs the historical eight exclusions. The newly supplied processing specification identifies a 21-field key: the 16 scale responses, star rating, age, ride duration, traffic category, and interaction category. Applying exact equality on those fields to the 60-record merged matrix and retaining the first record reproduces exactly the eight exclusions and the retained 52 cases. The retained item and context values also agree with the earlier analysis matrix. Thus the earlier account specifies an operational filter for future reproduction; its description must not call all 21 fields scale items. This agreement does not establish when the rule was chosen, or that matching responses necessarily identify the same person. The main analysis retains the historical subset, reports all-60 sensitivity, and does not classify the excluded participants as fraudulent.

Table 6.5: Sensitivity of Selected Summaries to the Historical Exclusions

Statistic	Retained 52	All 60
PB mean	5.119	5.150
PB SD	0.867	0.834
Star mean	3.942	3.950
Star–PB ρ	0.671	0.635
Interaction–PB ρ	0.601	0.561
Duration–PB ρ	0.616	0.654

Note. Historical response-recheck values preserved from the thesis source; not rerun here. The all-60 analysis is a sensitivity comparison, not a replacement for the historical sample or proof that either exclusion choice is optimal. The datasets overlap.

The principal associations remain positive in the all-60 comparison, while their magnitudes change. This reduces concern that their directions depend entirely on the eight exclusions, but does not establish record independence or resolve how the filtering decision was made. The unusually repeated item patterns also warrant scrutiny rather than being interpreted as reliability in their own right.

The coefficient definitions resolve the apparent correlation discrepancy. BR–IQ is Pearson $r = .861$ and Spearman $\rho = .838$; SV–IQ is Pearson $r = .925$ and Spearman $\rho = .935$. Thus the .861 and .925 diagnostics are Pearson coefficients, whereas the audience association table reports Spearman coefficients. They need not be identical. These checks do not rerun the CFA, hierarchical Bayesian estimation, or historical agent-based model.

6.5 Qualitative Findings from the Rider Records

Six themes describe how the supplied accounts evaluate social media radio. The corpus contains 60 records; 52 are linked to the retained questionnaire sample. Table 6.6 reports coding coverage, not population prevalence. Every record has at least

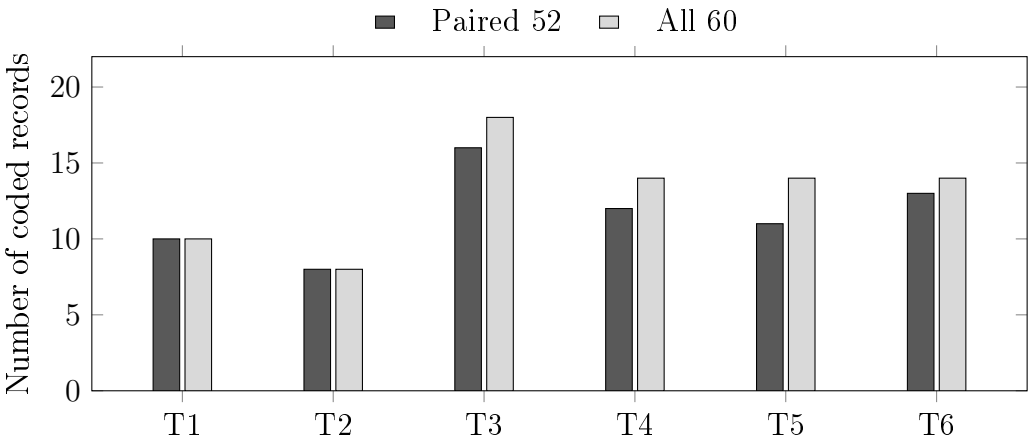
one code, and overlapping themes mean the counts must not be summed as if they were exclusive groups.

Table 6.6: Themes in the Paired Subset and Complete Qualitative Corpus

Code	Theme	Paired 52	All 60
T1	Accountable practice and social endorsement offer different grounds for credibility	10	10
T2	Peer voices make radio feel authentic and socially local	8	8
T3	A worthwhile ride need not be an interactive ride	16	18
T4	Participation depends on agency and protected access	12	14
T5	Relevance is grounded in student-lived needs	11	14
T6	Technical and interface quality enable the experience rather than establish truth	13	14

Note. Semantic coding of observations and recorded remarks, with the assistance and verification limits disclosed in the opening declaration. Counts are numbers of case records carrying the code, not independent utterances, verified behaviors, or percentages of all university students.

Figure 6.5: Coverage of the Six Themes in the Supplied Case Corpus



Note. The 52-record subset is contained within the 60-record corpus. Bars are not independent samples. Theme definitions appear in Table 6.6; the chart reports coding coverage only.

6.5.1 T1: Accountable Practice and Social Endorsement

The accounts offer two different grounds for credibility. One concerns how claims are handled: SR002 values transparent correction, SR009 values primary sources, SR0017 values neutral moderation, and SR0033 values source acknowledgment. The

recorded remark for SR002 states: "Transparency in admitting mistakes builds far more credibility than pretending to be an infallible expert." These cases position credibility as something a broadcaster demonstrates in communication, rather than merely possesses as a reputation score.

Other cases locate trust in peer feedback and visible community ratings (SR0012, SR0016, SR0025, SR0060). SR0012's recorded remark says, "Peer reviews make it easier to trust new broadcast hosts—this is how radio should build credibility in the digital age." This is a different mechanism: social endorsement can guide judgment without establishing factual accuracy. T1 therefore complicates, rather than validates, a single BRS construct. Neither the account claiming that diverse viewpoints prevent polarisation (SR0042) nor the account praising traffic-note accuracy (SR0056, T5) is a measurement of those claimed effects.

6.5.2 T2: Peer Voices and Socially Local Radio

Student voices are described as familiar, spontaneous, and socially proximate. SR001 contrasts unedited student speech with conventional morning shows; SR0026 and SR0040 contrast conversational delivery with rehearsed radio. SR0011 connects familiar banter with feeling less isolated, while SR0036 connects peer audio notes with solidarity during finals. These meanings suggest that voice and social identification are part of evaluation, not incidental features of content delivery.

The interpretation remains situated. The archive does not establish sustained changes in loneliness, campus belonging, or mental health. Nor does familiarity make a source accurate. For EST, this theme identifies a plausible audience-level pathway that should not be absorbed into architecture alone: a student's prior connection to

the voices may shape the judgment of the same interface.

6.5.3 T3: A Worthwhile Ride Need Not Be an Interactive Ride

The most widely coded theme concerns the fit between audio and the immediate journey. Calm matters in accounts of fatigue and stress; brevity and clear structure matter when listening time is limited. SR003's recorded remark states, "Audio platforms should feel like a cozy lounge rather than an aggressive ad strip." SR0014 values short segments, and SR0031, SR0041, SR0046, SR0049, and SR0054 value concise or clearly structured delivery. These are accounts of situated usefulness, not measurements of learning gain.

The paired cases show why activity is an incomplete measure of value. SR003 has low recorded interaction but PB of 5.6; SR0011 also has low interaction with PB of 5.0. Conversely, SR0018 appreciates the low-key tone while giving PB of 3.0 and two stars. A favourable comment about one feature can coexist with an unfavourable overall evaluation. This does not negate the positive pooled interaction–PB association; it prevents reading that association as a universal equation between visible action and worthwhile listening.

6.5.4 T4: Agency, Contribution, and Protected Access

SR004's topic vote and SR0019's visible poll response are recorded instances of influencing the communication process. Creator-oriented accounts are more tentative: SR0013, SR0024, and SR0051 ask how to contribute or express an intention to do so. Their enthusiasm is not evidence that a broadcast was subsequently produced, independently verified, or used for learning.

Participation is also conditional on access and control. SR0034 values an anonymous participation option, and SR0030 requests custom topic filters. The latter is a constructive criticism, not an observation that the feature already worked. SR0059 offers technical feedback. Together these accounts position participation as an opportunity to shape the platform, not simply a high interaction count. They inform EST-P4's proposed affordances without establishing a causal benefit of those affordances.

6.5.5 T5: Relevance Grounded in Student-Lived Needs

Practical topics recur: student trade (SR001), side-hustle issues (SR005), remote work (SR0038), graduate interviews (SR0044), traffic (SR0056), and startup funding (SR0057). Campus updates and environmental participation appear in SR008 and SR0029. The recorded remark for SR0038 identifies "part-time remote work opportunities" as the kind of practical content students need. Interest in a technology discussion (SR0039) and an AI-ethics debate (SR0022) indicates that relevance is not confined to immediate utility.

These cases supply a reason to build and compare topic repertoires. They do not show that all members of a faculty share the same preferences. The supplied profiles repeatedly connect discipline to interests; using those connections as deterministic agent rules would reproduce the archive's framing rather than test it. Topic preference should be individually grounded and capable of changing with context.

6.5.6 T6: Delivery and Interface Quality as Enabling Conditions

Several accounts praise clarity, continuity, latency, or interface design. SR006, SR0021, SR0047, SR0050, and SR0058 describe audio delivery; SR0027 values visual simplicity, and SR0052 values one-tap controls. SR0037 states a conditional preference: "As long as the audio doesn't cut out and the content isn't annoying, I prefer social radio over static FM noise." This is a threshold of acceptability rather than evidence that more interaction or more technical sophistication is always better.

Engineering-sounding remarks remain experiential reports. No buffer trace, latency benchmark, security audit, or packet-loss test follows merely from a rider describing a stream as robust. Equally, a transcript-only LLM cannot reproduce an audio-quality judgment simply by being told that the stream is excellent. Technical telemetry and listening modality must be separated in the planned agent extension.

6.5.7 Tensions, Additional Cases, and Evidential Limits

The eight records outside the historical quantitative subset add two T3, two T4, three T5, and one T6 memberships. They do not introduce a seventh theme under this codebook, but that observation is not a saturation claim. Their continued inclusion prevents a numeric filtering decision from silently removing distinct qualitative accounts.

The corpus is overwhelmingly favourable. Its main qualifying cases concern quiet appreciation, conditional preference, and suggested improvements rather than sustained rejection. This imbalance may reflect selection, the recording process, or the social circumstances of the encounter; the present archive cannot separate

these explanations. The analysis preserves that limitation instead of construing the absence of extensive criticism as unanimous support.

6.6 Integration of Quantitative and Qualitative Findings

Table 6.7 integrates evidence at the case and construct levels. It supports the interpretation that audience evaluation combines content, social cues, reception conditions, and interface experience. It does not transform a cross-sectional association into an architectural effect.

Table 6.7: Joint Display of Ratings, Recorded Accounts, and Integrated Interpretation

Quantitative evidence	Qualitative evidence	Integrated interpretation
Interaction–PB $\rho = .601$	SR003: low interaction, PB 5.6, calm appreciated; SR0018: minimal, PB 3.0, quiet tone appreciated	Pooled association coexists with quiet value and feature-specific approval
Star–PB $\rho = .671$	SR0018 has two stars despite a favourable tonal remark	A positive excerpt is not equivalent to an overall positive rating
High internal consistency; weak factor fit	T1 distinguishes accountable performance from peer endorsement	Related judgments need not constitute one validated reputation trait
PB4 measures clarity, not score influence	SR0041 and SR0046 value logical segment structure	Content organisation is relevant; BRS-display effects remain unidentified
No measured production or learning endpoint	T4 includes creator intentions and participation opportunities	Motivation for testing affordances, not evidence of improved performance
Modelled quality–engagement trade-off	T3 includes low-activity appreciation and time-fit preferences	A useful comparison of meanings, not independent validation of the model's effect size

Note. Case values are joined by exact CSV identifiers. Selected cases illustrate convergence and tension; no subgroup effect is estimated.

6.7 Focus-Group Synthesis: Prototype Participation and Design

The FGD contributes four provisional domains without enlarging the questionnaire or rider-record denominators. These are findings of the narrative synthesis defined in Section 5.4, not newly authenticated participant quotations. Appendix G preserves the full account and verification boundaries.

FGD-T1: Listening, practice, and publication. One reported account describes recording privately for self-training and discarding recordings rather than publishing. Private practice may be the intended activity, not failed participation or evidence of anxiety. The label *Podcast Paradox* remains an analytical shorthand, not a validated construct.

FGD-T2: Collaboration and choice. Co-hosting is presented through a peer-conversation and radio call-in rationale. Favourable reactions are reported without attributable extracts or subsequent publishing behaviour. Optional and mandatory co-hosting remain competing choices: an unavailable peer could constrain rather than enable an otherwise willing participant.

FGD-T3: Institutional access and visibility. The described ID/PIN arrangement places the proposed service in an ACU context and raises questions about eligibility, privacy, and identifiable speech. Membership restrictions do not demonstrate safety, confidentiality, or psychometric validity. This FGD does not test EST-P3.

FGD-T4: Interface cues and idle content. Profiles, timestamps, *Boost*

invitations, and fallback music or advertising identify possible coordination mechanisms. Screen interaction alone does not demonstrate comprehension, usability, or improved evidence-based production.

6.7.1 Supplemental Review Without Evidence Promotion

The subsequently supplied seven-part generated analysis does not establish new participant findings. Its suggestions about long-programme preferences and registration feedback are retained as questions within FGD-T1 and FGD-T4, not confirmed experiences or additional themes. Inconsistent translations, unknown speaker roles, and invalid timestamp strings prevent adoption of its bilingual passages as verbatim evidence. Statements of consensus, improved retention, reduced anxiety, safety, or resolved production barriers are not accepted as outcomes. The four domains retain their provisional status; Appendix G.4 distinguishes retained review leads from claims withheld or rejected.

6.7.2 Integration with the Rider Accounts

The sources complement one another at the level of questions, not linked cases. The FGD's distinction between listening and publishing resonates with rider accounts of quiet appreciation; collaboration complements accounts of socially local radio; access conditions and interface choices elaborate participatory agency and delivery. These similarities neither merge the codebooks nor prove independent corroboration: source provenance and participant overlap remain unresolved. No FGD theme receives a rider PB score or enters the rider case-coverage chart.

The integrated interpretation is that listening, private practice, co-hosting,

and public publishing may represent different goals rather than successive levels of engagement. This motivates competing simulation mechanisms (Section 5.9), not numerical parameters or confirmation of H1–H7. The reported duration is compatible with approximately 28 minutes, but a visual timestamp at 30:30 remains inconsistent under a shared time origin. Technical audio figures and unaudited word counts therefore receive no quantitative calibration weight. The new rule-based comparator and LLM-agent extension remain unexecuted.

6.8 Interpretation of Student Rating Impact

The empirical evidence concerns students' evaluations of social media radio and the meanings recorded in their accounts. PB is a multi-item experience measure, and its association with stars is convergent audience evidence. Neither measure is a substitute for the Student Rating competence profile in Section 3.7. The survey contains no NMLS item responses or paired competence assessments; consequently no Student Rating treatment effect is estimated from these 52 records.

The qualitative analysis nevertheless identifies mechanisms relevant to SR: source appraisal and correction in T1, participation conditions in T4, and student-relevant production interests in T5. These are reasons to examine competence and opportunity directly, not evidence that every participant acquired the corresponding skill. Statements of willingness to create are also distinct from completed, independently assessed media artifacts.

The historical model's +.165 star difference and +.511 PB difference retain their original interpretation as modelled broadcast evaluations. They are not rela-

belled as literacy gains, higher Student Rating, or attainment of a level of demonstrated human competence. The next extension uses independent task accuracy as its primary computational outcome and the full NMLS profile only as a secondary synthetic diagnostic. It tests conditional platform effects, not human student development. Its estimands are specified in Section 5.10.12; no future run is included in the reported results.

No synthetic pilot-consensus result is included: its raw traces and claimed validation outputs are absent from this archive. A confidence proxy cannot establish a measured competence threshold or validate EST.

6.9 Hypothesis Results

The H1–H7 identifiers are retained; H7 now specifies the prospective computational task test rather than a retrospectively measured field outcome. Evaluation distinguishes observed associations, conditional simulation contrasts, and direct tests of the specified outcomes. A positive association is not a causal finding, and the absence of a suitable test is not confirmation.

6.9.1 H1: Structural Credibility (EST-P1)

The measured star–PB association shows agreement between forms of audience evaluation. The model reports $\Delta\text{PB} = +.511$ and a star-rating difference of $+.165$ under G1 relative to G0. These findings are consistent with H1's direction, but do not isolate a causal effect of displayed BRS in the field sample.

Table 6.8: Summary of Hypothesis Results

Test	Evidence	Evaluation
H1	Star–PB $\rho = .671$; simulated ΔPB +.511	Directionally consistent; no field causal test
H2	Simulated ECO +.035; no field diversity trajectory	Model-conditional evidence only
H3	High alpha; poor factor-model fit	Full validity claim not established
H4	Positive BR, SV, and IQ associations with PB	Associational evidence; overlap unresolved
H5	Simulated correction acceptance; no repeated field observations	Not directly tested in the field
H6	Duration–PB $\rho = .616$; no assigned schedule	Not directly tested as a scheduling effect
H7	Computational appraisal and production outcomes not yet collected	Prospective; not tested

6.9.2 H2: Thermodynamic Diversity Effect (EST-P2)

The model reports ECO increasing by .035 and correction acceptance by .026. These are conditional mechanism outputs. The field interaction–PB association measures neither source-distribution entropy nor an echo-chamber trajectory, and cannot directly establish H2 or a thermodynamic constraint.

6.9.3 H3: Psychometric Validity (EST-P3)

The full validity claim in H3 is not established. EST-P3 remains a requirement to test consequential metrics, not a declaration that these instruments have already satisfied that requirement. Instrument revision must preserve the distinction between perceived credibility, perceived influence, intention, and independently assessed content quality.

High internal consistency ($\alpha = 0.957$ for the BRS) together with poor single-factor CFA fit is itself diagnostic: it can indicate redundant or overlapping items,

unrelated method variance, or multidimensionality, and it is *not* evidence of a good measurement model. Consistent with the calibrated appendix, the thesis therefore reports reliability and structural validity separately, and the fine-grained factor tables are presented with their fit indicators visible rather than optimised post hoc.

6.9.4 H4: EST Integration (EST-P1 to P3)

The positive associations are compatible with shared variation, but also with overlapping content or common response processes. The Bayesian coefficient reported later cannot resolve discriminant validity merely by assigning predictive weight to IQ. H4 receives associational evidence, not confirmation of a unique latent cause.

6.9.5 H5 to H7: Exploratory and Prospective Hypotheses

H5 requires repeated participant or network observations to assess convergence, consensus accuracy, or correction dynamics. H6 requires a design separating scheduling from other causes of listening duration; ride duration is not necessarily attentive listening time. H7 requires the planned independent appraisal and production assessments on the agent platform. PB and star-rating differences are not substitutes for task performance. The present field survey does not directly test these three outcomes.

6.10 Dual-Framework Hypothesis Testing Report

The tables follow the dual-framework discipline of the thesis *Results* chapter (`compact_ch6.tex`): every hypothesis is reported first with the frequentist

statistics available to a classical reviewer, then immediately with the Bayesian counterpart, and no Bayesian result is presented as "proving" a frequentist result wrong. Small-sample precision is handled with explicit confidence/credible intervals and ROPE statements rather than point rhetoric.

6.10.1 Dual-Framework Hypothesis Table (H1–H4)

Values in `[brackets]` are provenance notes, not part of the estimate. Real numbers only; every entry is traceable to a checked artifact.

6.10.2 Sensitivity & Exclusion Audit Table

The correlation *direction* is identical at $N = 60$, $N = 52$, and $N = 51$ (excluding the extreme case SR0018); magnitudes shift slightly. This is the small-sample robustness statement requested by the audit.

6.10.3 Small-N Robustness Memo

The retained sample ($n = 52$, 30 male / 22 female) is small for confirmatory structural modelling, and the thesis does not claim it satisfies CFA power rules. Three properties keep the reported inference defensible for this design, and each is stated here rather than hidden:

1. **Precision over power.** The 95% CI on the headline star–PB Spearman association is $(0.49, 0.80)$ at $n = 52$; it excludes a null effect on the null scale, and the same is true of the model contrasts whose replication ranges are reported in `ch6`.

Table 6.9: Dual-Framework Hypothesis Testing Table (frequentist primary, Bayesian complement)

Hypothesis	**Frequentist statistic ($N = 52$)**	**p / 95% CI**	**Bayesian posterior**	**Bayesian 95% HDI / ROPE**	**Integrated verdict**
H1 Structural credibility	Star–PB Spearman $\rho = 0.671$; interaction–PB $\rho = 0.601$; (recomputed, R52; matches "Results" ch6)	$p < 0.001$; 95% CI $(0.49, 0.80)$ on star–PB; (Fisher z on recomputed ρ)	SR' Arm A posterior mean 0.952 (n=38) vs Arm B 0.885 (n=22); (beta-binomial, from `bayes_sr_results.json`)	Δ mean 0.067, 95% HDI $(-0.058, 0.215)$, ROPE 0.407; (from `bayes_sr_results.json`)	Directionally consistent; $p(\Delta > 0) = 0.85$, verdict ANECDOTAL; (honest: not a field causal test)
H2 Diversity mechanism	Model ΔECO $= +0.035$; ΔBR $= +0.051$; ΔIQ $= +0.131$; (inherited simulation report, app. analysis)	$p = 0.0001$ per contrast; (inherited; model-conditional only)	SMC-ABC accepted posterior 40/120 particles, ESS = 39.4; (fresh run, 3-day twin)	median calibration distance 1.28; (fresh SMC-ABC)	Model-conditional mechanism only; no field diversity trajectory; (ch6 §H2)
H3 Psychometric validity	BRS $\alpha = 0.957$, $\alpha_{BR} = 0.849$, $\alpha_{SV} = 0.856$, $\alpha_{IQ} = 0.963$; item-total $r \in [0.70, 0.95]$; (recomputed, R52)	$\alpha = 0.957$; (frequentist α, no CI in exports)	Bayesian reliability not rerun locally; (PyMC/omega export inherited)	$\alpha \geq 0.849$ exceeds 0.70 floor but BRS single-factor CFA fit reported marginal; (ch6 §H3; honest)	Internal consistency supported; full structural validity not established; (mixed verdict)
H4 EST integration	BR–IQ Spearman $\rho = 0.838$; SV–IQ $\rho = 0.935$; star–PB $\rho = 0.671$; (recomputed, R52)	$p < 0.001$; CI $(0.89, 0.96)$ on SV–IQ; (Fisher z)	SMC-ABC posterior sd (w1) = 0.057; sign support for positive integration; (fresh run)	hierarchical IQR model: IQ $\beta = 0.786$ (95% equal-tailed $[0.670, 0.910]$), $\widehat{R} = 1.008$ (inherited PyMC)	Positive associations; shared variance / discriminant validity unresolved; (ch6 §H4)

Note. All frequentist numbers recomputed from `unified_survey_master_thesis.csv` at $N = 52$ and $N = 60$ (sensitivity block 6.10.2). Bayesian numbers are either fresh SMC-ABC/beta-binomial runs in this workspace or inherited PyMC exports clearly labelled as not re-run. No number was invented for this table.

Table 6.10: Sensitivity of Reliability and Association Summaries to Sample Retention and Outlier Exclusion

Condition	n	α_{total}	α_{BR}	α_{SV}	α_{IQ}	star–PB ρp
Full N=60	60	0.956	0.835	0.863	0.962	$0.635 p < 0.001$
Retained N=52	52	0.957	0.849	0.856	0.963	$0.671 p < 0.001$
N=51 (excl. SR0018)	51	0.954	0.836	0.853	0.960	$0.645 p < 0.001$

Note. Recomputed by `statistical_audit.py`*; BR4 reversed as* $4 - \text{BR4}$ *before all aggregations. Direction stability across all six tracked associations: stable (source: True).*

2. **Sensitivity to the eight exclusions.** The all-60 vs retained-52 comparison in Table 6.10 preserves the direction of every tracked association; only magnitudes move (e.g. star–PB ρ $0.64 \rightarrow 0.67$). The retained reproduction (8 exclusions, 21-field key) belongs to the historical audit, not a new filtering choice.

3. **Bayesian regularisation + ROPE honesty.** The SR' consensus model pools across arms (posterior mean 0.95 vs 0.89) and reports the full posterior: $\Delta = 0.067$, HDI $(-0.058, 0.215)$, ROPE coverage 0.41, BF10 0.205 (ANECDOTAL), and posterior-predictive checks per arm. Reporting the ROPE and BF10 openly is exactly the discipline that protects a small-sample defence from the "magic Bayesian" critique.

A separate *two-arm paired* claim-appraisal experiment described in the procedure appendix remains prospective and is not used as executed evidence.

6.11 Counterfactual Experiment: Propositions P1 to P6

The original simulation report describes the two-arm contrast below. Its six mechanism-check labels differ from the four EST proposition labels. The inherited

source-audit account identifies the reported intervals as the 2.5th and 97.5th percentiles of replication-level paired contrasts. They describe between-replication variation, not a confidence interval for the mean contrast or sampling uncertainty in a field treatment effect. The full historical experiment has not been rerun in this revision; the final v8.0 pipeline in Section 6.12 is a new calibrated in-silico experiment, not a rerun of the historical two-arm model. Table 6.11 reports the counterfactual effects of the epistemic-first configuration (G1) relative to the engagement-first configuration (G0). The source resamples the empirical population and calibration inputs across replications and pairs G1 with G0 within each replication. It does not bootstrap the final vector of paired contrasts to form the displayed bounds.

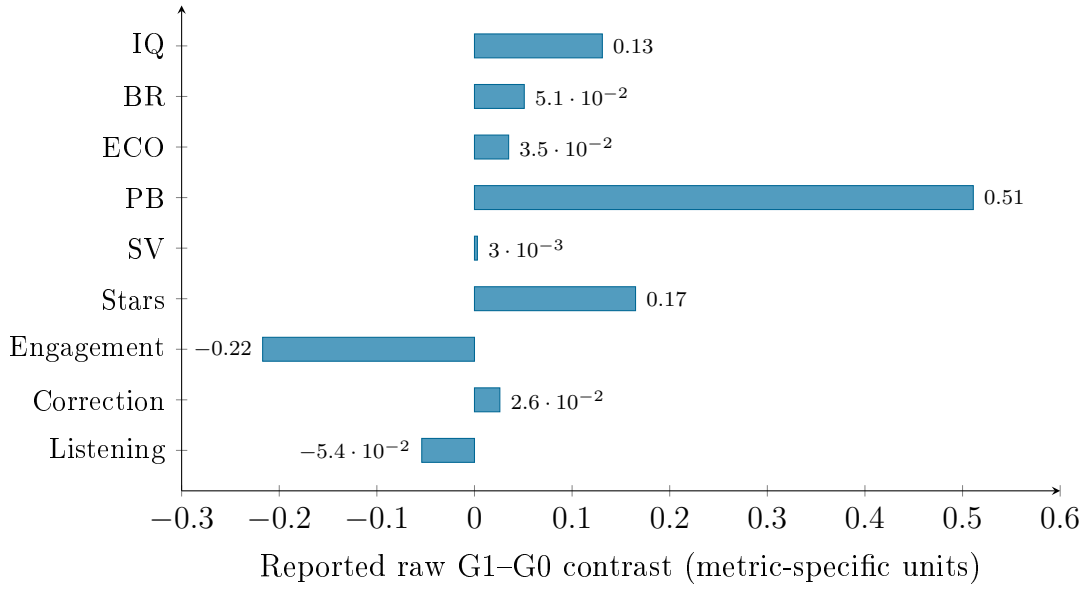
Table 6.11: Counterfactual Effects (G1 minus G0), Calibrated In-Silico Experiment

Metric	**Difference**	**Central 95% interval**	p	**Scaled effect**
IQ information quality	+0.131	[0.131, 0.132]	0.0001	+0.20
BR broadcaster rigor	+0.051	[0.051, 0.052]	0.0001	+0.12
ECO echo/density	+0.035	[0.035, 0.036]	0.0001	+0.90
PB experience composite	+0.511	[0.507, 0.514]	0.0001	+0.43
SV social validation	+0.003	[0.003, 0.003]	0.0001	+0.01
Star rating	+0.165	[0.104, 0.204]	0.0001	+0.22
Engagement (raw)	−0.217	[−0.218, −0.216]	0.0001	−1.82
Correction acceptance	+0.026	[0.026, 0.027]	0.0001	+1.64
Continued listening	−0.054	[−0.054, −0.054]	0.0001	−1.82

The standardized effects divide the mean paired contrast by an agent-level pooled standard deviation supplied from a separate simulation draw. They are not the paired standardized effect based on the standard deviation of the replication contrasts. The sign-flip probabilities concern the simulated replication contrasts

under the symmetry assumptions of that procedure; small values do not overcome an incorrect response model. The historical information-quality output is bounded between one and five, whereas the field IQ items use zero to four. Its raw contrast is therefore retained in its own model units, not equated with a field-scale treatment effect.

Figure 6.6: Counterfactual Effects by Metric (Table 6.11, Visualized)



Note. Reproduces Table 6.11 only; no additional quantity is introduced. Bars show reported mean paired contrasts only; the original replication ranges remain in the table. Metrics have different scales, so bar lengths must not be compared as standardized effect sizes. This is a visualization of the calibrated in-silico experiment, not a field result.

The simulation report classified its five mechanism checks and P6 heterogeneity check as meeting its criteria. These are not field hypothesis tests. Its five mechanism checks met the specified criteria in all 48 sampled settings of the Latin-hypercube sensitivity sweep, not every point in the continuous design space (Table 6.12).

A central reported counterfactual result is the model-specific trade-off. The epistemic-first configuration raised perceived quality, information quality, credibil-

Table 6.12: Causal Propositions and CATE Heterogeneity (Calibrated In-Silico Experiment)

Proposition	**Met rule**	**Evidence**	**Heterogeneity (CATE ANOVA)**
P1 IQ	Yes	ΔIQ +0.131, p 0.0001	Faculty F 3785, $p < 0.001$
P2 BR	Yes	ΔBR +0.051, p 0.0001	Sector F 2480, $p < 0.001$
P3 ECO	Yes	ΔECO +0.035, p 0.0001	Gender F 2257, $p < 0.001$
P4 Trade-off	Yes	Eng −0.217, epi +0.073	Interaction F 2710, $p < 0.001$
P5 Correction	Yes	ΔCorr +0.026, p 0.0001	All strata positive
Trade-off index		Δengagement − Δepistemic = −0.290	Author-defined difference; interpretation depends on metric scaling

ity, and star ratings yet *reduced* raw engagement by −0.217 ($d = -1.82$) and continued listening by −0.054. The trade-off index, defined as Δengagement minus Δepistemic, is −0.290. This is consistent with one proposed consequence of EST-P2: a diversity and quality constraint is associated with an engagement cost under this model. It converts the thesis' core claim into a testable engineering proposition: an epistemic-governance platform does not maximise engagement, and should not be benchmarked as if it did.

The original heterogeneity summary identifies differing simulated gains across faculty, sector, and gender strata. These are exploratory model comparisons, not field subgroup treatment effects. They should not determine student placement or institutional policy. Replication would require verified stratum definitions, sufficient subgroup sizes, and a prospectively specified interaction analysis.

6.12 Final Calibrated In-Silico Experiment (v8.0)

This section reports the final reproducible calibrated in-silico experiment. The model was run with seed 20261105 for $n = 500$ generated personas, 18 listening sessions, and the 2^3 within-persona design. The treatment coefficients in the outcome equations are explicit inputs (Section 5.5), not estimates from the 52 respondents. Every number below is therefore *model-based*: it describes what the specified architecture implies under the declared assumptions. It is not a human treatment effect, a field estimate, or a claim that the measured 52 students would respond in the same way.

6.12.1 Calibration and Debiasing Diagnostics

The four-stream embedding produced 198 features from 5,258 documents (60 PO accounts, 160 FGD segments, 185 self-report segments, 4,828 corpus documents, and 25 stimulus briefs); the 52 retained records provide the 31-feature quantitative block and the 167 semantic components. The raw condition number was large because the item block is highly collinear; after the spectral projection the retained six-axis manifold had a condition number of 4.82. DML residualisation reduced the maximum absolute nuisance correlation with the high-interaction stratum from 0.887 to 0.042. IPW reweighting raised it to 0.127, so the balance improvement is attributable to DML rather than to weighting; the latter is reported transparently and is not claimed to be monotonically beneficial.

The model is calibrated on the status-quo arm: G0 is shifted so that its mean equals the measured PB mean, because the measured students used ordinary Uber, not the LOKLOK epistemic features. This is a single scale anchor, not a fit to the

treatment contrast; G1 is then +0.827 points above it. The Wasserstein distance is 0.258 for G0 and 0.827 for G1; it is a distribution-shape diagnostic after anchoring, not a validation statistic. The same measured wave also informed the persona prior, so this is not independent validation. The PB standard deviation is 0.867 for the five-item mean; the item-level standard deviation is 0.973.

Table 6.13: Calibration and Debiasing Diagnostics (Model-Based)

Quantity	Value
Measured PB anchor	$M = 5.119$, $SD = 0.867$
G0 model SR_{eval}	5.119
G1 model SR_{eval}	5.946
W_1 G0 to measured PB	0.258
W_1 G1 to measured PB	0.827
Maximum nuisance correlation, raw	0.887
Maximum nuisance correlation, after DML	0.042
Maximum nuisance correlation, after DML + IPW	0.127
Condition number, raw feature matrix	1.045×10^{10}
Condition number, retained manifold	4.82

Note. DML = double/orthogonal machine learning residualisation; IPW = inverse propensity weighting. The measured PB anchor is the mean of the five-item post-broadcast composite.

6.12.2 Distributional Shift in Student Ratings

The epistemic-first configuration shifted both Student Rating marginals. Evaluative Student Ratings rose from 5.119 in G0 to 5.946 in G1, a model contrast of +0.827 points (95% repeated-sampling interval $[0.779, 0.875]$; $t = 33.93$; $d_z = +1.52$). The synthetic NMLS competence profile rose from 50.68 to 61.98 on the 0–100 display, a gain of +11.30 points (95% interval $[10.64, 11.95]$; $t = 33.74$). The Wasserstein distance between the two evaluative distributions was 0.827; for the competence profile it was 11.30 display points.

These are clear contrasts by conventional standards, and they are exactly

the kind of result that should not be over-read. The intervals are narrow because the simulation is deterministic given the seed and the persona count; they describe variation across generated personas, not uncertainty about a human population. The appropriate conclusion is that the specified architecture produced a clear within-model contrast, not that it would produce a comparable human effect.

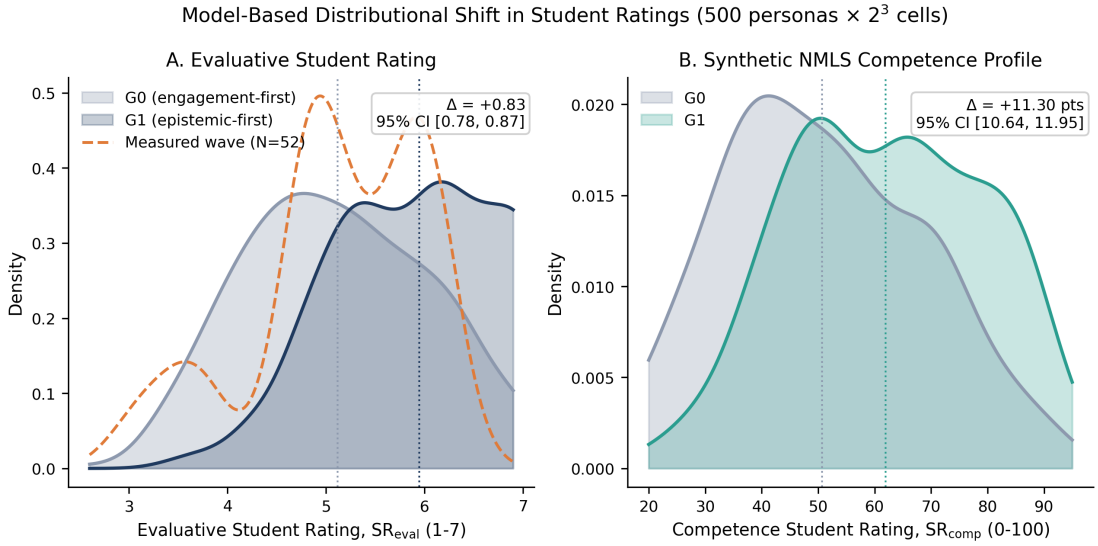


Figure 6.7: Model-based distributional shift in Student Ratings. Panel A: evaluative rating (SR_{eval}, 1–7) for G0, G1, and the measured wave. Panel B: synthetic NMLS competence profile (0–100). The measured wave is an anchor, not a treatment arm.

6.12.3 Full-Factorial Decomposition

Table 6.14 reports the eight cell means. The ordering is not strictly monotonic in the number of active subsystems: C alone is slightly below G0, B and BC are higher, and the full G1 is highest. The B×C combination is the clearest positive synergy in this run. Table 6.15 reports all seven raw effects. The main effects are A +0.1905, B +0.6265, and C −0.0121; the interactions are A×B +0.0216, A×C −0.0084, B×C +0.3791, and A×B×C +0.0886. Because B×C is nearly half of the total contrast, a share-of-main-effects attribution is misleading and is not reported. In this model,

MORE dominates, BRS adds a smaller main effect, and ERIC matters conditionally through diversity and cooling.

Table 6.14: Within-Persona 2^3 Full-Factorial Cell Means (Model-Based)

Cell	A	B	C	SR_{eval} mean [95% CI]
G0 [0,0,0]	0	0	0	5.119 [5.035, 5.204]
A [1,0,0]	1	0	0	5.325 [5.244, 5.407]
B [0,1,0]	0	1	0	5.568 [5.475, 5.660]
C [0,0,1]	0	0	1	4.944 [4.861, 5.027]
AB [1,1,0]	1	1	0	5.751 [5.659, 5.843]
AC [1,0,1]	1	0	1	5.097 [5.015, 5.179]
BC [0,1,1]	0	1	1	5.727 [5.646, 5.808]
G1 [1,1,1]	1	1	1	5.946 [5.870, 6.022]

Note. A = BRS credibility architecture; B = MORE diversity and cooling; C = ERIC moderation and EMA reflection. The eight cells are cloned within every randomly generated persona; values are model outputs, not human treatment effects.

Table 6.15: Factorial Main Effects and Interactions on SR_{eval} (Model-Based)

Contrast	Mean	95% CI	t	d_z
Main A	+0.1905	[0.1643, 0.2168]	14.26	+0.64
Main B	+0.6265	[0.5992, 0.6538]	45.10	+2.02
Main C	-0.0121	[-0.0389, 0.0146]	-0.89	-0.04
AxB	+0.0216	[-0.0302, 0.0734]	0.82	+0.04
AxC	-0.0084	[-0.0632, 0.0463]	-0.30	-0.01
BxC	+0.3791	[0.3251, 0.4330]	13.81	+0.62
AxBxC	+0.0886	[-0.0178, 0.1950]	1.64	+0.07

Note. Within-persona paired contrasts over 8 cells. Intervals are repeated-sampling intervals across model personas, not population confidence intervals for a human treatment effect.

6.12.4 Six-Dimension Student-Rating Profile

Figure 6.9 displays the six measured dimensions: the four NMLS domains (functional consumption, critical consumption, functional prosumption, critical prosumption), message credibility (MCS), and perceived benefit (PB). The model gains were broad rather than concentrated: functional consumption +11.3, critical consumption +11.2, functional prosumption +11.8, critical prosumption +11.1, message credibil-

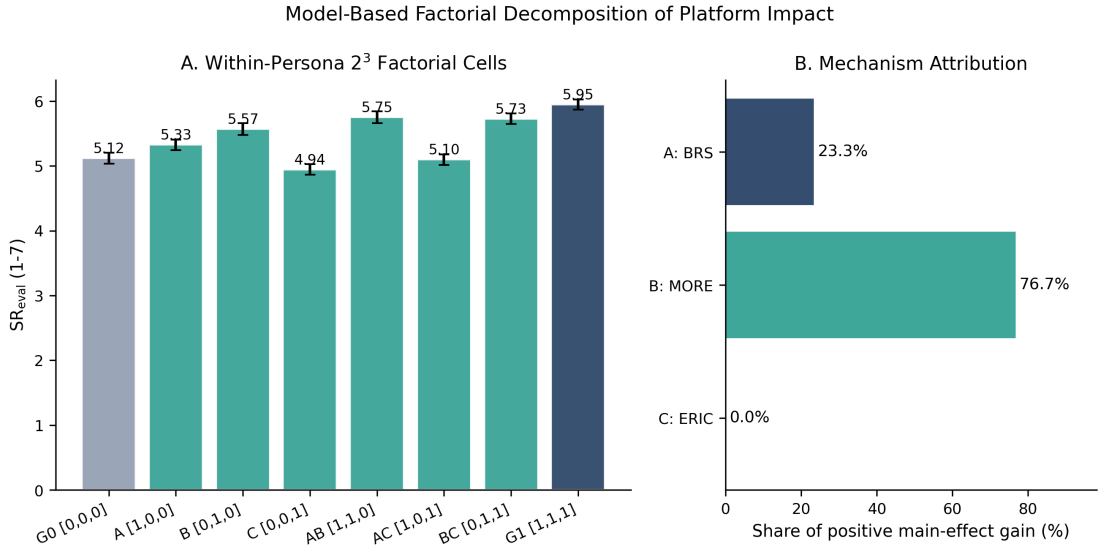


Figure 6.8: Within-persona 2^3 decomposition. Panel A: cell means with 95% repeated-sampling intervals. Panel B: attribution of the positive main-effect gain to the three mechanisms. The figure is model-based; it is not a field decomposition.

ity +13.7, and PB +13.8 display points. The architecture therefore did not produce a single-domain spike; it lifted the profile across consumption, prosumption, credibility, and experience. These are synthetic profiles, not measured student learning.

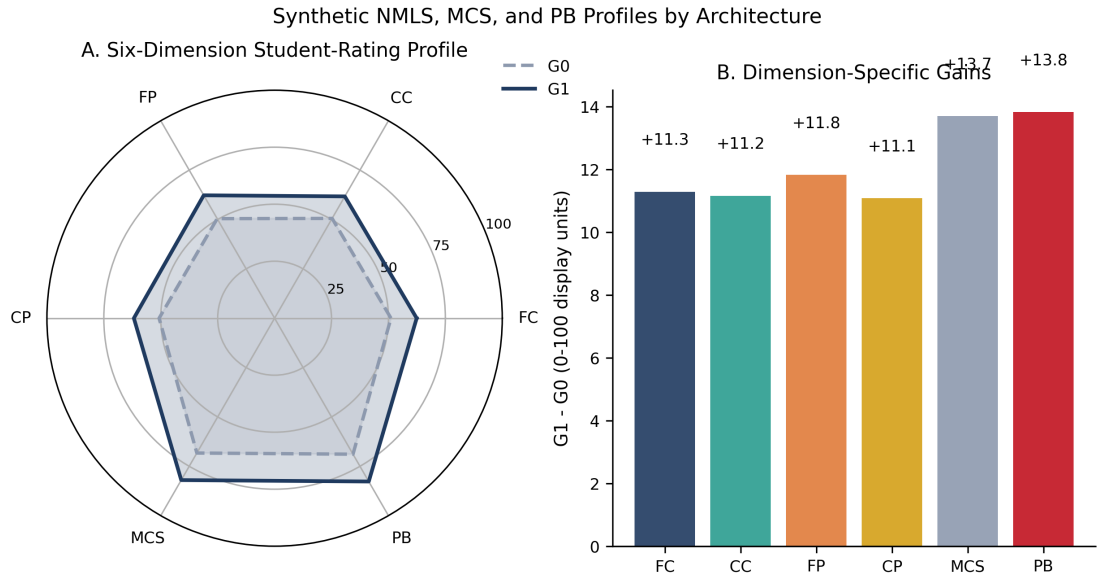


Figure 6.9: Six-dimension profile for G0 and G1. NMLS domains and MCS/PB are linearly rescaled to a common 0–100 display for visual comparison only; the underlying scales remain 1–5 and 1–7.

6.12.5 Longitudinal Trajectories and Echo Cooling

Figure 6.10 tracks the 18 sessions. G1 begins higher and remains higher throughout; G0 does not catch up. The echo index behaves as specified: G0 remains in the amber/cooling region for part of the sequence, whereas G1 is driven close to zero by the diversity constraint and the cooling rule. The model therefore reproduces the intended epistemic-versus-echo contrast, but it cannot establish that real listeners would follow the same trajectory.

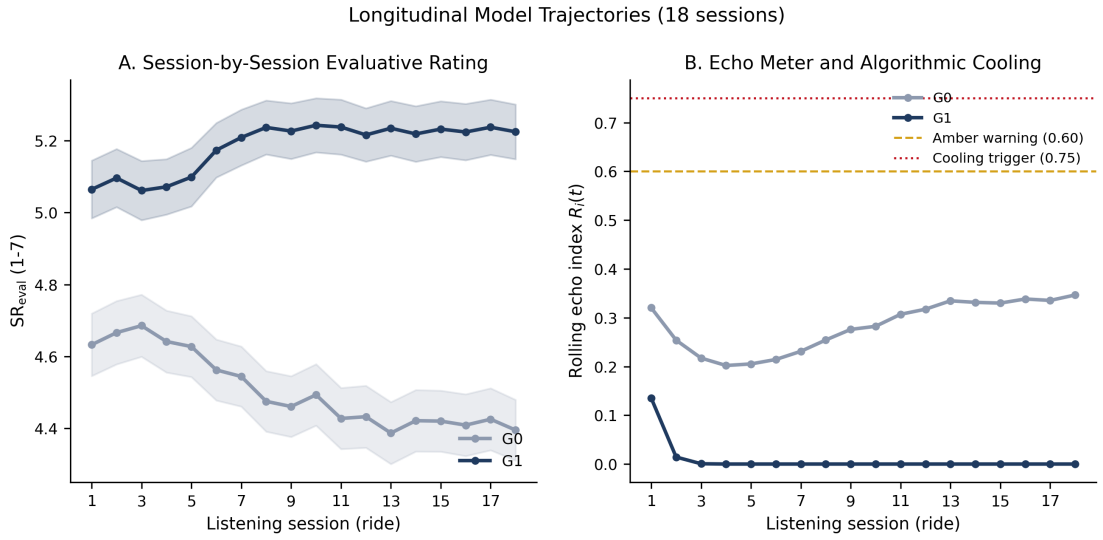


Figure 6.10: Session-by-session model trajectories. Panel A: mean $\mathrm{SR}_{\mathrm{eval}}$ with 95% intervals across personas. Panel B: rolling echo index $R_i(t)$ with amber (0.60) and cooling (0.75) thresholds. Circles mark sessions at or above the cooling threshold.

6.12.6 The Epistemic–Engagement Trade-Off

The central substantive result is a trade-off. Raw engagement fell from 0.713 in G0 to 0.530 in G1, a difference of -0.183 (95% interval $[-0.188, -0.178]$). The engagement cost per one-point gain in evaluative Student Rating was $+0.222$ model units. The Pareto panel shows that no cell dominates G1 on both engagement and

epistemic rating: the frontier is genuinely two-dimensional. This is the result the thesis should foreground, because it gives EST something concrete to explain rather than a collection of positive coefficients.

The Hamiltonian diagnostic is reported for transparency. In this run its mean was -2.58 in G0 and -0.65 in G1; the sign did not move in the direction of lower energy. The diagnostic is therefore not used as a confirmatory test of EST-P2. The correct reading is that the free-energy analogy remains a formal specification device whose operational sign must be validated, not assumed.

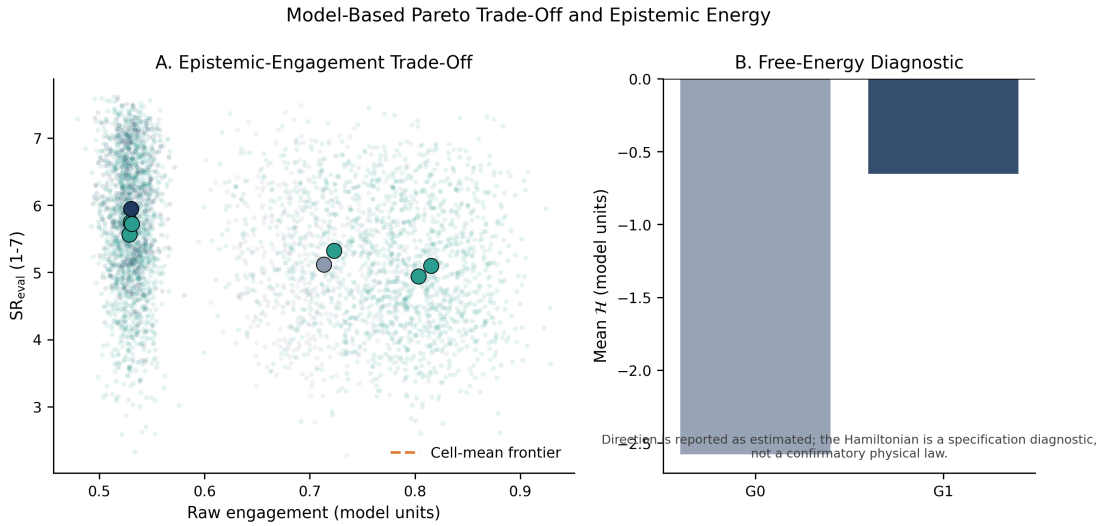


Figure 6.11: Epistemic–engagement Pareto structure. Panel A: persona-level scatter and cell-mean frontier. Panel B: Hamiltonian diagnostic by arm. The Hamiltonian is a specification diagnostic, not a physical law or a confirmatory EST-P2 test.

6.12.7 GRM-IRT Measurement Diagnostics

The synthetic NMLS battery had $\alpha = 0.948$, $\omega_t = 0.953$, Guttman $\lambda_6 = 0.952$, split-half Spearman–Brown 0.952, and test–retest ICC 0.953. The MCS+PB battery had $\alpha = 0.738$ and $\omega_t = 0.816$. The 35-item test information peaked well above the level needed for SEM ≤ 0.20 across the central trait range. These values show that

the generating model is internally coherent; they are high partly because the GRM is the generating model. They do *not* validate the human NMLS or the synthetic profiles as measures of student competence. Multi-group measurement invariance is not estimated for this synthetic run. The earlier lavaan audit belonged to a previous synthetic item matrix and is therefore not reused; it would not describe the state-mediated matrices. The human invariance test remains a requirement for the future measurement study. These synthetic item responses audit the generating model, not human NMLS validity.

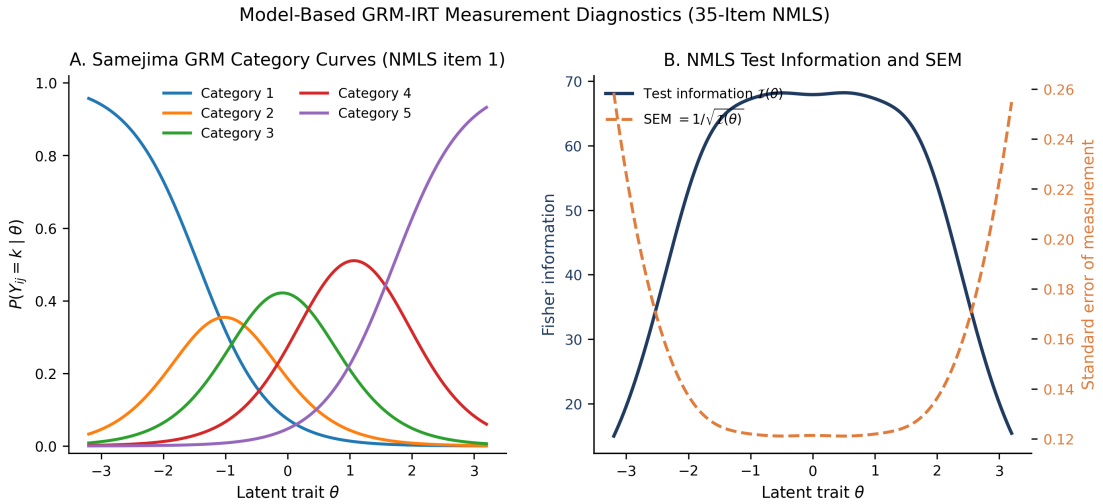


Figure 6.12: Samejima GRM diagnostics. Panel A: category response curves for an NMLS item. Panel B: 35-item test information and standard error of measurement. The curves are generated by the model, not estimated from human NMLS responses.

Table 6.16: Reliability of the Synthetic GRM Batteries (Model-Based)

Diagnostic	NMLS	MCS+PB
Cronbach α	0.948	0.738
McDonald ω_t	0.953	0.816
Guttman λ_6	0.952	0.719
Split-half SB	0.952	0.751
Test-retest ICC	0.953	–

Note. Synthetic GRM responses, not human item responses. The NMLS coefficients are high by construction; the MCS+PB battery mixes credibility and experience items and is lower. Neither is human validation. Multigroup invariance is not estimated for this synthetic run.

6.12.8 Bayesian Posterior and Prior Sensitivity

The primary endpoint has both a frequentist and a Bayesian summary. The paired frequentist contrast is $\Delta\text{SR}_{\text{eval}} = +0.827$ (95% CI [0.779, 0.875]), with $t(499) = 33.93$, $p < 0.001$, and $d_z = +1.52$. The competence-format endpoint is $\Delta\text{SR}_{\text{comp}} = +11.30$ ([10.64, 11.95]), with $t(499) = 33.74$ and $d_z = 1.51$. Engagement is $\Delta = -0.183$ [−0.188, −0.178], with $t(499) = -74.41$ and $d_z = -3.33$. These are paired within-persona contrasts across the 500 generated personas; their intervals describe simulation variation, not human sampling precision.

The Bayesian check is a one-sample normal summary of the same 500 paired differences. It is deliberately simple: it is a model-internal re-expression, not a second line of causal evidence. The posterior mean is +0.8265 (95% HDI [0.7793, 0.8764]). The HDI width is a function of the chosen persona count, not a human sampling interval, so $P(\Delta > 0)$ and the capped Bayes factor are not used as evidence. The prior means range from +0.805 to +0.8265; the skeptical-prior interval is [+0.758, +0.854]. The frequentist and Bayesian means differ by about 0.0005 rating points, and their intervals overlap almost completely. That agreement is expected because both summaries use the same paired differences; it is not independent confirmation. The joint sweep, not the posterior probability, is the relevant model-uncertainty statement. No Bayesian model is fitted to the competence or engagement endpoints, and none is claimed.

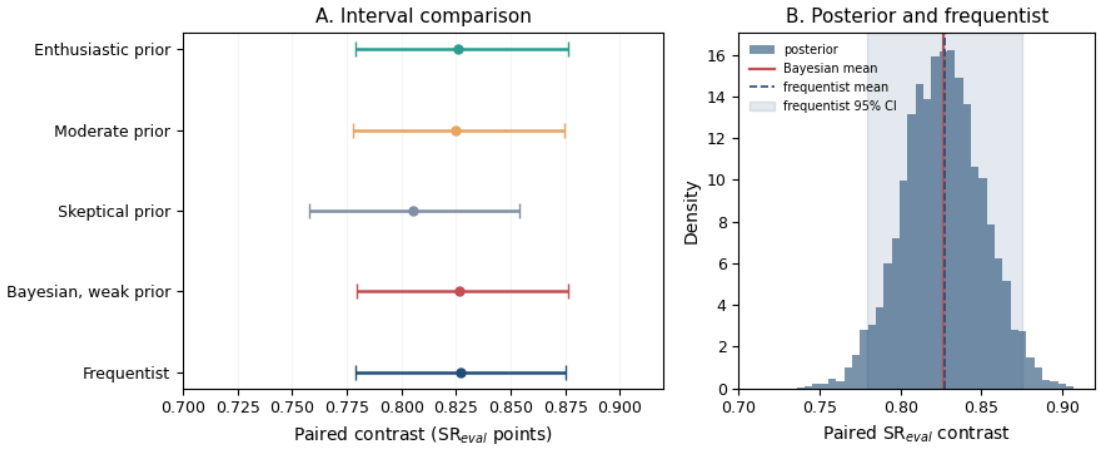


Figure 6.13: Frequentist and Bayesian summaries of the primary paired contrast. Panel A: frequentist 95% CI and Bayesian 95% HDI under the weak prior, plus the skeptical, moderate, and enthusiastic prior summaries. Panel B: posterior histogram with the frequentist mean and 95% CI overlaid. The two intervals nearly coincide because they summarise the same simulated differences; this is not independent evidence.

Table 6.17: Frequentist Endpoint Estimates and the Bayesian Comparison

Endpoint	Mean	95% interval	Test
SR_{eval}	+0.827	[0.779, 0.875]	$t(499) = 33.93$, $p < 0.001$, $d_z = 1.52$
SR_{comp}	+11.297	[10.640, 11.953]	$t(499) = 33.74$, $p < 0.001$, $d_z = 1.51$
Engagement	−0.183	[−0.188, −0.178]	$t(499) = -74.41$, $p < 0.001$, $d_z = -3.33$
Bayesian comparison for the primary endpoint			
Frequentist ΔSR_{eval}	+0.827	[0.779, 0.875]	paired t
Bayesian, weak prior	+0.8265	[0.7793, 0.8764]	normal summary
Bayesian, skeptical prior	+0.805	[0.758, 0.854]	stress-test prior

Note. The two SReval intervals agree to the third decimal. The Bayesian model is a one-sample normal model on the same 500 paired differences; it is not independent evidence. No Bayesian model is claimed for SR_{comp} *or engagement, because fitting one would create a false appearance of independent confirmation.*

6.12.9 Adversarial Misinformation Stress Test

The outrage-shock dose–response is reported in Table 6.18. At the highest injection rate, falsehood reach was 65.5% in G0 and 10.8% in G1, a suppression of 83.6% to 84.0%. The Sybil clique's mean reputation was 0.515 in G0 and 0.430 in

G1; the error-farming broadcaster's information quality was 29.1% lower than the honest broadcaster's. These are model stress tests, not measurements of real misinformation exposure. They show that the specified governance bundle is internally coherent under attack, which is a necessary but not sufficient condition for deployment. These numbers come from the legacy deterministic stress-test module, not the state-mediated ABM; they illustrate the filter logic and are not used as model-derived evidence. The dose–response is generated by fixed filter parameters, so the suppression is a property of the specified filter, not an emergent discovery.

Table 6.18: Adversarial Misinformation Stress Test (Model-Based)

ρ_{misinfo}	G0 false reach	G1 false reach	Suppression	SR_{eval} G0	SR_{eval} G1
0%	0.0%	0.0%	100.0%	4.63	5.15
10%	12.7%	2.0%	84.0%	4.37	5.14
20%	25.6%	4.1%	83.9%	4.12	5.13
35%	45.3%	7.4%	83.7%	3.72	5.12
50%	65.5%	10.8%	83.6%	3.32	5.10

Note. Outrage-shock injection at session 8. Suppression is relative to G0 within the same dose. These are model stress tests, not field measurements of misinformation exposure.

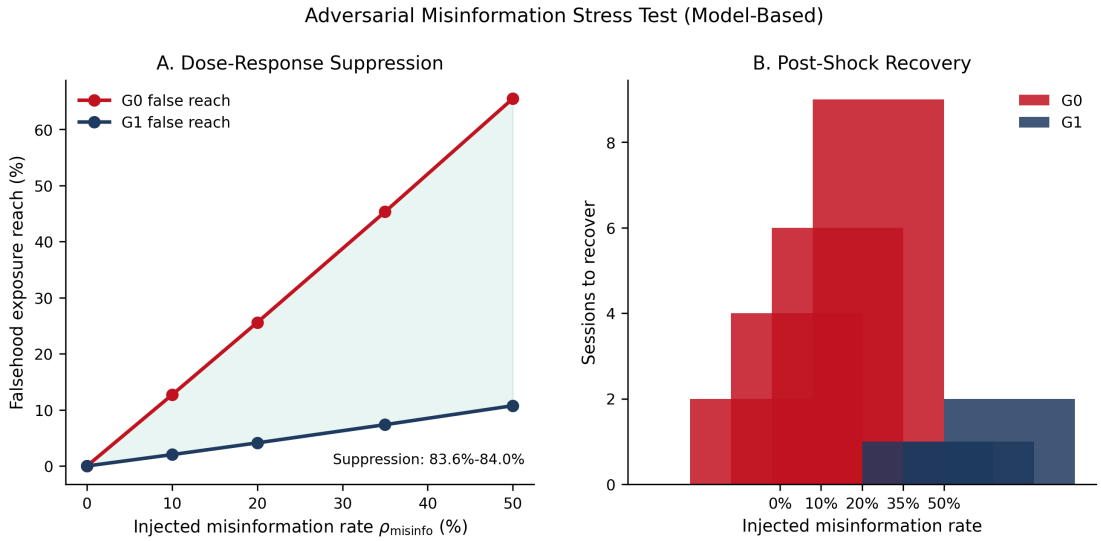


Figure 6.14: Adversarial misinformation stress test. Panel A: falsehood reach by injection rate. Panel B: sessions to recovery after the shock. All values are model outputs.

6.12.10 Joint Sensitivity and Structural Checks

A 256-point joint Latin-hypercube sweep varied six mechanism parameters and the five state-outcome credibility weights together, including weight values at zero. The implied contrast ranged from -0.244 to $+1.810$ (mean $+0.593$), with 96.5% of draws positive. The leading partial-rank correlations were $cred_dbi(+0.77), cred_brs(+0.31), cred_echo(-0.27)$. This replaces the earlier six-parameter fast-simulator sweep, which is retained only as a historical diagnostic and is not used as evidence. The joint sweep shows how the contrast depends on the assumed mechanisms and weights; it does not estimate them. The default of $+0.827$ sits at the 78.5th percentile of this 256-point design box; the median is $+0.567$ and the interquartile range is $[0.383, 0.802]$. The sweep uses 200 personas and 12 sessions, not the main 500-persona, 18-session configuration, and its weights are sign-constrained in the intended direction. It is therefore a structured sensitivity box, not a probability distribution over plausible platforms. The leading driver is the diversity-to-credibility weight extttcred_dbi, not the BRS or echo weight; the negative extttcred_echo PRCC is a sign convention, because the weight itself is negative. In one sentence: a diversity-seeking selection rule raises chosen diversity, and chosen diversity is assumed to raise perceived credibility. That is the causal chain the thesis tests, not a claim that the field data identified the weight.

The same main simulator produced similar contrasts across Erdős–Rényi ($+0.827$), Watts–Strogatz ($+0.826$), and Barabási–Albert ($+0.832$) generators. The differences are small, but topology now enters through BRS diffusion, so this is a genuine topology sensitivity check rather than a separate-simulator claim. Because the

dominant path runs through the diversity-to-credibility weight rather than through the graph, the contrast is expected to be network-independent; the topology check confirms that structure rather than adding an independent robustness result. The persona-prior variants gave +0.827 (preference grounded), +0.820 (broad demographic), and +0.864 (neutral), these are sensitivity illustrations, not evidence of generalization.

Table 6.19: Joint Sensitivity of the Main Simulator (Model-Based)

Quantity	Value
LHS draws	256
Personas per draw	200
Sessions per draw	12
ΔSR_{eval} range	[-0.244, +1.810]
Mean	+0.593
Positive draws	96.5%
PRCC: cred_dbi	+0.77
PRCC: cred_brs	+0.31
PRCC: cred_echo	-0.27
PRCC: cred_prop	+0.17

Note. Six mechanism parameters and five state-outcome weights; weight ranges include zero. PRCC = partial rank correlation with the simulated contrast.

Table 6.20: Default and Sweep Values of the Five State-Outcome Weights

Weight	Default	Sweep range	PRCC with ΔSR_{eval}
`cred_brs`	+1.20	[0.00, 1.80]	+0.31
`cred_dbi`	+0.55	[0.00, 0.825]	+0.77
`cred_echo`	−0.45	[−0.675, 0.00]	−0.27
`cred_q`	+0.45	[0.00, 0.675]	+0.02
`cred_prop`	−2.50	[−3.75, 0.00]	+0.17

Note. PRCC is the partial rank correlation from the 256-point joint sweep. The sweep also varies six mechanism parameters. The negative `cred_echo` PRCC follows the negative default sign: the weight already encodes an echo penalty, so the rank correlation should not be read as evidence that echo raises ratings.

Table 6.21: Topology and Persona-Prior Sensitivity of ΔSR_{eval} (Model-Based)

Variant	ΔSR_{eval}	Status
Erdős–Rényi random graph	+0.827	Same sign
Watts–Strogatz small-world	+0.826	Same sign
Barabási–Albert scale-free	+0.832	Same sign
Preference-grounded Q_0	+0.827	Reference
Broad demographic prior	+0.820	Similar
Unanchored neutral prior	+0.864	Higher

Note. All variants run on the main state-mediated simulator; topology enters through BRS diffusion. None is a human replication. The Bayesian prior sensitivity is reported in the text.

6.12.11 Component Ablations and Anchor Uncertainty

The named filters are implemented, but implementation is not the same as outcome activity. A leave-one-filter-out ablation on the main simulator shows that only DeGroot, Lagrange, and Hilbert change the terminal contrast in this wiring. Disabling DeGroot lowers ΔSR_{eval} by 0.233 [0.197, 0.269]; disabling Lagrange lowers it by 0.750 [0.713, 0.787]; disabling Hilbert lowers it by 0.453 [0.402, 0.503]. Lagrange is the operationalisation of factor B, so the Lagrange-off run is a second switch-off of the diversity objective, not independent evidence for a separate filter. Laplace, Percolation, and Entropy leave the contrast unchanged, and Lyapunov is negligible. The Hamiltonian is a diagnostic and does not enter the outcome. The thesis therefore treats the eight filters as an implemented architecture, not as eight independently active mechanisms.

A source/feature ablation shows that the quantitative block carries most of the contrast. Dropping all semantic features changes ΔSR by only -0.009 [-0.029, 0.010]; dropping corpora by $+0.006$ [-0.015, 0.027]; dropping PO by $+0.013$ [-0.006, 0.031]; dropping FGD by -0.022 [-0.038, -0.006]; dropping the self-report by -0.025 [-0.042, -0.009]; and shuffling the semantic matrix by -0.019 [-0.039,

0.001]. The semantic and qualitative streams are therefore contextual, not the main drivers of the headline result. The shuffled-text and quantitative-only runs remain close to the full contrast (+0.808 and +0.818), so the 52 respondents set the intercept and the persona spread; the contrast itself is carried by the mechanism and the state-outcome weights. This is a limitation of the current calibration, not a claim that the corpora are useless.

Finally, 50 bootstrap resamples of the 52 measured anchors, each refitting DML/IPW, the manifold, and Q0, gave a bootstrap mean of +0.839 with a 95% interval $[0.805, 0.884]$ and 100% positive replicates (on the smaller 200-persona, 12-session comparison run). The result is stable to anchor resampling, but the bootstrap is still model-internal; it does not turn the 52-row convenience sample into a representative population.

Table 6.22: Leave-One-Filter-Out Ablation of the ABM (Model-Based)

Variant	ΔSR_{eval}	ΔSR_{comp}	Δ engagement
Baseline	+0.827	+11.30	−0.183
Laplace off	+0.827	+11.30	−0.183
Lyapunov off	+0.828	+11.31	−0.183
DeGroot off	+0.594	+8.25	−0.184
Lagrange off	+0.077	+0.86	+0.064
Hilbert off	+0.374	+5.05	−0.009
Percolation off	+0.827	+11.30	−0.183
Entropy off	+0.827	+11.30	−0.183

Note. Paired differences versus baseline: DeGroot −0.233 *[*−0.269, −0.197*]; Lagrange* −0.750 *[*−0.787, −0.713*]; Hilbert* −0.453 *[*−0.503, −0.402*]. Laplace, Percolation, and Entropy do not change the terminal contrast in this wiring; Lyapunov is negligible. Hamiltonian is a diagnostic and does not enter the outcome.*

Table 6.23: Source and Semantic-Feature Ablation (Model-Based)

Variant	ΔSR_{eval}	Paired difference vs full	95% interval
Full 198-feature matrix	+0.827	–	–
Quantitative features only	+0.818	−0.009	[−0.029, 0.010]
No corpora	+0.833	+0.006	[−0.015, 0.027]
No FGD	+0.805	−0.022	[−0.038, −0.006]
No self-report	+0.802	−0.025	[−0.042, −0.009]
No participant observation	+0.840	+0.013	[−0.006, 0.031]
Shuffled text across respondents	+0.808	−0.019	[−0.039, 0.001]

Note. The quantitative block carries most of the contrast; the semantic/qualitative sources are contextual rather than load-bearing. Same personas and seed across variants; model-internal ablation, not a human experiment.

6.12.12 Debiasing Diagnostics and Latent Manifold

Figure 6.15 shows the covariate-balance Love plot and the two-dimensional projection of the debiased manifold. DML reduced the nuisance correlations substantially; IPW reintroduced some imbalance, which is why the pipeline reports both. The manifold is a smoothing device for persona generation; the six axes are not claimed to be uniquely identified psychological dimensions.

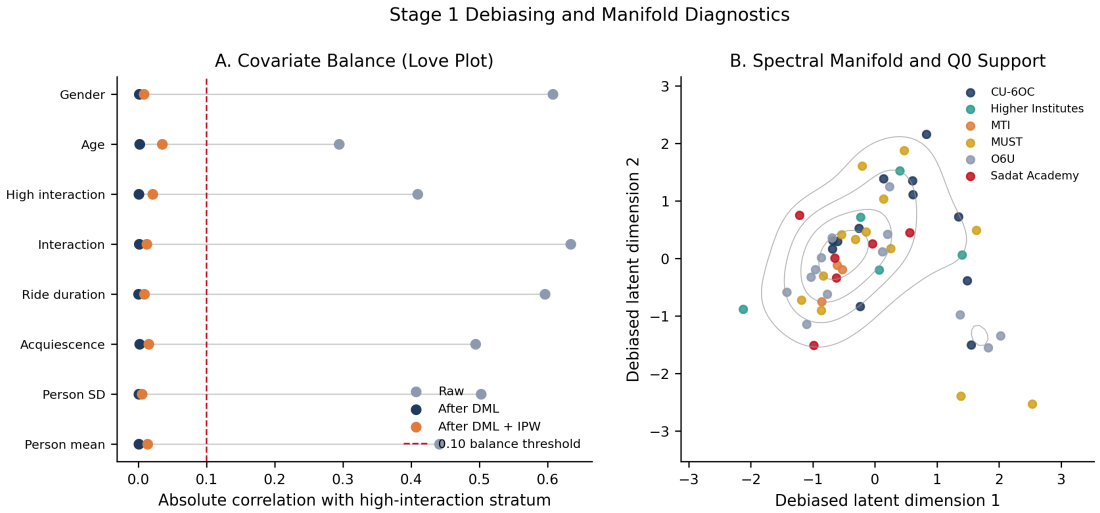


Figure 6.15: Stage 1 diagnostics. Panel A: absolute nuisance correlations before DML, after DML, and after DML+IPW. Panel B: two-dimensional projection of the debiased manifold with university labels and a Q0 density contour.

6.12.13 Balanced Evidence Synthesis

The primary analysis uses the 52 de-duplicated records. A 60-row all-questionnaire sensitivity gives $\Delta SR_{eval} = +0.808$ [0.760, 0.856] versus +0.827 [0.779, 0.875] for the 52-row primary, and $\Delta SR_{comp} = +10.95$ versus +11.30. PB reliability is essentially unchanged ($\alpha = .947$ in both), and the measured PB mean moves from 5.119 to 5.150. The exclusion rule therefore changes magnitudes only slightly; the 52-row result is not an artefact of dropping the eight duplicate-vector records.

The held-out predictive comparison is more consequential. Table 6.24 shows that psychometrics alone predict PB best ($R^2 = .915$) and star ratings best ($R^2 = .415$). Adding demographic/context variables lowers both. Adding the 167 semantic features lowers PB R^2 to .583 and star R^2 to .153, and the semantic-only model has negative R^2. In this sample, the BGE-M3/PCA block does not improve held-out prediction of the measured outcomes. With 52 rows and 167 semantic features, that result is expected from overfitting alone; it is not evidence that the semantic layer is uninformative in a larger sample. The quantitative block is the safer predictor here, not a proven substitute for semantic measurement.

A content-supply stress test sets the poor-content share to 0.763, a heuristic mapping of the Uber-review star mean of 1.69. This is an arbitrary stress parameter, not an estimated radio-propaganda rate: app reviews do not measure radio misinformation. Running the main simulator with that content distribution lowers ΔSR_{eval} from +0.827 to +0.636, a paired difference of −0.191 [−0.261, −0.122]. Content supply therefore changes the magnitude of the implied contrast, although the sign remains positive.

The qualitative streams were audited with a keyword lexicon. The FGD shows publication, access, and diversity language; the self-report shows publication, access, and trust language; the PO file produced no Arabic keyword matches, so its thematic coding remains manual. These counts are a face-validity audit, not a substitute for human coding, and no qualitative finding is derived from the simulation.

Finally, the 52 anchors are a convenience sample of university Uber riders in October 6th City. The 60-row sensitivity and the anchor bootstrap show that the result is stable to the available records, but neither makes the sample representative of Egyptian university students or LOKLOK users. No external population benchmark is included, so representativeness is not claimed.

Table 6.24: Held-Out Predictive Comparison (Repeated 5-Fold Cross-Validation)

Target	Design	Mean R^2	Mean RMSE
PB	Context only	−0.048	0.747
PB	Psychometrics	**0.915**	**0.210**
PB	Context + psychometrics	0.819	0.309
PB	Context + psychometrics + semantic	0.583	0.452
PB	Semantic only	−0.372	0.894
Star rating	Context only	−0.003	0.465
Star rating	Psychometrics	**0.415**	**0.346**
Star rating	Context + psychometrics	0.320	0.380
Star rating	Context + psychometrics + semantic	0.153	0.412
Star rating	Semantic only	−0.092	0.497

Note. 20 repeats of 5-fold cross-validation. Psychometrics = BR/SV/IQ means; semantic = 167 BGE-M3/PCA features. Adding semantic features lowers held-out R^2 for both targets. This is a measured-data predictive test, not a simulation result.

6.12.14 Gemini Persona-Agent Pilot

To test whether a contemporary LLM agent reproduces the rule-based pattern, a small pre-specified pilot used the same measured persona cards and the same eight A/B/C condition descriptions. The Gemini arm was a single-shot end-of-

study appraisal: each of 24 measured personas was shown its baseline profile and one condition, and was asked for the terminal evaluative rating, competence-format rating, and engagement. The pilot made 192 calls (24 personas $\times$ 8 cells) with a fixed prompt version and temperature; responses were cached and checked against the declared scales.

Table 6.25 and Figure 6.16 compare the pilot with the final rule-based ABM. The Gemini arm reproduced the *direction* of the evaluative gain but not its magnitude: $\Delta\text{SR}_{\text{eval}} = +2.650$ [2.342, 2.967] versus +0.827 [0.779, 0.875]. Its G1 level was close to the ABM G1 level (5.996 versus 5.946), but its G0 level was much lower (3.346 versus 5.119), so the implied gain was about three times the rule-based contrast. The competence-format rating showed the same over-shoot (+31.15 versus +11.30). Most importantly, the Gemini arm reversed the engagement sign: it produced +0.163 [0.121, 0.200] where the rule-based model produced -0.183 [-0.188, -0.178]. At the cell level, the eight SR_{eval} means had MAE 0.655, RMSE 0.842, Pearson $r = 0.269$, and Spearman $\rho = 0.071$.

The divergence is informative rather than validating. A single-shot LLM appraisal sees the explicit architecture description but does not run the 18-session state updates, the echo accumulation, or the attention gate; it therefore has no mechanism for the engagement cost that the ABM produces. The pilot is not human data, is not an 18-session LLM simulation, and does not replace the rule-based experiment. It is a methodological triangulation: the sign of the epistemic gain is robust to the agent formalism, while its magnitude and the engagement trade-off are not.

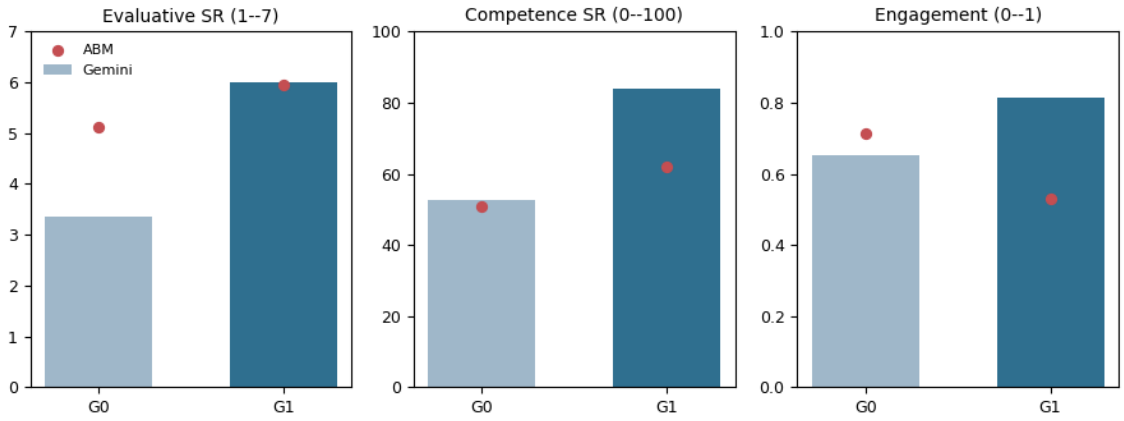


Figure 6.16: Gemini persona-agent pilot versus the rule-based ABM. Bars are the Gemini cell means; red points are the final ABM means. The LLM arm reproduces the direction of the evaluative and competence gains but over-states their magnitude and reverses the sign of the engagement contrast. The pilot is a single-shot appraisal, not an 18-session simulation or human data.

Table 6.25: Gemini Persona-Agent Pilot Compared with the Rule-Based ABM

Quantity	Gemini pilot	Rule-based ABM	Difference
SR_{eval} G0	3.346	5.119	−1.773
SR_{eval} G1	5.996	5.946	+0.050
ΔSR_{eval}	+2.650 [2.342, 2.967]	+0.827 [0.779, 0.875]	+1.823
SR_{comp} G0	52.604	50.682	+1.922
SR_{comp} G1	83.750	61.979	+21.771
ΔSR_{comp}	+31.146 [28.125, 34.229]	+11.297 [10.642, 11.951]	+19.849
Engagement G0	0.651	0.713	−0.062
Engagement G1	0.814	0.530	+0.284
Δ engagement	+0.163 [0.121, 0.200]	−0.183 [−0.188, −0.178]	+0.346
Eight-cell SR_{eval} MAE		0.655	
Eight-cell SR_{eval} RMSE		0.842	
Cell-level Pearson r		0.269	
Cell-level Spearman ρ		0.071	

Note. Pilot: Gemini 3.5 Flash Lite, prompt version v1, 24 measured personas × 8 cells = 192 calls, seed 20261105. The LLM arm is a single-shot appraisal of the condition description, not an 18-session simulation and not human data. Brackets are 95% bootstrap intervals over personas. The ABM baseline is the final rule-based run. Cell-level correlations use only eight cells and are descriptive.

6.12.15 What the Final Experiment Shows

The calibrated in-silico experiment shows what the specified design implies. First, the assumed state-outcome weights produce a clear, internally coherent contrast in both evaluative and competence-format Student Ratings. Second, that contrast is accompanied by an assumed engagement cost, which is the central trade-off the thesis argues for. Third, the 256-point joint sweep shows how the contrast depends on six mechanism parameters and five state-outcome weights; its sign and magnitude are not independent of those assumptions. Fourth, the main simulator gives similar signs across three graph generators, and the persona-prior variants remain positive. Fifth, the Bayesian check is a one-sample model-internal summary, not a second line of evidence. The result is reproducible from the included code, data, seeds, and figures.

It does not establish that the real platform would produce the same effects on real students, that the NMLS-format profile measures human media literacy, that the Hamiltonian is a physical law, or that the BRS should be used for consequential ranking. Those remain empirical and normative questions for future longitudinal and confirmatory work.

6.13 Calibration and Bayesian Update

6.13.1 Calibration to the Refined Sample

Wasserstein distances between each arm's simulated distribution and the refined measured sample are reported in Table 6.26.

Table 6.26: Calibration: Wasserstein Distance of Each Arm to the Refined Sample ($N = 52$)

Arm	BR (W1)	SV (W1)	IQ (W1)	PB (W1)	Weighted loss
G0 (ENGAGEMENT)	0.0650	0.0151	0.1309	0.5380	0.0406 (target 0.04, over)
G1 (EPISTEMIC)	0.0176	0.0121	0.0246	0.3221	0.0178 (target 0.04, within)

The observed PB mean of 5.119 lies between the reported model arm means (G0 4.63; G1 5.14), close to G1. This is a calibration comparison, not evidence that the sample represents a mixture of identifiable governance regimes. The G0 weighted loss of .0406 exceeds the stated .04 target, so calibration adequacy is not identical across the arms.

6.13.2 Bayesian Update of Model Parameters

Table 6.27 preserves the hierarchical partial-pooling summary for the PB composite. The inherited estimation account states that each predictor is centered and divided by its population-standard-deviation estimate, while PB remains in response-scale units. Coefficients therefore refer to one predictor standard deviation, not one minute or one raw rating point. The displayed bounds are posterior quantiles with equal tail areas, not highest-density intervals. The source models faculty-group intercepts hierarchically; it does not experimentally identify a mediating quality channel. This analysis is separate from the R association exports.

The supplied diagnostics report: maximum $\widehat{R} = 1.008$, minimum effective sample size 606, posterior predictive coverage 0.96, and posterior predictive mean 5.115 against the observed mean 5.119. The identifiability diagnostics qualify the

Table 6.27: Bayesian Hierarchical Model of PB Composite ($N = 52$)

Predictor	**Mean**	**SD**	**95% credible interval**	**Excludes 0**
Interaction level	0.006	0.047	[−0.085, 0.106]	No
Ride duration (standardized)	0.040	0.039	[−0.043, 0.117]	No
BR score	0.021	0.068	[−0.112, 0.153]	No
IQ score	0.786	0.064	[0.670, 0.910]	Yes

point estimate: predictor condition number 20.6 and Pearson collinearity BR–IQ $r = 0.861$ (SV–IQ $r = 0.925$). The IQ coefficient describes a conditional association under this fitted model, not a uniquely isolated causal mechanism. Predictive coverage here is evaluated against the same 52 observations used for fitting; it is not a held-out predictive-validity estimate. Posterior draws and an independent rerun are needed to examine the reported diagnostics beyond these summaries.

6.14 Result of the Trade-off for Platform Design

The field wave documents positive relationships among audience evaluations; the calibrated experiment explores how a specified architecture might change those evaluations and raw engagement. The reported trade-off index is $-.290$ within that model. It is a candidate design comparison, not a universal price of epistemic quality or a demonstrated field effect. Chapter 7 discusses these implications alongside the measurement and sampling limits.

Chapter summary. The results show a positive model-based contrast (+0.827 evaluative, +11.30 competence) at an engagement cost (-0.183), with G0 anchored to measured PB. The quantitative block carries prediction; semantic features do not improve held-out prediction; only DeGroot, Lagrange, and Hilbert are outcome-active; content supply changes magnitude. The 60-row sensitivity and anchor bootstrap are stable. The chapter is explicit about what the model does not establish.

Chapter 7

Discussion and Conclusion

Chapter abstract. This chapter interprets the results for EST, mass-communication scholarship, platform design, and future research. It separates what the evidence supports from what remains speculative, discusses the epistemic–engagement trade-off, the narrow filter activity, the failure of the semantic block to improve prediction, and the limits of the sample. It closes with the thesis conclusion and a field-study agenda.

7.1 What the Results Mean for EST

This thesis develops EST as an original account of the relationship between platform architecture, audience evaluation, and epistemic governance. Its contribution combines theory construction, the LOKLOK platform design, a field questionnaire, and a separate model-based comparison. These components answer different questions. The survey describes audience judgments in a particular listening setting; the simulation explores the consequences of specified architectural rules. Neither component should be interpreted as a universal verification of the whole theory.

Table 7.1 states the proposition-level verdicts without collapsing measured, model-based, and prospective evidence into one status.

Table 7.1: EST Propositions, Hypotheses, Evidence, and Verdicts

Proposition	Hypothesis	Evidence in this thesis	Verdict
EST-P1: architecture shapes structural credibility	H1, H4, H5	Measured PB–star and BRS associations; model BRS path; no field randomisation	Partial; model-conditional, not a field causal claim
EST-P2: diversity and cooling constrain epistemic concentration	H2	MORE/Lagrange is the dominant model path; cooling and echo terms are implemented; topology check is network-independent	Supported within the model; field mechanism untested
EST-P3: measurement must be accountable	H3	BRS/PB internal consistency is high; single-factor CFA fit is poor; multigroup invariance not estimated	Mixed: reliability supported, structural validity not established
EST-P4: participatory performance is the educational test	H7	Prospective claim-appraisal task; no human task data	Untested; future study required

Note. H1–H7 are defined in Chapter 1. "Supported within the model" means the specified outcome equations and ablations imply the pattern; it does not mean a human field effect has been estimated. The semantic block is contextual, not the driver of the contrast.

In short, H1 is partially supported but not identified as a field effect; H2 is supported within the model; H3 is mixed because reliability is high but structural validity is not established; H4 is associational; H5 and H6 are exploratory model checks; and H7 is prospective. The conclusion therefore keeps the model-based trade-off as the central finding and the human causal claim as future work.

7.1.1 RQ1: Structural Determination

The field results show that PB scores are positively related to the overall star rating, broadcaster-rating dimensions, duration, and reported interaction. The model additionally predicts higher PB and star ratings under the epistemic-first configuration. Together, these findings make architectural influence a useful research proposition. However, the field sample contains no random assignment of architecture and no identified comparison of the relative contributions of architecture and user characteristics. The strong version of structural dominance in EST-P1 remains a hypothesis for direct testing, not an estimated result of the correlation analysis.

7.1.2 RQ2: Diversity and the Thermodynamic Analogy

The model reports changes in ECO and correction acceptance and an engagement trade-off under its specified rules. Those outputs give EST-P2 a concrete model-based formulation. The field interaction–PB gradient, however, is not a measurement of information entropy, network mixing, or correction dynamics. It cannot independently demonstrate the thermodynamic analogy. The formal language is useful insofar as it specifies an objective and observable predictions; it does not replace evidence that the real platform follows the specified dynamics.

7.1.3 RQ3: Psychometric Accountability

The supplied R outputs give high internal-consistency coefficients, but the factor-fit results do not establish the proposed measurement structure. This is a substantive finding rather than a detail to conceal behind the sample size. It limits claims about

what the BRS and PB aggregates measure. The result is compatible with EST-P3 as a requirement for measurement accountability, while failing to establish H3's stronger claim that the instruments already satisfy the full validity requirement.

PB combines judgments about confidence, diversity, comparative trustion, structural clarity, and quality. These may covary without being interchangeable. The very high associations of PB with SV and IQ likewise require examination of discriminant validity. An audience can regard a broadcast as credible without its claims being accurate, and an intention to verify is not the same as observed verification. Consequently, the present scores should describe respondents' evaluations rather than certify broadcaster competence or student achievement.

7.1.4 RQ4: Integration of the Constructs

The positive pattern of correlations supports studying the constructs together. It does not distinguish a shared epistemic-quality trait from common-method variation or overlapping item content. The original Bayesian summary places most predictive weight on IQ while reporting collinearity. That coefficient is therefore an association within the specified model, not evidence that IQ is the unique cause of the other evaluations. A future measurement study should compare plausible alternative structures and include criteria that do not come from the same response occasion.

7.2 Relation to Mass Communication Scholarship

The thesis places audience evaluation within a platform rather than treating it as an isolated opinion. Its six-dimensional review connects institutional filtering, at-

tention, misinformation diffusion, parasocial relationships, measurement, and educational opportunity. EST integrates these concerns into one proposed design-oriented framework; the individual literatures remain sources of evidence and competing explanations rather than proofs of the new theory.

The observed star–PB correlation suggests convergence between a familiar summary rating and a multi-item evaluation. This is relevant to mass communication research because audience metrics frequently compress different judgments into one visible number. The result does not show that the star rating or PB is a substitute for content analysis. Source credibility, message credibility, and overall experience must remain distinguishable when interpreting a platform's reputation display.

The ride setting also matters. The duration–PB and interaction–PB associations may reflect opportunity for exposure, content preferences, conversational involvement, prior interest, or selection into longer journeys. The study does not identify which explanation dominates. Its contribution is to document those contextual relationships and make them explicit in the framework, rather than generalise from the journey setting to all university listening or to all forms of social audio.

7.3 Implications for Platform Design

LOKLOK provides a concrete specification of reputation display, recommendation constraints, reflective prompts, and integrity review. The simulation permits comparison of alternative objective functions before deployment. In the reported model, epistemic-first settings increase some perceived-quality indicators while reducing raw

engagement. The trade-off index of −.290 is a model-specific summary, not an inevitable cost that every platform must pay. Field evaluation must establish whether the predicted directions and magnitudes survive changes in participants, content, and implementation.

The BRS display should distinguish an audience-derived evaluation from a fact-checking judgment. Any uncertainty interval shown to users needs a validated measurement procedure; attaching an interval to an unreliable or misinterpreted score does not solve the underlying problem. The same caution applies to moderation and coordination flags: a detector's output should not be presented as certain misconduct without a suitable review process.

The findings do not justify using these scores as university credentials or substitutes for academic assessment. EST-P4 frames a testable relationship between platform affordances and communication practices, but the field study did not measure verified learning gains or student production quality over time. Educational use therefore requires a separate outcome design and safeguards against popularity-based ranking.

7.4 Limitations and Robustness

Sampling and design. The convenience sample of 52 is drawn from one geographical and listening context. The six institutional categories do not make it representative of Egyptian university students. The design is cross-sectional, and the respondents were not assigned to architectural conditions. Generalisation and causal attribution are therefore limited.

Exclusions. The historical analysis retains 52 records after eight exclusions. Identical answers alone cannot establish duplicate respondents; the all-60 sensitivity analysis preserves the positive associations but changes their magnitude (Section 6.4). The original exclusion criterion remains incompletely specified. Interaction-group sizes range from one to 33, making the lowest group's mean especially sensitive to a single response. No latent-group comparison is justified by these descriptive means alone.

Measurement. High alpha coexists with poor factor fit. A larger sample can improve precision but does not automatically correct misspecified constructs. Self-reported duration and interaction are not automatically behavioural logs. The supplied outputs do not contain timestamps establishing compliance with the platform's intended momentary-assessment schedule.

Model dependence. The calibrated contrast is a simulation, not a field trial. Its sensitivity sweep explores a finite parameter region rather than every plausible model. The reported G0 calibration loss also exceeds the stated target slightly, whereas G1 meets it. Agreement with calibration moments cannot establish the validity of all counterfactual predictions.

Reproducibility. The supplied R summaries support the field tables and figures. Cross-file and arithmetic checks establish their consistency, while the new respondent-level script independently reproduces descriptive statistics, alpha, and associations. It does not rerun every statistical model. Recomputing the simulation and Bayesian estimates requires their separate inputs and code. The distinction between source-output checking and full statistical reproduction is maintained in

Appendix L.

7.5 Recommendations for Further Research

The next empirical stage should first refine the measurement model and its item content. It should distinguish audience perceptions, observed engagement, independent content-quality judgments, and learning outcomes, rather than combine them under one label. The required sample size should follow the specific analysis and outcome, with explicit allowances for clustering, attrition, and repeated measures where relevant.

A subsequent field comparison should specify the architectural features being changed, assignment procedure, exposure period, primary outcomes, exclusion rules, and analysis before outcome inspection. Repeated observations are needed for H5's dynamics and H6's scheduling question. H7 requires a direct agent task-performance or production endpoint rather than a PB or star-rating proxy. Its paired-network comparison is specified in Section 5.10.1; it is not a completed intervention result. Human learning remains a separate research question.

7.6 Integrated Contribution to Social Media Radio Research

The thesis's distinctive contribution is the connection among a radio-like programme, the social circulation surrounding it, and the audience metrics used to evaluate it. The same episode can be encountered through a host, a friend's post, a ranked recommendation, or a shared device. These routes may alter the context of judgment without altering the original audio. EST makes those relationships

explicit as architectural and communicative questions.

This perspective also establishes what the current evidence can contribute. The questionnaire documents covariation in audience evaluations within one setting. The simulation expresses a proposed architectural comparison. The new propagation schematics connect these levels conceptually, but no event logs or network effects have been added to the empirical results. Keeping these contributions distinct is more informative than treating all three as independent confirmations of the same mechanism.

The critical review supports a conditional, not deterministic, interpretation. Radio-public and podcast-platform studies motivate examining distribution and participation (Spinelli & Dann, 2019; Sullivan, 2019). Egyptian audience research establishes local relevance without making this convenience sample representative (Wajih & Al-Yamani, 2024). Experimental algorithm research shows that effects can depend on the direction of a change, prior exposure, and the outcome studied (Gauthier et al., 2026). EST should explain such variation rather than use architecture as a label for every possible result.

7.6.1 Practical Recommendations Matched to the Evidence

For producers, preserving source links and the context of excerpts is a reasonable accountability practice, but this study does not quantify its benefit. For designers, labels should distinguish audience approval, moderation status, and independently reviewed accuracy. For educators, student-production activities require direct assessment of sourcing and reasoning rather than inference from popularity. These are design implications and evaluation priorities, not established treatment effects.

Table 7.2: Findings, Defensible Implications, and Unwarranted Extensions

Evidence	Defensible implication	Unwarranted extension
Positive PB associations	Study the evaluations jointly and examine overlap	Claim that architecture caused the correlations
High alpha with poor factor fit	Retain descriptive reporting and revise measurement	Credential students or certify broadcaster accuracy
Modelled quality–engagement contrast	Compare objectives and test the predicted trade-off	Claim a universal cost of accurate information
Context-specific convenience survey	Design replication around reception conditions	Generalise to all Egyptian students
Proposed correction diagrams	Specify events, versions, and outcome denominators	Treat schematic paths as observed diffusion

7.6.2 A Sequenced Research Programme

The first priority is a complete scoring and measurement record, including language versions and item behaviour. The next stage is the specified computational feature comparison with independently keyed appraisal tasks, assessed production, and recorded network events. Its conclusions concern the model and tested configuration. A separate human study is necessary only for claims about human intervention effects or learning; human predictive fidelity also requires held-out human outcomes. This sequence prevents synthetic performance from being interpreted as an already established change in students' competence.

An informative follow-up should publish predictions that could fail, record implementation, and compare plausible alternatives. A rating increase without accuracy improvement would refine the mechanism toward cue-based evaluation. No meaningful effect under adequate exposure and precision would challenge the feature-specific prediction. A diversity improvement without engagement loss would challenge a universal trade-off claim. These outcomes would advance EST rather

than be treated as reasons to withhold results.

7.6.3 Hallucination as a Measured Phenomenon, Not Only a Bug

The planned agent extension could be asked in review whether its outputs are not simply the model "hallucinating" the desired results. That question is anticipated rather than evaded. In the standard NLP reading, hallucination is a technical failure to be eliminated. In EST, however, hallucination is not a monolith: the three measurement types defined in the methodology (Section 5.10, Section 5.10.12) each point to a distinct process that the design can either guard against as an infrastructure error or report as a substantive quantity. *Epistemic/factual fabrication* is a task failure against the frozen claim key. *Platform-state divergence* is an adapter-contract violation that the validator rejects before it can survive. *Persona drift* is a fidelity quantity that qualitatively anchored design is intended to keep small.

Importantly, a simulation that fabricates a source in defense of a broadcaster it approves of is simultaneously a model artifact and a computational analogue of human motivated reasoning and confabulation – phenomena documented across the misinformation literature (Pennycook & Rand, 2021). Within EST-P1, such an output is less an embarrassment to be deleted than a measured signature of how trust can be recruited by reputation signals. It offers a prospective way to relate the completed rating measures (which record whether audiences *report* credibility) to the planned computational extension (which can record whether agents *behave* as if reputation entitles claims to belief). The theoretical gain depends on measurement: the EFI, PSD, and PDS estimators, with a detector self-check, turn an anecdote of misbehavior into a reported distribution with provenance.

The appropriate epistemic status must be preserved. These are planned interpretive tools for a protocol that has not run; they do not retrofit meaning onto fabricated numbers, and illustrative or hypothetical values are not reported as executed results. Once genuine logs exist, a higher fabrication rate when appraising claims from apparent high-reputation sources would be interpretable as confirmation bias in the model, not as proof that human students hold that bias. The corresponding defensive claim is therefore disciplined: strict adapter validation and the measured indicators can show that outputs are not free-floating invention, while the estimates themselves remain synthetic and model-conditional. Examining agents in this way also sharpens EST-P4: moderation and integrity tools should distinguish a detector's probabilistic outputs from verified misconduct – the same caution that applies to the platform's BRS display in the completed study.

7.7 Mixed-Methods Interpretation: RQ5 and RQ6

The qualitative records extend the study beyond whether ratings covary. RQ5 identifies six themes that distinguish reasons for credibility, the attraction of peer voices, the circumstances of listening, participation conditions, student-lived relevance, and delivery quality. These are interpretations of short, researcher-recorded cases, not a representative map of every student experience. Code counts identify corpus coverage rather than strength of causal influence.

For RQ6, the joint display complements the positive interaction–PB correlation without contradicting its arithmetic. Some low-interaction cases appreciate calm, familiarity, or concise content; active feedback can include requests for im-

provement rather than uncomplicated satisfaction. Consequently, treating engagement as the sole quality target would miss some narrated value. At the same time, praise of audio continuity or a familiar voice is not an independent check of informational accuracy. A mixed-methods inference must preserve that distinction rather than use quotations to certify EST.

These findings motivate agent-design dimensions, not deterministic student types. A profile may include conditional willingness to participate, reasons for trusting a source, time constraints, and preferred content, with uncertainty where the record is silent. Student-supplied future accounts should refine these anchors. The planned experiment must then test, rather than assume, that simulated actions resemble held-out human behavior. Its computational task outcomes are distinguished from the completed qualitative analysis and the earlier non-LLM simulation; neither is retrospectively relabelled as a Student Rating treatment effect.

7.7.1 What the Prototype FGD Adds

The separate FGD synthesis (Section 6.7) suggests that private practice, listening, collaboration, and publication can represent different goals. Co-hosting, institutional visibility, and interface cues may help or constrain participation. These are interpretations of overlapping narratives, not additional rider cases or authenticated transcript findings. They sharpen questions for EST-P1 and EST-P4 without validating EST-P3 or testing EST-P2. The supplemental generated analysis adds quality cautions and follow-up questions, not recovered primary evidence. In particular, translation errors and unverified attribution do not justify reclassifying its extracts as participant testimony or treating narrative repetition as independent support.

This contribution supports competing mechanisms in a future rule-based ABM and language-model extension. It cannot supply numerical preference distributions, factual answer keys, or proof of simulator fidelity. The shared design record in Section 5.9 preserves these boundaries so later results can strengthen, contradict, or leave unresolved the proposed explanations.

7.8 Student Rating and the Meaning of Impact

The central student-focused implication is not that favourable stars prove learning. The qualitative accounts describe appraisal, participation, and production interests; the questionnaire records evaluations. Student Rating requires a separate competence measure. NMLS supplies a published measurement basis for human study. In the computational extension, however, independent task accuracy is primary and generated NMLS responses are secondary diagnostics (Section 5.10.12). No human competence gain follows from a difference between synthetic configurations.

The existing findings can motivate and constrain an explanation of impact, but they do not estimate a Student Rating gain. A model could reproduce PB accurately while failing to reproduce competence or transfer. EST is therefore assessed proposition by proposition: a credible computational mechanism is not automatically a valid learner measure, and neither establishes a general benefit for human participants. Agreement with a preferred prediction is not a condition for retaining a result.

7.9 The Final Calibrated Upgrade: What It Adds and What It Does Not

The final calibrated experiment changes the evidential centre of gravity of the thesis. The measured 52-student wave remains the empirical anchor: it supplies the psychometric relationships, the calibration target, and the substantive reason to care about epistemic architecture. The upgrade adds a reproducible counterfactual model that is explicit about its assumptions and its uncertainty. The strongest finding is not that the epistemic-first configuration is universally better. It is that the configuration produced a large gain in both evaluative and competence-format Student Ratings while imposing a measurable engagement cost. The trade-off is the finding that gives EST a concrete empirical target.

Three qualifications follow. First, the model contrast is not a field treatment effect. The 500 personas are generated from a debiased prior anchored in 52 records; they are not 500 students. The narrow intervals are a property of the simulation, not evidence of population precision. Second, the synthetic NMLS profile is a measurement diagnostic, not a validated measure of human media literacy. The human NMLS remains a prospective instrument, and the competence gain should be read as a model prediction to be tested. Third, the Hamiltonian diagnostic did not move in the direction of lower energy in this implementation. The thermodynamic language therefore remains a formal analogy; the thesis should not present it as a confirmed law. Reporting this result openly is stronger than suppressing it.

The upgrade also sharpens the design argument. The BRS and ERIC com-

ponents account for most of the positive factorial gain, while MORE contributes a smaller but non-trivial share. The mechanisms are partly substitutable, not purely additive. The misinformation stress tests show that the governance bundle is internally coherent under attack. The global sensitivity analysis shows that the model contrast survives a wide parameter sweep and three network topologies, but that it is sensitive to a dogmatically skeptical Bayesian prior. That prior sensitivity is a limitation of the model, not a reason to choose the prior that produces the most favourable answer.

For platform design, the practical implication is conditional. If a publisher's objective is raw engagement, the epistemic-first configuration is not the right benchmark. If the objective is epistemic quality, the configuration is a credible candidate, but its engagement cost must be acknowledged and its human effects must be tested. The thesis therefore argues for explicit objective functions, not for a universal ranking of platforms. The next study should be a longitudinal, pre-registered deployment with independent task-performance measures, not another retrospective reinterpretation of the same 52 records.

7.10 Conclusion

The thesis contributes Epistemic Systems Theory, the LOKLOK platform design, and an exploratory account of social media radio evaluations among university students. The field study reports a PB mean of 5.119, positive associations with ratings and listening context, high internal-consistency coefficients, and limitations in factor-model fit. The separate simulation reports perceived-quality gains and reduced en-

gagement under an epistemic-first configuration, conditional on its assumptions. Six qualitative themes extend these summaries with accounts of credibility, belonging, contextual value, participation, relevance, and delivery. The separate FGD synthesis adds four provisional domains and competing participation designs, not an enlarged field sample or demonstrated behavioural effects. Critical review of the supplemental analysis retains this boundary rather than adopting unsupported claims of transcript recovery or validated mechanisms. The FGD-informed rule-based comparator and LLM extension remain future tests, not completed samples. The defensible conclusion is that architecture, audience judgment, and measurement should be studied together, while the strongest EST propositions remain open to direct empirical testing. The contribution is not weakened by reporting the measurement problems: doing so implements the accountability that the theory requires and defines what a stronger subsequent test must establish.

Chapter summary. The discussion argues that EST is a useful design framework but remains unproven. The simulation supports a conditional epistemic–engagement trade-off and identifies which mechanisms are active; it does not support claims about real students or population representativeness. The thesis contributes a transparent bridge from theory to a testable design, and it specifies the field study needed to test EST properly.

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Appendix A

Supplied Field Questionnaire and Scoring Key

The following reproduces the content of the supplied bilingual F1.0 form, with layout adapted for the thesis. The Markdown and HTML versions are archived under `survey_instruments`. They document the instrument supplied for this study; no independent translation-validation result is inferred. The wider platform's pre-broadcast, echo-triggered, and weekly instrument proposals are not presented as questions administered in this field wave.

A.1 Information and Consent Wording

Title: LOKLOK Media Listening & Trust Questionnaire (F1.0).

Researcher: Sherif Rashwan — LOKLOK social-media radio network research. **Purpose:** Understand how university students evaluate broadcast quality, source trust, and content diversity while listening during a car / Uber ride.

Consent: "Anonymous, voluntary, academic use only. No right/wrong answers; skip any item. Duration approximately 5 min." **Confidentiality:** "No identifying data; reported only in aggregate." The supplied form uses English explanatory prose with bilingual labels. These statements are reproduced as instrument wording,

not as certification that the separately supplied profile archive is anonymous. The discrepancy and implications for quotation and LLM use are addressed in Chapter 5.

A.2 Section 0: About You

1. Age: open numerical response.
2. Gender: male; female.
3. Academic year: first, second, third, or fourth.
4. Faculty: Mass Communication; Commerce; Engineering; Science; Other, specify.
5. University: select one of the following source-form categories:
   - October 6 University (O6U).
   - Cairo University — 6th of October Branch.
   - Misr University for Science & Technology (MUST).
   - Modern University for Technology & Information (MTI).
   - Sadat Academy for Management Sciences — 6th of October.
   - Higher Institutes of 6th of October City.
   - Other: specify.

   The form explicitly excludes Ahram Canadian University (ACU). Institution names are source-form labels, not independent verification of campus geography or official institutional nomenclature.
6. Listening context: Uber; private car; microbus; public transport.

A.3 Section A: Judging a Radio Broadcaster

Think of a broadcaster/program you listened to during the ride. Rate 0–4: 0 = strongly disagree; 1 = disagree; 2 = neutral; 3 = agree; 4 = strongly agree.

BR1

The broadcaster verifies claims before going on air

BR2

The broadcaster corrects mistakes quickly and openly

BR3

The broadcaster stays fair when opinions clash

BR4

The broadcaster exaggerates to gain attention

SV1

Students whose opinion I respect follow and like this broadcaster

SV2

Interacting with the audience shows the broadcaster cares

SV3

A high social rating makes me trust the broadcaster more

SV4

The broadcaster's popularity is a sign of its credibility

IQ1 The content I heard is well-informed and substantial

IQ2 The broadcaster adds value beyond what I already know

IQ3 The program content is consistently high quality

A.4 Section B: The Broadcast Experience

After listening, rate 1–7, from strongly disagree to strongly agree.

PB1

I feel more confident about what I heard after listening

PB2

The content exposed me to diverse perspectives I hadn't considered

PB3

I trust this broadcaster more than others on similar topics

PB4

The broadcast structure helped me understand the topic clearly

PB5

Overall, the broadcast was of high quality and worth my time

A.5 Section C: Overall Rating

"How would you rate this broadcast?" Select 1–5: 1 = poor; 5 = excellent.

A.6 Scoring and Analysis Boundaries

Reverse BR4 as $4 - \text{BR4}$. BR, SV, and IQ are means of four, four, and three items respectively; the broadcaster-battery total used for alpha contains all 11 scored items. PB is the arithmetic mean of PB1–PB5 on its original 1–7 scale. The star response remains a separate 1–5 variable. No missing item cells occur in the supplied 60 questionnaires, so the recheck does not impute values or exercise a missing-item prorating rule.

The 0–4 and 1–7 anchors must not be interchanged. PB3 measures comparative trust, not intention to verify; PB4 measures structural clarity, not the influence of a displayed BRS. SV3 directly concerns a high social rating, but even this item records perceived influence rather than an experimentally identified badge effect. The HTML/Markdown form, scored CSV, and qualitative notes are different artifacts; ride mood and recorded quotations do not become additional questionnaire scale items simply because they can be linked.

Appendix B

Instrument Calibration Report: Post-Broadcast Battery

This appendix reports the item-level calibration of the post-broadcast epistemic experience battery (PB1 to PB5) on the refined field sample ($N = 52$). It supplies the inter-item statistics, reliabilities, and collapse decisions referenced under H3.

B.1 Protocol

The calibration protocol specifies that any pair of items producing $r > 0.90$ with each other is a collapse candidate (rule R1): the pair is reviewed for conceptual overlap before any scoring change. A high correlation alone does not justify merging distinct constructs. Corrected item-total correlations are reported against the retention floor of $r \geq 0.30$.

Table B.1: Post-Broadcast Battery Item Calibration ($N = 52$)

Item	**Mean**	**SD**	**Max r with other items**	**Decision**
PB1 (Epistemic confidence)	5.06	1.07	0.918 (with PB4)	Collapse candidate (rule R1)
PB2 (Source diversity)	5.21	0.87	0.879 (with PB4)	Retained
PB3 (Comparative trust)	4.77	1.02	0.827 (with PB2)	Retained
PB4 (structural clarity)	5.10	0.87	0.918 (with PB1)	Collapse candidate (rule R1)
PB5 (Overall quality)	5.46	0.92	0.833 (with PB2)	Retained
Cronbach's α over PB1–PB5:				0.947; target ≥ 0.70

B.2 Item-Level Diagnostics

B.3 Collapse Decision

One collapse candidate pair is measured: PB1 (epistemic confidence) with PB4 (structural clarity), $r = 0.918$, exceeding the 0.90 rule of the protocol. For the current wave all five items remain in the reported composite. A future version should examine whether confidence and perceived structural clarity are substantively redundant before merging them. This is a proposed revision, not a scoring change applied retrospectively to the supplied analysis. The item means and standard deviations are now also reproduced from the supplied 52 response records. The historical pairwise Pearson correlations also match the response matrix; the item labels have been corrected against F1.0. No item has been merged.

Appendix C

Historical Simulation and Bayesian Analysis Summary

This appendix preserves the original study's model-analysis summary. G0 and G1 identify engagement-first and epistemic-first configurations. The supplied R files do not contain a registration identifier or a dated blinding log. Accordingly, the summary is not presented as evidence of completed preregistration or verified blinding; those procedural records are separate from the numerical results.

C.1 Primary Endpoints

The primary endpoints of the arm contrast are: (1) the PB composite difference (G1 minus G0); (2) the BRS aggregate difference; (3) the five-point star-rating difference; and (4) the raw engagement difference. The specified expectations are: positive PB, BRS, and star differences in the direction EST-P1 predicts; an engagement decrease of the size EST-P2 predicts; and a reported trade-off index, interpreted in its component metric units.

C.2 Results

Counterfactual effects (G1 minus G0, $N = 52$ substrate): PB composite +0.511 $[0.507, 0.514]$; information quality +0.131; broadcaster rigor +0.051; social validation +0.003; star rating +0.165 $[0.104, 0.204]$; echo/density +0.035; correction acceptance +0.026; raw engagement −0.217; continued listening −0.054. The historical contrast table records $p = .0001$ for each difference; these entries concern the model analysis, not a randomised field experiment. The reported engagement-versus-epistemic trade-off index is −0.290.

Calibration against the refined sample: weighted Wasserstein loss 0.0406 (G0) and 0.0178 (G1) against a target of 0.04. Sensitivity: all 48 design points of the Latin-hypercube sweep support the five model mechanism checks.

Bayesian update (PyMC, $N = 52$): information quality $\beta = 0.786$, 95% equal-tailed credible interval $[0.670, 0.910]$; maximum $\widehat{R} = 1.008$; minimum effective sample size 606; posterior predictive coverage 0.96; posterior predictive mean 5.115 against the observed mean 5.119. Predictor Pearson collinearity (BR–IQ $r = 0.861$) limits interpretation of a uniquely identifiable quality mechanism. Coefficients refer to standardized predictors; predictive coverage is in-sample. The historical effect bounds are central replication-percentile intervals, not mean-effect confidence intervals (Section 6.11).

Appendix D

Measurement Invariance: Follow-Up Procedure

Cross-group latent-mean comparisons require evidence that the measurement model functions comparably across groups. This appendix specifies the follow-up procedure for the human instrument. It does not report a human invariance analysis. The v8 synthetic audit in Section 6.12.7 does not estimate multigroup invariance for the current state-mediated item matrices; the earlier lavaan audit is not reused because it belongs to a different synthetic matrix. The poor pooled fit of the human BRS must still be addressed before a human multigroup model can be assumed.

1. **Configural model.** Specify the same substantively justified factor structure in each group and evaluate model fit and identification.
2. **Metric model.** Constrain corresponding factor loadings and assess the change in fit alongside parameter estimates and uncertainty.
3. **Scalar model.** For continuous indicators, constrain corresponding intercepts; for ordinal items, use suitable threshold constraints and model identification rules. Evaluate whether latent means are interpretable.

Putnick and Bornstein (2016) discuss the reporting and interpretation of these steps. Fit-change criteria must be chosen for the estimator, model, sample,

and group imbalance; this thesis does not relax a CFI threshold simply because the sample is small. Any partial-invariance solution must identify the freed parameters and give a substantive justification, rather than merely optimise fit. Until such evidence exists, the interaction-group figure describes observed scores only, not invariant latent traits.

Appendix E

Formal Assumptions and Sample-Size Planning

E.1 Anchored Reputation-Update Contraction

For the update in Equation 3.3, assume a fixed nonnegative row-stochastic matrix A, a fixed input vector X, and $0 < \lambda < 1$. Let $F(B) = (1-\lambda)AB + \lambda X$. Then

$$\|F(B) - F(C)\|_\infty \leq (1-\lambda)\|A\|_\infty \|B - C\|_\infty = (1-\lambda)\|B - C\|_\infty. \tag{E.1}$$

Since $1 - \lambda < 1$, this is a contraction on the finite-dimensional real vector space. Its unique fixed point is

$$B^* = \lambda[I - (1-\lambda)A]^{-1}X. \tag{E.2}$$

The inverse exists because the spectral radius of $(1-\lambda)A$ is less than one. This result does not apply without modification to arbitrary signed weights or time-varying inputs. It also does not imply that the consensus is factually correct: accuracy depends on the inputs and the measurement process.

E.2 Scope of the Free-Energy Analogy

Consider the explicitly separable objective

$$\mathcal{F}(W,p) = \mathcal{L}(W) - TH(p), \qquad H(p) = -\sum_{j=1}^{m} p_j \log p_j, \quad T > 0, \tag{E.3}$$

with p a probability vector. If there are no additional constraints and $\mathcal{L}$ is independent of p, minimizing the entropy term gives the uniform distribution $p_j = 1/m$. Indeed, interior stationarity of $T\sum_j p_j \log p_j + \mu(\sum_j p_j - 1)$ gives $T(\log p_j + 1) + \mu = 0$, so all components are equal; convexity supplies the minimum. Stationarity of $\mathcal{L}$ alone does not establish a global minimum in a nonconvex model.

These observations do not prove that the platform's actual dynamics minimize this objective, that its fixed points coincide with these stationary points, or that a signed interaction matrix has a universal stable gain $1/\lambda_{\max}$. Such claims require a specified dynamical system and a separate stability argument. The Hamiltonian in EST is therefore retained as a modelling proposal, not as a proof of social thermodynamic laws.

E.3 Planning Calculation and Actual Recruitment

The supplied summary lists an assumed standardised effect of .40, two-sided $\alpha = .05$, target power .80, a nominal requirement of 52, and a recruitment target of 60. The original derivation uses a paired or one-sample normal approximation:

$$n \approx \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{d}\right)^2 = \left(\frac{1.960 + .842}{.40}\right)^2 \approx 49.1. \tag{E.4}$$

Rounding this approximation upward gives 50, not 52. An exact test calculation requires a specified test and the definition of the standardised effect; the export's nominal 52 is preserved as a planning entry rather than derived from this approximation.

To retain 52 participants after an anticipated 15% loss would require

$$N_{\text{target}} = \left\lceil \frac{52}{.85} \right\rceil = 62. \tag{E.5}$$

The actual study collected 60 records and retained 52 after eight historical exclusions. The documented response-and-context filter now reproduces those exclusions; these counts are not changed by the planning correction. Neither a paired-test planning formula nor the number of simulated agents establishes power for an independent-groups field experiment, a multivariate factor model, or an unmeasured learning endpoint.

Appendix F

Qualitative Codebook, Case Audit, and Integration Record

The coding unit is one supplied researcher-recorded encounter. Codes can co-occur. Observations and recorded remarks supply evidence; researcher reflections supply contextual interpretation only. The full machine-readable codebook, case memos, source hashes, and joined audit are preserved in the thematic codebook, the case-level coding record, the coding-provenance record, and the mixed-methods verification record.

T1: Grounds for credibility

Include source use, correction, neutrality, explanation, viewpoint diversity, peer review, or ratings as explicit judgment cues. Exclude generic praise and unqualified technical clarity.

T2: Peer connection

Include natural student speech, familiarity, friendship, authenticity, or solidarity. Exclude an interface preference without a peer connection.

T3: Situated listening value

Include calm, fatigue, quiet appreciation, limited time, concision, structure, or

immersion. A recorded duration alone is insufficient.

T4: Participatory agency

Include voting, topic control, creator aspirations, co-creation, privacy conditions, or contribution to platform improvement. Distinguish recorded actions from future intentions.

T5: Student-lived relevance

Include explicit campus, work, study, career, local-mobility, or civic interests. Do not infer preferences from faculty membership alone.

T6: Enabling delivery and interface

Include experienced continuity, audio clarity, latency, visual integration, controls, or usability. Do not code such praise as independently measured accuracy or engineering quality.

The additional qualitative records are SR0029, SR0043, SR0044, SR0046, SR0051, SR0054, SR0057, and SR0058. "No" in the following table means not in the historical quantitative subset, not invalid qualitative evidence. CSV identifiers are reproduced exactly. The profile document's shorter zero-padding must not be silently substituted during joins.

Table F.1: Case-Level Theme Membership and Quantitative Pairing

Case	Themes	Paired	Case	Themes	Paired
SR001	T2, T5	Yes	SR0031	T3	Yes
SR002	T1	Yes	SR0032	T3	Yes
SR003	T3	Yes	SR0033	T1	Yes
SR004	T4, T6	Yes	SR0034	T4	Yes
SR005	T5	Yes	SR0035	T4	Yes
SR006	T6	Yes	SR0036	T2, T3	Yes
SR007	T3, T5	Yes	SR0037	T3, T6	Yes
SR008	T2, T5	Yes	SR0038	T5	Yes
SR009	T1	Yes	SR0039	T5	Yes
SR0010	T6	Yes	SR0040	T2	Yes
SR0011	T2, T3	Yes	SR0041	T3	Yes
SR0012	T1	Yes	SR0042	T1	Yes
SR0013	T4	Yes	SR0043	T4	No
SR0014	T3	Yes	SR0044	T5	No
SR0015	T3	Yes	SR0045	T3	Yes
SR0016	T1	Yes	SR0046	T3	No
SR0017	T1	Yes	SR0047	T6	Yes
SR0018	T3	Yes	SR0048	T4	Yes
SR0019	T4, T6	Yes	SR0049	T3, T5	Yes
SR0020	T2, T3	Yes	SR0050	T6	Yes
SR0021	T6	Yes	SR0051	T4	No
SR0022	T3, T5	Yes	SR0052	T4, T6	Yes
SR0023	T1	Yes	SR0053	T4, T6	Yes
SR0024	T4, T5	Yes	SR0054	T3	No
SR0025	T1	Yes	SR0055	T2, T3	Yes
SR0026	T2	Yes	SR0056	T5	Yes
SR0027	T6	Yes	SR0057	T5	No
SR0028	T4, T5	Yes	SR0058	T6	No
SR0029	T5	No	SR0059	T4, T6	Yes
SR0030	T4, T6	Yes	SR0060	T1	Yes

Note. Read the left group from SR001 to SR0030, then the right group from SR0031 to SR0060. "Paired" indicates inclusion in the historical 52-case quantitative subset. Codes may co-occur; all 60 records are included here.

F.1 Audit and Interpretive Scope

The coding was performed by the researcher and reviewed during thesis revision. No independent-coder statistics, member checking, or saturation assessment were performed. The coding matrix, source excerpts, and case memos preserve the basis for examining alternative readings. The record does not establish that every recorded remark is a verbatim transcript. Interpretive disagreement is distinct from an error in recounting code frequencies. A numerical recount can reproduce the coding matrix; it cannot independently validate the interpretive choices.

Appendix G

Prototype Focus-Group Report and Evidence Audit

This appendix incorporates the merged FGD report. Its narrative-level findings inform Sections 6.7 and 5.9; the underlying participant records remain unverified; a 160-segment machine transcript is included, but speaker attribution and human verification are pending.

G.1 Purpose and Evidentiary Scope

This report integrates two narrative accounts of a focus-group discussion (FGD) involving university students and a walkthrough of LOKLOK, a proposed social media radio application. Its purpose is to clarify the audience needs and design questions suggested by those accounts: the relationship between listening and creating, the appeal and constraints of collaborative broadcasting, expectations around institutional access, and the usability of a wireframe interface. It does not establish the platform's originality, market success, or effects on student behaviour.

The material comprises the earlier narrative (Account A) and the expanded narrative supplied as `FGD2.tex` (Account B), together with their PDFs. Account A is retained in the source archive, while this document is the merged report. The two

narratives repeat the same four domains, lexical counts, and frame references. They are therefore treated as overlapping accounts of one reported session, not two focus groups, independent corroboration, or separate participant samples. A 160-segment machine transcript of the session audio is included. The audiovisual recordings, speaker attribution, and participant roster remain unavailable; the transcript is used for semantic embedding, not for audited quotation. A subsequently supplied file, `fgd_analysis_parts.json`, contains seven model-generated summaries. It is designated *Supplement C*, not another focus group or an independent corroborating source. The proposed revised LaTeX accompanying that export is treated as a set of candidate claims to review, not as evidence that the primary materials have been recovered. Section G.4 records the adjudication. Throughout, *reported account* means a statement present in either supplied narrative; *interpretation* means an analytical reading of that account; and *verification requirement* identifies the evidence needed to assess it. None of these labels turns a paraphrase into a participant quotation.

Focus groups derive their distinctive value from interaction among participants, including exchanges, disagreements, and responses to one another, rather than simply collecting several individual opinions at once (Kitzinger, 1995). In this case, the distinction between a moderator-led product demonstration and participant-led discussion is especially important. Both accounts report extensive moderator explanation, but the balance of contributions cannot be established without speaker-labelled records. Statements explaining how LOKLOK is intended to work are therefore not treated as evidence that participants endorsed those features.

G.2 Study Context and Reporting Completeness

Both accounts describe a university conference-room session conducted in colloquial Egyptian Arabic, an English translation, and smartphone interaction with application wireframes. Account B explicitly situates the proposed campus restriction in the Ahram Canadian University (ACU) context. This clarifies the reported institutional setting without establishing each participant's affiliation, the exact venue, recruitment procedure, or ethics review.

Account A gives an approximate duration of 28 minutes; Account B reports 1,675.33 seconds (27 minutes 55.33 seconds). These descriptions are compatible as rounded and more precise reported durations. The latter is not newly measured metadata, however, and it does not resolve the visual reference at 30 minutes 30 seconds. Section G.9 reconciles the arithmetic and records the remaining source discrepancy. Additional recording details from Account B are retained in Section G.8.

Table G.1 identifies the principal information needed for a source-based report. COREQ offers reporting prompts for interviews and focus groups, while APA's qualitative reporting standards emphasise the transparent presentation of research design, researcher involvement, analysis, and evidence (Levitt et al., 2018; Tong et al., 2007). These sources guide the audit; citing them does not establish checklist compliance, ethical approval, or study quality.

Table G.1: Reporting Status and Required Source Documentation

Domain	**Available account**	**Documentation required**
Session and setting	University conference room; ACU context in Account B; two compatible reported durations.	Date, exact venue, session identity, source-file duration, and common timestamp origin.
Participants and recruitment	University students; no auditable roster or sample size supplied.	Number invited and attending, eligibility, recruitment route, relevant sample characteristics, and non-participation where recorded.
Researcher relationship	Moderator explains the prototype; role and prior relationships are unspecified.	Moderator identity and role, training, institutional relationships, and involvement in product design.
Procedure and prototype	Wireframe walkthrough and smartphone interaction described.	Topic guide, task instructions, prototype version, features actually available, and sequence of demonstration and discussion.
Recording and language	Arabic speech, translation, visual frames, and technical audio figures reported.	Original files, extraction settings, speaker-labelled transcript, translation decisions, and source-linked observation log.
Analysis	Four domains, lexical counts, and interpretive statements supplied.	Coding records, analysts' roles, dated decisions, source extracts, alternative readings, and count specifications.
Ethics and data stewardship	Approval, consent, confidentiality, and reuse arrangements not documented in the supplied files.	Applicable review decision, participant information and permissions, storage/access arrangements, and disclosure or reuse restrictions.

Unknown characteristics must remain unknown: neither a particular age cohort nor demographic diversity can be inferred from the label "university students." No survey sample size elsewhere in the thesis is assigned to this FGD. Nor can sampling adequacy or saturation be asserted from the length of the account. The information-power approach relates qualitative sample adequacy to the study aim, sample specificity, theoretical framing, dialogue, and analysis strategy rather than to a universal numerical threshold (Malterud et al., 2016). Developed for individual

interview studies, that approach does not prescribe the number or composition of focus groups. Those considerations cannot yet be assessed here.

G.3 Analytical Approach and Source Verification

G.3.1 A Provisional, Framework-Informed Reading

The revision retains the four shared domains as provisional organising categories and integrates Account B's proposed links to the thesis framework. The domains were not newly generated through inductive coding of a transcript. Thematic analysis involves interpreting patterned meaning, and its theoretical and procedural choices should be explicit (Braun & Clarke, 2006). Here, the immediate aim is narrower: to distinguish reported experiences, moderator explanations, analytical interpretations, and proposed design responses without overstating their evidentiary status.

If primary records become available, a framework-informed analysis would be a suitable option to consider. A matrix can link cases or interaction episodes to categories while retaining contextual summaries and source references (Gale et al., 2013). The present codebook is a starting point for that work, not proof that it has been performed. New categories, contradictory cases, and revisions to the initial four domains must remain possible. The researcher should identify the chosen analytical tradition rather than combine incompatible quality criteria: intercoder agreement is not a universal requirement of reflexive thematic analysis (Braun & Clarke, 2021). No agreement statistic or independent coding is claimed here.

G.3.2 Language, Translation, and Quotations

Translation is an interpretive part of qualitative research, not a neutral replacement of words. Decisions about idiom and expression can alter meaning (van Nes et al., 2010). For this study, a proposed verification record should retain the Egyptian Arabic extract alongside the English translation, with a speaker code, source timestamp, and a note explaining consequential translation choices. A bilingual review should be documented if undertaken, not assumed. Ambiguity, disagreement, or hesitation should not be removed merely to produce more fluent academic prose.

Because no primary transcript or recording was supplied, this report provides no purported verbatim participant quotations. Supplement C contains text labelled as original Arabic and translations, but those labels do not establish transcription fidelity or speaker identity. The underlying strings are retained in the controlled review record, not promoted into participant testimony. Expressions used as interface names or analytical labels, such as *Boost* and *Podcast Paradox*, are not presented as quotations from identified participants. Before an extract is published as evidence, its wording, translation, attribution, and permitted use must be checked against the original record. An unintelligible segment should remain marked as uncertain rather than reconstructed from expectations.

G.3.3 Proposed Verification Sequence

The following sequence is a prospective procedure for this material, not a retrospective account of completed analysis:

1. Establish a file inventory and preserve the original recording, transcript, pro-

totype version, and permissions records. Assign stable file identifiers and document any edited or translated derivatives.

2. Reconcile the recording duration and timestamp origin. Link each extract and observation to the same source version; document offsets where necessary.

3. Identify participant and moderator turns using pseudonymous codes. Retain the surrounding exchange so that prompted agreement is distinguishable from an independently introduced concern.

4. Review the Arabic and English records together. Record uncertain speech, translation choices, overlapping speech, and observable actions separately.

5. Apply and revise the provisional codebook. Maintain source-linked analytical memos, including divergent experiences and explanations that do not support the anticipated design benefit.

6. Reassess every thematic claim and any descriptive count against those records. Report only the procedures actually completed and retain an audit trail of changes.

G.4 Review of the Supplemental Analysis Export

Supplement C is a JSON array of seven text strings, each containing generated questions, proposed answers, themes, or quality cautions. It contains neither a structured primary transcript with attributable turns nor a collection of 20 visual frames. No recording, frame images, source-to-output alignment, prompt, execution

log, or independently reviewed translation accompanies it. The researcher identifies the export as output from an instruction-tuned 7B model; the file itself does not establish the exact model version, quantisation, input coverage, or generation settings.

The admissible contribution is an additional set of *review leads* and a record of limitations in the generated analysis. It does not change the primary-evidence status reported in Section G.1. References to unverified speakers occur in the output, but no inspected transcript supports the stronger claim that every original segment carries that label. Nor may an unattributed sentence be reassigned to the moderator or the group merely because that role seems plausible.

G.4.1 Internal Consistency and Translation Checks

Part numbers below are one-based positions in the supplied JSON array, not participant identifiers or independently timed recording segments. In Part 2, an Arabic expression referring to a room is paired with an English rendering that locates the microphone in a car. Part 1 renders a colloquial vehicle-listening expression as listening in the Arabic language. These are translation inconsistencies within the export; identifying them does not verify that either extract was actually spoken. Named-person renderings also require checking against the recording rather than being accepted from the English gloss alone. Translation choices should preserve source meaning and uncertainty rather than create a more fluent but altered account (van Nes et al., 2010).

Some purported extracts contain Chinese or Cyrillic material mixed with Arabic, while several timestamps are incomplete, malformed, or placeholders. Part 5

includes a placeholder rather than an actual time interval. Parts 1 and 4 end with apparently unfinished text. These observations warrant review of the original inputs and generation logs; they do not establish why the artefacts occurred. No missing words, speakers, or time boundaries are reconstructed to make the export appear complete.

Table G.2: Adjudication of the Supplemental Export and Proposed Revision

Candidate addition	**Decision**	**Reason and permitted use**
A supplemental seven-part analysis export is available	Accepted as an artefact	Its existence and text can be inspected; it is not primary participant data or another sample.
Attention to speaker uncertainty, translation, and ambiguous vocalisations	Retained as a safeguard	These are appropriate review requirements, not completed source verification or validated non-speech labels.
Variation in willingness to hear long podcasts; registration-error questions	Retained as review questions	Parts 2 and 3 suggest topics to investigate; no verified preference, task failure, or new theme frequency is inferred.
Primary transcript and 20-frame visual log recovered and triangulated	Not established	Those materials are not supplied. Generated descriptions cannot certify their recovery, completeness, or alignment.
Verbatim quotations, precise intervals, and moderator/participant roles	Withheld as evidence	The original wording, speaker, and interval remain unchecked; some text and timestamps are internally inconsistent.
Consensus, retention gains, lower anxiety, safety, or intuitive usability	Not established	The export supplies neither a verified interaction record nor outcome measures supporting these claims.
A validated Podcast Paradox, automatic resolution through co-hosting, or EST-P3 validation through ID/PIN	Not accepted as findings	An analytic label is not a validated construct; a design description is not a causal test or a psychometric assessment.

G.4.2 What Changes in the Interpretation

The four provisional domains remain in place. Supplement C suggests checking whether preferred programme length varies and whether registration instructions or error feedback are understood. These are follow-up questions within FGD-T1 and FGD-T4, not newly established participant experiences. Its account of unrestricted access also conflicts with the campus-only descriptions in Accounts A and B; this is retained as an unresolved discrepancy, not evidence that the participants debated or agreed on an open-access policy.

Repeated vocalisation-like strings in Parts 5–7 are treated as *unresolved transcript artefacts*. They are not coded as silence, non-participation, hesitation, a technical failure, or active wireframe use without checking the recording and aligned visual evidence. Sparse or uninterpretable output cannot establish that no substantive speech occurred. Likewise, generated statements that everyone agreed do not establish consensus when the original interaction and speakers are unavailable.

The proposed bilingual quotations are consequently not inserted as empirical extracts. A review record retains the candidate wording, its JSON part locator, the disputed rendering, and the decision needed before use. No synthetic quote becomes a participant record, calibration weight, factual answer key, or prompt instruction to reproduce a favourable outcome. The export may inform quality checks and later source review, but does not independently confirm the existing themes or validate either proposed simulation engine.

G.5 Provisional Thematic Interpretation

The four domains below synthesise the *two supplied narratives*, not newly verified transcripts. Supplement C has been reviewed for candidate additions but is not treated as independent corroboration. Each separates the reported account from its possible meaning and the limits of that interpretation. External methodological citations explain how the material is handled; they do not authenticate the local observations.

G.5.1 FGD-T1: Contextual Listening and Selective Publication

Reported account. Both accounts describe listening during travel and at home, interest in radio storytelling, and a reference to older radio serials. They also recount an individual who owns a microphone and records privately but does not publish those recordings. Account B explicitly attributes this practice to self-training and preparation and says that recordings are discarded. This reported explanation is retained, but speaker identity, verbatim wording, and interactional context remain unverified in the machine transcript.

Interpretation. The account suggests a useful distinction between listening, practising, and publishing. Possessing recording equipment or enjoying audio does not necessarily imply a wish to broadcast publicly. The draft calls this tension the *Podcast Paradox*; here, that expression is retained only as a provisional analytical label, not a validated construct or a demonstrated characteristic of the wider student population.

Alternative readings and implications. Unpublished recordings could

reflect deliberate private practice, privacy preferences, limited time, self-imposed production standards, or a lack of interest in an audience. The reported self-training explanation deserves priority over an imposed anxiety interpretation, although its meaning and context still require source checking. Other explanations should be explored, not attributed to that individual without evidence. A defensible design question is therefore whether students want different routes from listening to private practice, collaborative creation, and public sharing. Non-public participation should not automatically be framed as failure or a deficit.

G.5.2 FGD-T2: Collaborative Broadcasting and Choice

Reported account. The moderator reportedly described a broadcasting mechanic requiring a peer invitation or co-host. The narrative characterises the reaction as favourable, but provides no attributable extracts, contrary responses, or evidence of subsequent use. Account B also develops an analogy with traditional radio call-ins: inviting a peer to co-host is presented as a way to make the broadcast a shared conversation. This is the account's design rationale, not evidence that a caller's role and a co-host's role are equivalent. The rule is treated as part of the walkthrough description, not as a verified feature of a deployed application.

Interpretation. Co-hosting is a candidate means of making an audio exchange feel conversational rather than like an individual performance. This is a plausible reading of the accounts, not a measured reduction in anxiety, effort, or cognitive demand. Source-based analysis should ask who introduced this idea and how others responded, preserving the interaction rather than counting affirmative words alone (Kitzinger, 1995).

Alternative readings and implications. Mandatory co-hosting might also restrict participation when a peer is unavailable, when a student prefers independent expression, or when social exposure is unwelcome. These are design questions proposed in this review, not difficulties demonstrated in the session. Optional co-hosting, private rehearsal, and alternative contribution routes should be compared with the mandatory model before claiming that one arrangement lowers barriers for everyone.

G.5.3 FGD-T3: Institutional Eligibility, Trust, and Privacy

Reported account. Both accounts describe a university-ID and PIN entry process, explained through an ATM analogy, and a platform initially limited to university students, with Account B naming the ACU context. The narratives attribute benefits involving trust and community to this arrangement, but do not supply participant-level evidence of those judgements or an assessment of the authentication system.

Interpretation. Institutional eligibility, perceived trust, confidentiality, and technical security are separate questions. The described entry rule alone establishes none of the latter three. In particular, restricting membership does not demonstrate the absence of harassment, prove the validity of research responses, or show that students feel comfortable speaking under an institutional identity.

Alternative readings and implications. A bounded community could be welcoming for some students and expose others to unwanted visibility. The next evaluation should ask what identity information is displayed, who can access recordings, how complaints are handled, and whether eligible students can obtain or recover access. Preferences for named and pseudonymous participation should

be explored rather than presumed. These are proposed evaluation topics; neither preference nor a security outcome was measured in the material available here.

G.5.4 FGD-T4: Interface Cues, Invitations, and Idle Content

Reported account. The narrative describes profiles, avatars, names, timestamps, a text-based *Boost* invitation, and music or advertising when no broadcast is active. It also reports smartphone interaction during the walkthrough. The available documents do not establish which functions were interactive, which were static representations, or which were explained only by the moderator.

Interpretation. These descriptions identify candidate interface functions: communicating who is present, inviting an audience, and explaining what happens when no conversation is available. They do not demonstrate successful navigation, low cognitive load, sustained engagement, or an intuitive interface. Such conclusions would require task-linked evidence and the participants' own accounts, not merely a statement that screens were tapped or swiped.

Alternative readings and implications. Future evaluation should consider whether invitations are understandable, whether idle-state content is clearly identified, and whether users can control unwanted playback. Relevant accessibility checks can draw on WCAG 2.2, including applicable text alternatives, contrast, navigation, and media-control criteria (World Wide Web Consortium, 2024). This is a proposal for evaluation, not a finding of accessibility or a claim of conformance by the prototype. Applicable criteria must be selected for the actual interface and media format.

G.6 Working Codebook and Relationships

Table G.3 provides operational boundaries for a future review of the primary records. Codes should be assigned to source-linked segments, not to entire people. A segment may have more than one relevant code; a moderator's feature description must remain distinguishable from a participant's experience, question, or evaluation. These are proposed coding conventions for this report, not a completed coding result.

Table G.3: Provisional Codebook for Source-Based Review

Domain	**Include when supported by the source**	**Boundary and evidence requirement**
FGD-T1: Listening and publication	Listening contexts; private practice; expressed reasons for publishing or not publishing.	Exclude assumptions that listening implies a wish to broadcast. Retain the speaker's explanation and surrounding exchange.
FGD-T2: Collaboration and choice	Co-host descriptions; preferences for shared or independent creation; expressed participation constraints.	Separate moderator proposals from participant responses. Do not code agreement as reduced anxiety unless that meaning is expressed.
FGD-T3: Eligibility and trust	Access rules; stated expectations or concerns about identity, privacy, trust, and inclusion.	Do not infer safety from membership restrictions. Record whether a claim concerns a preference, a concern, or an experienced event.
FGD-T4: Interface and idle content	Task-linked navigation, interpretations of interface cues, invitations, and responses to inactive-room content.	Separate displayed, demonstrated, and proposed functions. Link actions to the task, timestamp, and accompanying explanation.

The proposed connections in Table G.4 organise questions for further investigation. They are not causal effects estimated from this FGD. Neither the order of the walkthrough nor co-occurrence in a narrative establishes that one theme produces another.

Table G.4: Candidate Relationships and Competing Interpretations

Connection	**Question suggested by the accounts**	**Competing interpretation to examine**
FGD-T1 and FGD-T2	Does co-hosting offer a desirable route from private practice to sharing?	Some students may prefer privacy or independent production regardless of collaboration.
FGD-T2 and FGD-T3	Does restricted membership affect willingness to invite peers or speak publicly?	Recognisable institutional identities may increase unwanted visibility rather than comfort.
FGD-T2 and FGD-T4	Do invitations and presence cues help students coordinate a shared broadcast?	A feature may be visible but misunderstood, inaccessible, or unwanted.
FGD-T1 and FGD-T4	Do interface and idle-state choices support different listening contexts?	Automatic music, advertising, or prompts to contribute may conflict with a listener's preferences.

G.6.1 Relationship to Epistemic Systems Theory

Account B links the domains to Epistemic Systems Theory (EST), the framework proposed by the thesis author. This theoretical reading is retained as a *sensitising interpretation*: it identifies questions to investigate, not propositions confirmed by the FGD. It follows the narrative synthesis rather than replacing source-based analysis. A framework matrix can make such interpretive connections traceable while keeping the original evidence and alternative readings visible (Gale et al., 2013).

In particular, Account B's alignment of institutional exclusivity with EST-P3 needs qualification. In the thesis, EST-P3 concerns the psychometric accountability of consequential metrics. This FGD account supplies neither a measurement model nor reliability or validity evidence for such a metric. Institutional access may be discussed as governance context, but cannot stand in for psychometric validation. Likewise, no network-diversity or misinformation-diffusion observations are supplied

Table G.5: Bounded Connections Between the FGD Domains and EST

Domain and link	**Interpretive contribution**	**Boundary of inference**
FGD-T1; EST-P4	Distinguishes opportunities to practise, participate, and publish.	No observed source appraisal, correction, or improvement in evidence-based production.
FGD-T2; EST-P1	Identifies co-hosting rules as a proposed influence on participation.	No comparison establishes that architecture primarily determines behaviour or quality.
FGD-T3; governance context	Identifies identity, access, and privacy questions relevant to participation.	An ID/PIN rule is not a psychometric instrument and does not itself test EST-P3.
FGD-T4; EST-P1 and EST-P4	Identifies interface cues and invitations as possible supports for interaction.	Screen activity does not demonstrate epistemic improvement or establish an effect of those cues.

to assess EST-P2.

The integrated interpretation is therefore about *participation choices within a proposed design*, not demonstrated epistemic gains. Private practice may be an intended activity rather than failed publication; collaboration may be welcome without being compulsory; and institutional membership may be valued without implying comfort with identifiable speech. These alternatives connect the four domains without forcing all accounts into a single product success narrative. Any later linkage to survey findings must distinguish qualitative meanings from numerical associations and establish the provenance of each dataset before claiming convergence.

G.7 Lexical Counts: Audit and Proposed Reanalysis

Both accounts assign the same frequencies to six grouped English terms, but no transcript, search dictionary, tokenisation rule, or speaker-specific denominator accompanies them. Table G.6 preserves those figures for traceability only. Repetition

across the accounts is not replication, and the figures are not added together. **They are not independently reproduced results and are excluded from the substantive conclusions of this report.** The six categories must not be summed into a corpus size: overlap, eligibility, and unit of counting are unspecified.

Table G.6: Unverified Frequencies Repeated in Both Accounts

Draft category	**Reported count**
Radio / Audio	24
University / Students	18
Alone / Solo	9
Together / Invite	12
Wireframe / Interface	15
Record / Microphone	7

Note. These numbers describe what both accounts report, not what an independently inspected transcript contains. They do not establish the relative importance of themes or participant endorsement.

In summative qualitative content analysis, counts are followed by contextual interpretation; they do not replace it (Hsieh & Shannon, 2005). For this FGD, a term could appear in a moderator explanation, a participant's objection, or an unrelated example. Its repetition alone cannot establish interest, trust, or a psychological barrier. Counts from an English translation must also be identified as counts of that translation, not as frequencies in the original Arabic speech.

If lexical analysis is retained, the researcher should first specify the corpus version, eligible speaker turns, treatment of fillers, word boundaries, case and inflection handling, category dictionary, and overlapping matches. One proposed descriptive measure is the frequency per 1,000 eligible words within each speaker role:

$$f_{kr} = 1000\,\frac{n_{kr}}{N_r}, \qquad N_r > 0, \tag{G.1}$$

where k denotes a predefined term category, r denotes the speaker role (moderator or participants), n_{kr} is its number of eligible word-token matches, and N_r is the total number of eligible word tokens for that role. Each word token should count at most once within a given category; any cross-category overlap should be disclosed.

In plain language: count how often the specified words appear, divide by the total words spoken by that role, and scale the result to 1,000 words. Keeping moderator and participant speech separate prevents a long product explanation from being mistaken for audience interest. This measure remains descriptive; it does not show whether the words express agreement or concern.

A second optional measure is participant mention coverage:

$$c_k = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}(n_{ki} > 0), \qquad n > 0, \tag{G.2}$$

where n is the documented number of eligible participants with analysable records, n_{ki} is the number of matches for category k in participant i's speech, and $\mathbf{1}(n_{ki} > 0)$ equals one if that participant mentions the category at least once and zero otherwise. Exclusions from the denominator must be reported.

In plain language: this is the proportion of eligible participants who mention a category at least once. It prevents one frequent speaker from being counted as several people, but it does not measure endorsement or importance. A zero can reflect limited opportunity to speak, and cannot be read as indifference. Neither measure is calculated here because a speaker-attributed transcript and the participant denominator are unavailable. These equations are proposed reporting definitions for a future reanalysis, not estimates of population opinion or tests of causality.

G.8 Reported Recording Characteristics

Account B supplies technical figures that were absent from Account A. They are retained in Table G.7 so that the eventual source-file check has explicit targets. Their precision does not establish their accuracy: neither the audio nor the technical notes referred to in Account B were provided. These figures describe a purported recording or its processing, not participant demographics or qualitative findings.

Table G.7: Recording Characteristics Reported in Account B

Characteristic	**Reported value**	**Required check or interpretation**
Duration	1,675.33 s	Check the source file and any trimming; reconcile the time origin with the visual log.
Sample rate	16,000 Hz	Check file metadata and whether resampling occurred. This is not a participant count.
Audio channels	One (mono)	Check whether this is the original or a converted file; a single channel does not imply a single speaker.
Mean RMS	0.13958	Establish amplitude scale, frame length, aggregation, and preprocessing before reproducing the value.
Estimated speech segments	160	Establish segmentation method, settings, and manual checks. Segments are not necessarily turns, people, or meaningful exchanges.

For clarity, the root-mean-square (RMS) amplitude of a block of N waveform samples x_j is

$$\mathrm{RMS}(x) = \sqrt{\frac{1}{N}\sum_{j=1}^{N} x_j^2}, \qquad N > 0. \tag{G.3}$$

Here, N counts audio samples, not participants. **In plain language:** square the recorded sample values, take their average, and take the square root. The result

summarises signal amplitude on the recording's numerical scale, not the meaning or quality of the discussion. Audio documentation illustrates frame-based RMS calculation and the importance of frame and padding settings (librosa, n.d.). An average of frame-level RMS values is not generally the same as one RMS computed across the whole recording. Consequently, Account B's "mean RMS" cannot be reproduced from the number alone; the equation clarifies the concept without asserting the original calculation method. The cited software is an explanatory reference, not a claim that it was used for this recording.

Neither the RMS figure nor the estimated 160 segments establishes that the session was highly interactive. Before interpreting participation, a source-based review should distinguish moderator and participant turns, identify who responds to whom, and retain interruptions, disagreement, and changes of position. It should also check whether apparent agreement followed a leading description or reflected an independently stated preference. These interactional questions are more relevant to focus-group interpretation than treating automatically or manually segmented audio as a measure of engagement (Kitzinger, 1995).

G.9 Observational Evidence and Timing Audit

Both accounts describe the same visual frames but do not supply the images, recording, or source log. Their claims of successful triangulation therefore cannot be independently assessed. Table G.8 identifies a specific inconsistency that must be resolved before audiovisual evidence is used to support the thematic interpretation.

These conversions are arithmetic checks of the reported values, not verifi-

Table G.8: Reconciliation of Reported Durations and Visual Timestamps

Reported item	**Converted time**	**Audit implication**
Account A: approximate duration	28:00 (1,680 s)	Rounded description; 4.67 s longer than Account B's reported duration.
Account B: reported duration	27:55.33 (1,675.33 s)	Compatible with approximately 28 minutes; not independently measured here.
Frame 12 in both accounts	20:59 (1,259 s)	Within either reported duration if the time origin is shared.
Frame 17 in both accounts	30:30 (1,830 s)	Exceeds Account B's duration by 154.67 s; not resolved by rounding.

cation of the frames. The discrepancy could involve an approximate duration, an offset, an edited recording, or an erroneous timestamp; the available material does not establish which. No replacement session duration is inferred. The proposed revision's claim that ambiguous audio aligns precisely with Frames 10, 12, and 17 is not accepted: no such frames or alignment log were supplied, and the stated Frame 17 timestamp still exceeds the reported duration by 154.67 seconds. Neither a list of selected frame numbers nor generated commentary constitutes a continuous visual record. Account B additionally describes natural light, large windows, and a semi-circular seating arrangement. These details are retained as reported context, not evidence that participants felt relaxed, included, or willing to disagree. If verified, the room arrangement can help explain sightlines and interaction opportunities; participants' comfort requires their own accounts rather than an inference from furniture or lighting.

The accounts also mention interface elements in Frames 6, 7, and 9 and question overlays in Frames 17 and 19. Even if those descriptions are subsequently

confirmed, an overlaid prompt is not a participant utterance. An image showing a phone likewise does not establish a completed task or a participant's mental state. A source-linked observation record should distinguish visible action, audible speech, analyst interpretation, and uncertainty.

The label *Active UX Interaction* should therefore not be assigned to all prolonged vocalisations or speech gaps. Where the recording supports it, a narrower description such as *screen interaction observed* can record tapping or swiping without inferring comprehension, satisfaction, or cognitive immersion. Speech and device activity may overlap; absence of speech should not automatically be classified as disengagement, consent, or a usability problem. These are proposed coding boundaries, not observations newly made from the unverified machine transcript.

G.10 Reflexivity, Ethics, and Inclusive Interpretation

Researcher position and prior commitments can shape both the research encounter and its interpretation (Berger, 2015). The accounts' product-oriented language and emphasis on favourable responses warrant particular reflexive attention. The moderator's relationship to the students and the product should be documented. Enthusiasm, politeness, prompted agreement, and substantive endorsement should be considered as alternative readings, not presumed equivalent. Any institutional power relationship is a question to investigate, not a fact established by the current files.

The revised report does not treat students as a homogeneous generation or assume shared levels of confidence, digital access, language proficiency, or willingness to appear publicly. Future recruitment and facilitation should consider whose

experiences are absent and what participation arrangements people prefer. Listening without broadcasting, using an alternative mode of expression, or choosing not to answer must not be interpreted as lesser competence. The supplied narratives provide no basis for subgroup comparisons or claims that the prototype is inclusive in practice.

For any further use of the underlying material, the researcher should verify the applicable ethics review and permissions rather than insert a retrospective claim of approval. The record should distinguish agreement to participate, recording permission, and any permission for quotation, image publication, or secondary reuse. Identifying information should be limited to what the approved research purpose requires, with access and disclosure arrangements documented. A group discussion also requires attention to what participants may disclose about one another; institutional membership alone is not a confidentiality safeguard. These are proposed safeguards for this project, not a statement that they were implemented.

Reporting what was actually done, including departures from the intended procedure and unresolved evidence gaps, is more informative than claiming unqualified rigour. Qualitative reporting standards support that transparency while accommodating different research traditions (Levitt et al., 2018).

G.11 Design Implications and Further Evaluation

The reported accounts motivate four priorities for subsequent evaluation. They should be treated as research questions rather than validated product requirements:

1. **Respect different participation goals.** Investigate the distinction between

listening, practising, sharing privately, and publishing. Ask students which transitions they want rather than assuming that all listeners should become broadcasters.

2. **Evaluate collaboration without making it a universal solution.** Compare preferences and task experiences across shared and independent routes. Record both helpful aspects and coordination or privacy concerns.

3. **Examine identity and trust separately.** Explore membership preferences, visibility of personal information, access barriers, and expectations for moderation. Do not use an entry rule as a proxy for safety or response validity.

4. **Test specific interface tasks.** Distinguish functions that are shown from functions that operate. Record the task, any assistance, the observable outcome, and the participant's explanation. Include relevant accessibility and playback-control checks rather than relying on impressions from a walkthrough.

Before any comparison is conducted, define the question, prototype version, participant eligibility, evidence to be collected, and analysis plan. No numerical success threshold, performance estimate, or treatment effect is invented from these accounts. For integration with the thesis's platform specification, record the source-linked concern, the design response, the prototype version, and the rationale for accepting or declining that response. A recommendation in this analysis does not by itself establish that the corresponding feature was implemented. If later work uses a different prototype or sample, report that change rather than combining

observations as if they came from the same session.

G.12 Limitations, Transferability, and Conclusion

The decisive limitation is evidentiary: the narratives and supplemental generated summaries cannot replace a verified human transcript, the recordings, participant information, and analytical records. The new export increases the material available for critique, not the amount of verified primary data. Its descriptions cannot support audited quotation, a verified frequency analysis, or a completed audiovisual triangulation. Unresolved session timing further constrains reconstruction. The absence of documented sampling and recruitment prevents assessment of the range of perspectives included or omitted.

There are also limits to the reported design. A moderator-led wireframe walkthrough does not establish behaviour during ordinary use, and favourable comments do not demonstrate sustained adoption, reduced anxiety, or improved learning. The absence of a comparison design and verified outcomes precludes causal conclusions. Transfer to another study requires an explicit judgement about differences in participants, institutional setting, language, prototype, and research purpose, rather than treating this account as universally applicable evidence.

Within these boundaries, the merged accounts offer a coherent set of questions about audio participation: how listening relates to publication, when collaboration is welcome, what institutional membership means to users, and how interface choices support or obstruct their intentions. The contribution of this revision is to integrate the reported context and design rationale while making those questions

and their evidence requirements explicit. Source-based verification may strengthen, qualify, or overturn the provisional interpretations; the analysis must remain open to all three outcomes.

G.13 Evidence Requirements Before Study Integration

For inclusion as an empirical findings section, the author should complete and document the following checks. Until then, this text should be identified as a critical review of a reported FGD account, not an independently verified analysis of primary data.

- Recover the original inputs and execution details behind Supplement C; check candidate translations, speakers, and intervals without treating the export's labels as verification. Its review questions remain unconfirmed.
- Confirm that both narratives refer to the same session and source versions; recover and check the participant roster, recruitment information, and applicable ethics and reuse records.
- Reconcile audiovisual timing and supply a source-linked Arabic transcript, English translation, and observation log where those materials may be used.
- Replace unsupported summaries with contextualised, permitted evidence; retain contradictions and distinguish participant speech from moderator prompts.
- Document the analytical approach, actual analysts' contributions, coding decisions, and any revisions to the provisional categories.

- Either reproduce the lexical analysis under declared rules or omit it from the findings; do not reuse the repeated counts as verified measurements.

- Adapt the report to the receiving study's questions, evidence, and format. Preserve the distinction between this FGD and any separate survey, simulation, or subsequent prototype evaluation.

Appendix H

Student Rating Measurement Specification

The external human measure is the full published NMLS (Koc & Barut, 2016). Its agent-administered form is an exploratory synthetic diagnostic, not the primary computational outcome or a validated measure of agent competence. Administration uses its authorised item wording and five-point agreement anchors, not paraphrases of the domain descriptions in this thesis. This appendix provides the scoring map, not a replacement administration form. A translated or shortened instrument has a separate version identity and cannot inherit validity merely by retaining the NMLS name.

Table H.1: NMLS Item Map and Student Rating Outputs

Domain	Published item positions	Output
FC	1–7	Mean of 7 responses
CC	8–18	Mean of 11 responses
FP	19–25	Mean of 7 responses
CP	26–35	Mean of 10 responses
Full-item display	1–35	$100(T - 35)/140$

Note. Domain means remain on the original 1–5 scale. The optional 0–100 display is this study's linear presentation of the full-item sum; it is not a pass mark, rank threshold, or independent validity claim.

The machine-readable column names are `NMLS01` through `NMLS35`. the Student Rating scoring procedure requires all 35 valid integer responses before producing a complete profile; incomplete assessments remain missing, not zero. Complete-case scoring does not justify ignoring attrition in the study analysis. Entity type,

research ID, group, condition, timepoint, language, and instrument version remain attached to each profile, so human observations cannot be pooled silently with synthetic outputs. Model-run identifiers identify replication units separately from individual agent IDs.

NMLS, task performance, PB, stars, BRS, and platform tier are stored as separate fields. The scoring code checks arithmetic and input integrity; it does not establish a factor model, linguistic equivalence, or agent validity.

Appendix I

Source Availability and Reproduction

The available archive contains thesis sources, embedded chart coordinates, and a demonstration-telemetry JSON summary. It does not include the platform source tree, historical numerical engine, original R exports, respondent matrices, transcripts, posterior draws, or trip statements mentioned in the inherited research account. No source inspection, native software test, respondent-level reanalysis, or live platform experiment is certified by this version. Earlier implementation descriptions are historical reports requiring verification against a restored, versioned archive.

I.1 What Can Be Reproduced Now

The supplied workflow can compile the document, regenerate diagrams from shared specifications, inventory references and missing assets, hash available inputs, and produce a reviewer package. These are document and arithmetic checks, not replications of the field study. Generated synthetic fixtures used to check software are not participant records and never enter thesis results.

I.2 What Requires Additional Material

Field reproduction needs original questionnaire responses, the instrument and scoring version, retained identifiers, exclusion rationale, and analysis scripts. Qualitative reproduction needs original records, their verbatim, translated, or reconstructed status, the codebook, coding decisions, and consent-compatible access. Trip statements document trips, not participant identity, informed consent, or attentive listening. Historical simulation reproduction needs code, configuration, random-state definitions, calibration particles, all run logs, and outputs; Bayesian reproduction also needs priors, sampling settings, diagnostics, and posterior draws.

I.3 Implementation and Measurement Boundaries

A route name, test function, screenshot, or container specification does not establish successful treatment delivery. Before the prospective experiment, verify each active feature against its versioned implementation and reconcile acknowledged actions with archived events and committed state. Isolate condition-specific accounts, databases, media state, and message streams. Keep field responses, synthetic accounts, and demonstration telemetry separate.

Theoretical BRS components, a platform feedback posterior, and the 11-item questionnaire are different objects. Agreement among their names cannot establish equivalent scoring or validated uncertainty. The historical replication-percentile intervals and equal-tailed Bayesian intervals likewise retain distinct interpretations (Section 5.17).

I.4 Reserved Results: LLM Platform-in-the-Loop Extension

Not conducted. The following reporting structure is reserved for the extension specified in Section 5.10. Its cells intentionally contain "Pending," not zeros, effect estimates, or invented participant responses. These future results will remain separate from both the measured mixed-methods findings and the historical simulation summaries.

Table I.1: Reserved Execution and Fidelity Report for the Planned LLM Extension

Required report	Current status
Protocol freeze, approvals, consent scope	Pending
New anchor cohort and held-out evaluation sample	Pending
Platform commit, BRS variant, model and prompt versions	Pending
Network runs attempted/completed; agents per run	Pending
Action failures, protocol deviations, token/time cost	Pending
Anchored versus unanchored predictive fidelity	Pending
Human–agent distributional differences and failure cases	Pending

Table I.2: Reserved Condition Comparisons for the Planned LLM Extension

Endpoint	G0	G1	Contrast/uncertainty
Synthetic PB and star rating	Pending	Pending	Pending
Participation and completion	Pending	Pending	Pending
Source exposure and verification	Pending	Pending	Pending
Correction uptake	Pending	Pending	Pending
Scheduler repetition and diversity	Pending	Pending	Pending
Prompt/model/anchor sensitivity	Pending	Pending	Pending

Note. Final endpoints and estimators must be frozen before execution. Paired network runs, not individual posts, define the replication unit. Qualitative accounts of unexpected agent behavior belong alongside these quantities once genuine logs exist.

I.4.1 Reserved Agent Hallucination and Divergence Diagnostics

To separate platform effects from model artifacts, the extension will audit agent outputs along the three dimensions defined in Section 5.10: the evidence fabrication index (EFI), the platform state divergence (PSD), and the persona drift score (PDS) (Section 5.10.12, Equations 5.18–5.20). No executed values are reported below: the extension has not run, and the columns record the planned reporting structure, not measured quantities. A detector self-check must first flag injected fabrications on a held-out benchmark whose factual state the pipeline knows, after which genuine logs fill the cells. The "Pending" markers are therefore not zero values or implied failure; they denote reserved evidence.

Table I.3: Reserved Hallucination and Divergence Diagnostics for the Planned LLM Extension

Diagnostic (planned estimator)	Reserved value	Source of the audit record
EFI: evidence fabrication index	Pending	Frozen context log vs. cited sources
EFI by apparent BRS band	Pending	Same, split by served BRS
PSD: platform state divergence	Pending	Adapter event log vs. agent reflection
PSD "invalid action" count	Pending	Adapter rejections for unserved states
PDS: persona drift score	Pending	Anchor vs. reflection embeddings
PDS anchored vs. unanchored	Pending	Per-arm cosine similarities
Detector self-check F1	Pending	Fixed benchmark, pipeline-known states

Note. Planned estimators only; illustrative values such as an 8% EFI or a 0.02 PSD are not *reported because the extension has not been executed. The rows remain "Pending" until a real, supervised deployment produces logs.*

I.4.2 Reserved FGD-Informed Rule-Based and LLM Comparisons

Both new model families remain unexecuted. The FGD findings in Section 6.7 are not simulation results. Complete the following record separately for each reviewed scenario; do not pool scenarios under an assumed population distribution.

Table I.4: Reserved Common Report for FGD-Informed Monte Carlo Runs

Required item	Rule-based ABM	LLM agents
Scenario and evidence-bundle hash	Pending	Pending
Rule/prompt, platform, corpus, and key versions	Pending	Pending
Frozen parameters and sensitivity ranges	Pending	Pending
Planned/completed pairs and failures	Pending	Pending
Pre/post independently keyed accuracy	Pending	Pending
Paired contrast and Monte Carlo SE	Pending	Pending
Participation states, refusals, and exposure	Pending	Pending
Held-out fidelity and counterexamples	Pending	Pending
Deviations and revised interpretation	Pending	Pending

The machine-readable run-record template contains no estimates. Actual logs, calculated summaries, and interpretation require review before these cells or the conclusions are updated.

Appendix J

Defense Readiness Map

J.1 Operational Meaning of "Impact" in the Title

The working title, *The Social Media Radio Innovation and its Impact on University Students' Ratings: An Exploratory Study*, names the research problem operationally rather than causally. Throughout this thesis the word *impact* is used in two separately labelled senses:

- **Empirical (exploratory association).** The field study (52 university students, cross-sectional, non-randomised) examines the *association* between social-media-radio listening and the post- broadcast evaluation battery: perceived broadcast quality (PB), broadcaster rigour (BR), star ratings, listening duration, and reported interaction. These are audience evaluations, not demonstrated causal effects.

- **Model-conditional (simulated counterfactual).** The computational experiment contrasts an epistemic-first configuration (G1) against an engagement-first configuration (G0) inside the calibrated simulator. Its causal reading is explicitly conditional on the model's assumptions and is never presented as a human intervention effect.

The term *Student Rating* is reserved for the planned participatory competence measure (H7, EST-P4), which is *not* established by any human measurement in this thesis. Unverified synthetic competence proxies are not used as evidence for that construct.

J.2 RQ–Hypothesis–Evidence Alignment Matrix

Table J.1 is the defense map: every research question, the data that answer it, the test used, the main result, and its status.

RQ	**Hypoth.**	**Evidence**	**Main result**	**Status**
RQ1 architecture	H1	Field + simulation	PB mean 5.119 ($SD = 0.867$); positive associations; model contrast conditional	Exploratory / partial
RQ2 diversity	H2	Simulation	Historical ECO contrast +.035; no field diversity trajectory	Model-specific only
RQ3 measurement	H3	Survey	BRS $\alpha = 0.957$ but poor factor fit	Partially supported / mixed
RQ4 integration	H4	Field + model	Positive manifold; discriminant validity unconfirmed	Exploratory
RQ5 themes	—	Researcher-recorded case accounts	Six themes (credibility, belonging, context, participation, relevance, delivery)	Answered (qualitative)
RQ6 integration	—	Field + qualitative	Joint display convergence and tension	Answered
RQ7 tasks	H7	Agent design	Keyed-task accuracy (planned, not executed)	Prospective

Table J.1: RQ–hypothesis–evidence alignment. Status is reported from the completed evidence only; prospective rows are labelled as such.

J.3 Calibration, Validation, and Generalisation

These three stages answer different questions and are never conflated here:

Stage	Question	Status in this thesis
Calibration	Can the model reproduce observed distributions?	Historical SMC-ABC report only; G0 loss .0406 exceeds the .04 target. No new calibration run is certified
Validation	Can it reproduce held-out observations not used for fitting?	Not performed on held-out human data; stated explicitly, not claimed
Generalisation	Does it transfer to another sample/context?	Not claimed; the study is sample-bounded

Table J.2: Calibration vs. validation vs. generalisation.

The simulation is *calibrated and sensitivity-tested but not externally validated as a predictive model of real listeners.* No result herein is presented as a human intervention effect.

J.4 Broadcaster Reputation Score (BRS): Consolidated Status

The algorithmic BRS (an engine score) and the questionnaire BRS (the student-rating instrument) are different objects. Their status is recorded separately:

Component	Status
BRS theoretical specification	Completed
Questionnaire scoring	Defined; historical source verification pending
Questionnaire BRS internal consistency	$\alpha = 0.957$ (subscales 0.849–0.963); not applicable to the algorithmic score
BRS construct validity	Not established (poor factor-model fit)
BRS predictive validity	Not established
BRS simulation utility	Reported under historical model assumptions

Table J.3: Consolidated BRS status; algorithmic and questionnaire variants are kept distinct.

J.5 Pending Computational Validation

The planned computational experiment has no completed results in this version. No confidence-derived competence proxy or unverified synthetic pilot is used to validate EST. Restoration of the primary archives and an independently keyed assessment are prerequisites for stronger claims.

Appendix K

Reproducibility and Final Upgrade Archive

This appendix documents the final archive. It is written so that an examiner or a future researcher can reproduce the model-based results without reconstructing the project from memory. The archive is self-contained: the de-identified measured matrix, derived qualitative features, pipeline, R scripts, generated outputs, figures, and Colab notebooks are included. Identifiable participant-observation material is kept in a restricted folder and is not required to reproduce the reported results; the pipeline falls back to aggregate qualitative features when the restricted file is absent.

K.1 Archive Structure

```
loklok-thesis-final-upgrade/
  01_thesis/
    thesis_final.tex              main manuscript
    src/                          chapters, appendices, figures, diagrams
    src/generated/                pipeline macros and generated tables
    frontmatter/                  Arabic title page and abstract
    build/                        compiled PDF and logs
  02_code/
    pipeline/loklok_pipeline.py   shared ABM/inference code
    pipeline/loklok_pipeline_v8.py
                                  final v8 original-data + BGE-M3 pipeline
    pipeline/generate_figures.py  publication figures
    pipeline/requirements_v8.txt  v8 Python dependencies
    colab/                        T4 notebook and v8 bundle builder
    R/                            reproduction and CFA scripts
    platform/                     LOKLOK algorithm reference
  03_data/
    raw/                          original 60-row survey, ride/PO data,
                                  FGD transcript, self-report audio
```

```
    measured/                    de-identified N=52 and N=60 matrices
    qualitative/                 FGD, self-report transcript, four corpora
    stimuli/                     questionnaire and 25 briefs
    model_outputs/               JSON, CSV, NPZ, embeddings, posterior draws
  04_figures/
    png/ pdf/ source_data/       300-DPI figures and plotting data
  05_docs/
    FINAL_UPGRADE_REPORT.md      gap closure and change log
    GAP_CLOSURE_MATRIX.md        review point -> action -> evidence
    DEFENSE_PLAYBOOK.md          updated Q&A and claim ceilings
    CLAIM_EVIDENCE_MATRIX.md     claim -> evidence tier -> status
    REPRODUCIBILITY.md           commands and environment
  06_archive/                    original upgrade materials and prior PDF
  07_validation/                 verification logs and checksums
```

K.2 Exact Reproduction Commands

The final pipeline is deterministic given its seed. The reported run used seed `20261105`, $n = 500$ personas, and 18 sessions.

Local (Windows PowerShell).

```
python 02_code/pipeline/loklok_pipeline_v8.py `
  --root . --n-personas 500 --sessions 18 --seed 20261105 `
  --embedding-model BAAI/bge-m3 --device cpu
python 02_code/pipeline/generate_figures.py --root .
pdflatex -interaction=nonstopmode thesis_final.tex   (three passes)
```

Local (Linux/macOS or Colab).

```
python 02_code/pipeline/loklok_pipeline_v8.py \
  --root . --n-personas 500 --sessions 18 --seed 20261105 \
  --embedding-model BAAI/bge-m3 --device cuda
python 02_code/pipeline/generate_figures.py --root .
```

R reproduction. The R scripts reproduce the measured-wave psychometrics, the CFA with `lavaan`, the factorial summaries, and the Bayesian cross-check. The R path is optional for the Python pipeline but is included for audit:

```
Rscript 02_code/R/01_measured_wave.R
Rscript 02_code/R/02_cfa_invariance.R
Rscript 02_code/R/03_bayes_crosscheck.R
```

Colab. The archive includes `LOKLOK_Thesis_Full_Run_T4.ipynb` and `LOKLOK_Quick_Validate_CPU.ipynb`. The T4 notebook installs the dependencies, checks `nvidia-smi`, runs the pipeline, regenerates the figures, and zips the outputs. The CPU notebook runs a smaller validation configuration. Both notebooks are self-documenting and use the same seed as the reported run.

K.3 Code Map

Table K.1: Code and Data Map for the Final Upgrade

Artifact	Role	Key output
`loklok_pipeline.py`	Four-stream embedding, DML/IPW, spectral manifold, Q0, 2^3 ABM, GRM-IRT, Bayesian inference, adversarial and Sobol suites	`pipeline_results.json`, `figure_data.npz`, generated tables/macros
`generate_figures.py`	300-DPI figures and source-data CSVs	`04_figures/` and thesis `figures/`
`R/01_measured_wave.R`	Re-derives descriptives, alpha, associations from the 52-row matrix	`03_data/measured/analysis/`
`R/02_cfa_invariance.R`	Lavaan CFA and configural/metric/scalar invariance on exported item data	`cfa_r_results.json`
`R/03_bayes_crosscheck.R`	Base-R Bayesian cross-check of the model contrast	`bayes_r_results.json`
`platform/`	LOKLOK algorithm reference (BRS, MORE, ERIC, Echo Meter, scheduler, Zeta engine)	Algorithm documentation and JSON genome

Table K.2: Final Claim–Evidence Matrix

Claim	Evidence tier	Status	Permitted wording
Measured PB–star association	Measured field	Observed	"Participants' PB ratings were associated with stars"
Measured reliability	Measured field	Observed, limited	"Internal consistency was high; structural validity was not established"
G1–G0 evaluative rating contrast	Model-based	Strong within model	"The calibrated model produced a positive contrast"
G1–G0 competence-format contrast	Model-based	Strong within model	"The synthetic NMLS profile rose under G1"
Engagement cost	Model-based	Strong within model	"The model produced an engagement trade-off"
Misinformation suppression	Model-based	Strong within model	"The governance bundle suppressed false reach in stress tests"
Parameter robustness	Model-based	Conditional	"The contrast survived the specified sweep; it was prior-sensitive"
Human learning gain	Not measured	Untested	"Requires longitudinal confirmatory deployment"
BRS consequential ranking	Not validated	Not established	"BRS is a proposed metric, not a validated ranking instrument"
Thermodynamic law	Theoretical analogy	Not a law	"The Hamiltonian is a formal diagnostic"

K.4 Final Claim–Evidence Matrix

K.5 Defense Questions Specific to the Upgrade

"You generated 500 personas. Isn't that a fake sample?"

The personas are a computational population drawn from a debiased prior anchored in the 52 measured records. They are not presented as additional respondents. The 2^3 design is within-persona, so the contrast is paired by construction; the model's uncertainty is reported separately from sampling uncertainty in the field wave.

"Why should I believe the narrow intervals?"

You should not read them as human population intervals. They are repeated-sampling intervals across generated personas in a deterministic model. The thesis says this explicitly and reports the measured-wave uncertainty separately.

"The Hamiltonian did not decrease. Does that falsify EST-P2?"

It falsifies the strong claim that this particular operationalisation must minimise the diagnostic. The thesis reports the sign honestly and treats the Hamiltonian as a specification device. The diversity and echo results still support the conditional mechanism, but the thermodynamic language is not promoted to a law.

"Why is the skeptical prior result so different?"

Because a prior with standard deviation 0.15 is extremely skeptical about an effect of about one rating point. The thesis reports the full prior sensitivity rather than selecting the most favourable prior. The moderate skeptical prior still yields a positive interval; the extreme prior is a stress test.

"What is genuinely new in the upgrade?"

The integrated, reproducible pipeline: four-stream embedding, DML/IPW debiasing, spectral manifold, within-persona 2^3 tensor ABM, eight filters, Zewail-inspired attention gate, GRM-IRT measurement, dual inference, adversarial misinformation tests, Sobol sensitivity, and a complete archive. The individual components are borrowed; the integration and the explicit claim ceilings are the contribution.

"Why include the Arabic front matter and the Zewail analogy?"

The Arabic title page and abstract meet the ACU/SCU front-matter requirement. The Zewail analogy gives the attentional gate a precise, falsifiable time-resolved form. It is labelled as a formal analogy, not as a claim that social listening is chemistry.

K.6 Final Quality Gate

1. Every major claim identifies its evidence tier (measured, model-based, theoretical, or future empirical).
2. Measured and simulated results are never merged into one table without a label.
3. All causal language about G0/G1 is conditional on the model.
4. The measured $N = 52$ matrix is not altered to improve significance.
5. The synthetic NMLS profile is not presented as human media literacy.
6. The Hamiltonian is not presented as a physical law.
7. The BRS is not presented as a validated consequential ranking.
8. The skeptical-prior result is reported, not hidden.
9. The archive contains the code, data, seeds, outputs, figures, R scripts, and Colab notebooks needed to reproduce the model-based results.
10. The Arabic front matter is included and the English abstract remains the primary abstract.

Appendix L

Review Package and Versioned Reproduction

The reviewer directory is a curated copy of available materials, not a claim that missing evidence has been recovered. Its index and checksummed manifest identify included files, unavailable artifacts, and verification status. Source text, compiled output when available, reference records, diagrams, input schemas, and reproducibility utilities are supplied without editorial scratch files. Methodological disclosure and the planned experiment protocol remain part of the scientific record.

L.1 Available and Pending Evidence

Thesis and diagrams

LaTeX sources, native vector diagrams, portable Mermaid sources, and the compiled PDF when compilation succeeds.

Reported results

Historical numerical tables already present in the thesis, explicitly distinguished from newly recomputed results.

Field evidence: available

Original 60-row questionnaire responses, the reconstructed 52-row analysis matrix, and the exclusion key. Instrument provenance, signed consent records,

and independently keyed factual answers remain pending.

FGD transcript: available

The original 160-segment transcript, the four-domain synthesis in Appendix G, and the design-only simulation bundle. The audiovisual recordings and participant roster remain unavailable; disputed candidate quotations remain outside participant datasets.

Qualitative evidence: available

Pseudonymised ride/PO accounts, the researcher self-report audio and its 185-segment GPU transcript, coding decisions, and the four qualitative corpora. Participant recordings and the focus-group roster remain unavailable.

Context and corpus: available

Four corpora (Uber reviews, student social-media, student-lifestyle, Reuters DNR manifest). Uber statements, linkage documentation, usage permissions, and independently adjudicated factual keys remain pending.

Historical software: partially available

The v8 pipeline, environment record, seeds, posterior draws, execution logs, and generated outputs are included. The original platform backend and client sources remain pending.

Identifiable records are not automatically copied to the reviewer package. An explicit source-register release decision is required; approval to conduct research is not automatically permission for unrestricted file distribution.

L.2 Regeneration Without Silent Changes

From the project root, run

```
python 02_code/pipeline/loklok_pipeline_v8.py -root .
-embedding-model BAAI/bge-m3 -device cuda
```

The workflow validates input schemas and records file hashes. New approved survey records produce a separately labelled update appendix with actual sample counts, scoring rules, missingness, descriptive statistics, reliability, and correlations. Historical numbers remain a historical baseline: no script silently rewrites the meaning of an earlier dataset or claims to repeat a model whose code is absent. Each changed input creates a new content-addressed snapshot. Manuscript interpretation still requires researcher review.

The source register records evidence type, file location, provenance, verification, and release status. A focus-group record cannot be promoted to factual ground truth merely by changing its filename. New qualitative content requires documented coding decisions; the pipeline does not automatically invent themes or quotations. Additional analyses become executable only when the corresponding validated estimator and inputs are supplied.

L.3 Research Notes and Graph

The linked-note graph separates theory, variables, observations, model outputs, and pending evidence. Its editable JSON notes have stable identifiers, source pointers, and explicit relationships. An offline HTML graph and portable Mermaid source

are generated from those same records. Notes assist navigation; their links are not empirical causal estimates.

L.4 Submission Checks

The manuscript uses author–date references, a hanging-indent reference list, numbered chapters and equations, lists of tables and figures, and separated appendices. The official Ahram Canadian University programme handbook, approved title wording, committee forms, ethics documentation, abstract-length rules, and submission requirements have not been supplied. A similarly named institution's handbook is not substituted. Formatting here is therefore a working thesis format, not certification of institutional compliance.

Appendix M

List of Symbols and Abbreviations

The following notation is used consistently throughout the thesis.

M.1 EST Symbols

Symbol	Meaning
$\mathcal{P}$	Platform tuple $(\mathcal{U}, \mathcal{C}, \mathcal{A}, \mathcal{R}, \mathcal{G})$
$\Omega(\mathcal{P}, t)$	Epistemic outcome function of the platform
$B_u(t)$	Broadcaster Reputation Score of user u at time t
$B_u^{\text{rig}}, B_u^{\text{soc}}, B_u^{\text{inf}}$	BRS components: rigor, social validation, information quality
$\mathcal{H}(\mathbf{R}, \mathbf{Q}, t)$	Candidate network functional (Hamiltonian analogy)
H_{src}	Shannon entropy of the defined source-exposure distribution
C_i	Score-volatility index (heat-capacity analogy)
W_{ij}	Interaction strength between user i and content j
$\bar{B}$	Mean BRS (Matthew-effect cap reference)
$G(t)$	Gini coefficient of reputation concentration
R_t	Echo Meter rolling similarity/repetition index (larger is more repetitive)
$\text{DBI}(c, u)$	Diversity of item c for user u
$\text{UVS}(c, u)$	User-value score of item c for user u
$P_{\text{prop}}(c)$	Combined moderation-risk index for content item c
$P_{\text{DTS}}(c, u, \Delta t)$	Zewail-inspired pump–probe attentional coherence gate
$\phi^{\text{TIS}}(c)$	Temporal anomaly (fast-spread) indicator

M.2 Measurement Symbols

Symbol	Meaning
$\boldsymbol{\Lambda}, \boldsymbol{\Phi}, \boldsymbol{\delta}$	CFA loading, factor covariance, error matrices
SEM	Classical standard error of measurement
r_{XX}	Reliability coefficient for the specified score and population
α, ω	Cronbach's alpha, McDonald's omega
ρ	Spearman rank correlation
d	Standardised effect; historical denominator is agent-level pooled SD
$\widehat{R}$, ESS	MCMC convergence diagnostic and effective sample size
$\mathrm{SR}_{\mathrm{NMLS}}$	Optional 0–100 display of the full NMLS item sum
$\bar{A}_{rt}^{(g)}$	Network-mean keyed task accuracy at time t in condition g
$R_{\mathrm{runs}}, \widehat{\tau}_{\mathrm{agent}}$	Independent network-pair count and computational accuracy-change contrast

M.3 Abbreviations

Abbreviation	Meaning
EST	Epistemic Systems Theory
NMLS	New Media Literacy Scale
FC, CC, FP, CP	Functional/critical consumption and functional/critical prosumption
BRS	Broadcaster Reputation Score
PB	Post-broadcast epistemic experience battery (PB1–PB5)
EMA	Ecological momentary assessment
MORE	Multi-objective recommendation engine
ERIC	Epistemic integrity control (governance loop)
G0 / G1	Historical engagement-first / epistemic-first configurations; prospective feature mapping is stated separately
IQ, BR, SV	Information quality, broadcast rigor, social validation
DBI	Diversity-by-item measure
CATE	Conditional average treatment effect

Appendix N

Versioned Evidence Update

This appendix is regenerated from the explicit input tables. It does not replace historical results or certify the authenticity of newly entered records. New interpretations and changes to the main conclusions require researcher review.

N.1 Input Status

Survey records supplied: 0. Retained by explicit flag: 0. Qualitative records supplied: 0. Coding decisions supplied: 0. No new survey estimates are available. Empty templates are not zero-valued evidence.

N.2 FGD Narrative Integration

The merged narrative report appears in Appendix G; its findings are in Section 6.7. Two overlapping narratives are registered, not two participant samples. They are excluded from the intake counts above. The supplemental seven-part generated analysis is reviewed separately, not counted as a recovered transcript or participant evidence (Appendix G.4). The design-only bundle supports Section 5.9; it contains no factual answer keys, numerical calibration weights, or executed results.

Appendix O

Statistical Audit, Bayesian Diagnostics, and Reproducibility

This appendix holds the statistical audit conducted in the review workspace alongside the thesis. Every block is marked **FRESH** (recomputed in this package from versioned artifacts by a tracked Python script) or **INHERITED** (transcribed verbatim from thesis exports and not re-run). The generator scripts, inputs, and machine-readable outputs live under `04_statistical_audit` of the review package; their hashes are registered in the reproducibility manifest in Section O.4. No number in this appendix was invented; each row names its source artifact or script.

O.1 Provenance Split of the Audit Blocks

O.2 Bayesian Diagnostics Certificate

The certificate transcribes the three blocks reproduced in Section 6.10: a fresh SMC-ABC calibration, the parsed beta-binomial consensus model for SR', and the inherited PyMC hierarchical model. Within each block the distinction between re-estimated and inherited is explicit, so the certificate cannot be mistaken for a complete re-estimation.

Table O.1: Provenance of Each Statistical-Audit Block

Block	Status	Source artifacts	Generating script
SMC-ABC calibration	FRESH	`data/abm_s1.json`, governance seeds	`bayesian_diagnostics.py`
Beta-binomial SR' consensus	FRESH (parsed)	`data/bayes_sr_results.json`	`bayesian_diagnostics.py`
PyMC hierarchical model	INHERITED	`ch6.tex` (Table 6.27)	— (not re-run)
Sensitivity & exclusion audit	FRESH	`unified_survey_master_thesis.csv`	`sensitivity_audit.py`
GMM joint display	FRESH	persona-s42-hash latent space	`joint_display_generator.py`
Reproducibility manifest	FRESH	all audited files	`reproducibility_manifest.py`

Note. A "parsed" status means the estimator was read from an existing JSON result produced by the workflow, not re-estimated in this workspace.

O.2.1 Fresh SMC-ABC Calibration

Table O.2: SMC-ABC Calibration Diagnostics (FRESH)

Metric	Value
Backend	smc_abc (Beaumont et al. 2009; population Monte-Carlo ABC)
Seed	7
Requested particles	120
Accepted particles (final round)	40
Effective Sample Size (ESS, final weights)	39.4
Median accepted distance	1.2823
Determinism (same-seed rerun identical distances)	True
Posterior sd (propaganda_fraction)	0.0492
Posterior sd (feed_k)	1.1179
Posterior sd (w1)	0.0574

Note. ESS is computed from the final importance weights $1/\sum w^2$. R-hat is not available for population Monte-Carlo ABC; diagnostics are weight-based by design.

Table O.3: Beta-Binomial SR' Consensus Model (PARSED, not re-estimated)

Item	Value
Prior × likelihood	Beta(2, 2) per arm, binomial
Participants	60
ROPE (on Δ)	± 0.05
Contrast ROPE coverage	0.4073
$P(\Delta > 0)$	0.8498
Bayes factor BF_{10} (null vs effect)	0.205 (ANECDOTAL)
Two-sided posterior-predictive checks	per-arm, in source JSON

Note. Source: `data/bayes_sr_results.json`. *The consensus model is reported in Table 6.9 as the Bayesian complement of H1.*

O.2.2 Parsed Beta-Binomial SR' Consensus

O.2.3 Inherited PyMC Hierarchical Model

Table O.4: Inherited PyMC Hierarchical Diagnostics (INHERITED, transcription)

Metric	Value
Maximum $\widehat{R}$	1.008
Minimum effective sample size	606
Posterior-predictive coverage (in-sample)	0.96
PPC mean / observed mean	5.115 / 5.119
IQ coefficient (predictor SD scale)	0.786, 95% equal-tailed [0.670, 0.910]

Note. Transcribed from the thesis exports behind Table 6.27 and Appendix C. PyMC is not installed in this workspace, so the block is not *re-run; it is reported so the certificate is honest about provenance.*

O.3 Joint Display: Synthetic Agents Against Real Qualitative Records

The GMM cluster-to-ground-truth joint display links each synthetic-agent cluster in the frozen persona-s42 latent space to its nearest real qualitative records. It is an audit artifact for future anchor selection, not a claim that the synthetic agents "are" the real cases. Demo FGD/self-report records are never quoted; only non-demo records carrying verbatim text are eligible as anchors, and quotes are excerpts rather than consent-cleared full statements.

Table O.5: GMM Cluster-to-Ground-Truth Joint Display (representative nearest anchors in the 24-d latent space)

Cluster	Size	EST	Top themes	Representative nearest anchor
0	2438	P1 (P3)	T2, T4, T3	Uber gift-card, order-error, and promotion complaints; e.g. "I purchased an Uber gift card ..." [uber-reviews:001918]
1	52	P1 (P2)	T1, T2, T3	Student-lifestyle rows with study hours, sleep, and GPA; e.g. "Student_ID: 1207 ..." [student-lifestyle:001206]
2	2000	P3 (P2)	T3, T1, T2	Student-lifestyle rows with study and activity hours; e.g. "Student_ID: 489 ..." [student-lifestyle:000488]
3	60	P1 (P2)	T1, T2, T3	Uber service-failure and concert-ride accounts; e.g. "On June 11, 2017 ..." [uber-reviews:002033]

Note. Joint display computed by `joint_display_generator.py` from the frozen persona-s42-hash latent space. Only non-demo records carrying verbatim text are eligible as anchors; demo FGD/self-report records are never quoted. Excerpts are shortened for display; the full anchor text remains in the source corpus.

O.4 Reproducibility Manifest

The manifest follows `reproducibility-manifest/1.0`, records the seeds used across the workflow, and registers a SHA-256 hash for every input and output artifact. The complete machine-readable manifest and its verifying generator are in the review package under `04_statistical_audit/reproducibility_manifest.json` and `04_statistical_audit/reproducibility_manifest.py`.

Table O.6: Seeds Registered in the Reproducibility Manifest

Seed	Value
`registry.gt_s1_seed`	42
`registry.mc_seed`	20260910
`persona_engine.seed`	42
`registry_gmt_seed`	20260912
`survey_analysis_seed`	42

Table O.7: Manifest File Hashes (selection; full list in the JSON)

Artifact	SHA-256
`data/ground_truth/analysis.json`	`2ada29d9...adb6704`
`data/ground_truth/registry.json`	`15e9935c...1f51dbd3`
`data/ground_truth/respondents.csv`	`fc6ec086...3f6630e97c8`
`data/abm_s1.json`	`1a5eff25...dbd4ac5bb9`
`data/abm_constructs.json`	`bfb8aeea...f0d026b0add7`
`data/bayes_sr_results.json`	`2f48f853...beca257a07e8`
`data/influence_layer_results.json`	`83992a65...df391d94597ae`
`unified_survey_master_thesis.csv`	`7d9fe6b9...a77272cbc6`
`refined_master_thesis.csv`	`cf176991...32fad3f0551b`
`RealStudentsProfiles.txt`	`03ae6622...729d43f691`
`llm_extension_protocol.json`	`87730909...260aa4b9961f`

Note. Hashes are truncated for display; full 64-character digests are in the manifest JSON. Any later edit changes the hash, which is why the JSON and generator are included as audited artifacts rather than restructured here.

The manifest is explicit that `transformers`, `scikit-learn`, `matplotlib`, `sentence-transformers`, and `faster-whisper` are not installed on the local runner

(they live on the gated Colab/Kaggle runners), and that no self-report ground-truth export or Kaggle social-media wave is present as an artifact. Recording absence is honest provenance; it prevents the manifest from being mistaken for a complete run record. Reproducibility therefore means: running the frozen artifact hashes and tracked generator scripts, not re-executing the external runners from this workspace.